What Survives Diffusion Noise?

A compendium: every concept, from scratch

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Abstract

This compendium accompanies the manuscript *What Survives Diffusion Noise?* and the meeting report on the EM chapters. It defines and derives, from first principles, every mathematical and statistical object those documents use: Hermite polynomials and cumulants, the Ornstein–Uhlenbeck channel and Tweedie's identity, Fisher information and the Cramér–Rao bound, belief propagation on a chain and its quadrature on a grid, expectation–maximisation and Fisher's identity, quasi-likelihood and moment estimators, score matching in its denoising and implicit forms, and the statistics of the experimental protocol. Each chapter states where the object appears in the paper and in the code, and works one small example by hand. Nothing here is new; everything here is what the reader needs in order to check the paper line by line.

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Chapter 1

Conventions and notation

This chapter fixes the symbols, the typographical rules and the index conventions that the rest of the compendium and the paper share. Everything here is a decision, not a result. The decisions were taken once, in the file `paper/notation.tex`, and every chapter of this compendium is written against that file. A reader who wants to check the paper line by line should read this chapter first, because several of the paper's choices differ from the habits of the diffusion literature, and the differences are deliberate.

1.1 How to read this compendium

Every chapter is self-contained at the level of a chapter: it defines the objects it uses, derives every formula it states (the scalar case in full, then the multivariate statement), and works one small example by hand. Four typographical devices recur.

- A **term** in bold at its first use is a definition; the term is also entered in the index at the end of the document.
- A paragraph opening with *Where it is used* closes each chapter and names the paper sections, the experiment identifiers (E01 to E21) and the code paths in which the chapter's objects appear.
- A paragraph opening with *Why it matters here* explains, inside a derivation, why a particular hypothesis or constant is the one the paper needs.
- A path set in typewriter, such as `src/markov_score/theory.py`, is a file of the repository. Every such path was checked to exist when the chapter was written.

Theorems, propositions, lemmas and corollaries are numbered within chapters, as are definitions, examples, assumptions and remarks. "The paper" always means the ICML manuscript *What Survives Diffusion Noise?*; its sections, theorems and figures are referred to in words ("the paper, Section 3, Theorem 2"), never by cross-reference, because the two documents are compiled separately. Experiments are referred to by their registry identifier (E01, ..., E21), which names an entry of `experiments/registry/`. No measured number from the experiments is typed in this compendium: those numbers live in generated macros of the paper, and the compendium says "see the paper, Figure 2" instead. Analytic constants such as $\frac{1}{6}$, $\frac{1}{24}$ or e^{-4t} are typed freely.

1.2 The symbol table

Table 1.1 lists every macro of `paper/notation.tex` and every macro added in the compendium's preamble, with its rendering, its meaning and the chapter in which the object is defined from scratch. The macro names are given so that a reader of the sources can search for them.

Table 1.1: Every macro of the paper's notation file and of the compendium preamble.

macro	renders as	meaning	defined in
Diffusion			
`\Pz`	P_0	the clean law of a sequence, a density on $\mathbb{R}^L$	Markov chains
`\Pt`	P_t	the noised law at diffusion time t, the law of $\mathbf{x}_t$	The OU channel and Tweedie
`\Dt`	Δ_t	$1 - e^{-2t}$, the conditional variance of one noised coordinate	The OU channel and Tweedie
`\score`	S	the diffusion score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$	The OU channel and Tweedie
`\gwn`	η	standardised Gaussian white noise, $\langle \eta_i(t)\eta_j(t')\rangle = \delta_{ij}\delta(t - t')$	The OU channel and Tweedie
`\tmin`, `\tmax`	$t_{\min}$, $t_{\max}$	the ends of the noise schedule	The protocol
`\pmean{·}`	$\langle \mathbf{a} \rangle_{x,t}$	posterior mean $\mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$, in bracket notation	The OU channel and Tweedie
`\pmeansite{i}`	$\langle a_i \rangle_{x,t}$	posterior mean of one site	The OU channel and Tweedie
Vectors and matrices			
`\vect{v}`, `\mat{A}`	$\mathbf{v}$, $\mathbf{A}$	bold lowercase for vectors, bold uppercase for matrices and tensors	this chapter
`\va`	$\mathbf{a}$	the clean chain, a point of $\mathbb{R}^L$; its sites are the scalars a_i	Markov chains
`\vx`	$\mathbf{x}$	the noised chain, $\mathbf{x} = \mathbf{x}_t \in \mathbb{R}^L$	The OU channel and Tweedie
`\vz`	$\mathbf{z}$	a standard normal draw in $\mathbb{R}^L$	Gaussian integrals and Hermite
`\Sig`	$\mathbf{\Sigma}$	a covariance matrix; $\mathbf{\Sigma}_0$ is the clean chain's, $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$ the noised one's	Markov chains
`\Id`	$\mathbf{I}$	the identity matrix	this chapter
Chain and graphical model			
`\nsites`	L	chain length, equal to the ambient dimension	Markov chains
`\nseq`	M	number of observed sequences	Markov chains
`\ngrid`	N_{g}	number of quadrature grid points	Grid quadrature
`\gridpt{k}`	u_k	the k-th grid point	Grid quadrature
`\gridh`	h	the grid spacing	Grid quadrature
`\halfwidth`	A	the grid half-width; the domain is $[-A, A]$	Grid quadrature
`\kernel`	K	the Markov transition kernel $K(a' \mid a)$	Markov chains
`\kernelth`	K_θ	a transition kernel with parameters θ	Maximum likelihood and EM
`\like`	ℓ_t	the unary likelihood factor $\ell_t(x_i \mid a_i)$	Factor graphs and BP
`\lmsg`, `\rmsg`	$\overrightarrow{m}$, $\overleftarrow{m}$	forward and backward messages	Factor graphs and BP
`\innov`	φ	the innovation density	Markov chains
`\corr`	ρ	the autoregression coefficient	Markov chains
`\ivar`	q	the innovation variance, $q = 1 - \rho^2$ in the paper	Markov chains
`\ncomp`	C	number of mixture components of the fitted innovation	Maximum likelihood and EM
Estimation			
`\params`	θ	a parameter vector	Fisher information
`\Qfun`	$\mathcal{Q}$	the EM auxiliary function	Maximum likelihood and EM
`\Xis`	$\mathbf{\Xi}$	the expected transition statistic, a matrix on the grid	Maximum likelihood and EM

macro	renders as	meaning	defined in
`\Lik`	$\mathcal{L}$	the marginal log-likelihood	Maximum likelihood and EM
`\mmse`, `\lmmse`	$\widehat{\mathbf{a}}_{\mathrm{MMSE}}$, $\widehat{\mathbf{a}}_{\mathrm{LMMSE}}$	the posterior mean and the linear MMSE estimator	Gaussian closure and LMMSE
Operators			
`\E`, `\Prob`	$\mathbb{E}$, $\mathbb{P}$	expectation, probability	Moments and cumulants
`\Reals`	$\mathbb{R}$	the real line	this chapter
`\Cov`, `\Var`	Cov, Var	covariance, variance	Moments and cumulants
`\KL`	D_{KL}	Kullback–Leibler divergence	Maximum likelihood and EM
`\normal{m}{v}`	$\mathcal{N}(m, v)$	the Gaussian law with mean m and variance v	Gaussian integrals and Hermite
`\Pgauss`	Π_{G}	Gaussian moment projection: the Gaussian with the argument's mean and variance	Gaussian closure and LMMSE
`\dd`	d	the differential in integrals, $\mathrm{d}x$	this chapter
Pending-data guard and repository pointers			
`\needsdata`	red bracket	with an argument naming the slot: marks a place in the paper still waiting for a frozen number; the build fails if any survives	this chapter
`\codepath`, `\codefile`	typewriter	a repository path	this chapter
`\coderef{p}{f}`	path → name	a path with a function or target inside it	this chapter
Macros the paper defines in `paper/main.tex`			
`\Frob{T}`	$\lVert\mathbf{T}\rVert_F$	Frobenius norm, the root of the sum of squares over ordered indices	Moments and cumulants
`\Hen{k}`	He_k	the k-th probabilists' Hermite polynomial	Gaussian integrals and Hermite
`\kfour`	$\boldsymbol{\kappa}_4$	the fourth-cumulant tensor of the clean law	Moments and cumulants
`\Sgauss`	S_{G}	the score of the covariance-matched Gaussian twin, $-\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$	Gaussian closure and LMMSE
`\tstar`	t_*	the crossover time $\frac{1}{2}\log\lambda_{\max}(\boldsymbol{\Sigma}_0)$	Markov chains; Attenuation theorems
`\lmax`	$\lambda_{\max}$	the largest eigenvalue of a symmetric matrix	Markov chains
`\Pgaussl`	P_{G}	the Gaussian law with the clean chain's covariance (the twin)	Gaussian closure and LMMSE
Compendium-only macros			
`\tr`, `\diag`	tr, diag	trace; diagonal matrix built from a vector	this chapter
`\argmax`, `\argmin`	arg max, arg min	maximiser and minimiser	this chapter
`\term{·}`	bold	a defined term, also indexed	this chapter
`\where{·}`, `\why{·}`	italic lead-in	the two closing paragraphs described above	this chapter
`\code{·}`	typewriter	a repository path or identifier	this chapter

Three macros are typeset slightly differently in the paper and in this compendium, and the reader should not be surprised by the difference. The paper writes the fourth-cumulant tensor as κ_4 in light face, the compendium as $\boldsymbol{\kappa}_4$ in bold, because the compendium's rule is that tensors are bold. The paper writes the twin's score with a superscript, S^{G}, the compendium with a subscript, S_{G}. The paper writes the crossover as t_* and the compendium as t_*. The objects are

the same.

1.3 Indices: sites, sequences, grid points, and why n and d are not used

Sites are the coordinates of a sequence. A clean sequence is $\mathbf{a} = (a_1, \dots, a_L)$, and i, j always index sites, $i \in \{1, \dots, L\}$. The **chain length** L is therefore also the **ambient dimension** of the space in which the diffusion runs: the noised sequence $\mathbf{x}_t$ lives in $\mathbb{R}^L$, the score is a map $\mathbb{R}^L \times [0, \infty) \to \mathbb{R}^L$, and every covariance matrix is $L \times L$. The paper writes this identification once, as $n = L$, and then never uses n again.

Sequences are the independent draws. The data are M sequences $\mathbf{a}^1, \dots, \mathbf{a}^M$ (or their noised versions), and μ indexes sequences, $\mu \in \{1, \dots, M\}$. Fisher information is stated *per sequence*, and the Cramér–Rao and local asymptotic minimax statements are in the limit $M \to \infty$ at fixed L and fixed t.

Grid points are the abscissae on which messages are stored. There are N_g of them, $u_1, \dots, u_{N_\mathrm{g}}$, uniformly spaced by h on $[-A, A]$, and k, l index them. A matrix on the grid, such as the discretised kernel $K(u_k \mid u_l)$ or the expected transition statistic $\mathbf{\Xi}_{kl}$, is $N_\mathrm{g} \times N_\mathrm{g}$ and is indexed *child first, parent second*: k is the site a_i, l is the site a_{i-1}. This orientation is the one thing a self-consistent recursion cannot catch if it is wrong, which is why `tests/test_fisher_bruteforce.py` checks it against an enumeration that shares nothing with the recursion.

Multi-indices $\nu = (\nu_1, \dots, \nu_L) \in \mathbb{N}^L$ label Hermite products; the **degree** is $|\nu| = \sum_i \nu_i$ and $\nu! = \prod_i \nu_i!$.

Why n and d are avoided. Much of the diffusion literature writes n for the number of samples and d for the dimension of the data, and much of the statistics literature writes n for the sample size as well. In this laboratory the dimension *is* the chain length, and the reader should never have to remember that a statement "in dimension d" is a statement about a chain of d sites; so the dimension is called L and the number of sequences M. The letter n is kept for the degree of a scalar Hermite polynomial He_n and for nothing else. Two further collisions were removed at the same time: the transition kernel is K and only K, and the number of grid points is N_g, whereas an earlier draft used K for the grid size and W for the kernel.

Levels. The noise schedule is a finite set of diffusion times, twelve of them log-spaced on $[t_\mathrm{min}, t_\mathrm{max}]$ in the frozen protocol; "level" means one such t. A statistic "pooled over the schedule" sums over levels before taking a root or a ratio; see the chapter on the statistics of the experiments.

1.4 Vectors, matrices, tensors and scalars

Vectors are bold lowercase ($\mathbf{a}$, $\mathbf{x}$, $\mathbf{z}$, $\mathbf{s}$), matrices and tensors bold uppercase ($\mathbf{\Sigma}$, $\mathbf{I}$, $\mathbf{\Xi}$, $\boldsymbol{\kappa}_4$). Anything indexed down to a scalar loses its bold: a_i is the i-th site of $\mathbf{a}$, Σ_{ij} an entry of $\mathbf{\Sigma}$, $\kappa_4[ijkl]$ an entry of $\boldsymbol{\kappa}_4$. The Euclidean norm of a vector is $|\mathbf{a}|$ or $\|\mathbf{a}\|$; the Frobenius norm of a matrix or of a symmetric tensor is $\|\mathbf{T}\|_F$, with

$$\|\mathbf{T}\|_F^2 = \sum_{i_1, \dots, i_r} T_{i_1 \cdots i_r}^2, \tag{1.1}$$

the sum running over *ordered* index tuples. For a symmetric matrix this counts each off-diagonal entry twice, and for a symmetric 4-tensor it counts an entry with distinct indices 24 times. The paper's constants $\frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$ and $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$ are stated with this convention, and `src/markov_score/theory.py` sums the full (L, L, L, L) array; the chapter on moments and cumulants explains how the ordered-index sum turns into a sum over multisets with weights $4!/\nu!$, which is where the paper's $\frac{1}{24}$ comes from.

$\mathbf{I}$ is the identity, tr the trace, $\mathrm{diag}(\mathbf{v})$ the diagonal matrix with $\mathbf{v}$ on the diagonal, $\mathbf{A} \succ 0$ positive definiteness, $\lambda_{\max}(\mathbf{A})$ the largest eigenvalue. $\mathcal{N}(m, v)$ is the Gaussian law with mean m and variance (or covariance matrix) v. The symbol γ is reserved for the standard Gaussian density on $\mathbb{R}^L$, and γ_{Δ_t} for $\mathcal{N}(0, \Delta_t \mathbf{I})$.

1.5 Hermite polynomials: probabilists' versus physicists'

Two families of polynomials carry the name Hermite, and they differ by a rescaling that changes every constant. The **probabilists' Hermite polynomials** are

$$\mathrm{He}_n(x) = (-1)^n e^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2}, \tag{1.2}$$

so that $\mathrm{He}_0 = 1$, $\mathrm{He}_1 = x$, $\mathrm{He}_2 = x^2 - 1$, $\mathrm{He}_3 = x^3 - 3x$, $\mathrm{He}_4 = x^4 - 6x^2 + 3$; they are monic and orthogonal under the standard Gaussian, with $\mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m] = n!\, \delta_{nm}$. The **physicists' Hermite polynomials** are $H_n(x) = (-1)^n e^{x^2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2}$, with leading coefficient 2^n and orthogonal under e^{-x^2} with norm $2^n n! \sqrt{\pi}$. The two are related by

$$H_n(x) = 2^{n/2} \mathrm{He}_n(\sqrt{2}\, x), \qquad \mathrm{He}_n(x) = 2^{-n/2} H_n(x/\sqrt{2}). \tag{1.3}$$

The paper and this compendium use the probabilists' family throughout, because the noise is standardised and the orthogonality weight is then exactly the law of the noise. One notational hazard must be named: the paper writes $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$ for the *multivariate product* of probabilists' polynomials. That H_ν is not a physicists' polynomial, despite the capital letter; it is the product of probabilists' ones, and $\mathbb{E}_\gamma[H_\nu H_\mu] = \nu!\, \delta_{\nu\mu}$. The chapter on Gaussian integrals and Hermite polynomials derives all of this.

1.6 Standardised white noise and the explicit $\sqrt{2}$

The forward process is written, in the paper and in the code (`src/markov_score/noising.py`), as the Ornstein–Uhlenbeck equation

$$\mathrm{d}\mathbf{x}_t = -\mathbf{x}_t \, \mathrm{d}t + \sqrt{2} \, \mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 = \mathbf{a} \sim P_0, \tag{1.4}$$

with $\mathbf{W}$ a standard Brownian motion, equivalently with a **standardised white noise** η satisfying $\langle \eta_i(t) \eta_j(t') \rangle = \delta_{ij} \delta(t - t')$ and $\mathrm{d}\mathbf{W}_t = \eta(t)\, \mathrm{d}t$. The factor $\sqrt{2}$ is carried explicitly for one reason: with it, the stationary variance of each coordinate is exactly one. Solving Equation (1.4) by variation of constants,

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{2} \int_0^t e^{-(t-s)} \, \mathrm{d}\mathbf{W}_s, \qquad \mathrm{Var}\Big(\sqrt{2} \int_0^t e^{-(t-s)} \, \mathrm{d}W_s\Big) = 2 \int_0^t e^{-2(t-s)} \, \mathrm{d}s = 1 - e^{-2t}, \tag{1.5}$$

so that, conditionally on $\mathbf{a}$,

$$\mathbf{x}_t \mid \mathbf{a} \;\sim\; \mathcal{N}\Big(e^{-t}\mathbf{a}, \Delta_t \mathbf{I}\Big), \qquad \Delta_t = 1 - e^{-2t}, \qquad \text{that is} \qquad \mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\, \mathbf{z}, \;\; \mathbf{z} \sim \mathcal{N}(0, \mathbf{I})\,. \tag{1.6}$$

Since $e^{-2t} + \Delta_t = 1$, a coordinate of unit variance keeps unit variance at every t, and $P_t \to \mathcal{N}(0, \mathbf{I})$. Two other conventions appear in the literature and are *not* used here. The physicists' convention absorbs the $\sqrt{2}$ into the noise by demanding $\langle \eta(t) \eta(t') \rangle = 2\delta(t - t')$; the formulas are the same but the noise is no longer standardised. The "variance-preserving" convention of the score-based literature writes $\mathrm{d}\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\, \mathrm{d}t + \sqrt{\beta(t)}\, \mathrm{d}\mathbf{W}$ with a schedule $\beta(t)$; Equation (1.4) is the case $\beta \equiv 2$, and the noise level is then parameterised by t directly. The decay factor is written e^{-t} in every displayed equation of the paper and is never abbreviated to α_t; inside proofs the paper allows the local abbreviation $\alpha = e^{-t}$, always introduced in the sentence that uses it, and this compendium does the same. The **signal-to-noise ratio** of the channel is e^{-2t}/Δ_t, which is $e^{-2t}(1 + O(e^{-2t}))$ at large t; this is why the paper's expansions are in powers of e^{-2t}.

1.7 What "the score" means in the paper

The word **score** is used in two different senses in statistics and in diffusion modelling, and the paper is careful to use it in one of them. The **diffusion score** is the gradient of the log-density of the *noised data law* with respect to the *data*,

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}), \tag{1.7}$$

a vector field on $\mathbb{R}^L$ for each t. The **statistical score** is the gradient of a log-likelihood with respect to a *parameter*, $\partial_\theta \log p_\theta(\mathbf{x})$, a function of θ for fixed data. The chapter on Fisher information and the Cramér–Rao bound defines both and keeps them apart; the paper writes "the score" for the diffusion score and "the likelihood score" or "per-sequence scores obtained by Fisher's identity" for the statistical one.

Within the diffusion sense, four objects have to be distinguished, and the paper's sentence "throughout, 'the score' means this tree-exact inference, evaluated by validated quadrature" names one of them. They are:

(L1) exact functional sum-product on the posterior factor graph, which for this model is a chain and therefore a tree, so that the forward and backward messages are exact and Tweedie's identity $S(\mathbf{x}, t) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$ turns the exact posterior mean into the exact score;

(L2) Gaussian-projected sum-product, in which every message is replaced by the Gaussian with its first two moments; the paper's Proposition 1 shows this returns the linear MMSE estimator of the covariance twin;

(L3) the numerical implementation of (L1) on a grid of N_{g} points on $[-A, A]$ with trapezoidal weights, at the frozen setting $N_{\mathrm{g}} = 401$, $A = 8$, validated by refinement and by an independent brute-force enumeration on short chains;

(L4) the LMMSE estimator of the true covariance, $e^{-t}\boldsymbol{\Sigma}_0 \boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, and its score $S_{\mathrm{G}} = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, which is what (L2) computes.

"The score" in the paper is (L1) computed as (L3). It is the reference against which every estimator, whether the EM-fitted kernel or a trained network, is scored; the estimand of the experiments is the relative error to this reference, pooled over the schedule. It is never the output of a learned model; those are "estimates", "fits" or "arms". The grid setting is not inherited from the literature but measured: the chapter on grid quadrature explains what truncation and quadrature errors are and how the paper controls them, and why a heavy-tailed innovation tests the domain A rather than the spacing h.

1.8 Other fixed choices

- The clean chain has $a_1 \sim \mathcal{N}(0, 1)$, $|\rho| < 1$, innovation variance $q = 1 - \rho^2$, so that $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation law. The frozen configuration is $\rho = 0.85$, $L = 32$, Laplace innovations; every departure from it is stated where it occurs.
- Innovation families are always *variance-matched*: each is parameterised by its variance q, and the remaining parameter, if any, is a shape parameter (the Student-t degrees of freedom ν, the mixture separation κ, the generalised-Gaussian exponent β).
- Cumulants are written κ_m; the excess kurtosis of a scalar law is κ_4/κ_2^2; the fourth-cumulant tensor of a vector is $\boldsymbol{\kappa}_4$.
- log is the natural logarithm; a "log-slope" is a derivative or a regression coefficient of $\log y$ against t, so that $y \propto e^{-8t}$ has log-slope -8.
- British spelling and the serial "and" are used; no number is typed from a measurement.

Where it is used. Every chapter. The symbol file is `paper/notation.tex`, read verbatim by `paper/main.tex` and by `reports/compendium/compendium.tex`; the channel convention is implemented in `src/markov_score/noising.py` (the functions `alpha_delta` and `ou_noise_sample`); the Hermite convention in `src/markov_score/theory.py` (the function `he`); the frozen configuration in `experiments/frozen_config.py` and the comparison protocol in `experiments/protocol.py`. The paper's Section 2 fixes the chain and the channel, its Section 3 the Hermite notation, and its Appendix B the grid.

References. Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon and B. Poole (2021), "Score-based generative modeling through stochastic differential equations", ICLR. M. Mézard and A. Montanari (2009), *Information, Physics, and Computation*, Oxford University Press. G. Szegő (1975), *Orthogonal Polynomials*, 4th edition, American Mathematical Society (Chapter V for the two Hermite conventions). C. W. Gardiner (2009), *Stochastic Methods*, 4th edition, Springer (Chapter 4 for the Ornstein–Uhlenbeck process and the white-noise conventions). B. Efron (2011), "Tweedie's formula and selection bias", Journal of the American Statistical Association 106, 1602–1614.

Part I

Probability and statistics

Chapter 2

Moments and cumulants

The paper's laboratory is built on one arithmetic fact: a linear recursion with independent innovations has cumulants that add. That fact fixes the covariance of the chain independently of the innovation law, places every non-Gaussian feature in the cumulants of order four and higher, and gives the fourth-cumulant tensor $\boldsymbol{\kappa}_4$ that appears in the constants $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$ and $\frac{1}{24}\|\partial_\theta \boldsymbol{\kappa}_4\|_F^2$. This chapter builds cumulants from moments, derives their three properties (shift, homogeneity, additivity), extends them to tensors of a random vector, computes the cumulants of the stationary AR(1) marginal, derives the excess kurtosis of every innovation family the paper uses, and ends with the exponential-moment assumption under which the paper's Theorem 2 is proved.

2.1 Moments and central moments

Let X be a real random variable. Its **moment** of order n is $m_n = \mathbb{E}[X^n]$, defined whenever $\mathbb{E}|X|^n < \infty$. Its **central moment** of order n is $\mu_n = \mathbb{E}[(X - m_1)^n]$. If $\mathbb{E}|X|^n < \infty$ then $\mathbb{E}|X|^k < \infty$ for every $k \le n$, because $|x|^k \le 1 + |x|^n$; so the moments that exist form an initial segment of the orders. The variance is $\mu_2 = \mathrm{Var}(X) = m_2 - m_1^2$.

A law is **symmetric** if $-X$ has the law of X; then every odd moment that exists vanishes, $m_{2k+1} = 0$, because $\mathbb{E}[X^{2k+1}] = \mathbb{E}[(-X)^{2k+1}] = -\mathbb{E}[X^{2k+1}]$. Every innovation law in the paper is symmetric, which is why the first non-Gaussian cumulant is the fourth and not the third.

The **kurtosis** of a law with finite fourth moment is μ_4/μ_2^2 and its **excess kurtosis** is $\mu_4/\mu_2^2 - 3$. The subtraction of 3 makes the excess kurtosis of every Gaussian zero, which the next section explains.

2.2 Generating functions and cumulants

Definition 2.1 (Moment- and cumulant-generating functions)**.** The **moment-generating function** of X is $M(s) = \mathbb{E}[e^{sX}]$, $s \in \mathbb{R}$, an extended-real quantity in $(0, \infty]$ with $M(0) = 1$. The **cumulant-generating function** is $K(s) = \log M(s)$ wherever $M(s) < \infty$.

The set where M is finite is an interval containing 0, because e^{sx} is convex in s. Two cases matter for the paper.

Lemma 2.2 (Exponential moment and analyticity)**.** *Suppose $\mathbb{E}[e^{\lambda|X|}] < \infty$ for some $\lambda > 0$. Then $M(s) < \infty$ for $|s| < \lambda$, every moment m_n is finite with $m_n \le n!\,\lambda^{-n}\mathbb{E}[e^{\lambda|X|}]$, and $M(s) = \sum_{n\ge 0} m_n s^n/n!$ for $|s| < \lambda$.*

Proof. For $|s| < \lambda$, $e^{sX} \le e^{\lambda|X|}$, so $M(s) < \infty$. From $e^{\lambda|x|} \ge \lambda^n|x|^n/n!$ (one term of the exponential series) we get $|x|^n \le n!\lambda^{-n}e^{\lambda|x|}$ and the moment bound. Finally $\sum_n |s|^n|X|^n/n! = e^{|s||X|} \le e^{\lambda|X|}$ is integrable, so the series $e^{sX} = \sum_n s^n X^n/n!$ may be integrated term by term by dominated convergence. □

If the exponential moment fails, $M(s)$ may be $+\infty$ for every $s \neq 0$ even though many moments are finite; the Student-t law of Section 2.6 is the paper's example. In that case K is not a function whose Taylor series one can take, but cumulants of every order for which the moments exist are still defined by the polynomial relations that follow, read as identities of formal power series.

Definition 2.3 (Cumulants)**.** The **cumulant** of order n is $\kappa_n = K^{(n)}(0)$, so that $K(s) = \sum_{n\geq 1} \kappa_n s^n/n!$ near the origin when Lemma 2.2 applies. In general κ_n is the polynomial in $m_1, \dots, m_n$ obtained by expanding $\log(1+\sum_{k\geq 1} m_k s^k/k!)$ formally and reading off the coefficient of $s^n/n!$.

The first four cumulants. Centre X so that $m_1 = 0$ (the effect of a shift is treated below); then $M(s) = 1 + \mu_2 s^2/2 + \mu_3 s^3/6 + \mu_4 s^4/24 + O(s^5)$ and, using $\log(1+u) = u - u^2/2 + O(u^3)$ with $u = O(s^2)$,

$$K(s) = \frac{\mu_2 s^2}{2} + \frac{\mu_3 s^3}{6} + \frac{\mu_4 s^4}{24} - \frac{1}{2}\Big(\frac{\mu_2 s^2}{2}\Big)^2 + O(s^5) = \frac{\mu_2}{2}s^2 + \frac{\mu_3}{6}s^3 + \frac{\mu_4 - 3\mu_2^2}{24}s^4 + O(s^5). \tag{2.1}$$

Reading off coefficients of $s^n/n!$,

$$\kappa_1 = m_1, \qquad \kappa_2 = \mu_2, \qquad \kappa_3 = \mu_3, \qquad \kappa_4 = \mu_4 - 3\mu_2^2. \tag{2.2}$$

So the fourth cumulant is the fourth central moment minus three times the squared variance, and the excess kurtosis is exactly κ_4/κ_2^2. The factor 3 is the number of ways to pair four objects, which Section 2.3 and the chapter on Gaussian integrals explain.

Proposition 2.4 (Three properties of cumulants)**.** *Let X, Y be random variables with the relevant moments finite, and $c \in \mathbb{R}$.*

(i) Shift: $\kappa_1(X + c) = \kappa_1(X) + c$ and $\kappa_n(X + c) = \kappa_n(X)$ for $n \geq 2$.

(ii) Homogeneity: $\kappa_n(cX) = c^n \kappa_n(X)$.

(iii) Additivity: if X and Y are independent, $\kappa_n(X + Y) = \kappa_n(X) + \kappa_n(Y)$.

Proof. When the generating functions exist near the origin the three statements are read off $K_{X+c}(s) = cs + K_X(s)$, $K_{cX}(s) = K_X(cs)$ and, by independence, $M_{X+Y}(s) = M_X(s)M_Y(s)$, hence $K_{X+Y} = K_X + K_Y$. Each statement is a polynomial identity in finitely many moments, so it holds whenever those moments are finite, by the formal-series definition. □

Proposition 2.5 (Gaussian cumulants)**.** *If $X \sim \mathcal{N}(m, \sigma^2)$ then $K(s) = ms + \frac{1}{2}\sigma^2 s^2$, so $\kappa_1 = m$, $\kappa_2 = \sigma^2$ and $\kappa_n = 0$ for every $n \geq 3$.*

Proof. For $Z \sim \mathcal{N}(0, 1)$, completing the square in the exponent, $\mathbb{E}[e^{sZ}] = (2\pi)^{-1/2} \int e^{sz - z^2/2}\, \mathrm{d}z = e^{s^2/2}(2\pi)^{-1/2} \int e^{-(z-s)^2/2}\, \mathrm{d}z = e^{s^2/2}$; then $X = m + \sigma Z$ and Proposition 2.4 give the rest. □

A theorem of Marcinkiewicz says the converse: if K is a polynomial then its degree is at most two and the law is Gaussian. So "non-Gaussian" means "some cumulant of order at least three is non-zero", and for a symmetric law "some cumulant of order at least four". This is the sense in which the paper's families "differ only through cumulants of order four and higher".

Why it matters here. Additivity and homogeneity are what make the channel act on cumulants diagonally. If $x_t = e^{-t}a + \sqrt{\Delta_t}\, z$ with z Gaussian and independent of a, then $\kappa_m(x_t) = e^{-mt}\kappa_m(a) + \Delta_t^{m/2}\kappa_m(z) = e^{-mt}\kappa_m(a)$ for $m \geq 3$. This is the paper's first "exact consequence" of Theorem 1, and it needs nothing but this section.

2.3 Cumulant tensors of a random vector

Let $\mathbf{X} = (X_1, \dots, X_L)$ be a random vector. The joint generating functions are $M(\mathbf{s}) = \mathbb{E}[e^{\mathbf{s}^\top \mathbf{X}}]$ and $K(\mathbf{s}) = \log M(\mathbf{s})$, and the **joint cumulant** of the coordinates $X_{i_1}, \dots, X_{i_n}$ is

$$\operatorname{cum}(X_{i_1}, \dots, X_{i_n}) = \frac{\partial^n K}{\partial s_{i_1} \cdots \partial s_{i_n}}(\mathbf{0}), \tag{2.3}$$

a quantity symmetric in its arguments and multilinear in them. Collected over all index tuples it forms the **cumulant tensor** of order n, an array with L^n entries. The order-two tensor is the covariance matrix, since $\partial_i \partial_j K(\mathbf{0}) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j]$. The order-four tensor is the object the paper writes $\boldsymbol{\kappa}_4$.

Lemma 2.6 (Fourth joint cumulant of a zero-mean vector)**.** *If $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ and fourth moments are finite, with $\Sigma_{ij} = \mathbb{E}[X_i X_j]$,*

$$\kappa_4[ijkl] = \mathbb{E}[X_i X_j X_k X_l] - \Sigma_{ij}\Sigma_{kl} - \Sigma_{ik}\Sigma_{jl} - \Sigma_{il}\Sigma_{jk}. \tag{2.4}$$

Proof. Expand

$$M(\mathbf{s}) = 1 + \tfrac{1}{2}\sum_{ij} \Sigma_{ij} s_i s_j + \tfrac{1}{6}\sum_{ijk} \mathbb{E}[X_i X_j X_k] s_i s_j s_k + \tfrac{1}{24}\sum_{ijkl} \mathbb{E}[X_i X_j X_k X_l] s_i s_j s_k s_l + O(|\mathbf{s}|^5)$$

and take $\log(1+u) = u - u^2/2 + \dots$; the only fourth-order contribution of $-u^2/2$ is

$$-\tfrac{1}{8}\Big(\sum_{ij} \Sigma_{ij} s_i s_j\Big)^2 = -\tfrac{1}{8}\sum_{ijkl} \Sigma_{ij}\Sigma_{kl} s_i s_j s_k s_l.$$

The coefficient of $s_i s_j s_k s_l$ in K, symmetrised, is therefore $\frac{1}{24}\big(\mathbb{E}[X_i X_j X_k X_l] - 3\,\mathrm{sym}(\Sigma_{ij}\Sigma_{kl})\big)$, where the symmetrisation of $\Sigma_{ij}\Sigma_{kl}$ over the 24 permutations of (i,j,k,l) produces each of the three pairings 8 times, that is

$$\mathrm{sym}(\Sigma_{ij}\Sigma_{kl}) = \tfrac{1}{3}\big(\Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}\big).$$

Multiplying by 4! gives Equation (2.4). On the diagonal $i = j = k = l$ it reduces to Equation (2.2). □

The three properties of Proposition 2.4 have tensor versions, proved the same way from K. Shifts change only the first-order tensor. Independence gives additivity: if $\mathbf{X}$ and $\mathbf{Y}$ are independent then the cumulant tensors of $\mathbf{X}+\mathbf{Y}$ are the sums of those of $\mathbf{X}$ and of $\mathbf{Y}$. Homogeneity becomes the **transformation rule** for a linear map $\mathbf{A}$:

$$\operatorname{cum}((\mathbf{A}\mathbf{X})_{i_1}, \dots, (\mathbf{A}\mathbf{X})_{i_n}) = \sum_{j_1, \dots, j_n} A_{i_1 j_1} \cdots A_{i_n j_n} \operatorname{cum}(X_{j_1}, \dots, X_{j_n}), \tag{2.5}$$

which follows from $K_{\mathbf{A}\mathbf{X}}(\mathbf{s}) = K_{\mathbf{X}}(\mathbf{A}^\top \mathbf{s})$ and the chain rule. A Gaussian vector has $K(\mathbf{s}) = \mathbf{m}^\top \mathbf{s} + \frac{1}{2}\mathbf{s}^\top \boldsymbol{\Sigma} \mathbf{s}$ and no cumulant tensor of order three or more.

The Frobenius norm over ordered indices. For a tensor $\mathbf{T}$ of order r, $\|\mathbf{T}\|_F^2 = \sum_{i_1 \cdots i_r} T_{i_1 \cdots i_r}^2$, the sum over all ordered tuples. This is the Euclidean norm of $\mathbf{T}$ viewed as a vector with L^r entries, and it is invariant under orthogonal changes of coordinates, because by Equation (2.5) an orthogonal $\mathbf{O}$ acts on the tensor by $\mathbf{O} \otimes \cdots \otimes \mathbf{O}$, which is orthogonal on $\mathbb{R}^{L^r}$. For a symmetric tensor the ordered sum can be regrouped by **multisets**: an ordered tuple with multiplicities $\nu = (\nu_1, \dots, \nu_L)$, $|\nu| = r$, has $r!/\nu!$ rearrangements, all carrying the same entry T_ν, so

$$\|\mathbf{T}\|_F^2 = \sum_{|\nu| = r} \frac{r!}{\nu!} T_\nu^2, \qquad \text{equivalently} \qquad \sum_{|\nu| = r} \frac{T_\nu^2}{\nu!} = \frac{1}{r!}\|\mathbf{T}\|_F^2. \tag{2.6}$$

For $r = 4$ the right-hand side is $\|\mathbf{T}\|_F^2/24$. This is the origin of the paper's constant $\frac{1}{24}\|\partial_\theta \boldsymbol{\kappa}_4\|_F^2$: the general-order theorem gives $\sum_{|\nu|=4}(\partial_\theta \mathbb{E}H_\nu)^2/\nu!$, the degree-four Hermite moments of a variance-matched family are the entries of $\boldsymbol{\kappa}_4$ (chapter on Gaussian integrals and Hermite polynomials), and Equation (2.6) converts the multiset sum into the ordered-index norm. For a symmetric matrix ($r = 2$), $\sum_{i\le j}$ with weights 2 off the diagonal gives $\frac{1}{2}\|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2 = \sum_i \frac{1}{2}(\partial\Sigma_{ii})^2 + \sum_{i<j}(\partial\Sigma_{ij})^2$, the paper's constant for a covariance coordinate.

2.4 Cumulants of the stationary AR(1) marginal

Let $(\varepsilon_k)_{k\in\mathbb{Z}}$ be independent copies of a law with mean zero, variance q and finite cumulants up to the order considered, and let $|\rho| < 1$. The **moving-average representation**

$$X = \sum_{k\ge 0} \rho^k \varepsilon_{-k} \tag{2.7}$$

converges in mean square (its partial sums are Cauchy in L^2 because $\sum_k \rho^{2k} < \infty$) and almost surely, and X has the **invariant law** of the recursion $a_i = \rho a_{i-1} + \varepsilon_i$: if a_{i-1} has the law of X then so does a_i, since $\rho X + \varepsilon$ is again of the form Equation (2.7). The chapter on Markov chains returns to this.

Proposition 2.7 (Cumulants of the AR(1) marginal)**.** *If* $\mathbb{E}|\varepsilon|^n < \infty$ *then* $\mathbb{E}|X|^n < \infty$ *and*

$$\kappa_n(X) = \frac{\kappa_n(\varepsilon)}{1-\rho^n}, \qquad n \ge 1. \tag{2.8}$$

Proof. For the finite sum $X_N = \sum_{k=0}^N \rho^k \varepsilon_{-k}$, additivity and homogeneity give $\kappa_n(X_N) = \kappa_n(\varepsilon)\sum_{k=0}^N \rho^{nk}$. Since $\mathbb{E}|X_N - X|^n \to 0$ (Minkowski's inequality and the tail $\sum_{k>N}|\rho|^k$), every moment of order at most n of X_N converges to that of X, so the cumulants, polynomials in those moments, converge too. Summing the geometric series gives Equation (2.8). □

Two consequences fix the paper's laboratory. With $q = 1 - \rho^2$,

$$\operatorname{Var}(X) = \frac{q}{1-\rho^2} = 1, \tag{2.9}$$

so the choice of innovation variance is exactly the one that gives unit marginal variance, whatever the innovation law. And the fourth cumulant and excess kurtosis of the marginal are

$$\kappa_4(X) = \frac{\kappa_4(\varepsilon)}{1-\rho^4}, \qquad \frac{\kappa_4(X)}{\operatorname{Var}(X)^2} = \frac{\kappa_4(\varepsilon)}{q^2}\cdot\frac{q^2}{1-\rho^4} = \gamma_2(\varepsilon)\,\frac{(1-\rho^2)^2}{(1-\rho^2)(1+\rho^2)} = \gamma_2(\varepsilon)\,\frac{1-\rho^2}{1+\rho^2}, \tag{2.10}$$

where $\gamma_2(\varepsilon) = \kappa_4(\varepsilon)/q^2$ is the innovation's excess kurtosis. For Laplace innovations, $\gamma_2(\varepsilon) = 3$ (next section), the marginal's excess kurtosis is $3(1-\rho^2)/(1+\rho^2)$, which is the expression the paper uses in its regime sweep (Section 5) and in the hypothesis of E14. It falls from 3 at $\rho = 0$ to 0 as $\rho \to 1$: a strongly correlated chain has a nearly Gaussian marginal, because each site is the sum of many independent innovations with slowly decaying weights, and cumulants of order four add with weights ρ^{4k} while the variance adds with weights ρ^{2k}.

Remark 2.8 (The paper's chain is not started in its invariant law). The paper's chain has $a_1 \sim \mathcal{N}(0,1)$, which is the invariant law only for Gaussian innovations. For every other family the first site has no fourth cumulant, and the site-i marginal has

$$\kappa_4(a_i) = \kappa_4(\varepsilon)\sum_{m=2}^{i}\rho^{4(i-m)} = \kappa_4(\varepsilon)\,\frac{1-\rho^{4(i-1)}}{1-\rho^4},$$

which rises geometrically to the stationary value Equation (2.10). The covariance is unaffected (chapter on Markov chains); the fourth-cumulant tensor of `src/markov_score/theory.py` (function `chain_kappa4`) carries the exact site-dependent form, and the moment estimator of E19 (`experiments/e19_moment_estimators.py`) uses the site-dependent factor $(1-\rho^{4(i-1)})/(1-\rho^4)$ when it converts a per-site fourth cumulant into an estimate of $\kappa_4(\varepsilon)$. The paper's sentence "its excess kurtosis falling as $3(1-\rho^2)/(1+\rho^2)$" refers to the stationary marginal.

2.5 The innovation families at variance q

Every family below is symmetric, so $\kappa_3 = 0$, and is parameterised so that its variance is exactly q. The remaining freedom, if any, is a shape parameter. For each we derive the fourth moment by hand and record the excess kurtosis $\gamma_2 = \mu_4/q^2 - 3$ and the fourth cumulant $\kappa_4 = \gamma_2 q^2$. The parameterisations are those of `src/markov_score/priors.py` and `src/markov_score/shape_kernel.py`.

Gaussian. $\varepsilon \sim \mathcal{N}(0, q)$. By Proposition 2.5, $\kappa_4 = 0$ and $\gamma_2 = 0$; $\mu_4 = 3q^2$.

Laplace. Density $\varphi(e) = \frac{1}{2b} e^{-|e|/b}$. The absolute moments are $\mathbb{E}|\varepsilon|^n = \frac{1}{b}\int_0^\infty e^n e^{-e/b}\,\mathrm{d}e = n!\,b^n$ (the Gamma integral), so $\mu_2 = 2b^2$ and $\mu_4 = 24b^4$. Variance matching sets $2b^2 = q$, i.e. $b = \sqrt{q/2}$, and then

$$\gamma_2 = \frac{24b^4}{(2b^2)^2} - 3 = 6 - 3 = 3, \qquad \kappa_4 = 3q^2. \tag{2.11}$$

Student-t_ν. $\varepsilon = sT$ with T a Student-t variable with ν degrees of freedom, density proportional to $(1 + u^2/\nu)^{-(\nu+1)/2}$. Writing $T = Z/\sqrt{V/\nu}$ with $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_\nu$ independent, $\mathbb{E}[T^{2k}] = \mathbb{E}[Z^{2k}]\,\nu^k \mathbb{E}[V^{-k}]$, and $\mathbb{E}[V^{-k}] = \prod_{j=1}^{k}(\nu - 2j)^{-1}$ for $\nu > 2k$ (the inverse moments of a chi-square). Hence $\mathbb{E}[T^2] = \nu/(\nu-2)$ for $\nu > 2$ and $\mathbb{E}[T^4] = 3\nu^2/((\nu-2)(\nu-4))$ for $\nu > 4$. Variance matching sets $s^2\nu/(\nu-2) = q$, and

$$\gamma_2 = \frac{3\nu^2}{(\nu-2)(\nu-4)} \cdot \frac{(\nu-2)^2}{\nu^2} - 3 = \frac{3(\nu-2)}{\nu-4} - 3 = \frac{6}{\nu-4}, \qquad \kappa_4 = \frac{6}{\nu-4} q^2. \tag{2.12}$$

At $\nu = 5$, the paper's choice, $\gamma_2 = 6$. Moments of order r exist only for $r < \nu$: the t_5 law has finite moments up to order four and an infinite sixth moment, so its cumulants beyond the fourth do not exist. Its tail is polynomial, $\varphi(e) \sim c|e|^{-\nu-1}$, which is what Section 2.6 is about.

Uniform. $\varepsilon \sim \mathrm{Uniform}(-h, h)$, density $1/(2h)$ on $[-h, h]$. Then $\mu_2 = \frac{1}{2h}\int_{-h}^{h} e^2\,\mathrm{d}e = h^2/3$ and $\mu_4 = h^4/5$. Variance matching sets $h = \sqrt{3q}$, and

$$\gamma_2 = \frac{h^4/5}{h^4/9} - 3 = \frac{9}{5} - 3 = -\frac{6}{5} = -1.2, \qquad \kappa_4 = -\frac{6}{5} q^2. \tag{2.13}$$

This is the most platykurtic law of the paper, and it is compactly supported: the conditional law of a_i given a_{i-1} is zero outside an interval, which is a different failure of Gaussian closure from heavy tails.

Symmetric two-component mixture. $\varepsilon \sim \frac{1}{2}\mathcal{N}(m, s^2) + \frac{1}{2}\mathcal{N}(-m, s^2)$ with $m^2 = \kappa q$ and $s^2 = (1-\kappa)q$, $\kappa \in [0, 1)$; κ is the separation fraction and the frozen value is $\kappa = 0.6$. For the $+m$ component, $\mathbb{E}[(m + sZ)^4] = m^4 + 6m^2s^2 + 3s^4$ (binomial expansion and Gaussian moments $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, odd ones zero); the $-m$ component gives the same, so $\mu_4 = m^4 + 6m^2s^2 + 3s^4$ and $\mu_2 = m^2 + s^2 = q$. Substituting,

$$\gamma_2 = \frac{\kappa^2 + 6\kappa(1-\kappa) + 3(1-\kappa)^2}{1} - 3 = \kappa^2 + 6\kappa - 6\kappa^2 + 3 - 6\kappa + 3\kappa^2 - 3 = -2\kappa^2, \qquad \kappa_4 = -2\kappa^2 q^2. \tag{2.14}$$

Table 2.1: The innovation families at variance q: parameter, excess kurtosis γ_2 (the fourth cumulant is $\kappa_4 = \gamma_2 q^2$), and the range of λ for which the exponential moment of Section 2.6 is finite.

family	parameter at variance q	γ_2	$\mathbb{E}e^{\lambda\lvert\varepsilon\rvert} < \infty$
Gaussian	—	0	all λ
Laplace	$b = \sqrt{q/2}$	3	$\lambda < 1/b$
Student-t_ν	$s = \sqrt{q(\nu-2)/\nu}$	$6/(\nu-4)$	no λ
uniform	$h = \sqrt{3q}$	$-6/5$	all λ
mixture, separation κ	$m^2 = \kappa q$, $s^2 = (1-\kappa)q$	$-2\kappa^2$	all λ
generalised Gaussian, β	$s_\beta = \sqrt{q\Gamma(1/\beta)/\Gamma(3/\beta)}$	Equation (2.16)	all if $\beta > 1$; $\lambda < 1/s_1$ if $\beta = 1$

At $\kappa = 0.6$, $\gamma_2 = -0.72$, the value in the paper's oracle table. The law is bimodal for κ large enough, and at $\kappa = 0$ it is the Gaussian.

Generalised Gaussian with exponent β. Density $\varphi_\beta(e) = \dfrac{\beta}{2s\,\Gamma(1/\beta)} \exp(-(|e|/s)^\beta)$, $\beta > 0$. The substitution $u = (e/s)^\beta$, $e = su^{1/\beta}$, $\mathrm{d}e = (s/\beta)u^{1/\beta-1}\,\mathrm{d}u$ turns the absolute moments into Gamma integrals:

$$\mathbb{E}|\varepsilon|^k = \frac{\beta}{s\,\Gamma(1/\beta)} \int_0^\infty e^k e^{-(e/s)^\beta}\,\mathrm{d}e = \frac{s^k}{\Gamma(1/\beta)} \int_0^\infty u^{(k+1)/\beta-1} e^{-u}\,\mathrm{d}u = s^k\,\frac{\Gamma\big((k+1)/\beta\big)}{\Gamma(1/\beta)}. \quad (2.15)$$

(At $k = 0$ this confirms the normalisation.) Hence $\mu_2 = s^2\Gamma(3/\beta)/\Gamma(1/\beta)$, so variance matching sets $s_\beta = \sqrt{q\,\Gamma(1/\beta)/\Gamma(3/\beta)}$, and

$$\gamma_2(\beta) = \frac{\Gamma(5/\beta)\,\Gamma(1/\beta)}{\Gamma(3/\beta)^2} - 3. \quad (2.16)$$

Three values by hand. At $\beta = 1$: $\Gamma(5)\Gamma(1)/\Gamma(3)^2 - 3 = 24/4 - 3 = 3$, the Laplace law (indeed $s_1 = \sqrt{q \cdot 1/2} = b$). At $\beta = 2$: $\Gamma(5/2)\Gamma(1/2)/\Gamma(3/2)^2 - 3 = (\frac{3}{4}\sqrt{\pi})(\sqrt{\pi})/(\frac{1}{2}\sqrt{\pi})^2 - 3 = 3 - 3 = 0$, the Gaussian. At $\beta = 6$: $\Gamma(5/6)\Gamma(1/6)/\Gamma(1/2)^2 - 3$, and by the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$, $\Gamma(1/6)\Gamma(5/6) = \pi/\sin(\pi/6) = 2\pi$ while $\Gamma(1/2)^2 = \pi$; so $\gamma_2(6) = 2 - 3 = -1$ exactly. As $\beta \to \infty$ the law tends to the uniform and $\gamma_2 \to -6/5$. The exponent β therefore moves the excess kurtosis continuously from $+3$ through 0 to -1 on the range $\beta \in [1, 6]$ that the paper's protocol declares for E02b, at fixed variance and (for the chain) fixed covariance; the paper's Section 2 quotes the platykurtic end as -1.1, a value that Equation (2.16) reaches only near $\beta \approx 9.5$. Along the family q is fixed and β is a pure **shape coordinate**: $\partial_\beta \boldsymbol{\Sigma}_0 = 0$ and $\partial_\beta \boldsymbol{\kappa}_4 \neq 0$, which is exactly what the paper's Theorem 2 needs for its e^{-8t} statement.

2.6 The exponential-moment assumption

Assumption 2.9 (Exponential moment)**.** A law has an **exponential moment** if $\mathbb{E}[e^{\lambda|X|}] < \infty$ for some $\lambda > 0$. For a random vector the condition is $\mathbb{E}[e^{\lambda|\mathbf{X}|}] < \infty$ with $|\cdot|$ the Euclidean norm.

This is the paper's Assumption 1, the one hypothesis under which its Theorem 2 is proved. Its consequences, all used in that proof, are collected here.

Proposition 2.10. *Under Assumption 2.9:*

(i) all moments are finite, and the cumulant-generating function $K(\mathbf{s})$ is finite and real-analytic on $\{|\mathbf{s}| < \lambda\}$;

(ii) the law is determined by its moments;

(iii) for the scalar Student-t_ν law the assumption fails for every $\lambda > 0$, and $K(s) = +\infty$ for every $s \neq 0$.

Proof. (i) is Lemma 2.2 applied to $\mathbf{s}^\top \mathbf{X} \leq |\mathbf{s}||\mathbf{X}|$; the power series of M converges on the ball and log of a positive analytic function is analytic. (ii) The characteristic function $\phi(\mathbf{s}) = \mathbb{E}[e^{\mathrm{i}\mathbf{s}^\top \mathbf{X}}]$ extends to an analytic function of complex $\mathbf{s}$ on the strip $|\operatorname{Im} \mathbf{s}| < \lambda$, because $\mathbb{E}[e^{\lambda|\mathbf{X}|}] < \infty$ dominates the integrand there; an analytic function on a strip containing the real axis is determined by its derivatives at the origin, which are the moments up to powers of i; and the characteristic function determines the law. (iii) The density of T satisfies $\varphi(u) \geq c\, u^{-\nu-1}$ for $u \geq u_0$, so for $s > 0$, $\mathbb{E}[e^{sT}] \geq c \int_{u_0}^{\infty} e^{su} u^{-\nu-1}\, \mathrm{d}u = \infty$, because $e^{su} u^{-\nu-1} \to \infty$; by symmetry the same holds for $s < 0$. Then $M(s) = \infty$ for all $s \neq 0$, so $\mathbb{E}[e^{\lambda|T|}] \geq \mathbb{E}[e^{\lambda T}] = \infty$. □

What fails without it. For the Student-t law the moments up to order $\lfloor \nu \rfloor$ exist and so do the cumulants up to that order, as polynomials in them; but K is not the sum of the series $\sum \kappa_n s^n/n!$, because K is infinite off the origin. A Taylor expansion of K with a Lagrange remainder therefore does not exist, and this is the reason the paper's Theorem 2 excludes Student-t and treats it separately as a heavy-tailed regime with a rate of its own, supported by quadrature and not claimed as a theorem. The paper's damped generating function $\Lambda(\mathbf{s}, b) = \log \mathbb{E}_{P_0} \exp(\mathbf{s}^\top \mathbf{a} - b|\mathbf{a}|^2/2)$ is the cumulant-generating function of the linear form $\mathbf{s}^\top \mathbf{a}$, damped by a Gaussian factor; for $b > 0$ the damping makes it finite for every $\mathbf{s}$ and every law, but the expansion the theorem needs is at $b = 0$, where $\Lambda(\mathbf{s}, 0) = K(\mathbf{s})$, and Assumption 2.9 is what makes that expansion legitimate on $|\mathbf{s}| < \lambda$ uniformly in $b \geq 0$.

Which families satisfy it. The Gaussian, the uniform and the mixture have all exponential moments (the Gaussian by $\mathbb{E}e^{\lambda|Z|} \leq 2e^{\lambda^2/2}$, the others by bounded support or by being mixtures of Gaussians). The Laplace law with scale b has $\mathbb{E}e^{\lambda|\varepsilon|} = \frac{1}{b}\int_0^\infty e^{(\lambda - 1/b)e}\, \mathrm{d}e = (1 - \lambda b)^{-1}$ for $\lambda < 1/b$ and $+\infty$ for $\lambda \geq 1/b$: the assumption holds, with a finite range of λ. The generalised Gaussian with $\beta > 1$ has all exponential moments, since $\lambda e - (e/s)^\beta \to -\infty$; at $\beta = 1$ it is the Laplace case. Student-t fails, by (iii). This is the last column of Table 2.1.

From the innovation to the chain. The chain $\mathbf{a}$ has an exponential moment if and only if its innovation does. If it does, write $a_i = \rho^{i-1} a_1 + \sum_{m=2}^{i} \rho^{i-m} \varepsilon_m$; then $|\mathbf{a}| \leq \sum_i |a_i| \leq C|a_1| + C \sum_m |\varepsilon_m|$ with $C = \sum_{k \geq 0} |\rho|^k = (1 - |\rho|)^{-1}$, so by independence

$$\mathbb{E}e^{\lambda|\mathbf{a}|} \leq \mathbb{E}e^{\lambda C|a_1|} \prod_{m=2}^{L} \mathbb{E}e^{\lambda C|\varepsilon_m|} < \infty$$

for λC below the innovation's threshold. Conversely, if the chain has one then so does a_2, and from $|\varepsilon_2| \leq |a_2| + |\rho||a_1|$ and the Cauchy–Schwarz inequality,

$$\mathbb{E}e^{\lambda|\varepsilon_2|} \leq \big(\mathbb{E}e^{2\lambda|a_2|}\big)^{1/2} \big(\mathbb{E}e^{2\lambda|\rho||a_1|}\big)^{1/2} < \infty$$

for λ small enough. So the assumption is a property of the innovation family, and Table 2.1 settles it for the chain as well.

2.7 Worked example: the Laplace chain at $\rho = 0.85$

Take the frozen configuration. The innovation variance is $q = 1 - \rho^2 = 1 - 0.7225 = 0.2775$, so the Laplace scale is $b = \sqrt{q/2} = \sqrt{0.13875} \approx 0.3725$. The innovation's fourth cumulant is $\kappa_4(\varepsilon) = 3q^2 = 3 \times 0.07701 \approx 0.2310$, and its excess kurtosis is 3 whatever ρ is.

The stationary marginal has, by Equation (2.8), $\kappa_4(X) = \kappa_4(\varepsilon)/(1 - \rho^4)$ with $\rho^4 = 0.7225^2 \approx 0.5220$, so $\kappa_4(X) \approx 0.2310/0.4780 \approx 0.4833$; since $\operatorname{Var}(X) = 1$ this is also the marginal excess

kurtosis, and Equation (2.10) gives the same number directly: $3 \times 0.2775/1.7225 \approx 0.4833$. The marginal is six times less kurtotic than the innovation: most of the fourth-order structure of the innovation is averaged away by the correlation, which is the direction of the paper's regime sweep.

Along the paper's chain, which starts at a Gaussian site, the site-i fourth cumulant is $\kappa_4(\varepsilon)(1-\rho^{4(i-1)})/(1-\rho^4)$: it is 0 at site 1, $\kappa_4(\varepsilon) \approx 0.231$ at site 2, $\kappa_4(\varepsilon)(1+\rho^4) \approx 0.352$ at site 3, ≈ 0.415 at site 4, and within one per cent of the stationary value from site 9 on, since $\rho^{32} \approx 0.0056$. The exponential moment holds for $\lambda < 1/b \approx 2.68$.

The scalar leading constant of the paper's closure gap, $\kappa_4^2/6$ for a unit-variance law, is $9/6 = 1.5$ for the unit-variance Laplace law (excess kurtosis 3, so $\kappa_4 = 3$); the chain constant $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$ is a sum of squares of the site-dependent tensor of the chapter on Markov chains, and its numerical value at $L = 32$ is a generated macro of the paper, not typed here.

Where it is used. The paper's Section 2 (variance matching, the families and their excess kurtoses), Section 3 (the two exact consequences of Theorem 1, the constants $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$ and $\frac{1}{24}\|\partial_\theta \boldsymbol{\kappa}_4\|_F^2$, Assumption 1 and the heavy-tailed paragraph), Section 5 (the marginal excess kurtosis in the regime sweep) and Appendix A (the grading of Λ by cumulants). Experiments E01, E02b, E13, E14, E18, E19, E21. Code: `src/markov_score/priors.py` (the families and their `innovation_excess_kurtosis` properties), `src/markov_score/shape_kernel.py` (the generalised Gaussian, its scale at fixed variance and `generalized_gaussian_excess_kurtosis`), `src/markov_score/theory.py` (`chain_kappa4`, `score_gap_constant`, `excess_kurtosis_to_kappa4`), `tests/test_theory_hermite.py` (the fourth-cumulant tensor against brute-force cumulants of sampled chains).

References. P. McCullagh (1987), *Tensor Methods in Statistics*, Chapman and Hall (Chapters 2 and 3 for cumulant tensors and the transformation rule). P. Billingsley (1995), *Probability and Measure*, 3rd edition, Wiley (Sections 21 and 26 for moments, generating functions and the moment problem). E. Lukacs (1970), *Characteristic Functions*, 2nd edition, Griffin (analytic characteristic functions; Marcinkiewicz's theorem). P. J. Brockwell and R. A. Davis (1991), *Time Series: Theory and Methods*, 2nd edition, Springer (the moving-average representation of AR(1)). S. Nadarajah (2005), "A generalized normal distribution", Journal of Applied Statistics 32, 685–694 (moments of the generalised Gaussian).

Chapter 3

Gaussian integrals and Hermite polynomials

The paper's first theorem says that the Ornstein–Uhlenbeck channel multiplies the Hermite moment of degree m by e^{-mt}, exactly, for every law with finite moments of that order. Its proof is four lines long once three classical facts are in place: the Gaussian generating function, the generating function of the Hermite polynomials, and Mehler's formula. This chapter builds those facts from the Gaussian integral upwards, proves Mehler's formula and the paper's Theorem 1 in full, derives the two exact consequences the paper states after it, works out the Wick (Isserlis) rule and the second-moment lemma that produce the constants $\frac{1}{6}$ and $\frac{1}{24}$, explains when a density ratio belongs to $L^2(\gamma)$ and why the paper does not need it to, and defines the first Edgeworth correction that the chapter on misspecification uses.

3.1 Gaussian integrals

Lemma 3.1 (The scalar Gaussian integral and its moments)**.** $\int_{\mathbb{R}} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$. *Writing* $\gamma(x) = (2\pi)^{-1/2}e^{-x^2/2}$ *for the* ***standard Gaussian density*** *and* $Z \sim \gamma$*,*

$$\mathbb{E}[e^{sZ}] = e^{s^2/2}, \qquad \mathbb{E}[Z^{2k}] = (2k-1)!! = \frac{(2k)!}{2^k k!}, \qquad \mathbb{E}[Z^{2k+1}] = 0. \tag{3.1}$$

Proof. Squaring the integral and passing to polar coordinates,

$$\Big(\int e^{-x^2/2}\,\mathrm{d}x\Big)^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2} r\;\mathrm{d}r\;\mathrm{d}\phi = 2\pi.$$

Completing the square, $sz - z^2/2 = -(z-s)^2/2 + s^2/2$, so

$$\mathbb{E}[e^{sZ}] = e^{s^2/2}(2\pi)^{-1/2}\int e^{-(z-s)^2/2}\,\mathrm{d}z = e^{s^2/2}.$$

Expanding both sides in s, $\sum_n \mathbb{E}[Z^n]s^n/n! = \sum_k s^{2k}/(2^k k!)$, and comparing coefficients of s^{2k} gives $\mathbb{E}[Z^{2k}] = (2k)!/(2^k k!)$, which equals $(2k-1)!! = 1\cdot 3\cdots(2k-1)$; odd coefficients vanish. The exchange of sum and expectation is justified by $\sum_n |s|^n|Z|^n/n! = e^{|s||Z|}$, which is integrable. □

So $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, $\mathbb{E}Z^6 = 15$. The number $(2k-1)!!$ is the number of ways of pairing $2k$ objects, which is the content of the next section.

Lemma 3.2 (The multivariate Gaussian integral)**.** *For a symmetric positive definite* $\mathbf{A}$ *on* $\mathbb{R}^L$ *and* $\mathbf{b} \in \mathbb{R}^L$*,*

$$\int_{\mathbb{R}^L} \exp\!\big(-\tfrac{1}{2}\mathbf{x}^{\top}\mathbf{A}\mathbf{x} + \mathbf{b}^{\top}\mathbf{x}\big)\,\mathrm{d}\mathbf{x} = (2\pi)^{L/2}(\det \mathbf{A})^{-1/2}\exp\!\big(\tfrac{1}{2}\mathbf{b}^{\top}\mathbf{A}^{-1}\mathbf{b}\big). \tag{3.2}$$

Consequently $\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})$ has density $(2\pi)^{-L/2}(\det \mathbf{\Sigma})^{-1/2} \exp(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\mathbf{m}))$ and generating function $\mathbb{E}[e^{\mathbf{s}^\top \mathbf{X}}] = \exp(\mathbf{s}^\top \mathbf{m} + \frac{1}{2}\mathbf{s}^\top \mathbf{\Sigma} \mathbf{s})$.

Proof. Complete the square,

$$-\tfrac{1}{2}\mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{b}^\top \mathbf{x} = -\tfrac{1}{2}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})^\top \mathbf{A}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) + \tfrac{1}{2}\mathbf{b}^\top \mathbf{A}^{-1}\mathbf{b},$$

shift, and diagonalise $\mathbf{A} = \mathbf{O}\,\mathrm{diag}(\lambda_k)\mathbf{O}^\top$ with $\mathbf{O}$ orthogonal; in the rotated coordinates $\mathbf{y} = \mathbf{O}^\top \mathbf{x}$ the integral factorises into

$$\prod_k \int e^{-\lambda_k y_k^2/2}\, \mathrm{d}y_k = \prod_k \sqrt{2\pi/\lambda_k} = (2\pi)^{L/2}(\det \mathbf{A})^{-1/2}.$$

The density and the generating function follow with $\mathbf{A} = \mathbf{\Sigma}^{-1}$. □

The standard Gaussian on $\mathbb{R}^L$, written γ throughout, is the case $\mathbf{\Sigma} = \mathbf{I}$; its coordinates are independent standard Gaussians, which is the fact that reduces every multivariate Hermite statement below to the scalar one.

3.2 Isserlis' theorem (Wick's rule)

Theorem 3.3 (Isserlis)**.** *Let $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$. For any indices $i_1, \dots, i_{2k}$ (repetitions allowed),*

$$\mathbb{E}[X_{i_1} \cdots X_{i_{2k}}] = \sum_{\text{pairings } \pi} \prod_{\{a,b\} \in \pi} \Sigma_{i_a i_b}, \tag{3.3}$$

the sum over the $(2k-1)!!$ ways of partitioning $\{1, \dots, 2k\}$ into k unordered pairs; products of an odd number of coordinates have expectation zero.

Proof sketch. By Lemma 3.2, $\mathbb{E}[e^{\mathbf{s}^\top \mathbf{X}}] = \exp(\frac{1}{2}\sum_{ab} \Sigma_{ab} s_a s_b)$. The moment $\mathbb{E}[X_{i_1} \cdots X_{i_{2k}}]$ is the mixed derivative $\partial_{s_{i_1}} \cdots \partial_{s_{i_{2k}}}$ at $\mathbf{s} = \mathbf{0}$ of the left side. On the right side, expand the exponential: the term of order k is $\frac{1}{k!}(\frac{1}{2}\sum_{ab} \Sigma_{ab} s_a s_b)^k$, a homogeneous polynomial of degree $2k$ in $\mathbf{s}$; differentiating it $2k$ times and setting $\mathbf{s} = \mathbf{0}$ picks out, for each way of assigning the $2k$ differentiations to the k factors in pairs, the product of the corresponding Σ entries. Each unordered pairing arises $k!\,2^k$ times (order of the factors, order within each pair), which cancels the $\frac{1}{k!}\frac{1}{2^k}$ in front. Odd derivatives vanish because the right side is even in $\mathbf{s}$. □

For $k = 2$ this is $\mathbb{E}[X_i X_j X_k X_l] = \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$, the three pairings that appeared in the fourth-cumulant formula of the chapter on moments and cumulants: for a Gaussian vector the fourth moment is exactly the sum over pairings, so the fourth cumulant, which subtracts that sum, vanishes.

3.3 Probabilists' Hermite polynomials

Definition 3.4 (Rodrigues formula)**.** The **probabilists' Hermite polynomial** of degree n is

$$\mathrm{He}_n(x) = (-1)^n e^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2}, \qquad n = 0, 1, 2, \dots \tag{3.4}$$

Differentiating $e^{-x^2/2}$ once gives $-xe^{-x^2/2}$, twice $(x^2-1)e^{-x^2/2}$, and so on; each derivative is a polynomial times $e^{-x^2/2}$, and the sign $(-1)^n$ makes the polynomial monic. The first seven are

$$\begin{aligned} &\mathrm{He}_0(x) = 1, && \mathrm{He}_1(x) = x, && \mathrm{He}_2(x) = x^2 - 1, \\ &\mathrm{He}_3(x) = x^3 - 3x, && \mathrm{He}_4(x) = x^4 - 6x^2 + 3, && \mathrm{He}_5(x) = x^5 - 10x^3 + 15x, \\ &\mathrm{He}_6(x) = x^6 - 15x^4 + 45x^2 - 15. \end{aligned} \tag{3.5}$$

He_n has the parity of n. The coefficient pattern is the Gaussian moment sequence with alternating signs: $\mathrm{He}_n(x) = \sum_k (-1)^k \binom{n}{2k} (2k-1)!!\, x^{n-2k}$, which the generating function below explains.

Proposition 3.5 (Generating function)**.** *For all real x and s,*

$$e^{sx - s^2/2} = \sum_{n \geq 0} \mathrm{He}_n(x) \frac{s^n}{n!}. \tag{3.6}$$

Proof. Write $e^{-(x-s)^2/2} = e^{-x^2/2} e^{sx - s^2/2}$. The left side, as a function of s, is entire, and its Taylor expansion at $s = 0$ is $\sum_n \frac{s^n}{n!} \frac{\mathrm{d}^n}{\mathrm{d}s^n} e^{-(x-s)^2/2}\big|_{s=0}$. Since $\frac{\mathrm{d}}{\mathrm{d}s} f(x-s) = -f'(x-s)$, the n-th derivative at $s = 0$ is $(-1)^n \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2} = \mathrm{He}_n(x) e^{-x^2/2}$ by Equation (3.4). Dividing by $e^{-x^2/2}$ gives Equation (3.6). □

Two identities follow by differentiating Equation (3.6). Differentiating in x, $s\, e^{sx - s^2/2} = \sum_n \mathrm{He}_n'(x) s^n / n!$, and comparing coefficients,

$$\mathrm{He}_n'(x) = n \mathrm{He}_{n-1}(x). \tag{3.7}$$

Differentiating in s, $(x - s) e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^{n-1}/(n-1)!$, and comparing coefficients of $s^n/n!$,

$$\mathrm{He}_{n+1}(x) = x \mathrm{He}_n(x) - n \mathrm{He}_{n-1}(x), \tag{3.8}$$

the three-term recursion that `src/markov_score/theory.py` (function `he`) uses to evaluate the polynomials. Expanding $e^{sx} e^{-s^2/2}$ as a product of two series and collecting s^n gives the explicit coefficient formula stated above.

Proposition 3.6 (Orthogonality)**.** $\mathbb{E}_\gamma[\mathrm{He}_n(Z) \mathrm{He}_m(Z)] = n!\, \delta_{nm}$.

Proof. Multiply two copies of Equation (3.6) with parameters s and u and take the expectation under γ:

$$\mathbb{E}_\gamma\big[e^{sZ - s^2/2} e^{uZ - u^2/2}\big] = e^{-(s^2+u^2)/2} \mathbb{E}_\gamma[e^{(s+u)Z}] = e^{-(s^2+u^2)/2} e^{(s+u)^2/2} = e^{su} = \sum_{n \geq 0} \frac{s^n u^n}{n!}.$$

On the other side the product of the two series is $\sum_{n,m} \mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m]\, s^n u^m / (n!\, m!)$. Comparing the coefficient of $s^n u^m$ gives $\mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m]/(n!\, m!) = \delta_{nm}/n!$, which is the claim. The exchange of expectation and double sum is justified as in Lemma 3.1, the double series being dominated by $e^{(|s|+|u|)|Z|}$. □

So $\{\mathrm{He}_n/\sqrt{n!}\}_{n \geq 0}$ is an orthonormal system in $L^2(\gamma)$, the space of functions f with $\mathbb{E}_\gamma[f(Z)^2] < \infty$. It is also complete: the polynomials are dense in $L^2(\gamma)$ (one proof: if $f \in L^2(\gamma)$ is orthogonal to every polynomial then the entire function $s \mapsto \mathbb{E}_\gamma[f(Z) e^{sZ}]$ has every derivative equal to zero at the origin, hence the Fourier transform of $f\gamma$ vanishes, hence $f = 0$). A function $f \in L^2(\gamma)$ therefore has the **Hermite expansion**

$$f = \sum_{n \geq 0} \frac{\mathbb{E}_\gamma[f \mathrm{He}_n]}{n!} \mathrm{He}_n, \qquad \mathbb{E}_\gamma[f^2] = \sum_{n \geq 0} \frac{\mathbb{E}_\gamma[f \mathrm{He}_n]^2}{n!}, \tag{3.9}$$

with convergence in $L^2(\gamma)$ and the Parseval identity on the right. The **Hermite moment** of degree n of a law P is $\mathbb{E}_P[\mathrm{He}_n]$; when P has a density and $P/\gamma \in L^2(\gamma)$, these are the expansion coefficients of the density ratio: $P/\gamma = \sum_n \mathbb{E}_P[\mathrm{He}_n] \mathrm{He}_n / n!$.

3.4 Products of Hermite polynomials and Wick products

Definition 3.7 (Multivariate Hermite products)**.** For a multi-index $\nu \in \mathbb{N}^L$,

$$H_\nu(\mathbf{x}) = \prod_{i=1}^{L} \mathrm{He}_{\nu_i}(x_i), \qquad |\nu| = \sum_i \nu_i, \qquad \nu! = \prod_i \nu_i!,$$

and $|\nu|$ is the **degree** of H_ν.

Under γ on $\mathbb{R}^L$ the coordinates are independent, so the expectation of a product factorises and Proposition 3.6 gives at once

$$\mathbb{E}_\gamma[H_\nu H_\mu] = \prod_i \mathbb{E}_\gamma[\mathrm{He}_{\nu_i}\mathrm{He}_{\mu_i}] = \prod_i \nu_i!\,\delta_{\nu_i\mu_i} = \nu!\,\delta_{\nu\mu}. \tag{3.10}$$

This is the normalisation the paper states in Section 3; $\{H_\nu/\sqrt{\nu!}\}$ is an orthonormal basis of $L^2(\gamma)$ on $\mathbb{R}^L$ (completeness by the same argument coordinatewise), and the **chaos expansion** of $f \in L^2(\gamma)$ is $f = \sum_\nu \mathbb{E}_\gamma[fH_\nu]H_\nu/\nu!$. The paper writes H_ν with a capital H; it is the product of probabilists' polynomials, not a physicists' polynomial (see the chapter on conventions).

The same objects appear in the paper's proofs in a second notation, the **Wick product** $:x_{i_1}\cdots x_{i_n}:$, defined by the multivariate generating function

$$e^{\mathbf{s}^\top\mathbf{x} - |\mathbf{s}|^2/2} = \sum_{n\ge 0} \frac{1}{n!} \sum_{i_1,\dots,i_n} s_{i_1}\cdots s_{i_n}\, :x_{i_1}\cdots x_{i_n}:. \tag{3.11}$$

Since the left side is $\prod_i e^{s_i x_i - s_i^2/2} = \prod_i \sum_k \mathrm{He}_k(x_i)s_i^k/k!$, comparing the coefficient of a monomial $\prod_i s_i^{\nu_i}$ (which appears $n!/\nu!$ times among the ordered tuples $(i_1,\dots,i_n)$) shows

$$:x_{i_1}\cdots x_{i_n}: = H_\nu(\mathbf{x}) \qquad \text{where } \nu_i \text{ is the number of times } i \text{ occurs in } (i_1,\dots,i_n). \tag{3.12}$$

The Wick product is symmetric in its indices and depends only on the multiset. Two degrees are needed explicitly. For degree two, $:x_ix_j: = x_ix_j - \delta_{ij}$, so that $:\mathbf{x}\mathbf{x}^\top: = \mathbf{x}\mathbf{x}^\top - \mathbf{I}$. For degree four, expanding Equation (3.11) or multiplying out He_k's,

$$\begin{aligned} :x_ix_jx_kx_l: = {} & x_ix_jx_kx_l - \big(\delta_{ij}x_kx_l + \delta_{ik}x_jx_l + \delta_{il}x_jx_k + \delta_{jk}x_ix_l + \delta_{jl}x_ix_k + \delta_{kl}x_ix_j\big) \\ & + \big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big), \end{aligned} \tag{3.13}$$

the inclusion–exclusion over pairs and double pairs; on the diagonal $i=j=k=l$ it gives $x^4 - 6x^2 + 3 = \mathrm{He}_4(x)$, as it must. Equation (3.13) is what the paper means by "the cumulant contractions with δ_{ii} are exactly the terms that convert monomials into Hermite polynomials".

Lemma 3.8 (Second moment of a symmetric contraction)**.** *Let* $\mathbf{T}$ *be a symmetric tensor of order* n *on* $\mathbb{R}^L$*. Then*

$$\mathbb{E}_\gamma\Big[\Big(\sum_{i_1\cdots i_n} T_{i_1\cdots i_n}\, :x_{i_1}\cdots x_{i_n}:\Big)^2\Big] = n!\sum_{i_1\cdots i_n} T_{i_1\cdots i_n}^2 = n!\,\|\mathbf{T}\|_F^2. \tag{3.14}$$

Proof. Group the ordered tuples by their multiset ν: each ν with $|\nu| = n$ collects $n!/\nu!$ tuples, all with the same entry T_ν (symmetry) and the same Wick product H_ν (Equation (3.12)). So the sum is $\sum_{|\nu|=n}(n!/\nu!)T_\nu H_\nu$, and by Equation (3.10) its second moment is $\sum_\nu (n!/\nu!)^2 T_\nu^2\,\nu! = n!\sum_\nu (n!/\nu!)T_\nu^2 = n!\sum_{\text{ordered}} T^2$. □

Why it matters here. This lemma is where the paper's two constants come from. The leading term of the score gap has m-th component $\frac{1}{6}\sum_{jkl}\kappa_4[mjkl]\,:x_jx_kx_l:$ times e^{-4t}; with $n=3$ and the symmetric tensor $T_{jkl} = \kappa_4[mjkl]$, Equation (3.14) gives a γ-second moment of $\frac{1}{36}\cdot 3!\sum_{jkl}\kappa_4[mjkl]^2$, and summing over m gives $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$. The leading term of a shape score is $\frac{1}{24}\sum \partial_\theta\kappa_4[ijkl]\,:x_ix_jx_kx_l:$ times e^{-4t}; with $n=4$, its second moment is $\frac{4!}{24^2}\|\partial_\theta\boldsymbol{\kappa}_4\|_F^2 = \frac{1}{24}\|\partial_\theta\boldsymbol{\kappa}_4\|_F^2$. The paper's Appendix A states both computations in one line each; this lemma is that line.

3.5 When is a density ratio in $L^2(\gamma)$, and why Theorem 1 does not care

The Hermite expansion Equation (3.9) of a density ratio P/γ requires $P/\gamma \in L^2(\gamma)$, that is

$$\int \frac{P(\mathbf{x})^2}{\gamma(\mathbf{x})}\,\mathrm{d}\mathbf{x} < \infty, \qquad \text{equivalently} \qquad \chi^2(P\,\|\,\gamma) = \int \frac{P^2}{\gamma} - 1 < \infty. \tag{3.15}$$

This is a strong condition: it asks P to have tails no heavier than $\sqrt{\gamma}$, that is $P(\mathbf{x}) \lesssim e^{-|\mathbf{x}|^2/4}$ in every direction.

Proposition 3.9 (Gaussian laws)**.** *Let $P = \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$. Then $P/\gamma \in L^2(\gamma)$ if and only if $\lambda_{\max}(\mathbf{\Sigma}) < 2$, and in that case $\int P^2/\gamma = \prod_k (\lambda_k(2-\lambda_k))^{-1/2}$ over the eigenvalues λ_k of $\mathbf{\Sigma}$.*

Proof. $P^2/\gamma = (2\pi)^{-L/2}(\det \mathbf{\Sigma})^{-1} \exp(-\frac{1}{2}\mathbf{x}^\top(2\mathbf{\Sigma}^{-1} - \mathbf{I})\mathbf{x})$. By Lemma 3.2 the integral is finite if and only if $2\mathbf{\Sigma}^{-1} - \mathbf{I} \succ 0$, i.e. $\mathbf{\Sigma} \prec 2\mathbf{I}$, and then equals $(\det \mathbf{\Sigma})^{-1} \det(2\mathbf{\Sigma}^{-1} - \mathbf{I})^{-1/2} = \det(\mathbf{\Sigma}(2\mathbf{I} - \mathbf{\Sigma}))^{-1/2}$. □

Two applications, both stated in the paper's Section 3. First, the clean Gaussian chain at $\rho = 0.85$ has $\lambda_{\max}(\mathbf{\Sigma}_0) > 2$ (the chapter on Markov chains bounds $\lambda_{\max}$ between the row sums of the Toeplitz matrix and $(1+\rho)/(1-\rho)$, and already the 2×2 block $\left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ has eigenvalue $1 + \rho > 2$ when $\rho > 1$, while for $L = 32$ the Rayleigh quotient of the constant vector exceeds 2 as soon as $\rho \gtrsim 0.35$), so $P_0/\gamma \notin L^2(\gamma)$ for the Gaussian chain itself. Second, the noised law $P_t = \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_t)$ with $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$ has eigenvalues $e^{-2t}\lambda_k + 1 - e^{-2t}$, which are all below 2 if and only if $e^{-2t}(\lambda_{\max} - 1) < 1$, i.e.

$$P_t/\gamma \in L^2(\gamma) \iff e^{2t} > \lambda_{\max}(\mathbf{\Sigma}_0) - 1, \tag{3.16}$$

the exact statement in the paper's paragraph "Where the asymptote begins".

Proposition 3.10 (The Laplace law)**.** *If P is the scalar Laplace density $\frac{1}{2b}e^{-|x|/b}$, then $P/\gamma \notin L^2(\gamma)$ for every $b > 0$. The same holds for the Laplace chain on $\mathbb{R}^L$.*

Proof. $P(x)^2/\gamma(x) = \frac{\sqrt{2\pi}}{4b^2}\exp(x^2/2 - 2|x|/b)$, and $x^2/2 - 2|x|/b \to +\infty$, so the integrand is unbounded and not integrable. For the chain, $P_0(\mathbf{a}) = \mu(a_1)\prod_{i\geq 2}\varphi(a_i - \rho a_{i-1})$ with φ Laplace; along the ray $\mathbf{a} = u\mathbf{e}_L$ (all sites zero except the last) the ratio P_0^2/γ is proportional to $\exp(u^2/2 - 2|u|/b)$, and by continuity the same growth holds on a tube around the ray, so the integral over $\mathbb{R}^L$ diverges. □

Exponential tails are lighter than Gaussian ones in the sense of moments, and the Laplace law has an exponential moment; but they are heavier than $e^{-x^2/4}$, and that is what $L^2(\gamma)$ asks. The same is true of every family in the paper other than the uniform and the mixture.

Why the paper's Theorem 1 does not need it. A tempting route to attenuation is spectral: the Ornstein–Uhlenbeck semigroup $(\mathcal{P}_t f)(a) = \mathbb{E}[f(e^{-t}a + \sqrt{\Delta_t}Z)]$ is self-adjoint on $L^2(\gamma)$ with eigenfunctions He_n and eigenvalues e^{-nt}, so one could expand P_0/γ in the chaos basis and let the semigroup act coefficient by coefficient. That route needs $P_0/\gamma \in L^2(\gamma)$, which the two propositions above show fails for the paper's chain, Gaussian or Laplace. Theorem 1 is instead an identity for one expectation, $\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)]$, proved by Fubini for each fixed ν; it needs only $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$, so that $H_\nu(\mathbf{a})$ is integrable. No expansion of a density is ever summed. This is why the paper can state the theorem for every law with finite moments, and why its proofs of the asymptotic theorems go through the damped cumulant-generating function Λ rather than through the chaos expansion of P_t/γ.

3.6 Mehler's formula and the paper's Theorem 1

Theorem 3.11 (Mehler's formula, moment form)**.** *Let $Z \sim \mathcal{N}(0,1)$, $a \in \mathbb{R}$, $\alpha \in [0,1]$ and $\beta = \sqrt{1-\alpha^2}$. Then for every $n \geq 0$,*

$$\mathbb{E}_Z\big[\mathrm{He}_n(\alpha a + \beta Z)\big] = \alpha^n \mathrm{He}_n(a). \tag{3.17}$$

Proof. Put $x = \alpha a + \beta Z$ in the generating function Equation (3.6) and take the expectation over Z:

$$\mathbb{E}_Z\big[e^{s(\alpha a+\beta Z)-s^2/2}\big] = e^{s\alpha a - s^2/2}\,\mathbb{E}_Z\big[e^{s\beta Z}\big] = e^{s\alpha a - s^2/2}\,e^{s^2\beta^2/2} = e^{s\alpha a - s^2\alpha^2/2},$$

using Equation (3.1) and $\beta^2 = 1-\alpha^2$. The right-hand side is the generating function Equation (3.6) evaluated at $x = a$ and parameter αs: $e^{(\alpha s)a-(\alpha s)^2/2} = \sum_n \mathrm{He}_n(a)\alpha^n s^n/n!$. The left-hand side is $\mathbb{E}_Z[\sum_n \mathrm{He}_n(\alpha a+\beta Z)s^n/n!]$, and the expectation may be taken inside the sum: the series $\sum_n |\mathrm{He}_n(x)||s|^n/n!$ is dominated, for $|s| \leq 1$, by a polynomial in $|x|$ times $e^{|x|}$ (from the explicit coefficient formula, $|\mathrm{He}_n(x)| \leq \sum_k \binom{n}{2k}(2k-1)!!|x|^{n-2k} \leq (1+|x|)^n\, n!!$), which is integrable against the Gaussian law of Z, so dominated convergence applies. Comparing the coefficients of $s^n/n!$ on the two sides gives Equation (3.17). □

The name refers to Mehler's 1866 kernel identity,

$$\sum_{n\geq 0} \mathrm{He}_n(x)\mathrm{He}_n(y)\frac{\alpha^n}{n!} = \frac{1}{\sqrt{1-\alpha^2}}\exp\Big(-\frac{\alpha^2(x^2+y^2)-2\alpha xy}{2(1-\alpha^2)}\Big),$$

which is the transition density of the Ornstein–Uhlenbeck process divided by $\gamma(y)$; Equation (3.17) is its moment form, and says that He_n is an eigenfunction of the channel with eigenvalue $\alpha^n = e^{-nt}$.

Theorem 3.12 (Hermite attenuation; the paper's Theorem 1)**.** *Let $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ with $\mathbf{a} \sim P_0$ and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ independent. For every multi-index ν with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$,*

$$\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)] = e^{-|\nu|t}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]. \tag{3.18}$$

Proof. Condition on $\mathbf{a}$. Given $\mathbf{a}$, the coordinates $x_{t,i} = e^{-t}a_i + \sqrt{\Delta_t}z_i$ are independent, each of the form $\alpha a_i + \beta z_i$ with $\alpha = e^{-t}$, $\beta = \sqrt{1-\alpha^2} = \sqrt{\Delta_t}$. So

$$\mathbb{E}\big[H_\nu(\mathbf{x}_t)\,\big|\,\mathbf{a}\big] = \prod_i \mathbb{E}_{z_i}\big[\mathrm{He}_{\nu_i}(\alpha a_i + \beta z_i)\big] = \prod_i \alpha^{\nu_i}\mathrm{He}_{\nu_i}(a_i) = \alpha^{|\nu|}H_\nu(\mathbf{a})$$

by Theorem 3.11 applied coordinate by coordinate. Taking the expectation over $\mathbf{a} \sim P_0$ gives Equation (3.18), provided $H_\nu(\mathbf{a})$ is integrable, which holds because $|H_\nu(\mathbf{a})| \leq C(1+|\mathbf{a}|)^{|\nu|}$ and $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$; the tower property $\mathbb{E}[\mathbb{E}[\cdot \mid \mathbf{a}]] = \mathbb{E}[\cdot]$ is Fubini's theorem, which also needs only that integrability. □

The theorem is exact at every t, needs no assumption on the dependence between sites, and needs no $L^2(\gamma)$ condition. It is verified by Monte Carlo in `tests/test_theory_hermite.py` on a Laplace sample, and the cases $n \leq 4$ are checked in Lean as polynomial identities under the first four Gaussian moments (`lean/README.md`).

3.7 Two exact consequences

Corollary 3.13 (Cumulants of order three and higher)**.** *For every $m \geq 3$ with $\mathbb{E}|\mathbf{a}|^m < \infty$, the order-$m$ cumulant tensor of $\mathbf{x}_t$ is e^{-mt} times that of $\mathbf{a}$. The covariance is* $\mathrm{Cov}(\mathbf{x}_t) = e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t\mathbf{I}$.

Proof. $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ is a sum of two independent vectors. By additivity and the transformation rule of the chapter on moments and cumulants, the order-m cumulant tensor of $\mathbf{x}_t$ is e^{-mt} times that of $\mathbf{a}$ plus $\Delta_t^{m/2}$ times that of $\mathbf{z}$; the Gaussian $\mathbf{z}$ has no cumulant tensor of order $m \geq 3$. For $m = 2$ the Gaussian contributes $\Delta_t \mathbf{I}$. □

This corollary uses no Hermite polynomials, only the arithmetic of cumulants; the paper's parenthesis "Gaussian noise has none" is the whole proof. The Hermite form of the theorem is nevertheless the one the paper needs, because Fisher information is built from moments, and the next corollary says how the Hermite moments of degree four see the fourth cumulant.

Corollary 3.14 (Degree-four Hermite moments and the fourth-cumulant tensor)**.** *Let* $\mathbf{a}$ *have mean zero, covariance* $\mathbf{\Sigma}_0$ *and finite fourth moments. For any indices, with* ν *the multiset of* (i, j, k, l)*,*

$$\mathbb{E}[H_\nu(\mathbf{a})] = \mathbb{E}[{:}a_i a_j a_k a_l{:}] = \kappa_4[ijkl] + (\mathbf{\Sigma}_0 - \mathbf{I})_{ij}(\mathbf{\Sigma}_0 - \mathbf{I})_{kl} + (\mathbf{\Sigma}_0 - \mathbf{I})_{ik}(\mathbf{\Sigma}_0 - \mathbf{I})_{jl} + (\mathbf{\Sigma}_0 - \mathbf{I})_{il}(\mathbf{\Sigma}_0 - \mathbf{I})_{jk}. \tag{3.19}$$

Consequently, if two laws have the same covariance $\mathbf{\Sigma}_0$*, their degree-four Hermite moments differ exactly by the difference of their fourth-cumulant tensors, and after noising that difference is* $e^{-4t}(\boldsymbol{\kappa}_4 - \boldsymbol{\kappa}_4')$*.*

Proof. Take the expectation of Equation (3.13). By the fourth-cumulant formula of the chapter on moments and cumulants, $\mathbb{E}[a_i a_j a_k a_l] = \kappa_4[ijkl] + \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$ with $\Sigma = \mathbf{\Sigma}_0$; the six pair terms give $-\delta_{ij}\Sigma_{kl} - \dots$; the three double-pair terms give $+\delta_{ij}\delta_{kl} + \dots$. Collecting by pairing, $\Sigma_{ij}\Sigma_{kl} - \delta_{ij}\Sigma_{kl} - \Sigma_{ij}\delta_{kl} + \delta_{ij}\delta_{kl} = (\Sigma - \mathbf{I})_{ij}(\Sigma - \mathbf{I})_{kl}$, and likewise for the other two pairings. The second statement follows because the product terms depend only on $\mathbf{\Sigma}_0$, and the attenuation by e^{-4t} is Theorem 3.12 with $|\nu| = 4$. □

For the paper's variance-matched families, $\mathbf{\Sigma}_0$ is the same for all, so the degree-four Hermite moment of the chain differs from that of its covariance twin by exactly $\boldsymbol{\kappa}_4$; on the diagonal, $\mathbb{E}[\mathrm{He}_4(a_i)] = \kappa_4(a_i)$ because $\mathrm{Var}(a_i) = 1$. Degree two works the same way: $\mathbb{E}[{:}a_i a_j{:}] = (\mathbf{\Sigma}_0 - \mathbf{I})_{ij}$, attenuated by e^{-2t}, which is the "correlation coordinate" and is consistent with Corollary 3.13. This is the paper's second exact consequence, checked in Lean as target 2 (`lean/README.md`).

3.8 The Edgeworth expansion: the first correction

The paper's E16 asks how much of the gap between the true score and the twin's is explained by "adding the fourth-order term back". The object added is the first term of the Edgeworth (Gram–Charlier A) expansion, defined here; the chapter on misspecification and Edgeworth corrections uses it.

Scalar form. Let X have mean 0, variance 1 and cumulants $\kappa_3, \kappa_4, \dots$. Formally, its generating function is $\exp(\frac{1}{2}s^2 + \sum_{n\geq 3} \kappa_n s^n/n!)$, that of the standard Gaussian times $\exp(\sum_{n\geq 3} \kappa_n s^n/n!)$; since multiplying a generating function by s^n corresponds to applying $(-\,\mathrm{d}/\,\mathrm{d}x)^n$ to the density, and $(-\,\mathrm{d}/\,\mathrm{d}x)^n\gamma = \mathrm{He}_n\gamma$ by the Rodrigues formula Equation (3.4),

$$p(x) \text{ “=” } \exp\Big(\sum_{n\geq 3} \frac{\kappa_n}{n!}\big(-\tfrac{\mathrm{d}}{\mathrm{d}x}\big)^n\Big)\gamma(x) = \gamma(x)\Big[1 + \frac{\kappa_3}{6}\mathrm{He}_3(x) + \frac{\kappa_4}{24}\mathrm{He}_4(x) + \dots\Big], \tag{3.20}$$

where the dots collect $\kappa_3^2\mathrm{He}_6/72$ and all higher terms. The **first Edgeworth correction** is the truncation $p \approx \gamma(1 + \kappa_3\mathrm{He}_3/6 + \kappa_4\mathrm{He}_4/24)$, and for a symmetric law it is $p \approx \gamma(1 + \kappa_4\mathrm{He}_4/24)$. The series need not converge (it does not for the Laplace law, whose κ_n grow like $n!$), and even when it does the truncation can be negative for large $|x|$; the expansion is an asymptotic device, exact order by order in the cumulants, and the paper reports where it helps and where it diverges rather than assuming either.

Multivariate form around a Gaussian with covariance $\boldsymbol{\Sigma}$. Replace γ by $\gamma_{\boldsymbol{\Sigma}}$, the density of $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, and the Hermite polynomials by the **$\boldsymbol{\Sigma}$-Hermite tensors**

$$h_{i_1 \cdots i_n}(\mathbf{x}) = (-1)^n \, \frac{\partial_{i_1} \cdots \partial_{i_n} \gamma_{\boldsymbol{\Sigma}}(\mathbf{x})}{\gamma_{\boldsymbol{\Sigma}}(\mathbf{x})}.$$

With $\mathbf{y} = \boldsymbol{\Sigma}^{-1}\mathbf{x}$, $\partial_i \gamma_{\boldsymbol{\Sigma}} = -y_i \gamma_{\boldsymbol{\Sigma}}$ and $\partial_i y_j = (\boldsymbol{\Sigma}^{-1})_{ij}$, so the fourth tensor is

$$\begin{aligned} h_{ijkl}(\mathbf{x}) = {} & y_i y_j y_k y_l - \big[(\boldsymbol{\Sigma}^{-1})_{ij} y_k y_l + \text{five more}\big] \\ & + \big[(\boldsymbol{\Sigma}^{-1})_{ij}(\boldsymbol{\Sigma}^{-1})_{kl} + (\boldsymbol{\Sigma}^{-1})_{ik}(\boldsymbol{\Sigma}^{-1})_{jl} + (\boldsymbol{\Sigma}^{-1})_{il}(\boldsymbol{\Sigma}^{-1})_{jk}\big], \end{aligned} \tag{3.21}$$

Equation (3.13) with $\mathbf{x} \mapsto \mathbf{y}$ and $\delta \mapsto \boldsymbol{\Sigma}^{-1}$. For a symmetric law with fourth-cumulant tensor $\mathbf{K}$ the first correction is $p \approx \gamma_{\boldsymbol{\Sigma}}(1+\epsilon)$ with $\epsilon = \frac{1}{24}\sum_{ijkl} K_{ijkl} h_{ijkl}$; contracting a symmetric $\mathbf{K}$ with the three groups of terms gives, with $A_{kl} = \sum_{ij} K_{ijkl}(\boldsymbol{\Sigma}^{-1})_{ij}$,

$$\epsilon(\mathbf{x}) = \frac{1}{24}\Big[\mathbf{K}[\mathbf{y},\mathbf{y},\mathbf{y},\mathbf{y}] - 6\,\mathbf{y}^\top \mathbf{A}\mathbf{y} + 3\langle \mathbf{A}, \boldsymbol{\Sigma}^{-1}\rangle\Big], \qquad \nabla\epsilon = \frac{1}{6}\boldsymbol{\Sigma}^{-1}\mathbf{K}[\cdot,\mathbf{y},\mathbf{y},\mathbf{y}] - \frac{1}{2}\boldsymbol{\Sigma}^{-1}\mathbf{A}\mathbf{y}, \tag{3.22}$$

where $\mathbf{K}[\mathbf{y},\mathbf{y},\mathbf{y},\mathbf{y}] = \sum K_{ijkl} y_i y_j y_k y_l$ and $\langle \mathbf{A}, \boldsymbol{\Sigma}^{-1}\rangle = \sum_{kl} A_{kl}(\boldsymbol{\Sigma}^{-1})_{kl}$; the six pair terms each contract to $\mathbf{y}^\top\mathbf{A}\mathbf{y}$ and the three double-pair terms each to $\langle \mathbf{A}, \boldsymbol{\Sigma}^{-1}\rangle$ by the symmetry of $\mathbf{K}$, and the gradient uses $\nabla_{\mathbf{x}} = \boldsymbol{\Sigma}^{-1}\nabla_{\mathbf{y}}$. The score of the corrected density is $\nabla \log[\gamma_{\boldsymbol{\Sigma}}(1+\epsilon)] = -\boldsymbol{\Sigma}^{-1}\mathbf{x} + \nabla\epsilon/(1+\epsilon)$. This is exactly the formula in the paper's Appendix E for E16, with $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_t$ and $\mathbf{K} = e^{-4t}\boldsymbol{\kappa}_4$: by Corollary 3.13 the noised chain has covariance $\boldsymbol{\Sigma}_t$ and fourth-cumulant tensor $e^{-4t}\boldsymbol{\kappa}_4$, so the first Edgeworth correction around the covariance twin is the degree-four Hermite term that the paper's Theorem 2 identifies as the whole of the leading gap.

3.9 Worked example: a unit-variance Laplace site at $t = \log 2$

Let a have the unit-variance Laplace law, $b = 1/\sqrt{2}$, with $\kappa_4 = 3$ (chapter on moments and cumulants), and let $x_t = e^{-t}a + \sqrt{\Delta_t}z$ at $t = \log 2$, so $\alpha = e^{-t} = \frac{1}{2}$ and $\Delta_t = \frac{3}{4}$. Theorem 1 predicts

$$\mathbb{E}[\mathrm{He}_4(x_t)] = \alpha^4 \mathbb{E}[\mathrm{He}_4(a)] = \tfrac{1}{16}(\mathbb{E}a^4 - 6\mathbb{E}a^2 + 3) = \tfrac{1}{16}(6 - 6 + 3) = \tfrac{3}{16},$$

using $\mathbb{E}a^4 = 24b^4 = 6$. Directly: $\mathbb{E}x_t^2 = \frac{1}{4} + \frac{3}{4} = 1$ and $\mathbb{E}x_t^4 = \alpha^4\mathbb{E}a^4 + 6\alpha^2\Delta_t\mathbb{E}a^2\mathbb{E}z^2 + \Delta_t^2\mathbb{E}z^4 = \frac{6}{16} + 6\cdot\frac{1}{4}\cdot\frac{3}{4} + \frac{9}{16}\cdot 3 = \frac{6+18+27}{16} = \frac{51}{16}$, so $\mathbb{E}[\mathrm{He}_4(x_t)] = \frac{51}{16} - 6 + 3 = \frac{3}{16}$, as predicted. The fourth cumulant of x_t is $\kappa_4(x_t) = \mathbb{E}x_t^4 - 3 = \frac{3}{16} = e^{-4t}\kappa_4(a)$, illustrating Corollary 3.13; and the degree-two moment $\mathbb{E}[\mathrm{He}_2(x_t)] = \mathbb{E}x_t^2 - 1 = 0$ equals $e^{-2t}\mathbb{E}[\mathrm{He}_2(a)] = 0$, as unit variance requires.

For the constant of Lemma 3.8 in the scalar case: the leading term of the score gap is $(\kappa_4/6)\mathrm{He}_3(x)e^{-4t}$, whose γ-second moment is $(\kappa_4/6)^2 \cdot 3!\, e^{-8t} = (\kappa_4^2/6)e^{-8t}$, that is $\frac{9}{6}e^{-8t} = 1.5\, e^{-8t}$ for the Laplace law; this is the value that `tests/test_theory_hermite.py` compares with a fine quadrature of the scalar noised density.

Where it is used. The paper's Section 3 (Theorem 1 and its two consequences, the Hermite normalisation $\nu!$, the $L^2(\gamma)$ remarks, the Edgeworth paragraph "The gap is the fourth-order term"), Appendix A (the proofs of Theorems 1 and 2, Wick's theorem for the constants, the crossover $e^{2t} > \lambda_{\max} - 1$) and Appendix E (E16). Experiments E01, E16, E18, E21. Code: the functions `he`, `hermite_moment_attenuation`, `score_gap_constant` and `shape_information_constant` of `src/markov_score/theory.py`; the Edgeworth oracle `experiments/e16_edgeworth_oracle.py`; the tests in `tests/test_theory_hermite.py`; and the Lean targets listed in `lean/README.md`.

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distribution in any number of variables", Biometrika 12, 134–139. F. G. Mehler (1866), "Ueber die Entwicklung einer Function von beliebig vielen Variablen nach Laplaceschen Functionen höherer Ordnung", Journal für die reine und angewandte Mathematik 66, 161–176. S. Janson (1997), *Gaussian Hilbert Spaces*, Cambridge University Press (Wick products, chaos expansion, the Ornstein–Uhlenbeck semigroup on $L^2(\gamma)$). P. Hall (1992), *The Bootstrap and Edgeworth Expansion*, Springer (Chapter 2: the Edgeworth expansion and its first terms). P. McCullagh (1987), *Tensor Methods in Statistics*, Chapman and Hall (Chapter 5: Edgeworth series in several variables, Hermite tensors).

Chapter 4

Markov chains and the AR(1) laboratory

The paper's data are sequences drawn from a first-order Markov chain with a linear autoregressive transition and a non-Gaussian innovation. The choice is not decorative: it is the one family in which the covariance is fixed by two numbers and every non-Gaussian feature lives in the cumulants of order four and higher, so that "second-order structure" and "higher-order structure" are separated by construction and not by an approximation. This chapter defines Markov chains and transition kernels, distinguishes strict from second-order stationarity (the paper's chain has the second and, for non-Gaussian innovations, not the first), derives the covariance $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation law, studies the covariance matrix $\mathbf{\Sigma}_0$ and its spectrum, derives the chain's fourth-cumulant tensor, and fixes what the two sizes L and M mean.

4.1 The Markov property and transition kernels

Definition 4.1 (Markov chain)**.** A sequence of real random variables $a_1, a_2, \ldots, a_L$ is a **Markov chain** if, for every i, the conditional law of a_i given $(a_1, \ldots, a_{i-1})$ depends only on a_{i-1}:

$$\mathbb{P}(a_i \in B \mid a_1, \ldots, a_{i-1}) = \mathbb{P}(a_i \in B \mid a_{i-1}) \qquad \text{for every Borel set } B. \tag{4.1}$$

When the conditional laws have densities, the chain is described by an **initial density** μ and **transition kernels** $K_i(a' \mid a)$, the density of a_i at a' given $a_{i-1} = a$. The kernel is a density in its first argument for every value of the second, $\int K_i(a' \mid a)\,\mathrm{d}a' = 1$. Multiplying the conditional densities along the sequence gives the joint density

$$p(a_1, \ldots, a_L) = \mu(a_1) \prod_{i=2}^{L} K_i(a_i \mid a_{i-1}), \tag{4.2}$$

and conversely any density of this product form is Markov, because conditioning on $(a_1, \ldots, a_{i-1})$ leaves a function of a_i that depends on the past only through the factor $K_i(a_i \mid a_{i-1})$. The chain is **time-homogeneous** if every K_i is the same kernel K; the paper's chain is. The pair (μ, K) then determines the law of the whole sequence, which is why "learning the dynamics" in the paper means learning K (the initial law $\mu = \mathcal{N}(0, 1)$ is held fixed and known).

Two consequences of Equation (4.2) are used constantly. The **Chapman–Kolmogorov equation** says that the two-step kernel is the composition $K^{(2)}(a'' \mid a) = \int K(a'' \mid a')K(a' \mid a)\,\mathrm{d}a'$, and more generally the k-step kernel is the k-fold composition. And conditioning on a site splits the chain: given a_i, the past $(a_1, \ldots, a_{i-1})$ and the future $(a_{i+1}, \ldots, a_L)$ are independent, which is what makes the posterior factor graph of the paper a tree and the forward–backward recursion exact (chapter on factor graphs and belief propagation).

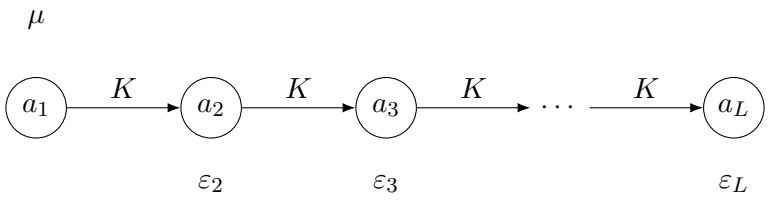


Figure 4.1: The chain: the initial law μ at site 1, one transition kernel K per edge, and one independent innovation ε_i entering at each site $i \geq 2$. The posterior factor graph of the paper's Figure 1 is this chain with one observation attached to each site.

4.2 Stationarity: strict and second-order

Definition 4.2 (Invariant law and stationarity)**.** A density π is **invariant** for K if $\int K(a' \mid a)\pi(a)\,\mathrm{d}a = \pi(a')$ for every a'. A time-homogeneous chain started from an invariant law, $\mu = \pi$, is **strictly stationary**: the joint law of $(a_i, \ldots, a_{i+k})$ does not depend on i. A sequence is **second-order stationary** if its means are constant and $\mathrm{Cov}(a_i, a_j)$ depends only on $|i - j|$.

Strict stationarity implies second-order stationarity when second moments exist; the converse fails, and the paper's chain is the example. Its first site is $\mathcal{N}(0, 1)$, which is the invariant law of the transition only when the innovation is Gaussian; for every other family the chain is second-order stationary (next section) but not strictly stationary, and its fourth cumulant relaxes toward the invariant value along the chain, as the chapter on moments and cumulants computed. The paper's proof of Proposition 1 says explicitly that "nothing assumes stationarity", and its fourth-cumulant tensor (Section 4.6) carries the site dependence exactly. The word "stationary" in this compendium always says which sense is meant.

4.3 The AR(1) chain

Definition 4.3 (First-order autoregression)**.** Let φ be a density on $\mathbb{R}$ with mean zero and variance q, and let $|\rho| < 1$. The **AR(1) chain** with innovation φ is

$$a_1 \sim \mu, \qquad a_i = \rho a_{i-1} + \varepsilon_i \quad (i \geq 2), \qquad \varepsilon_2, \varepsilon_3, \ldots \text{ i.i.d. with density } \varphi, \text{ independent of } a_1. \tag{4.3}$$

Its transition kernel is $K(a' \mid a) = \varphi(a' - \rho a)$.

The kernel is a function of the residual $a' - \rho a$ alone, which is what makes the grid representation of the chapter on grid quadrature a matrix of residuals; the helper `_outer_residual` in `src/markov_score/priors.py` builds that matrix once for every family. Because ε_i is independent of $(a_1, \ldots, a_{i-1})$, the law of a_i given the whole past depends only on a_{i-1}: the recursion is Markov.

The invariant law. Iterating Equation (4.3), $a_i = \rho^{i-1} a_1 + \sum_{m=2}^{i} \rho^{i-m} \varepsilon_m$. As $i \to \infty$ the contribution of a_1 vanishes and the sum converges to the moving-average series $\sum_{k \geq 0} \rho^k \varepsilon_{-k}$ of the chapter on moments and cumulants; its law is the invariant law of the kernel, whatever μ was. For Gaussian innovations it is $\mathcal{N}(0, q/(1 - \rho^2))$, which at $q = 1 - \rho^2$ is $\mathcal{N}(0, 1)$: the paper's Gaussian chain is strictly stationary. For a Laplace innovation the invariant law is the law of $\sum_k \rho^k \varepsilon_{-k}$, which has no closed form but has all the cumulants of the chapter on moments and cumulants; it is not the law of a_1.

4.4 Unit variance and geometric covariance for every innovation law

Proposition 4.4 (Second-order structure of the chain)**.** *Let $\mu = \mathcal{N}(0,1)$ and $q = 1-\rho^2$ in Equation* (4.3)*. Then, for every innovation density of mean zero and variance q,*

$$\mathbb{E}[a_i] = 0, \qquad \operatorname{Var}(a_i) = 1, \qquad \operatorname{Cov}(a_i, a_j) = \rho^{|i-j|} \quad \textit{for all } i, j. \tag{4.4}$$

Proof. Means: $\mathbb{E}a_1 = 0$, and inductively $\mathbb{E}a_i = \rho\mathbb{E}a_{i-1} + \mathbb{E}\varepsilon_i = 0$. Variances: $\operatorname{Var}(a_1) = 1$, and inductively, since ε_i is independent of a_{i-1},

$$\operatorname{Var}(a_i) = \rho^2 \operatorname{Var}(a_{i-1}) + \operatorname{Var}(\varepsilon_i) = \rho^2 + (1-\rho^2) = 1.$$

Covariances: for $i > j$, write $a_i = \rho^{i-j} a_j + \sum_{m=j+1}^{i} \rho^{i-m}\varepsilon_m$, where the sum is independent of a_j and has mean zero, so $\operatorname{Cov}(a_i, a_j) = \rho^{i-j}\operatorname{Var}(a_j) = \rho^{i-j}$. Only the mean and variance of φ were used. □

Why it matters here. This is the sentence "the variance choice makes $\operatorname{Var}(a_i) = 1$ and $\operatorname{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly for every innovation law" of the paper's Section 2, and it is the whole reason the laboratory separates orders. Every innovation family of the paper is symmetric, so its odd cumulants vanish; Proposition 4.4 pins the cumulants of order two; therefore two chains with different innovation families share every cumulant tensor of order one, two and three, and differ only from order four on. A Gaussian-innovation kernel fitted by maximum likelihood can at best recover (ρ, q), hence $\mathbf{\Sigma}_0$; whatever it then leaves on the table is, by this proposition, fourth order or higher. That is what the paper's Proposition 1 turns into a computable floor.

The code parameterises every family by ρ alone, computing $q = 1-\rho^2$ and the family's scale from it (Table 4.1); the theory module `src/markov_score/theory.py` (function `chain_covariance`) keeps q free, because the two-parameter family (ρ, q) is what the Fisher information of the chapter on Fisher information is computed on. For $q \neq 1-\rho^2$ the induction gives $\operatorname{Var}(a_i) = \rho^{2(i-1)} + q(1-\rho^{2(i-1)})/(1-\rho^2)$, drifting from 1 toward $q/(1-\rho^2)$, and $\operatorname{Cov}(a_i, a_j) = \rho^{|i-j|}\operatorname{Var}(a_{\min(i,j)})$; this is what makes $\partial_q \mathbf{\Sigma}_0 \neq 0$, so that q is a second-order coordinate in the paper's sense.

4.5 The covariance matrix and its spectrum

At $q = 1-\rho^2$ the covariance matrix of the chain is the symmetric Toeplitz matrix

$$\mathbf{\Sigma}_0 = \left[\rho^{|i-j|}\right]_{i,j=1}^{L} = \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{L-1} \\ \rho & 1 & \rho & \cdots & \rho^{L-2} \\ \vdots & & \ddots & & \vdots \\ \rho^{L-1} & \rho^{L-2} & \cdots & \rho & 1 \end{pmatrix}, \tag{4.5}$$

known as the **Kac–Murdock–Szegő matrix**. Three of its properties matter.

Determinant and inverse. By the chain rule for densities, $\det \mathbf{\Sigma}_0 = \operatorname{Var}(a_1)\prod_{i\geq 2}\operatorname{Var}(a_i \mid a_{i-1}) = 1 \cdot q^{L-1} = (1-\rho^2)^{L-1}$. The inverse (the **precision matrix**) is tridiagonal,

$$\mathbf{\Sigma}_0^{-1} = \frac{1}{q}\begin{pmatrix} 1 & -\rho & & & \\ -\rho & 1+\rho^2 & -\rho & & \\ & \ddots & \ddots & \ddots & \\ & & -\rho & 1+\rho^2 & -\rho \\ & & & -\rho & 1 \end{pmatrix}, \tag{4.6}$$

which one verifies by multiplying Equation (4.5) by Equation (4.6) row by row (the interior row i of the product has entries $\rho^{|i-j|}(-\rho\cdot\rho^{\mp 1} + (1+\rho^2) - \rho\cdot\rho^{\pm 1})/q$ for $j \neq i$, which vanish, and $(1-$

$\rho^2)/q = 1$ on the diagonal). A banded precision matrix is the linear-algebraic face of the Markov property: for a Gaussian vector, $(\mathbf{\Sigma}^{-1})_{ij} = 0$ means a_i and a_j are conditionally independent given the rest. This is the matrix `GaussianLongRange.precision` in `src/markov_score/priors.py` builds before perturbing it.

Eigenvalue bounds.

Proposition 4.5 (Spectrum of the KMS matrix)**.** *Every eigenvalue λ of $\mathbf{\Sigma}_0$ satisfies*

$$\frac{1-|\rho|}{1+|\rho|} \le \lambda \le \frac{1+|\rho|}{1-|\rho|}, \tag{4.7}$$

and $\lambda_{\max}(\mathbf{\Sigma}_0)$ increases with L to the upper bound.

Proof. Upper bound. For a symmetric matrix the largest eigenvalue is at most the largest absolute row sum (the operator norm is bounded by the ∞-norm), and every row sum of Equation (4.5) is at most $1 + 2\sum_{k\ge1} |\rho|^k = (1+|\rho|)/(1-|\rho|)$. Lower bound. $\mathbf{\Sigma}_0$ is the covariance of the vector $\mathbf{a}$, so for any unit vector $\mathbf{v}$, $\mathbf{v}^\top \mathbf{\Sigma}_0 \mathbf{v} = \mathrm{Var}(\mathbf{v}^\top \mathbf{a})$. Writing $\mathbf{a}$ in terms of the innovations, $\mathbf{v}^\top \mathbf{a} = \sum_m c_m \xi_m$ with $\xi_1 = a_1$, $\xi_m = \varepsilon_m$ independent, and one checks from the moving-average coefficients that this variance is the quadratic form of the Toeplitz matrix with symbol $g(\omega) = (1-\rho^2)/(1 - 2\rho\cos\omega + \rho^2)$; a symmetric Toeplitz form is bounded below by the minimum of its symbol, which is $(1-\rho^2)/(1+|\rho|)^2 = (1-|\rho|)/(1+|\rho|)$. The monotonicity in L holds because the $L \times L$ matrix is a principal submatrix of the $(L+1)\times(L+1)$ one (Cauchy interlacing), and the limit is the maximum of the symbol, $g(0) = (1+\rho)/(1-\rho)$ for $\rho > 0$, by the Szegő theory of Toeplitz forms. □

The expansion parameter and the crossover. The paper's asymptotic theorems expand in e^{-2t}, but for the Gaussian chain the natural parameter is $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0)$: the noised covariance $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I} = \mathbf{I} + e^{-2t}(\mathbf{\Sigma}_0 - \mathbf{I})$ is a perturbation of the identity whose size in operator norm is $e^{-2t}(\lambda_{\max} - 1)$. The **crossover time**

$$t_* = \tfrac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0) \le \tfrac{1}{2}\log\frac{1+\rho}{1-\rho} = \operatorname{artanh}\rho \tag{4.8}$$

is the time at which $e^{-2t}\lambda_{\max} = 1$; below it the longest-wavelength covariance mode is still of order one and the asymptotic rates are not yet visible, above it they are. The bound $\operatorname{artanh}\rho$ is exact as $L \to \infty$; at $L = 32$ the paper's value, a generated macro, is a little smaller. The paper uses t_* as an exact scale for the Gaussian chain (where the chapter on Fisher information shows how it enters the closed-form information) and as a heuristic for the non-Gaussian ones, and reads every "shallower than predicted" slope of the earlier literature as a measurement below t_*. The neighbouring time $\frac{1}{2}\log(\lambda_{\max} - 1)$, at which P_t/γ enters $L^2(\gamma)$ (chapter on Gaussian integrals and Hermite polynomials), is smaller by a little; the two coincide in spirit and the paper names only t_*.

4.6 The fourth-cumulant tensor of the chain

Proposition 4.6 (Fourth-cumulant tensor)**.** *For the chain Equation (4.3) with $\mu = \mathcal{N}(0,1)$ and an innovation with fourth cumulant $\kappa_4(\varepsilon)$, for sites $i, j, k, l \ge 1$,*

$$\kappa_4[ijkl] = \kappa_4(\varepsilon) \sum_{m=2}^{\min(i,j,k,l)} \rho^{\,i+j+k+l-4m}, \tag{4.9}$$

with the empty sum equal to zero (any index equal to 1).

Table 4.1: Variance-q parameterisation of the innovation families, as sampled.

family	draw ε as	parameter from q
Gaussian	$\sqrt{q}\,Z$	—
Laplace	Laplace$(0, b)$	$b = \sqrt{q/2}$
Student-t_ν	$s\,T_\nu$	$s = \sqrt{q(\nu-2)/\nu}$
uniform	Uniform$(-h, h)$	$h = \sqrt{3q}$
mixture, separation κ	$\pm m + sZ$, sign fair	$m = \sqrt{\kappa q}$, $s = \sqrt{(1-\kappa)q}$
generalised Gaussian, β	density $\propto \exp(-\lvert e/s_\beta\rvert^\beta)$	$s_\beta = \sqrt{q\,\Gamma(1/\beta)/\Gamma(3/\beta)}$

Proof. Write each site as a linear combination of the independent variables $\xi_1 = a_1$, $\xi_m = \varepsilon_m$ $(m \geq 2)$: $a_i = \sum_{m=1}^{i} c_{im}\xi_m$ with $c_{i1} = \rho^{i-1}$ and $c_{im} = \rho^{i-m}$ for $2 \leq m \leq i$. By the transformation rule and additivity of cumulant tensors (chapter on moments and cumulants), and because the ξ_m are independent so that only "diagonal" joint cumulants $\operatorname{cum}(\xi_m, \xi_m, \xi_m, \xi_m) = \kappa_4(\xi_m)$ survive,

$$\kappa_4[ijkl] = \sum_m c_{im}c_{jm}c_{km}c_{lm}\,\kappa_4(\xi_m).$$

The Gaussian ξ_1 has $\kappa_4 = 0$; each innovation contributes $\kappa_4(\varepsilon)\rho^{(i-m)+(j-m)+(k-m)+(l-m)}$ for $m \leq \min(i, j, k, l)$, which is Equation (4.9). □

On the diagonal, $\kappa_4[iiii] = \kappa_4(\varepsilon)(1-\rho^{4(i-1)})/(1-\rho^4)$, the site-dependent value already met, which tends to the stationary $\kappa_4(\varepsilon)/(1-\rho^4)$. The Frobenius norm $\|\boldsymbol{\kappa}_4\|_F^2$ grows linearly in L (each site far from the start contributes the same amount), so the paper's score-gap constant $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$ is proportional to the chain length and the paper's theorems are stated at fixed L. The function `chain_kappa4` of `src/markov_score/theory.py` evaluates Equation (4.9) as a geometric sum, and `tests/test_theory_hermite.py` checks six of its entries against brute-force fourth cumulants of sampled Laplace chains, including $\kappa_4[1111] = 0$.

4.7 Sampling a chain

Sampling is the recursion itself: draw $a_1 \sim \mathcal{N}(0, 1)$, then for $i = 2, \ldots, L$ draw ε_i from the innovation law with variance $q = 1 - \rho^2$ and set $a_i = \rho a_{i-1} + \varepsilon_i$. Table 4.1 lists how each family's parameter is derived from q. The samplers are the classes `GaussianAR1`, `LaplaceAR1`, `StudentTAR1`, `UniformAR1` and `GaussianMixtureAR1` of `src/markov_score/priors.py`, and the generalised Gaussian lives in `src/markov_score/shape_kernel.py`. Each class also exposes a method `log_transition_matrix` that evaluates the kernel on the residual matrix of a grid, indexed child first and parent second.

Two details of the implementation are part of the experimental design. First, random streams are keyed by strings and integers through `rng_for` in `src/markov_score/utils.py`, so that the same clean chains and the same noise draws are used by every method compared at a setting; this is what makes the paper's comparisons paired (chapter on the statistics of the experiments). Second, the sampler draws the innovations and the recursion exactly, with no truncation; the grid on which inference runs truncates the domain to $[-A, A]$, and a sampled site beyond it is the mechanism the paper's Appendix B discusses for the heavy-tailed family.

4.8 The chain length L and the number of sequences M

The **chain length** L is the number of sites of one sequence and, since a sequence is a point of $\mathbb{R}^L$, the dimension of the space in which the diffusion runs. The frozen value is $L = 32$; the length study evaluates kernels fitted at 32 on chains of length up to 256, and the regime sweep uses 32 and 128. The **number of sequences** M is the number of independent draws of the

chain in a data set; the sizes are $M \in \{32, 128, 512, 2048\}$ in the comparisons, and the theory's sample-size statements (Fisher information per sequence, the local asymptotic minimax bound) are in the limit $M \to \infty$ at fixed L.

A data set therefore has LM scalar observations, but they are not LM independent numbers: within a sequence the sites are dependent, with the correlation length of $\rho^{|i-j|}$, and the independent unit is the sequence. This is why every information statement in the paper is "per sequence" and why the constant $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$ carries a factor of order L: a longer chain carries more information about its kernel, in proportion to its length, and the same estimator applies to any length because the kernel is a function on $\mathbb{R}^2$ and inference at length L' is the same recursion. The chapter on conventions explains why these two numbers are not called n and d.

4.9 Worked example: the frozen chain at $\rho = 0.85$

Take $\rho = 0.85$, so $q = 1 - 0.7225 = 0.2775$. The first four rows and columns of $\boldsymbol{\Sigma}_0$ are, from Equation (4.5),

$$\begin{pmatrix} 1 & 0.85 & 0.7225 & 0.6141 \\ 0.85 & 1 & 0.85 & 0.7225 \\ 0.7225 & 0.85 & 1 & 0.85 \\ 0.6141 & 0.7225 & 0.85 & 1 \end{pmatrix},$$

with determinant $q^3 = 0.2775^3 \approx 0.02137$ for the 4×4 block. The precision entries are $1/q \approx 3.604$, $(1 + \rho^2)/q \approx 6.207$ and $-\rho/q \approx -3.063$; the interior diagonal 6.207 is the conditional precision of a site given both neighbours, and 3.604 that of an end site given its one neighbour.

The row-sum bound of Proposition 4.5 for the infinite chain is $(1+\rho)/(1-\rho) = 1.85/0.15 \approx 12.33$, and the lower bound is $0.15/1.85 \approx 0.081$; the 4×4 block's largest row sum is $1 + 0.85 + 0.7225 + 0.6141 \approx 3.19$, so at length 4 the largest eigenvalue is at most 3.19, and it grows toward 12.33 with the length. The crossover bound is $\operatorname{artanh}(0.85) = \frac{1}{2}\log 12.33 \approx 1.256$; the paper's t_* at $L = 32$ lies below this and is reported in its Section 3. Already the 2×2 block has eigenvalues $1 \pm \rho$, i.e. 1.85 and 0.15, so $\lambda_{\max} > 2$ from length 3 on, and the Gaussian chain's P_0/γ is outside $L^2(\gamma)$ for every length the paper uses.

For Laplace innovations, $\kappa_4(\varepsilon) = 3q^2 \approx 0.2310$, and Equation (4.9) gives on the diagonal $\kappa_4[2222] \approx 0.231$, $\kappa_4[3333] \approx 0.231(1 + 0.522) \approx 0.352$, $\kappa_4[4444] \approx 0.231(1 + 0.522 + 0.2725) \approx 0.415$, and off the diagonal, for instance, $\kappa_4[2233] = \kappa_4(\varepsilon)\rho^2 \approx 0.167$ (only $m = 2$ contributes, with exponent $2 + 2 + 3 + 3 - 8 = 2$).

Where it is used. The paper's Section 2 (the chain, variance matching, the covariance twin), Section 3 (the crossover t_*, the bound $\lambda_{\max} \le (1 + \rho)/(1 - \rho)$, the $L^2(\gamma)$ remark), Section 5 (the regime sweep in ρ and L, the length study), Appendix A (the proof of Proposition 1, "nothing assumes stationarity") and Appendix B (the kernel on the grid). Experiments E01, E07, E11, E13, E14, E15 and the length rows of E03. Code: `src/markov_score/priors.py` (the chain classes and their samplers), `src/markov_score/shape_kernel.py` (the generalised-Gaussian kernel), `src/markov_score/theory.py` (`chain_covariance`, `chain_kappa4`, `crossover_time`), `experiments/frozen_config.py` (the frozen ρ, L and family), `tests/test_theory_hermite.py`.

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Chapter 5

Fisher information and the Cramér–Rao bound

The paper's information theorems are statements about Fisher information: how much one noised sequence tells an estimator about a coordinate of the clean law, and how that amount decays with the noise level. Its learning section then compares an estimator's dispersion with the Cramér–Rao reference built from that information, and its corollary on sample complexity invokes the local asymptotic minimax theorem of Hájek and Le Cam. This chapter defines the statistical score and keeps it apart from the diffusion score, derives Fisher information and its properties, proves the Cramér–Rao bound, works out the efficient information under nuisance parameters and why a nuisance at another Hermite degree costs nothing to leading order, derives the Gaussian information formula the paper uses in closed form, states local asymptotic normality and the Hájek–Le Cam theorem in plain words and in the form the paper's corollary uses, explains what "maximum likelihood attains it" does and does not say, and derives Fisher's identity for latent-variable models, which is how the paper computes the gradient of its marginal likelihood from one forward–backward pass.

5.1 Two scores

The word **score** has two meanings in this project, and confusing them makes the paper unreadable.

Definition 5.1 (Statistical score)**.** For a parametric family of densities $p_\theta(\mathbf{x})$ with $\theta \in \Theta \subset \mathbb{R}^d$, the **statistical score** at $\mathbf{x}$ is the gradient of the log-likelihood with respect to the *parameter*,

$$\mathbf{s}_\theta(\mathbf{x}) = \nabla_\theta \log p_\theta(\mathbf{x}) \in \mathbb{R}^d. \tag{5.1}$$

Definition 5.2 (Diffusion score)**.** For a density P_t on $\mathbb{R}^L$, the **diffusion score** (also Stein score) at $\mathbf{x}$ is the gradient of the log-density with respect to the *data*,

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) \in \mathbb{R}^L. \tag{5.2}$$

Both are gradients of a log-density; they differ in the variable of differentiation, in their dimension (d against L), and in their use. The statistical score is a random vector whose covariance is the Fisher information and whose sum over a sample is set to zero by maximum likelihood. The diffusion score is a vector field that drives the reverse process and that a network is trained to approximate. The paper writes "the score" for the diffusion score throughout, and "the likelihood score", "per-sequence scores obtained by Fisher's identity" or "$\partial_\theta \log P_t$" for the statistical one. The two meet in one place: the paper's Theorem 2 computes the Fisher information of a parameter of P_0 from the noised law P_t, so the *statistical* score $\partial_\theta \log P_t(\mathbf{x})$ is a function on the same space as the *diffusion* score $\nabla_{\mathbf{x}} \log P_t(\mathbf{x})$. A third, harmless usage, "an arm is scored on the test bundle", means evaluated, and is avoided here.

5.2 The score function and Fisher information

Fix a family p_θ, $\theta \in \Theta$ open in $\mathbb{R}^d$, and write $\mathbb{E}_\theta$ for the expectation under p_θ.

Assumption 5.3 (Regularity)**.** (R1) The support $\{\mathbf{x} : p_\theta(\mathbf{x}) > 0\}$ does not depend on θ. (R2) $\theta \mapsto p_\theta(\mathbf{x})$ is continuously differentiable for every $\mathbf{x}$. (R3) Differentiation with respect to θ may be taken under the integral sign in $\int p_\theta \,\mathrm{d}\mathbf{x}$ and in $\int T(\mathbf{x}) p_\theta(\mathbf{x}) \,\mathrm{d}\mathbf{x}$ for the statistics T considered (a dominating integrable bound on $\|\nabla_\theta p_\theta\|$ locally in θ suffices).

Proposition 5.4 (The score has mean zero)**.** *Under Assumption 5.3,* $\mathbb{E}_\theta[\mathbf{s}_\theta(\mathbf{x})] = \mathbf{0}$.

Proof. $\mathbb{E}_\theta[\mathbf{s}_\theta] = \int \frac{\nabla_\theta p_\theta}{p_\theta} p_\theta \,\mathrm{d}\mathbf{x} = \int \nabla_\theta p_\theta \,\mathrm{d}\mathbf{x} = \nabla_\theta \int p_\theta \,\mathrm{d}\mathbf{x} = \nabla_\theta 1 = \mathbf{0}$, using (R1) so that the integrand is defined on the support and (R3) to exchange ∇_θ and $\int$. □

Definition 5.5 (Fisher information)**.** The **Fisher information matrix** of the family at θ is the covariance of the score,

$$\mathbf{I}(\theta) = \mathbb{E}_\theta\big[\mathbf{s}_\theta(\mathbf{x})\mathbf{s}_\theta(\mathbf{x})^\top\big], \qquad I_{jk}(\theta) = \mathbb{E}_\theta\big[\partial_j \log p_\theta \, \partial_k \log p_\theta\big]. \tag{5.3}$$

For a scalar parameter it is the number $I(\theta) = \mathbb{E}_\theta[(\partial_\theta \log p_\theta)^2]$.

The paper's $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ is this definition for the family $\{P_t^\theta\}$ of noised laws at fixed t, one sequence being one observation.

Proposition 5.6 (Four properties)**.** *Under Assumption 5.3, with second derivatives where stated:*

(i) (second-derivative form) if $\theta \mapsto p_\theta$ is twice differentiable and (R3) holds for the second derivatives, $I_{jk}(\theta) = -\mathbb{E}_\theta[\partial_j \partial_k \log p_\theta]$;

(ii) (additivity) for M independent observations with a common θ the information is $M\,\mathbf{I}(\theta)$; for independent observations from different families it is the sum of their informations;

(iii) (reparametrisation) if $\theta = g(\psi)$ with g differentiable and Jacobian $\mathbf{J} = \partial\theta/\partial\psi$, then $\mathbf{I}_\psi(\psi) = \mathbf{J}^\top \mathbf{I}_\theta(g(\psi))\mathbf{J}$; for scalars, $I_\psi = (g'(\psi))^2 I_\theta$;

(iv) (data processing) if $\mathbf{y} = T(\mathbf{x})$ is a statistic, the information carried by $\mathbf{y}$ about θ is at most that carried by $\mathbf{x}$, in the Loewner order.

Proof. (i) Differentiate $\partial_j \log p = \partial_j p / p$ once more: $\partial_j \partial_k \log p = \partial_j \partial_k p / p - (\partial_j p)(\partial_k p)/p^2$, and $\mathbb{E}_\theta[\partial_j \partial_k p / p] = \int \partial_j \partial_k p \,\mathrm{d}\mathbf{x} = \partial_j \partial_k 1 = 0$ by (R3); the second term's expectation is $-I_{jk}$. (ii) The log-likelihood of independent observations is a sum, so the score is a sum of independent mean-zero vectors and its covariance is the sum of their covariances. (iii) By the chain rule $\nabla_\psi \log p = \mathbf{J}^\top \nabla_\theta \log p$, and covariances transform as $\mathbf{J}^\top \operatorname{Cov} \mathbf{J}$. (iv) Write $p_\theta(\mathbf{x}) = p_\theta(\mathbf{y})\, p_\theta(\mathbf{x} \mid \mathbf{y})$; the score of $\mathbf{x}$ is the score of $\mathbf{y}$ plus the score of the conditional law, and the two are uncorrelated because the conditional score has conditional mean zero given $\mathbf{y}$ (Proposition 5.4 applied to the conditional family); so $\mathbf{I}_\mathbf{x} = \mathbf{I}_\mathbf{y} + \mathbb{E}[\mathbf{I}_{\mathbf{x}|\mathbf{y}}] \succeq \mathbf{I}_\mathbf{y}$. □

Why it matters here. Three of these are used by the paper without comment. Additivity is why the bound for M sequences is $1/(MI_t)$. Reparametrisation is how the paper's Cramér–Rao reference for the excess kurtosis is obtained "through the chain rule at the Laplace member of the one-parameter shape family": the information about β is computed, and the information about $\gamma_2 = \kappa_4/q^2$ as a function of β is $I_\beta / (\,\mathrm{d}\gamma_2/\,\mathrm{d}\beta)^2$, so the reference standard deviation of an unbiased estimator of γ_2 is $|\,\mathrm{d}\gamma_2/\,\mathrm{d}\beta| / \sqrt{M I_\beta}$. Data processing is why the information of the noised sequence is at most that of the clean one, and why the "clean-data floor" of the paper's Figure 3 is a floor.

5.3 The Cramér–Rao bound

Theorem 5.7 (Cramér–Rao, scalar)**.** *Let θ be scalar, let Assumption 5.3 hold, let $0 < I(\theta) < \infty$, and let $T(\mathbf{x})$ be an estimator with $\mathbb{E}_\theta[T^2] < \infty$ and bias $b(\theta) = \mathbb{E}_\theta[T] - \theta$ differentiable with (R3). Then*

$$\operatorname{Var}_\theta(T) \;\geq\; \frac{\big(1 + b'(\theta)\big)^2}{I(\theta)}, \qquad \text{and in particular} \qquad \operatorname{Var}_\theta(T) \geq \frac{1}{I(\theta)} \text{ if } T \text{ is unbiased.} \tag{5.4}$$

Proof. Compute the covariance of T with the score. Since the score has mean zero,

$$\operatorname{Cov}_\theta(T, s_\theta) = \mathbb{E}_\theta[T\, s_\theta] = \int T(\mathbf{x})\, \frac{\partial_\theta p_\theta(\mathbf{x})}{p_\theta(\mathbf{x})}\, p_\theta(\mathbf{x})\, \mathrm{d}\mathbf{x} = \partial_\theta \int T(\mathbf{x}) p_\theta(\mathbf{x})\, \mathrm{d}\mathbf{x} = \partial_\theta \mathbb{E}_\theta[T] = 1 + b'(\theta),$$

using (R3) for the exchange. The Cauchy–Schwarz inequality for covariances, $\operatorname{Cov}(T, s)^2 \leq \operatorname{Var}(T) \operatorname{Var}(s)$, with $\operatorname{Var}_\theta(s_\theta) = I(\theta)$, gives $(1 + b'(\theta))^2 \leq \operatorname{Var}_\theta(T)\, I(\theta)$, which is Equation (5.4). □

Equality in Cauchy–Schwarz holds if and only if $T - \mathbb{E}T$ is a multiple of the score, $T(\mathbf{x}) - \theta = c(\theta)\, s_\theta(\mathbf{x})$; an estimator that achieves the bound at every θ is called **efficient**, and exists only for exponential families in their mean parameterisation. The bound is a property of the family, not of any estimator; that is why the paper calls it a *reference* and not a target.

Theorem 5.8 (Cramér–Rao, vector)**.** *Let $\theta \in \mathbb{R}^d$, let Assumption 5.3 hold with $\mathbf{I}(\theta)$ finite and non-singular, and let $\mathbf{T}(\mathbf{x}) \in \mathbb{R}^d$ be unbiased for θ with finite second moments. Then $\operatorname{Cov}_\theta(\mathbf{T}) \succeq \mathbf{I}(\theta)^{-1}$ in the Loewner order, i.e. $\mathbf{v}^\top \operatorname{Cov}_\theta(\mathbf{T})\mathbf{v} \geq \mathbf{v}^\top \mathbf{I}(\theta)^{-1}\mathbf{v}$ for every $\mathbf{v}$.*

Proof. As in the scalar case, $\operatorname{Cov}_\theta(\mathbf{T}, \mathbf{s}_\theta) = \mathbb{E}_\theta[\mathbf{T}\mathbf{s}_\theta^\top] = \nabla_\theta \mathbb{E}_\theta[\mathbf{T}]^\top = \mathbf{I}$. The joint covariance of the stacked vector $(\mathbf{T}, \mathbf{s}_\theta)$ is $\left(\begin{smallmatrix} \operatorname{Cov}(\mathbf{T}) & \mathbf{I} \\ \mathbf{I} & \mathbf{I} \end{smallmatrix}\right) \succeq 0$, and a block matrix $\left(\begin{smallmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{smallmatrix}\right)$ with $\mathbf{C} \succ 0$ is positive semidefinite if and only if its Schur complement $\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\top \succeq 0$; here that is $\operatorname{Cov}(\mathbf{T}) - \mathbf{I}^{-1} \succeq 0$. □

5.4 Nuisance parameters and the efficient information

Partition $\theta = (\psi, \eta)$ into a coordinate of interest ψ (scalar, say) and nuisance parameters η, and partition the information accordingly, $\mathbf{I} = \left(\begin{smallmatrix} I_{\psi\psi} & \mathbf{I}_{\psi\eta} \\ \mathbf{I}_{\eta\psi} & \mathbf{I}_{\eta\eta} \end{smallmatrix}\right)$. By Theorem 5.8 the variance of an unbiased estimator of ψ that must also estimate η is bounded below by $(\mathbf{I}^{-1})_{\psi\psi}$, and by the block inverse formula

$$(\mathbf{I}^{-1})_{\psi\psi} = \frac{1}{I_\psi^{\mathrm{eff}}}, \qquad I_\psi^{\mathrm{eff}} = I_{\psi\psi} - \mathbf{I}_{\psi\eta}\mathbf{I}_{\eta\eta}^{-1}\mathbf{I}_{\eta\psi}, \tag{5.5}$$

the Schur complement. I_ψ^{eff} is the **efficient information** about ψ: the information that survives after the nuisance has been estimated. It is at most $I_{\psi\psi}$, with equality if and only if the scores of ψ and of η are uncorrelated. The ratio $I_{\psi\psi}/I_\psi^{\mathrm{eff}} \geq 1$ is the **nuisance cost**, the quantity `summarise` in `src/markov_score/information.py` reports. Equivalently, I_ψ^{eff} is the variance of the **efficient score**, the residual of the ψ-score after its linear regression on the η-score.

Proposition 5.9 (Nuisance at another Hermite degree)**.** *Suppose that, as $t \to \infty$ and with $\alpha = e^{-t}$, the scores of two coordinates of the noised family have expansions $s_\psi = \alpha^m A(\mathbf{x}) + \alpha^{m+2} A_2(\mathbf{x}) + \dots$ and $s_\eta = \alpha^{m'} B(\mathbf{x}) + \alpha^{m'+2} B_2(\mathbf{x}) + \dots$ with A, B polynomials that are γ-orthogonal, $\mathbb{E}_\gamma[AB] = 0$, and with $I_{\psi\psi} = \alpha^{2m}\mathbb{E}_\gamma[A^2](1 + O(\alpha^2))$, $I_{\eta\eta} = \alpha^{2m'}\mathbb{E}_\gamma[B^2](1 + O(\alpha^2))$, $\mathbb{E}_\gamma[B^2] > 0$, and $P_t = \gamma(1 + O(\alpha^2))$ on the relevant moments. Then $I_{\psi\eta} = O(\alpha^{m+m'+2})$ and*

$$I_\psi^{\mathrm{eff}} = I_{\psi\psi}(1 + O(\alpha^4)) = \alpha^{2m}\mathbb{E}_\gamma[A^2](1 + O(\alpha^2)) : \tag{5.6}$$

the efficient information has the same rate and the same leading constant as the raw information. If instead $m = m'$ and $\mathbb{E}_\gamma[AB] \neq 0$, the correction is of the same order as $I_{\psi\psi}$ and the efficient information is smaller by an $O(1)$ factor.

Proof. $I_{\psi\eta} = \mathbb{E}_{P_t}[s_\psi s_\eta] = \alpha^{m+m'}\mathbb{E}_{P_t}[AB] + O(\alpha^{m+m'+2})$, and $\mathbb{E}_{P_t}[AB] = \mathbb{E}_\gamma[AB] + O(\alpha^2) = O(\alpha^2)$. Hence $I_{\psi\eta}^2/I_{\eta\eta} = O(\alpha^{2m+2m'+4})/\Theta(\alpha^{2m'}) = O(\alpha^{2m+4})$, which is $O(\alpha^4)$ relative to $I_{\psi\psi} = \Theta(\alpha^{2m})$. In the same-degree case the cross term is $\alpha^{2m}\mathbb{E}_\gamma[AB] \neq 0$ at leading order and the Schur complement subtracts a fixed fraction. □

Why it matters here. This is the paper's sentence "the leading orders of a degree-2 and a degree-4 coordinate do not couple, because He_2 and He_4 are γ-orthogonal, so the efficient information of a shape coordinate after lower-degree nuisance coordinates are estimated has the same rate and constant as its raw information; a nuisance of the same degree costs an $O(1)$ factor". For the generalised-Gaussian chain, $\psi = \beta$ has $m = 4$ with A a degree-four Hermite polynomial, and $\eta = (\rho, q)$ have $m' = 2$ with B a degree-two one. The proposition says the e^{-8t} law and the constant $\frac{1}{24}\|\partial_\beta \boldsymbol{\kappa}_4\|_F^2$ of the paper's Theorem 2 hold for the efficient information that the paper's Figure 2b actually plots, not only for the diagonal entry. It also says why the eight-component mixture's shape coordinates, which are all of degree four, cost each other an $O(1)$ factor, one reason the paper calls its one-parameter reference a yardstick.

The paper measures the information matrix of (ρ, q, β) by the **outer-product estimator**: with per-sequence scores $\mathbf{g}_\mu = \nabla_\theta \log P_t(\mathbf{x}^\mu)$ at the true θ, $\hat{\mathbf{I}} = \frac{1}{M}\sum_\mu \mathbf{g}_\mu \mathbf{g}_\mu^\top$, which is unbiased for $\mathbf{I}$ by Equation (5.3) (it is not the observed information $-\nabla^2 \log p$, which would need second derivatives). The function `outer_product_information` of `src/markov_score/information.py` forms it, and `efficient_information` inverts it coordinate by coordinate as in Equation (5.5). Its own sampling error is what limits the paper's chain verification of the shape constant; the scalar constant is verified by quadrature instead (E21).

5.5 The Gaussian family in closed form

Proposition 5.10 (Fisher information of a zero-mean Gaussian family)**.** *Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}(\theta))$ on $\mathbb{R}^L$ with $\boldsymbol{\Sigma}(\theta)$ positive definite and differentiable in $\theta \in \mathbb{R}^d$. Then*

$$I_{jk}(\theta) = \tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma}\,\boldsymbol{\Sigma}^{-1}\partial_k\boldsymbol{\Sigma}). \tag{5.7}$$

Proof. The log-density is $\log p_\theta(\mathbf{x}) = -\frac{L}{2}\log 2\pi - \frac{1}{2}\log\det\boldsymbol{\Sigma} - \frac{1}{2}\mathbf{x}^\top\boldsymbol{\Sigma}^{-1}\mathbf{x}$. Using $\partial_j \log\det\boldsymbol{\Sigma} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma})$ and $\partial_j\boldsymbol{\Sigma}^{-1} = -\boldsymbol{\Sigma}^{-1}(\partial_j\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}$,

$$\partial_j \log p_\theta(\mathbf{x}) = -\tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma}) + \tfrac{1}{2}\mathbf{x}^\top\boldsymbol{\Sigma}^{-1}(\partial_j\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}\mathbf{x} = \tfrac{1}{2}\big(\mathbf{x}^\top\mathbf{B}_j\mathbf{x} - \mathbb{E}[\mathbf{x}^\top\mathbf{B}_j\mathbf{x}]\big),$$

with $\mathbf{B}_j = \boldsymbol{\Sigma}^{-1}(\partial_j\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}$ symmetric, since $\mathbb{E}[\mathbf{x}^\top\mathbf{B}\mathbf{x}] = \operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma}) = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma})$. The information is the covariance of these centred quadratic forms. For $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ and symmetric $\mathbf{B}$, $\mathbf{C}$, Isserlis' theorem (chapter on Gaussian integrals and Hermite polynomials) gives $\mathbb{E}[\mathbf{x}^\top\mathbf{B}\mathbf{x}\,\mathbf{x}^\top\mathbf{C}\mathbf{x}] = \operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma})\operatorname{tr}(\mathbf{C}\boldsymbol{\Sigma}) + 2\operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma}\mathbf{C}\boldsymbol{\Sigma})$, so $\operatorname{Cov}(\mathbf{x}^\top\mathbf{B}\mathbf{x}, \mathbf{x}^\top\mathbf{C}\mathbf{x}) = 2\operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma}\mathbf{C}\boldsymbol{\Sigma})$. With the factors $\frac{1}{2}\cdot\frac{1}{2}$, $I_{jk} = \frac{1}{4}\cdot 2\operatorname{tr}(\mathbf{B}_j\boldsymbol{\Sigma}\mathbf{B}_k\boldsymbol{\Sigma}) = \frac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1}\partial_k\boldsymbol{\Sigma})$. □

The noised Gaussian chain. For the Gaussian chain of the chapter on Markov chains, $\mathbf{x}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t(\theta))$ exactly, with $\boldsymbol{\Sigma}_t = e^{-2t}\boldsymbol{\Sigma}_0(\theta) + \Delta_t\mathbf{I}$ and $\theta = (\rho, q)$, so $\partial_\theta\boldsymbol{\Sigma}_t = e^{-2t}\partial_\theta\boldsymbol{\Sigma}_0$ and Equation (5.7) gives the paper's closed form

$$I_t(\theta) = \tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_t\,\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_t) = \tfrac{1}{2}e^{-4t}\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_0\,\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_0), \tag{5.8}$$

exact at every t and every ρ; this is the curve of the paper's Figure 2a and the function `gaussian_chain_fisher` of `src/markov_score/theory.py`, which evaluates $\partial_\theta\boldsymbol{\Sigma}_0$ by central differences. Its large-t limit is immediate: $\boldsymbol{\Sigma}_t \to \mathbf{I}$, so

$$e^{4t}I_t(\theta) \;\longrightarrow\; \tfrac{1}{2}\operatorname{tr}(\partial_\theta\boldsymbol{\Sigma}_0\,\partial_\theta\boldsymbol{\Sigma}_0) = \tfrac{1}{2}\|\partial_\theta\boldsymbol{\Sigma}_0\|_F^2, \tag{5.9}$$

the constant of the paper's equation for a covariance coordinate, checked in Lean as target 3 (`lean/README.md`). The approach to the limit is governed by the spectrum of $\mathbf{\Sigma}_0$: writing $\mathbf{\Sigma}_t = \mathbf{I} + e^{-2t}(\mathbf{\Sigma}_0 - \mathbf{I})$, the Neumann series $\mathbf{\Sigma}_t^{-1} = \mathbf{I} - e^{-2t}(\mathbf{\Sigma}_0 - \mathbf{I}) + e^{-4t}(\mathbf{\Sigma}_0 - \mathbf{I})^2 - \dots$ converges in operator norm exactly when $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0 - \mathbf{I}) < 1$, i.e. $e^{2t} > \lambda_{\max}(\mathbf{\Sigma}_0) - 1$, and its first correction is of relative size $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0)$: this is the origin of the paper's $\epsilon_t = O(e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0))$ and of the crossover $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ below which the measured slope is shallower than -4. The same formula gives the off-diagonal entry $I_{\rho q}$, hence the correlation between the two scores and the efficient information of each, which the paper's Figure 2a uses; the efficient slope is already close to -4 below t_* where the raw diagonal is not, because the nuisance coupling of ρ with q is strong there.

5.6 Local asymptotic normality and the Hájek–Le Cam theorem

Fisher information bounds the variance of *unbiased* estimators. Nothing in the Cramér–Rao bound stops a biased estimator from having a smaller mean squared error at some θ (the constant estimator has zero variance). The statement that $1/I$ is nevertheless the right yardstick for *every* estimator is asymptotic and local, and it is the theorem the paper's corollary on sample complexity invokes.

In plain words. Take M independent sequences from p_θ and look at parameter values within a $1/\sqrt{M}$ neighbourhood of θ, the scale on which the data can distinguish them. As $M \to \infty$ the whole experiment, restricted to that neighbourhood, looks like observing one Gaussian vector with mean $\mathbf{h}$ (the rescaled parameter offset) and covariance $\mathbf{I}(\theta)^{-1}$. In that Gaussian experiment no estimator of $\mathbf{h}$ has worst-case risk below that of the observation itself, whose covariance is $\mathbf{I}^{-1}$. So no estimator sequence can beat $\mathbf{I}(\theta)^{-1}/M$ uniformly over a shrinking neighbourhood, however biased, and the maximum-likelihood estimator, which becomes that Gaussian observation, attains it.

Definition 5.11 (Local asymptotic normality)**.** The family is **locally asymptotically normal** (LAN) at θ with information $\mathbf{I}(\theta)$ if, for every $\mathbf{h} \in \mathbb{R}^d$, the log-likelihood ratio of M independent observations satisfies

$$\log \prod_{\mu=1}^{M} \frac{p_{\theta+\mathbf{h}/\sqrt{M}}(\mathbf{x}^\mu)}{p_\theta(\mathbf{x}^\mu)} = \mathbf{h}^\top \mathbf{\Delta}_M - \tfrac{1}{2}\mathbf{h}^\top \mathbf{I}(\theta)\mathbf{h} + o_p(1), \qquad \mathbf{\Delta}_M = \frac{1}{\sqrt{M}}\sum_\mu \mathbf{s}_\theta(\mathbf{x}^\mu) \Rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I}(\theta))\,, \tag{5.10}$$

under p_θ. A sufficient condition is that the family be **differentiable in quadratic mean** at θ: $\sqrt{p_{\theta+\mathbf{h}}} = \sqrt{p_\theta} + \frac{1}{2}\mathbf{h}^\top \mathbf{s}_\theta \sqrt{p_\theta} + r_\mathbf{h}$ with $\int r_\mathbf{h}^2 = o(|\mathbf{h}|^2)$; a continuously differentiable log-density with continuous finite information suffices for that.

Theorem 5.12 (Hájek–Le Cam local asymptotic minimax)**.** *Let the family be LAN at θ with non-singular $\mathbf{I}(\theta)$, let $\ell : \mathbb{R}^d \to [0, \infty)$ be bowl-shaped (its sublevel sets convex and symmetric), and let $\mathbf{T}_M$ be any sequence of estimators. Then*

$$\lim_{c\to\infty} \liminf_{M\to\infty} \sup_{|\mathbf{h}|\le c} \mathbb{E}_{\theta+\mathbf{h}/\sqrt{M}}\, \ell\big(\sqrt{M}(\mathbf{T}_M - \theta - \mathbf{h}/\sqrt{M})\big) \;\ge\; \mathbb{E}\,\ell(\mathbf{Z}), \qquad \mathbf{Z} \sim \mathcal{N}\Big(\mathbf{0}, \mathbf{I}(\theta)^{-1}\Big)\,. \tag{5.11}$$

With the quadratic loss $\ell(\mathbf{u}) = (v^\top \mathbf{u})^2$ for a coordinate direction, $\mathbb{E}\ell(\mathbf{Z}) = (\mathbf{I}^{-1})_{\psi\psi} = 1/I_\psi^{\mathrm{eff}}$, and Equation (5.11) reads: for any estimator sequence $\hat{\psi}_M$,

$$\lim_{c\to\infty} \liminf_{M\to\infty} \sup_{|\theta'-\theta|\le c/\sqrt{M}} M\,\mathbb{E}_{\theta'}(\hat{\psi}_M - \psi')^2 \;\ge\; \frac{1}{I_\psi^{\mathrm{eff}}(\theta)}. \tag{5.12}$$

This is the statement the paper's corollary uses, with I^{eff} the efficient information of the paper's general-order theorem at fixed t, so that the right-hand side is $\Theta(e^{2mt})$ for a coordinate of lowest

Hermite degree m. Two features of the exact statement deserve attention. The supremum over a shrinking neighbourhood is essential: without it a superefficient estimator can beat $1/I$ at isolated points. And the outer limit $c \to \infty$ is part of the theorem; for a fixed c the local minimax risk of the limiting Gaussian experiment over $|\mathbf{h}| \le c$ is strictly below $\mathbb{E}\ell(\mathbf{Z})$ (a bounded normal-mean problem has a shrinkage estimator with smaller worst-case risk), and the bound is approached as c grows. The paper's corollary writes the bound with a fixed c and without the outer limit; read literally that is slightly stronger than Equation (5.12), and the intended statement is Equation (5.12).

What "maximum likelihood attains it" means. Under the regularity that gives LAN plus conditions for consistency, the maximum-likelihood estimator satisfies $\sqrt{M}(\hat{\theta}_M - \theta) \Rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I}(\theta)^{-1})$ under p_θ, locally uniformly, and is **regular** in the sense of the convolution theorem; so its rescaled risk converges to $\mathbb{E}\ell(\mathbf{Z})$ and equality holds in Equation (5.11). Three things this does *not* say, all named by the paper. (i) It is a statement at fixed t as $M \to \infty$; the LAN approximation needs $M \gg 1/I_t$, i.e. $M \gg e^{2mt}$, so the limits $M \to \infty$ and $t \to \infty$ do not commute and nothing is claimed uniformly in t. (ii) It is a statement about *local* precision, on the scale $1/\sqrt{MI_t}$; once that scale exceeds the range of a bounded coordinate (the excess kurtosis of the mixture family is bounded below by -2 and the paper's shape coordinate has range 3), the neighbourhood $|\theta' - \theta| \le c/\sqrt{M}$ is no longer inside the parameter space and the statement is empty; the paper reads its estimator's collapse at that level as the information running out, not as a failure of the estimator. (iii) It says nothing about an optimiser stopped at a finite budget, nor about an estimator whose iteration count is selected on validation.

5.7 Why "at the bound" is a yardstick and not an efficiency theorem

The paper compares, level by level, the dispersion across sixteen replicates of its fitted correlation and fitted excess kurtosis with the Cramér–Rao reference $1/\sqrt{MI_t^{\mathrm{eff}}}$. Four facts make that comparison a yardstick.

1. The reference is computed in a *one-parameter* shape family (the generalised Gaussian at its Laplace member, with (ρ, q) as lower-degree nuisances that cost nothing to leading order, Proposition 5.9). The estimator fits an *eight-component mixture*, whose shape coordinates are all of degree four and cost each other $O(1)$ factors. The two families share the Laplace law as a point but not their tangent spaces, so the reference is not the bound for the estimator's own family.

2. The Cramér–Rao bound is for *unbiased* estimators, and Theorem 5.7 shows that a bias with $b'(\theta) < 0$ lowers the bound by $(1 + b')^2$. Early stopping selected on validation is a shrinkage device: it pulls the fitted shape toward the Gaussian innovation as the information falls. A dispersion *below* the unbiased reference, which the paper reports at some levels, is therefore not a contradiction; it is the signature of shrinkage, and the paper reports bias and dispersion separately for that reason.

3. $M = 512$ is fixed and moderate, whereas the theorem is asymptotic. At the levels where $1/\sqrt{MI_t}$ is a sizeable fraction of the coordinate's range, the local statement is empty (Section 5.6, point (ii)).

4. The reported dispersion is a sixteen-replicate standard deviation, whose own relative sampling error is about $1/\sqrt{2 \cdot 15} \approx 18\%$ under normality (chapter on the statistics of the experiments); "within the sampling error of a sixteen-replicate standard deviation" is the most that can be claimed.

The defensible statement, and the one the paper makes, is that at the low levels the fits are unimodal and their dispersion is consistent with the one-parameter reference, that collapse to a Gaussian innovation begins at a stated level, and that it is total at the level where the reference crosses the coordinate's range.

5.8 Fisher's identity for latent-variable models

The paper's estimator maximises the marginal likelihood of the noised sequences, $\mathcal{L}(\theta) = \sum_\mu \log P_t[t_\mu, \theta](\mathbf{x}^\mu)$, where each term is an integral over the unobserved clean chain. Its gradient is obtained without differentiating through the integral, by an identity due to Fisher.

Proposition 5.13 (Fisher's identity)**.** *Let $p_\theta(\mathbf{a}, \mathbf{x})$ be a joint density of a latent $\mathbf{a}$ and an observation $\mathbf{x}$, with marginal $p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{a}, \mathbf{x})\,\mathrm{d}\mathbf{a}$, differentiable in θ with (R3). Then*

$$\nabla_\theta \log p_\theta(\mathbf{x}) = \mathbb{E}_\theta\big[\nabla_\theta \log p_\theta(\mathbf{a}, \mathbf{x}) \,\big|\, \mathbf{x}\big], \tag{5.13}$$

the posterior expectation of the complete-data score.

Proof.

$$\nabla_\theta \log p_\theta(\mathbf{x}) = \frac{\nabla_\theta \int p_\theta(\mathbf{a}, \mathbf{x})\,\mathrm{d}\mathbf{a}}{p_\theta(\mathbf{x})} = \int \frac{\nabla_\theta p_\theta(\mathbf{a}, \mathbf{x})}{p_\theta(\mathbf{a}, \mathbf{x})} \cdot \frac{p_\theta(\mathbf{a}, \mathbf{x})}{p_\theta(\mathbf{x})}\,\mathrm{d}\mathbf{a} = \int \nabla_\theta \log p_\theta(\mathbf{a}, \mathbf{x})\, p_\theta(\mathbf{a} \mid \mathbf{x})\,\mathrm{d}\mathbf{a},$$

using (R3) to move the gradient inside the integral and $p_\theta(\mathbf{a} \mid \mathbf{x}) = p_\theta(\mathbf{a}, \mathbf{x})/p_\theta(\mathbf{x})$. □

Equivalently, with the EM auxiliary function $\mathcal{Q}(\theta \mid \theta') = \mathbb{E}_{\theta'}[\log p_\theta(\mathbf{a}, \mathbf{x}) \mid \mathbf{x}]$, Equation (5.13) says $\nabla_\theta \mathcal{Q}(\theta \mid \theta')|_{\theta=\theta'} = \nabla_\theta \log p_{\theta'}(\mathbf{x})$: the gradient of the EM surrogate at the current point is the gradient of the marginal log-likelihood, which is why EM's ascent directions are likelihood ascent directions (chapter on maximum likelihood and EM).

For the chain. The complete-data density is Equation (4.2) of the chapter on Markov chains times the channel, $p_\theta(\mathbf{a}, \mathbf{x}) = \mu(a_1) \prod_{i \geq 2} K_\theta(a_i \mid a_{i-1}) \prod_i \ell_t(x_i \mid a_i)$, and only the transition factors depend on θ. So

$$\begin{aligned} \nabla_\theta \log p_\theta(\mathbf{x}) &= \sum_{i \geq 2} \mathbb{E}_\theta\big[\nabla_\theta \log K_\theta(a_i \mid a_{i-1}) \,\big|\, \mathbf{x}\big] \\ &= \sum_{i \geq 2} \iint \nabla_\theta \log K_\theta(a' \mid a)\, p_\theta(a_i = a', a_{i-1} = a \mid \mathbf{x})\,\mathrm{d}a\,\mathrm{d}a', \end{aligned} \tag{5.14}$$

a sum of integrals against the **pairwise posterior marginals** of consecutive sites, which the forward–backward recursion returns in the same pass that computes the site marginals. On the grid, with $\boldsymbol{\Xi}_{kl} = \sum_\mu \sum_{i \geq 2} p(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu)$ the expected transition mass (child first, parent second), the gradient of the whole marginal log-likelihood is the inner product $\langle \boldsymbol{\Xi}, \nabla_\theta \log K_\theta \rangle$ of the paper's equation (4). The same $\boldsymbol{\Xi}$ is all the M-step of EM needs. The identity is verified in the paper against finite differences of the grid evidence to 10^{-9}, and, independently of the recursion, `tests/test_fisher_bruteforce.py` enumerates every latent path on a five-point grid and checks the evidence, the orientation of $\boldsymbol{\Xi}$ (a transposed $\boldsymbol{\Xi}$ paired with a transposed kernel would pass a finite-difference check), and the gradient against a finite difference of the enumerated evidence. The per-sequence scores $\mathbf{g}_\mu$ that feed the outer-product information estimator of Section 5.4 are computed by the batched version of the same identity in `src/markov_score/information.py` (`per_chain_scores`), one term per edge, without materialising a per-sequence $N_\mathrm{g} \times N_\mathrm{g}$ matrix.

5.9 Worked example: the variance of one noised Gaussian coordinate

Let $x \sim \mathcal{N}(0, \theta)$, one scalar observation, $\theta > 0$. The log-density is $-\frac{1}{2}\log(2\pi\theta) - x^2/(2\theta)$, the score is $-1/(2\theta) + x^2/(2\theta^2)$, and since $\mathbb{E}x^2 = \theta$ and $\mathrm{Var}(x^2) = 2\theta^2$,

$$I(\theta) = \mathrm{Var}\Big(\frac{x^2}{2\theta^2}\Big) = \frac{2\theta^2}{4\theta^4} = \frac{1}{2\theta^2},$$

which agrees with Equation (5.7): $\frac{1}{2}\,\mathrm{tr}(\theta^{-1}\cdot 1\cdot\theta^{-1}\cdot 1) = 1/(2\theta^2)$. The Cramér–Rao bound for M unbiased observations is $2\theta^2/M$, attained by the sample second moment $\frac{1}{M}\sum x_\mu^2$, whose variance is exactly $2\theta^2/M$: an efficient estimator, as the exponential-family remark after Theorem 5.7 predicts.

Now noise the coordinate: $x_t = e^{-t}a + \sqrt{\Delta_t}z$ with $a \sim \mathcal{N}(0,\theta)$, so $x_t \sim \mathcal{N}(0, v_t)$ with $v_t = e^{-2t}\theta + 1 - e^{-2t}$ and $\partial_\theta v_t = e^{-2t}$. By Equation (5.7),

$$I_t(\theta) = \frac{(\partial_\theta v_t)^2}{2v_t^2} = \frac{e^{-4t}}{2\big(1 + e^{-2t}(\theta - 1)\big)^2}.$$

At $\theta = 1$ the denominator is exactly 1 at every t, so $I_t(1) = \frac{1}{2}e^{-4t}$ *exactly*: the constant $\frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2 = \frac{1}{2}$ of the paper's covariance-coordinate law, with no correction, because a unit-variance coordinate keeps unit variance under the channel. This is the "$I_t e^{4t}$ equal to one half" that the paper's Appendix E checks for the variance coordinate of the scalar law, and that `tests/test_theory_hermite.py` checks by quadrature for the Laplace law as well (the paper's Theorem 2 says the constant is the same for every law with an exponential moment). At $\theta \neq 1$ the factor $(1+e^{-2t}(\theta-1))^{-2}$ is the scalar version of the crossover: for $\theta = 5$ the local log-slope of I_t is $-4+4\cdot 4e^{-2t}/(1+4e^{-2t})$, about -1.6 at $t = 0.5$, -3.3 at $t = 1.5$ and -4.0 at $t = 3$, with the crossover $\frac{1}{2}\log(\theta - 1) = \frac{1}{2}\log 4 \approx 0.69$ at which the correction term equals one. With $M = 512$ sequences at $t = 1$ and $\theta = 1$, the reference standard deviation is $\sqrt{2e^4/512} \approx 0.46$, more than the range of a coordinate confined to an interval of length one: at that level the local theory has nothing to say about such a coordinate, which is the content of point (ii) in Section 5.6.

Where it is used. The paper's Section 3 (the definition of $I_t(\theta)$, the closed-form Gaussian information, the efficient-information sentence, Theorem 3 on the general order, Corollary 4 and the three caveats after it), Section 4 (the Cramér–Rao reference, "a yardstick, not an efficiency theorem", Fisher's identity and equation (4)), Appendix A (the proof of the corollary and the nuisance-projection argument), Appendix C (the finite-difference verification) and Appendix E (E21, the scalar constants). Experiments E01, E06, E12b, E19, E21. Code: `src/markov_score/theory.py` (`gaussian_chain_fisher`, `correlation_information_constant`, `shape_information_constant`), `src/markov_score/information.py` (`per_chain_scores`, `outer_product_information`, `efficient_information`, `summarise`), `src/markov_score/shape_kernel.py` (the shape coordinate and its gradient), `tests/test_fisher_bruteforce.py`, `tests/test_theory_hermite.py`, `tests/test_shape_kernel.py`.

References. H. Cramér (1946), *Mathematical Methods of Statistics*, Princeton University Press (Chapter 32: the information inequality). C. R. Rao (1945), "Information and the accuracy attainable in the estimation of statistical parameters", Bulletin of the Calcutta Mathematical Society 37, 81–91. E. L. Lehmann and G. Casella (1998), *Theory of Point Estimation*, 2nd edition, Springer (Chapter 2: Fisher information, nuisance parameters, efficiency). A. W. van der Vaart (1998), *Asymptotic Statistics*, Cambridge University Press (Chapters 7 and 8: differentiability in quadratic mean, local asymptotic normality, the convolution and local asymptotic minimax theorems). J. Hájek (1972), "Local asymptotic minimax and admissibility in estimation", Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability

1, 175–194. O. Cappé, E. Moulines and T. Rydén (2005), *Inference in Hidden Markov Models*, Springer (Chapter 10: Fisher's identity, Louis' identity and the EM algorithm for state-space models).

Chapter 6

The statistics of the experiments

Every number in the paper's tables and figures is a statistic of a small number of replicates, sixteen at the frozen configuration and eight in the sweeps, and every comparison between two arms is paired on identical data. This chapter defines the quantities the paper reports (RMSE, bias, dispersion, standard errors, bootstrap intervals, paired differences and ratios, log-slopes), explains why a local slope over a finite window is not an asymptotic exponent, describes the bundle structure (training, validation, test), the two kinds of seeds (development and confirmatory), and what "pre-registered" and "pre-data amendment" mean in this project. The numbers themselves are in the paper; here are the definitions of what they are numbers *of*.

6.1 Replicates, seeds and the inferential unit

A **replicate** of an experiment is one complete run with one value of the random **seed**: the seed determines the clean sequences of the training, validation and test bundles, the noise draws that corrupt them, the initialisation of the estimator, and any Monte Carlo inside the fit. Two arms run at the same seed see the same data, which is what makes their comparison paired (Section 6.4). The **inferential unit** is the replicate: every standard error, interval and sign count below is over seeds, never over sequences or sites within a seed, because sequences within a seed share the bundle and sites within a sequence are dependent.

Let $e_1, \dots, e_n$ be the values of a statistic (an error, a fitted coordinate) over n seeds. The **mean over seeds** is $\bar{e} = \frac{1}{n}\sum_s e_s$, the **sample standard deviation** is $\hat{\sigma} = \big(\frac{1}{n-1}\sum_s (e_s - \bar{e})^2\big)^{1/2}$, and the **standard error of the mean** is $\hat{\sigma}/\sqrt{n}$. With $n = 16$ the standard error is a quarter of the standard deviation; with $n = 8$ it is 0.35 of it.

6.2 RMSE, bias and dispersion

Let $\hat{\theta}_1, \dots, \hat{\theta}_n$ be estimates of a known truth θ over n seeds (the paper's fitted correlation and fitted excess kurtosis at each level of E06 are of this kind). The **root mean squared error** over replicates is

$$\mathrm{RMSE} = \Big(\frac{1}{n}\sum_{s=1}^{n}(\hat{\theta}_s - \theta)^2\Big)^{1/2}, \tag{6.1}$$

and it decomposes into a **bias** and a **dispersion**:

$$\frac{1}{n}\sum_s (\hat{\theta}_s - \theta)^2 = \underbrace{(\bar{\theta} - \theta)^2}_{\text{bias}^2} + \underbrace{\frac{1}{n}\sum_s (\hat{\theta}_s - \bar{\theta})^2}_{\text{dispersion}^2}, \qquad \bar{\theta} = \frac{1}{n}\sum_s \hat{\theta}_s, \tag{6.2}$$

because the cross term $\frac{2}{n}\sum_s(\hat{\theta}_s - \bar{\theta})(\bar{\theta} - \theta)$ vanishes. The dispersion is the population-form standard deviation (divisor n); the sample form (divisor $n-1$) differs by the factor $\sqrt{n/(n-1)}$,

which is 1.03 at $n = 16$. The paper reports RMSE, bias and dispersion separately at each level because they answer different questions: the Cramér–Rao reference of the chapter on Fisher information bounds the dispersion of an unbiased estimator, so an estimator whose bias is not zero can have a dispersion below the reference, and only the decomposition says which is happening.

The sampling error of a standard deviation. A dispersion estimated from n replicates is itself random. If the replicates were normal, $(n-1)\hat{\sigma}^2/\sigma^2$ would be χ^2_{n-1}, whose relative standard deviation is $\sqrt{2/(n-1)}$, so the relative standard error of $\hat{\sigma}$ is about $1/\sqrt{2(n-1)}$: $1/\sqrt{30} \approx 18\%$ at $n = 16$ and $1/\sqrt{14} \approx 27\%$ at $n = 8$. This is what the paper means by "within the sampling error of a sixteen-replicate standard deviation": two dispersions that differ by less than about a fifth are not distinguishable at this replicate count.

6.3 Confidence intervals by the bootstrap

The paper's intervals are **percentile bootstrap** intervals over seeds. The principle is the plug-in principle: the unknown law of the statistic under resampling of the true population is approximated by its law under resampling of the observed replicates.

Definition 6.1 (Percentile bootstrap over seeds)**.** Given per-seed values $e_1, \ldots, e_n$ and a statistic $T(e_1, \ldots, e_n)$ (a mean, a mean of paired differences, a slope), draw B **resamples** $(e_{s_1^*}, \ldots, e_{s_n^*})$ with the indices s_j^* drawn independently and uniformly from $\{1, \ldots, n\}$ with replacement, compute T_b^* for each, and report the interval $[T^*_{(0.025)}, T^*_{(0.975)}]$ of the 2.5th and 97.5th percentiles of $\{T_b^*\}_{b=1}^B$.

The code uses $B = 20{,}000$ resamples with a fixed bootstrap seed; the function is `boot_ci` in `tools/make_capacity_numbers.py`, and the other number generators carry copies of it. Two things must be said about such an interval. It carries *Monte Carlo error* from the finite B, which is largest in the tails: the 97.5th percentile of 20,000 draws rides on the 500 largest values, and with $n = 8$ seeds the bootstrap distribution is discrete with large atoms, so the percentile can jump between neighbouring atoms from one stream to another. And it is not an exact confidence interval: with $n = 8$ or 16 the percentile interval tends to under-cover, and it inherits any skewness of the statistic. The paper therefore reports the interval as the outcome of a declared procedure and reads it as such, never as a p-value.

The exact enumerated bootstrap. A resample is a multiset of n draws from n seeds, and the statistic depends only on the multiset (the counts $c_1, \ldots, c_n$ with $\sum_s c_s = n$) when it is symmetric in the replicates. The number of distinct multisets is the number of weak compositions of n into n parts,

$$\binom{2n-1}{n}, \qquad \text{which is 6435 at } n = 8, \quad 92{,}378 \text{ at } n = 10, \quad 300{,}540{,}195 \text{ at } n = 16, \tag{6.3}$$

and the multiset with counts (c_s) has probability $\dfrac{n!}{\prod_s c_s!}\, n^{-n}$ under resampling with replacement (the multinomial law). When the count is small enough, the bootstrap distribution is therefore *enumerable*: compute T on every multiset, weight it by its multinomial probability, and read the weighted percentiles. The interval then has no Monte Carlo error at all. The tool `tools/make_sample_complexity.py` does this whenever $\binom{2n-1}{n} \le 200{,}000$ (that is, $n \le 10$) and falls back to Monte Carlo with a reported B above that limit. This is the **exact enumerated bootstrap** the paper's provenance appendix refers to, and it entered the project because a Monte Carlo interval for a fitted slope was found to have its upper limit on one side or the other

of the value being tested depending on the bootstrap stream (`docs/SCIENCE_AUDIT.md`, finding 3 of 18 September); enumeration settled it. At $n = 16$ the enumeration is out of reach and $B = 20{,}000$ Monte Carlo draws are used, which is adequate for a mean over sixteen seeds.

6.4 Paired differences and paired ratios

Two arms A and B run at the same seed are fitted on the same training bundle, selected on the same validation bundle and scored on the same test bundle. Their errors e_s^A and e_s^B are then **paired**, and the comparison is made on the per-seed differences or ratios, not on the two means.

Paired differences. With $d_s = e_s^A - e_s^B$, the mean paired difference is $\bar{d} = \bar{e}^A - \bar{e}^B$, the same number as the difference of means; what changes is its uncertainty:

$$\mathrm{Var}(\bar{d}) = \frac{1}{n}\big(\mathrm{Var}(e^A) + \mathrm{Var}(e^B) - 2\,\mathrm{Cov}(e^A, e^B)\big), \tag{6.4}$$

and the covariance is large and positive when both arms suffer from the same hard seeds (a bundle with an unusual chain, a heavy-tailed draw). Pairing removes that shared variation. The paper reports $\bar{d}$ with a percentile bootstrap of the d_s over seeds, and beside it the **sign count**, "in favour of A in k of n seeds", a distribution-free summary that needs no scale.

Paired ratios. When the question is a factor ("the best network is a factor r behind"), the paper forms the per-seed ratio $r_s = e_s^A / e_s^B$ and reports the **mean of ratios** $\bar{r} = \frac{1}{n}\sum_s r_s$ with a percentile bootstrap. This is not the same as the **ratio of means** $\bar{e}^A/\bar{e}^B$: for positive quantities the mean of ratios exceeds the ratio of means when the denominator varies across seeds (Jensen's inequality applied to $1/e^B$), and the two can differ by a few per cent at $n = 16$. The paper states which it uses ("ratios are formed within seed and averaged") and, for the headline factor, reports both, so that a reader can see the size of the difference. A paired ratio is only meaningful when both arms are scored on the same test bundle; the generator that joins rows from different experiments checks the SHA-256 of the bundles and raises if they differ (`docs/EXPERIMENTAL_CONTRACT.md`, Section 3).

6.5 Log-slopes and why a local slope is not an asymptotic exponent

Many of the paper's theoretical statements are power laws in e^{-t}: an information decays as e^{-4t} or e^{-8t}, an RMSE grows as e^{2t} or e^{4t}. A measured curve $y(t)$ is compared with such a law through its **log-slope**, the derivative of $\log y$ with respect to t, so that $y \propto e^{-8t}$ has log-slope -8.

The OLS log-slope over a window. Given values $y_1, \ldots, y_K$ at levels $t_1 < \cdots < t_K$ in a declared window, the ordinary least-squares slope of $\log y$ on t is

$$\hat{b} = \frac{\sum_k (t_k - \bar{t})(\log y_k - \overline{\log y})}{\sum_k (t_k - \bar{t})^2}, \tag{6.5}$$

which for $K = 2$ reduces to the two-point slope $(\log y_2 - \log y_1)/(t_2 - t_1)$. The window and the estimator are declared in the registry entry before the data are seen (Section 6.8); the two-point and the OLS estimators differ, and the audit of 18 September replaced a two-point estimate by the pre-declared OLS one for exactly this reason. Its uncertainty is a bootstrap over seeds of the whole fit (Section 6.3), because the levels within a seed share the seed's chains.

Local versus asymptotic. An asymptotic law $y(t) = Ce^{-\beta t}(1+\epsilon_t)$ with $\epsilon_t \to 0$ has log-slope

$$\frac{\mathrm{d}\log y}{\mathrm{d}t} = -\beta + \frac{\mathrm{d}}{\mathrm{d}t}\log(1+\epsilon_t), \tag{6.6}$$

and the second term is not small until ϵ_t is. For the paper's Gaussian chain $\epsilon_t = O(e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0))$, and the scalar example of the chapter on Fisher information makes the effect explicit: for a noised Gaussian coordinate of variance θ, $I_t = \frac{1}{2}e^{-4t}(1+(\theta-1)e^{-2t})^{-2}$ has local log-slope

$$-4 + \frac{4(\theta-1)e^{-2t}}{1+(\theta-1)e^{-2t}}, \tag{6.7}$$

which for $\theta = 5$ is about -1.6 at $t = 0.5$, -3.3 at $t = 1.5$ and -4.0 at $t = 3$. A window that ends near the crossover $\frac{1}{2}\log(\theta-1) \approx 0.7$ measures a slope of -2 or so and says nothing against the asymptotic -4; a window well above it measures -4. The OLS slope over a window is a weighted average of the local slopes in it, so it lies between the shallowest and the steepest local values. This is the whole content of the paper's remark that slopes of about -3 and -6 measured over $t \le 1.6$ at $\rho = 0.85$ are pre-asymptotic values below t_*, and of the pre-data amendment of E12b, which replaced "compare the fitted slope with 4 and 8" by "compare $M^*(t)$ with the exact information curve, whose local slope on the swept levels is what the corollary predicts there".

6.6 Bundles: training, validation and test

Each seed owns three **bundles** of sequences drawn from three disjoint random streams keyed by a tag, the seed and, for training, the size M (`rng_for` in `src/markov_score/utils.py`; the tags are in `experiments/protocol.py`):

- the **training bundle**, $M \in \{32, 128, 512, 2048\}$ sequences, the only data the estimator's fit sees; under the one-view protocol each sequence contributes one noised view at one level, the sequences being split evenly across the twelve levels;
- the **validation bundle**, 1024 sequences, used only to *select*: the EM iteration count on its checkpoint ladder, the network checkpoint, and, for the architecture screens, nothing (those use development seeds, Section 6.7);
- the **test bundle**, 2048 sequences, used only to *report*.

Selection on validation and reporting on test are separated so that the reported number is not optimistically biased by the choice it depended on. The regions differ too: selection uses the **bulk** (sites 5 to $L-4$), reporting the headline uses all sites, and the centre site is a diagnostic. The SHA-256 of every bundle is stored with every row, so that pairing across arms and across experiments is checkable after the fact.

The estimand. The headline quantity is the **schedule-pooled relative score error**,

$$\Big(\frac{\sum_t \sum_{j,i} \big(\hat{s}_t(\mathbf{x}_j)_i - s_t(\mathbf{x}_j)_i\big)^2}{\sum_t \sum_{j,i} s_t(\mathbf{x}_j)_i^2}\Big)^{1/2}, \tag{6.8}$$

the sum over the twelve levels t of the schedule, the test sequences j and the sites i of the squared error between an estimator's score $\hat{s}$ and the tree-exact reference s, divided by the same sum of the reference's squared norm, and then rooted. "Pooled" means that the sums are formed before the ratio and the root: the low levels, where the score is large, dominate the pooled energy, and a per-level statement (such as the Edgeworth correction of E16 or the LMMSE floor per level of E07) must not be confused with a pooled one. A relative error of x means the estimator's squared error is x^2 of the reference's energy over the schedule.

6.7 Development and confirmatory seeds

Two kinds of seed never mix. **Development seeds** (10000 to 10003) are used for every choice that a human or a script makes among alternatives: architectures, learning rates, output parameterisations, the mixture capacity ladder's shape. Numbers computed on them appear in no reported table. **Confirmatory seeds** (0 to 15, or 0 to 7 in the sweeps) are used for every reported number and for no choice. The reason is selection bias: a configuration chosen because it did best on a set of seeds has an error on those seeds that is biased downward (the winner's curse), and only seeds that played no part in the choice give an unbiased reading. The architecture screen of the paper's Appendix C is therefore a table on development seeds whose winners are then run once on the confirmatory seeds. The per-seed validation bundle is a third thing: it belongs to a confirmatory seed but is used only to pick the checkpoint within an arm, which is part of the arm's definition rather than a choice among arms.

6.8 Pre-registration and pre-data amendments

A **registered experiment** is an entry of `experiments/registry/` (one JSON file per identifier, E01 to E21) that records, before the experiment runs, the hypothesis it tests with a stated prediction, the design (the one factor varied and everything held fixed), the pre-declared analysis (which script computes which statistic from which file, with which estimator and window), the seeds, and the status. When the run is complete, its output files are **frozen**: their SHA-256 digests are written into the entry (`tools/freeze.py`), any later byte change fails the build (`tools/freeze.py -check`, run by `tools/check_manuscript.py`), and a generator refuses to emit a number from an output that is not frozen or whose provenance records a dirty code tree. Every number in the paper is a macro emitted by such a generator from a frozen file, and an independent tool (`tools/audit_claims.py`) recomputes the headline numbers from the same frozen files by a different code path. The contract is written out in `docs/EXPERIMENTAL_CONTRACT.md`.

A **pre-data amendment** is a change to a registered entry made *before any cell of the experiment has run*, recorded in the entry's `amendments` list with a timestamp, what changed, why, and what did not change. It is the one kind of change the contract allows to an entry; once a confirmatory cell has run, the entry is never changed, and a fix is a new experiment with a new identifier or a versioned directory, with the old run kept and reported for what it is. Two examples fix the idea.

- E14 (`experiments/registry/E14.json`) was pre-registered on 18 September and submitted to the cluster; while its jobs were still pending and no cell had run, the entry was amended so that each network row records the whole held-out objective curve rather than only its minimum, so that a diverged fit can be reported as a divergence and not as the network's error. The amendment states that nothing in the design, the arms, the seeds or the analysis changed, and the jobs were resubmitted.

- E12b (`experiments/registry/E12b.json`) was amended, before any cell ran, to compare the measured sample-complexity curve with the exact information curve computed by Fisher's identity (E21) rather than with the asymptotic exponents 4 and 8, because a computation made before the data showed that the local slope of the information curve on the swept levels is between 6 and 7 for the shape coordinate; the original plan would have recorded a confirmation as a refutation.

The value of the discipline is that a reader can tell, for every number, whether it was predicted or observed, and whether the analysis that produced it was chosen before or after the data existed. Where the paper reports something it did not pre-register (the audit computations E15 to E21 were registered on the day they were run), it says so.

6.9 Worked example: eight seeds

Take $n = 8$ confirmatory seeds, as in the regime sweeps. The exact bootstrap enumerates $\binom{15}{8} = 6435$ multisets. The multiset in which every seed appears once has probability $8!/8^8 = 40320/16777216 \approx 0.0024$; a given seed is absent from a resample with probability $(7/8)^8 \approx 0.344$, so a typical resample contains about $8(1 - 0.344) \approx 5.2$ distinct seeds, which is why bootstrap intervals at $n = 8$ are wide and discrete. The relative standard error of an eight-replicate standard deviation is about $1/\sqrt{14} \approx 27\%$.

For a paired comparison, suppose four seeds gave errors $e^A = (0.10, 0.12, 0.11, 0.13)$ and $e^B = (0.08, 0.10, 0.09, 0.11)$. The unpaired means are 0.115 and 0.095, each with a sample standard deviation of about 0.013, and the unpaired standard error of their difference is about 0.009, so the two arms look barely distinguishable; the paired differences are all exactly 0.02, with zero dispersion, and A is behind in 4 of 4 seeds. The per-seed ratios are 1.25, 1.20, 1.22 and 1.18, with mean of ratios 1.214; the ratio of means is $0.115/0.095 \approx 1.211$, a little smaller, as Section 6.4 predicts. All of these are properties of the four invented numbers, not of any experiment.

Where it is used. The paper's Section 5 (paired seeds as the inferential unit, ratios formed within seed, sixteen and eight seeds), Section 4 (RMSE, dispersion and bias per level, the sixteen-replicate standard deviation, fitted log-slopes on declared windows), Appendix C (bundles, seeds, the estimand, the checkpoint ladders, percentile bootstraps over seeds, provenance) and Appendix E. Experiments: every one of E01 to E21; the examples above are E06, E12, E12b, E14 and E19. Code: `experiments/protocol.py` (sizes, seeds, bundle tags, the estimand and regions), `experiments/frozen_config.py`, `src/markov_score/utils.py` (`rng_for`), `tools/make_capacity_numbers.py` (`boot_ci`), `tools/make_sample_complexity.py` (the enumerated bootstrap and the OLS slope), `tools/make_learning_numbers.py`, `tools/freeze.py`, `tools/registry.py`, `tools/check_manuscript.py`, `tools/audit_claims.py`, `docs/EXPERIMENTAL_CONTRACT.md`, `docs/SCIENCE_AUDIT.md`, `experiments/registry/E14.json`, `experiments/registry/E12b.json`.

References. B. Efron and R. J. Tibshirani (1993), *An Introduction to the Bootstrap*, Chapman and Hall (Chapters 6 and 13: the bootstrap standard error and the percentile interval). A. C. Davison and D. V. Hinkley (1997), *Bootstrap Methods and Their Application*, Cambridge University Press (Chapter 2: the enumeration of resamples and the multinomial weights; Chapter 5: percentile intervals and their coverage). P. Hall (1992), *The Bootstrap and Edgeworth Expansion*, Springer (Chapter 3: the accuracy of percentile intervals). L. Wasserman (2004), *All of Statistics*, Springer (Chapters 6 to 9: bias, variance, mean squared error, standard errors and the bootstrap). B. A. Nosek, C. R. Ebersole, A. C. DeHaven and D. T. Mellor (2018), "The preregistration revolution", Proceedings of the National Academy of Sciences 115, 2600–2606.

Part II

Diffusion

Chapter 7

The Ornstein–Uhlenbeck channel and Tweedie's identity

This chapter builds the forward process of the paper from nothing. It writes down the stochastic differential equation that adds the noise, solves it, and derives the three facts on which every later chapter rests: the noised law is a Gaussian convolution of the clean law; its per-site variance is one at every noise level; and its score is a rescaled posterior mean (Tweedie's identity). It ends with the observation protocol of the paper, one noised view per sequence on a twelve-level schedule, written out level by level. Throughout, $\mathbf{a} \in \mathbb{R}^L$ is a clean sequence with law P_0, $\mathbf{x}_t$ its noised version at level t with law P_t, and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ a standard Gaussian vector independent of $\mathbf{a}$. In derivations we abbreviate $\alpha = e^{-t}$, as the paper's appendix does; in displayed statements we keep e^{-t}, as the paper's body does.

7.1 Stochastic differential equations in one paragraph

A **Brownian motion** $(W_t)_{t \geq 0}$ is a random continuous path with $W_0 = 0$ whose increments over disjoint intervals are independent and Gaussian, $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $s < t$. Its formal time derivative $\eta(t) = \mathrm{d}W_t / \mathrm{d}t$ is **white noise**. It is **standardised** in the sense that $\mathbb{E}[\eta(t)\eta(t')] = \delta(t - t')$, which is the same as saying that $\int_s^t \eta(u)\,\mathrm{d}u = W_t - W_s$ has variance $t - s$. A **stochastic differential equation**

$$\mathrm{d}x_t = f(x_t, t)\ \mathrm{d}t + g(t)\ \mathrm{d}W_t \tag{7.1}$$

is shorthand for the recursion $x_{t+h} - x_t \approx f(x_t, t)\,h + g(t)\,(W_{t+h} - W_t)$ taken to the limit $h \to 0$; f is the **drift** (the deterministic push) and g the **diffusion coefficient** (the size of the random kick). The limit defines the Itô integral, and only two of its properties are needed here. First, for a deterministic integrand ϕ the random variable $\int_0^t \phi(u)\,\mathrm{d}W_u$ is Gaussian with mean zero and variance $\int_0^t \phi(u)^2\,\mathrm{d}u$ (the **Itô isometry**). Second, a linear equation can be solved with an integrating factor exactly as an ordinary differential equation can, because multiplying by a smooth deterministic function introduces no Itô correction term. In L dimensions W_t has L independent coordinates and Equation (7.1) is read coordinate by coordinate.

7.2 The variance-preserving Ornstein–Uhlenbeck process

Definition 7.1 (The channel)**.** The **Ornstein–Uhlenbeck channel** of the paper is the coordinatewise process

$$\mathrm{d}\mathbf{x}_t = -\mathbf{x}_t\ \mathrm{d}t + \sqrt{2}\ \mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 = \mathbf{a} \sim P_0, \tag{7.2}$$

with $\mathbf{W}$ an L-dimensional Brownian motion independent of $\mathbf{a}$. In white-noise form, $\mathrm{d}\mathbf{x}_t / \mathrm{d}t = -\mathbf{x}_t + \sqrt{2}\,\eta(t)$ with η standardised.

The drift $-\mathbf{x}_t$ pulls every coordinate towards zero at unit rate; the kick $\sqrt{2}\,\mathrm{d}\mathbf{W}_t$ is the same for every coordinate and independent across coordinates. Nothing couples the sites: the channel acts on each a_i separately, and this is what keeps the posterior factor graph a tree in the paper's Section 2.

Proposition 7.2 (Solution of the channel)**.** *For every $t \geq 0$,*

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \qquad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I})\,, \tag{7.3}$$

with $\mathbf{z}$ independent of $\mathbf{a}$. Conditionally on $\mathbf{a}$, $\mathbf{x}_t \sim \mathcal{N}\big(e^{-t}\mathbf{a}, \Delta_t \mathbf{I}\big)$.

Proof. Take one coordinate and drop the index. Put $y_t = e^t x_t$. The factor e^t is deterministic and smooth, so the ordinary product rule applies:

$$\mathrm{d}y_t = e^t x_t\,\mathrm{d}t + e^t\,\mathrm{d}x_t = e^t x_t\,\mathrm{d}t + e^t(-x_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t) = \sqrt{2}\,e^t\,\mathrm{d}W_t.$$

Integrating from 0 to t gives $y_t = a + \sqrt{2}\int_0^t e^u\,\mathrm{d}W_u$, hence

$$x_t = e^{-t}a + \sqrt{2}\int_0^t e^{-(t-u)}\,\mathrm{d}W_u.$$

The integrand $\sqrt{2}\,e^{-(t-u)}$ is deterministic, so by the Itô isometry the integral is Gaussian with mean zero and variance

$$\int_0^t 2\,e^{-2(t-u)}\,\mathrm{d}u = 2\cdot\frac{1-e^{-2t}}{2} = 1 - e^{-2t} = \Delta_t.$$

It is independent of a because W is. Writing the integral as $\sqrt{\Delta_t}\,z$ with $z \sim \mathcal{N}(0,1)$ gives the scalar claim; the coordinates use independent Brownian motions, so the vector claim follows with $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$. □

Remark 7.3 (Why $\sqrt{2}$). The variance $v_t = \mathrm{Var}(x_t)$ of one coordinate obeys $\mathrm{d}v_t/\,\mathrm{d}t = -2v_t + 2$, whose fixed point is $v = 1$. With the kick $\mathrm{d}W_t$ instead of $\sqrt{2}\,\mathrm{d}W_t$ the fixed point would be $\frac{1}{2}$ and one would have $\Delta_t = (1 - e^{-2t})/2$. The paper's convention (`paper/notation.tex`, item 4) fixes the noise as standardised and carries the $\sqrt{2}$ explicitly so that $\Delta_t = 1 - e^{-2t}$ throughout, in the forward and in the reverse equation. The variance-preserving SDE of Song and co-authors, $\mathrm{d}\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}\mathbf{W}$, is the same process when $\beta \equiv 2$; for a general schedule β the paper's t is $\frac{1}{2}\int_0^{t'}\beta(u)\,\mathrm{d}u$, and nothing below changes.

Remark 7.4 (The semigroup property). Noising for time t and then for time s is noising for time $t + s$: if $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ and $\mathbf{x}_{t+s} = e^{-s}\mathbf{x}_t + \sqrt{\Delta_s}\mathbf{z}'$ with $\mathbf{z}'$ fresh, then $\mathbf{x}_{t+s} = e^{-(t+s)}\mathbf{a} + (e^{-s}\sqrt{\Delta_t}\mathbf{z} + \sqrt{\Delta_s}\mathbf{z}')$, and the bracket is Gaussian with variance $e^{-2s}(1 - e^{-2t}) + 1 - e^{-2s} = 1 - e^{-2(t+s)} = \Delta_{t+s}$ per coordinate. This is the Markov property of the channel, used in Chapter 9.

7.3 Variance preservation and the Gaussian limit

Proposition 7.5 (Second moments of the noised law)**.** *Let $\mathbf{\Sigma}_0 = \mathrm{Cov}(\mathbf{a})$. Then $\mathbb{E}[\mathbf{x}_t] = e^{-t}\mathbb{E}[\mathbf{a}]$ and*

$$\mathrm{Cov}(\mathbf{x}_t) = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I} =: \mathbf{\Sigma}_t. \tag{7.4}$$

*In particular, if $\mathrm{Var}(a_i) = 1$ for every site, then $\mathrm{Var}(x_{t,i}) = e^{-2t} + \Delta_t = 1$ at every t: the channel **preserves the variance**.*

Proof. From Equation (7.3), $\mathbf{a}$ and $\mathbf{z}$ are independent, so covariances add: $\mathrm{Cov}(\mathbf{x}_t) = e^{-2t}\,\mathrm{Cov}(\mathbf{a}) + \Delta_t\,\mathrm{Cov}(\mathbf{z})$. □

The paper's chain (Section 2) has $a_1 \sim \mathcal{N}(0,1)$ and innovation variance $q = 1-\rho^2$, which gives $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation law; hence every family shares the same $\mathbf{\Sigma}_0$ and the same $\mathbf{\Sigma}_t$.

Proposition 7.6 (The Gaussian limit)**.** *As $t \to \infty$, P_t converges in distribution to $\gamma = \mathcal{N}(0, \mathbf{I})$, whatever P_0 is, provided $\mathbb{E}|\mathbf{a}|^2 < \infty$.*

Proof. For a bounded continuous ϕ, $\mathbb{E}\phi(\mathbf{x}_t) = \mathbb{E}\phi(e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z})$. The term $e^{-t}\mathbf{a}$ tends to zero in probability by Chebyshev's inequality, $\mathbb{P}(e^{-t}|\mathbf{a}| > \epsilon) \le e^{-2t}\mathbb{E}|\mathbf{a}|^2/\epsilon^2$, and $\sqrt{\Delta_t}\mathbf{z} \to \mathbf{z}$ almost surely. Dominated convergence then gives $\mathbb{E}\phi(\mathbf{x}_t) \to \mathbb{E}\phi(\mathbf{z})$. □

The limit is the stationary law of the process: if $\mathbf{a} \sim \gamma$ then $\mathbf{x}_t \sim \mathcal{N}(0, e^{-2t}\mathbf{I} + \Delta_t\mathbf{I}) = \gamma$ for every t. The rate of convergence is the subject of Chapter 10: the Hermite moment of degree m of P_t is exactly e^{-mt} times that of P_0, so the memory of P_0 fades order by order rather than all at once.

7.4 The noised law as a Gaussian convolution

Write γ_Δ for the density of $\mathcal{N}(0, \Delta\mathbf{I})$ on $\mathbb{R}^L$,

$$\gamma_\Delta(\mathbf{w}) = (2\pi\Delta)^{-L/2} \exp\Big(-\frac{|\mathbf{w}|^2}{2\Delta}\Big). \tag{7.5}$$

Proposition 7.7 (Density of the noised law)**.** *For every $t > 0$ the law P_t has the density*

$$P_t(\mathbf{x}) = \int_{\mathbb{R}^L} P_0(\,\mathrm{d}\mathbf{a})\, \gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a}) = \mathbb{E}_{P_0}\big[\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})\big], \tag{7.6}$$

which is strictly positive and infinitely differentiable in $\mathbf{x}$, whatever P_0 is (it need not have a density). Every derivative may be taken under the integral sign.

Proof. Conditionally on $\mathbf{a}$, $\mathbf{x}_t$ has density $\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})$ by Proposition 7.2; integrating over $\mathbf{a}$ gives Equation (7.6). Positivity is clear. For differentiation under the integral, every $\mathbf{x}$-derivative of $\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})$ is a polynomial in $(\mathbf{x} - e^{-t}\mathbf{a})/\Delta_t$ times the same Gaussian, and is therefore bounded uniformly in $\mathbf{a}$ on every bounded set of $\mathbf{x}$ by a constant, which is integrable against the probability measure P_0; dominated convergence applies. □

Two readings of Equation (7.6) are used. As an operator statement, P_t is the convolution of the dilated law (the law of $e^{-t}\mathbf{a}$) with the Gaussian γ_{Δ_t}. As a factor-graph statement, the integrand factorises over sites,

$$\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a}) = \prod_{i=1}^{L} \ell_t(x_i \mid a_i), \qquad \ell_t(x_i \mid a_i) \propto \exp\Big[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\Big], \tag{7.7}$$

so conditioning on $\mathbf{x}$ attaches one **unary likelihood factor** ℓ_t to each site and adds no edge among the latent variables. The code builds exactly this table: `log_likelihood_matrix` in `src/markov_score/noising.py` returns $\log \ell_t(x_i \mid u_k)$ on the quadrature grid, shifted per row by a constant so that no row underflows; the shift cancels in every normalised message and changes nothing.

7.5 The signal-to-noise ratio

Definition 7.8 (Signal-to-noise ratio)**.** The **signal-to-noise ratio** of the channel at level t is the ratio of the signal variance to the noise variance in one coordinate of Equation (7.3) for unit-variance data,

$$\mathrm{SNR}_t = \frac{e^{-2t}}{\Delta_t} = \frac{e^{-2t}}{1 - e^{-2t}} = \frac{1}{e^{2t} - 1}. \tag{7.8}$$

Table 7.1: The twelve-level schedule of the paper (`experiments/frozen_config.py`, $t_k = 0.05 \cdot r^{k-1}$ with $r = 60^{1/11} \approx 1.4509$), with the analytic values of the decay factor, the noise variance and the signal-to-noise ratio at each level. All entries are computed from the definitions; none is measured.

k	t_k	e^{-t_k}	Δ_{t_k}	SNR_{t_k}
1	0.0500	0.9512	0.0952	9.51
2	0.0725	0.9300	0.1350	6.40
3	0.1053	0.9001	0.1898	4.27
4	0.1527	0.8584	0.2632	2.80
5	0.2216	0.8012	0.3580	1.79
6	0.3215	0.7250	0.4743	1.11
7	0.4665	0.6272	0.6066	0.648
8	0.6769	0.5082	0.7417	0.348
9	0.9821	0.3745	0.8597	0.163
10	1.4250	0.2405	0.9422	0.061
11	2.0676	0.1265	0.9840	0.016
12	3.0000	0.0498	0.9975	0.0025

For small t, $\Delta_t = 2t + O(t^2)$ and $\mathrm{SNR}_t \approx 1/(2t)$, which diverges: the observation is almost clean. For large t, $\mathrm{SNR}_t \approx e^{-2t}$: the signal is a small perturbation of pure noise, and e^{-2t} is the natural small parameter of the expansion in Chapter 10. The same quantity appears there under the name $b = e^{-2t}/\Delta_t$, together with $s = e^{-t}\mathbf{x}/\Delta_t$; the reader should keep in mind that b is the SNR. Inverting Equation (7.8), $t = \frac{1}{2}\log(1 + 1/\mathrm{SNR}_t)$, so the diffusion time is a logarithmic clock on the SNR.

7.6 Tweedie's identity

The **score** of the noised law is the gradient of its log-density, $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$. Tweedie's identity says that it is a rescaled residual of the posterior mean of the clean sequence. We write $\langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$ for that posterior mean, in the bracket notation of the paper.

Theorem 7.9 (Tweedie's identity)**.** *For every $t > 0$ and every $\mathbf{x} \in \mathbb{R}^L$,*

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) = -\frac{\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t}}{\Delta_t}, \qquad \langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]. \tag{7.9}$$

Equivalently, $\langle \mathbf{a} \rangle_{x,t} = e^{t}\big(\mathbf{x} + \Delta_t\, S(\mathbf{x}, t)\big)$. No assumption on P_0 is needed beyond $\mathbb{E}|\mathbf{a}| < \infty$.

Proof. Scalar case first. By Proposition 7.7, $P_t(x) = \int P_0(\mathrm{d}a)\, \gamma_{\Delta_t}(x - \alpha a)$ with $\alpha = e^{-t}$, and the derivative may be taken inside:

$$\frac{\mathrm{d}}{\mathrm{d}x}\gamma_{\Delta_t}(x - \alpha a) = -\frac{x - \alpha a}{\Delta_t}\, \gamma_{\Delta_t}(x - \alpha a).$$

Hence

$$P_t'(x) = -\frac{1}{\Delta_t}\int P_0(\mathrm{d}a)\, (x - \alpha a)\, \gamma_{\Delta_t}(x - \alpha a) = -\frac{1}{\Delta_t}\Big[x\, P_t(x) - \alpha \int P_0(\mathrm{d}a)\, a\, \gamma_{\Delta_t}(x - \alpha a)\Big].$$

The posterior of a given $x_t = x$ is, by Bayes' rule, the probability measure $P_0(\mathrm{d}a)\, \gamma_{\Delta_t}(x - \alpha a)/P_t(x)$, so the last integral equals $P_t(x)\, \mathbb{E}[a \mid x_t = x]$. Dividing by $P_t(x) > 0$,

$$\frac{P_t'(x)}{P_t(x)} = -\frac{x - \alpha\, \mathbb{E}[a \mid x_t = x]}{\Delta_t}.$$

The integrability of $|a|$ under P_0 is what makes the last integral finite. In L dimensions, $\nabla_{\mathbf{x}}\gamma_{\Delta_t}(\mathbf{x}-\alpha\mathbf{a}) = -\Delta_t^{-1}(\mathbf{x}-\alpha\mathbf{a})\gamma_{\Delta_t}(\mathbf{x}-\alpha\mathbf{a})$ and the same three lines go through coordinate by coordinate. □

Remark 7.10 (The classical forms). Robbins (1956) wrote the identity for an empirical Bayes problem in which an observation $z \sim \mathcal{N}(\mu, \sigma^2)$ has an unknown prior on μ: $\mathbb{E}[\mu \mid z] = z + \sigma^2\,(\log f)'(z)$ with f the marginal density of z. Efron (2011) named it Tweedie's formula. The channel is this setting with $\mu = e^{-t}\mathbf{a}$ and $\sigma^2 = \Delta_t$: $\mathbb{E}[e^{-t}\mathbf{a} \mid \mathbf{x}] = \mathbf{x} + \Delta_t S(\mathbf{x}, t)$, which is the second form in Theorem 7.9. The code uses it in both directions: `score_from_mean` in `src/markov_score/theory.py` turns a posterior mean into a score, and `denoising_readout` in `src/markov_score/reverse.py` turns a score into a posterior mean.

Proposition 7.11 (Second-order Tweedie identity)**.** *For $t > 0$ the posterior covariance is also a function of the noised law:*

$$\operatorname{Cov}(\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}) = \frac{\Delta_t}{e^{-2t}}\Big(\mathbf{I} + \Delta_t\,\nabla_{\mathbf{x}}^2 \log P_t(\mathbf{x})\Big), \tag{7.10}$$

so that $\operatorname{div} S(\mathbf{x}, t) = \operatorname{tr}\nabla^2 \log P_t(\mathbf{x}) = -L/\Delta_t + (e^{-2t}/\Delta_t^2)\operatorname{tr}\operatorname{Cov}(\mathbf{a} \mid \mathbf{x}_t = \mathbf{x})$.

Proof. Differentiate Equation (7.9) once more. With $m(\mathbf{x}) = \langle\mathbf{a}\rangle_{x,t}$, $\nabla^2 \log P_t = -\mathbf{I}/\Delta_t + (\alpha/\Delta_t)\nabla_{\mathbf{x}} m$. Writing $m_i(\mathbf{x}) = \int a_i\, P_0(\mathrm{d}\mathbf{a})\gamma_{\Delta_t}(\mathbf{x}-\alpha\mathbf{a})/P_t(\mathbf{x})$ and differentiating the quotient with respect to x_j,

$$\partial_j m_i = -\frac{1}{\Delta_t}\Big(\mathbb{E}[a_i(x_j - \alpha a_j) \mid \mathbf{x}] - m_i\,(x_j - \alpha m_j)\Big) = \frac{\alpha}{\Delta_t}\big(\mathbb{E}[a_i a_j \mid \mathbf{x}] - m_i m_j\big) = \frac{\alpha}{\Delta_t}\operatorname{Cov}(a_i, a_j \mid \mathbf{x}),$$

because the terms in x_j cancel. Substituting gives $\nabla^2 \log P_t = -\mathbf{I}/\Delta_t + (\alpha^2/\Delta_t^2)\operatorname{Cov}(\mathbf{a} \mid \mathbf{x})$, which is Equation (7.10) rearranged. □

Why it matters here. The divergence of the score is what implicit score matching (Chapter 8) has to compute, and Equation (7.10) says what it is: up to constants, the trace of the posterior covariance. It also shows that the score is a gradient field whose Jacobian is symmetric, a property a network's output need not have.

Corollary 7.12 (The Gaussian case)**.** *If $P_0 = \mathcal{N}(0, \mathbf{\Sigma}_0)$ then $P_t = \mathcal{N}(0, \mathbf{\Sigma}_t)$ with $\mathbf{\Sigma}_t$ as in Equation* (7.4)*, the score is linear, $S(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$, and the posterior mean is $\langle\mathbf{a}\rangle_{x,t} = e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$.*

Proof. A Gaussian pushed through Equation (7.3) stays Gaussian, with the covariance of Proposition 7.5. Then $\log P_t(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^\top\mathbf{\Sigma}_t^{-1}\mathbf{x} + \text{const}$ and $S = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Inverting Equation (7.9),

$$\langle\mathbf{a}\rangle_{x,t} = \frac{\mathbf{x} + \Delta_t S}{\alpha} = \frac{(\mathbf{I} - \Delta_t\mathbf{\Sigma}_t^{-1})\mathbf{x}}{\alpha} = \frac{(\mathbf{\Sigma}_t - \Delta_t\mathbf{I})\mathbf{\Sigma}_t^{-1}\mathbf{x}}{\alpha} = \frac{\alpha^2\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}}{\alpha} = \alpha\,\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}. \qquad \square$$

This is the **linear minimum-mean-square estimator** (LMMSE) of $\mathbf{a}$ from $\mathbf{x}$, and the paper's Proposition 1 (Section 2) shows that it is exactly what Gaussian message closure computes for *any* innovation law with the same $\mathbf{\Sigma}_0$. The score $S_{\mathrm{G}}(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$ of this **covariance twin** is the reference against which the paper measures the price of second-order closure (Chapter 10).

7.7 The posterior mean as a denoiser

Proposition 7.13 (The posterior mean is the best denoiser)**.** *For every measurable $f\colon \mathbb{R}^L \to \mathbb{R}^L$ with $\mathbb{E}|f(\mathbf{x}_t)|^2 < \infty$,*

$$\mathbb{E}\big|\mathbf{a} - f(\mathbf{x}_t)\big|^2 = \mathbb{E}\big|\mathbf{a} - \langle\mathbf{a}\rangle_{x,t}\big|^2 + \mathbb{E}\big|\langle\mathbf{a}\rangle_{x,t} - f(\mathbf{x}_t)\big|^2. \tag{7.11}$$

The first term on the right, written MMSE_t*, is the smallest mean squared error any denoiser can reach; it is reached only by $f = \langle\mathbf{a}\rangle_{x,t}$.*

Proof. Expand $|\mathbf{a}-f|^2 = |(\mathbf{a}-m)+(m-f)|^2$ with $m = \langle \mathbf{a} \rangle_{x,t}$. The cross term is $2\,\mathbb{E}\langle \mathbf{a}-m,\, m-f\rangle = 2\,\mathbb{E}\langle \mathbb{E}[\mathbf{a}-m \mid \mathbf{x}_t],\, m-f\rangle = 0$, because $m-f$ is a function of $\mathbf{x}_t$ and $\mathbb{E}[\mathbf{a}-m \mid \mathbf{x}_t] = 0$. □

Two consequences organise the paper's measurements. First, any estimate $\widehat{m}$ of the posterior mean converts into an estimate of the score and the errors are proportional:

$$\widehat{S} = -\frac{\mathbf{x} - e^{-t}\widehat{m}}{\Delta_t} \quad \Longrightarrow \quad \widehat{S} - S = \frac{e^{-t}}{\Delta_t}(\widehat{m} - \langle \mathbf{a} \rangle_{x,t}), \tag{7.12}$$

an exact identity that `score_mean_identity_residual` in `src/markov_score/metrics.py` checks to machine precision. Every estimator in the paper, the tree-exact recursion, the fitted kernel, the LMMSE twin and the networks, produces a posterior mean and is scored through Equation (7.12). Second, the energy of the score is tied to the MMSE.

Proposition 7.14 (Score energy)**.** $\mathbb{E}_{P_t}|S(\mathbf{x}_t, t)|^2 = L/\Delta_t - (e^{-2t}/\Delta_t^2)\,\mathrm{MMSE}_t$.

Proof. By Equation (7.3), $-(\mathbf{x}_t - e^{-t}\mathbf{a})/\Delta_t = -\mathbf{z}/\sqrt{\Delta_t}$, and by Theorem 7.9 the score is its conditional expectation, $S(\mathbf{x}_t, t) = \mathbb{E}[-\mathbf{z}/\sqrt{\Delta_t} \mid \mathbf{x}_t]$. Apply Equation (7.11) to the random vector $-\mathbf{z}/\sqrt{\Delta_t}$ in place of $\mathbf{a}$ and $f = S$: $\mathbb{E}|\mathbf{z}|^2/\Delta_t = \mathbb{E}|S|^2 + \mathbb{E}|-\mathbf{z}/\sqrt{\Delta_t} - S|^2$. The last term is $\mathbb{E}|\, e^{-t}(\mathbf{a} - \langle \mathbf{a} \rangle_{x,t})/\Delta_t\,|^2 = (e^{-2t}/\Delta_t^2)\mathrm{MMSE}_t$ by Equation (7.12) applied to $\widehat{m} = \mathbf{a}$; and $\mathbb{E}|\mathbf{z}|^2 = L$. □

At low noise the energy is close to L/Δ_t and large; at high noise it tends to L, the energy of the score of γ. This is why the paper's pooled estimand (Chapter 8) is dominated by its lowest levels.

7.8 One noised view per sequence and the twelve-level schedule

The paper observes M sequences $\mathbf{a}^1, \ldots, \mathbf{a}^M$ and, in its own estimator's interface (interface C of Section 5), sees each of them exactly once, at one level t_μ, as $\mathbf{x}^\mu = e^{-t_\mu}\mathbf{a}^\mu + \sqrt{\Delta_{t_\mu}}\mathbf{z}^\mu$; the clean sequence, a second noised view, and the noise draw itself are never seen. The sequences are split evenly across the twelve levels of Table 7.1, so each level receives $M/12$ views (`_one_view_groups` in `experiments/e03_baselines.py`; `one_view=True` in `experiments/protocol.py`). The likelihood of one view is then $P_{t_\mu,\theta}(\mathbf{x}^\mu)$, the density of Equation (7.6) under the parameterised kernel, and the marginal log-likelihood of the paper's Section 4 is the sum over views. The schedule is log-spaced so that the eleven ratios t_{k+1}/t_k are equal; on the SNR clock of Equation (7.8) the levels run from an almost clean observation (level 1) to almost pure noise (level 12), and the crossover scale t_* of Chapter 10 at the frozen correlation lies between the ninth and the tenth level. The networks of the paper draw the level of each training example uniformly from the same twelve values (`t_levels` in `src/markov_score/nn_torch.py`), so that their population objective is an average over the schedule (Chapter 8).

7.9 Worked example

Example 7.15 (A two-point prior at $t = \log 2$)**.** Let $L = 1$ and let $a = \pm 1$ with probability $\frac{1}{2}$ each, so that $\mathrm{Var}(a) = 1$. At $t = \log 2$ we have $e^{-t} = \frac{1}{2}$, $\Delta_t = 1 - \frac{1}{4} = \frac{3}{4}$ and $\mathrm{SNR} = \frac{1}{3}$. By Equation (7.6),

$$P_t(x) = \tfrac{1}{2}\gamma_{3/4}(x - \tfrac{1}{2}) + \tfrac{1}{2}\gamma_{3/4}(x + \tfrac{1}{2}).$$

The posterior of a given x puts mass proportional to $\exp[-(x \mp \frac{1}{2})^2/(2 \cdot \frac{3}{4})]$ on ± 1; the ratio of the two weights is $\exp(2 \cdot \frac{1}{2}x/\frac{3}{4}) = \exp(4x/3)$, so

$$\langle a \rangle_{x,t} = \tanh(\tfrac{2}{3}x), \qquad S(x,t) = -\frac{x - \frac{1}{2}\tanh(\frac{2}{3}x)}{3/4} = -\tfrac{4}{3}\Big(x - \tfrac{1}{2}\tanh\tfrac{2x}{3}\Big).$$

At $x = 1$: $e^{4/3} = 3.7937$, so $\tanh(2/3) = (3.7937 - 1)/(3.7937 + 1) = 0.5828$, $\langle a \rangle_{x,t} = 0.5828$ and $S(1,t) = -\frac{4}{3}(1 - 0.2914) = -0.9448$. The covariance twin of this prior is $\mathcal{N}(0,1)$, whose noised law is again $\mathcal{N}(0,1)$; its score at $x = 1$ is $S_{\mathrm{G}} = -1$ and its posterior mean is $e^{-t}x = 0.5$. The true denoiser pulls harder towards $+1$ than the linear one, by 0.083, and the score gap $S - S_{\mathrm{G}} = +0.0552$ is, by Equation (7.12), exactly $(e^{-t}/\Delta_t)(0.5828 - 0.5) = \frac{2}{3} \cdot 0.0828$. As a check of Corollary 7.12 on the twin: $-(x - \frac{1}{2} \cdot \frac{1}{2}x)/\frac{3}{4} = -x$, as it must be.

Example 7.16 (Reading the schedule)**.** At level 8 of Table 7.1, $t = 0.6769$, a site with clean value $a_i = 2$ is observed as $x_i = 0.5082 \cdot 2 + 0.8612\, z_i = 1.016 + 0.861\, z_i$: the signal has half its amplitude and the noise has standard deviation $\sqrt{0.7417} = 0.861$. At level 12 the same site is observed as $0.0996 + 0.9988\, z_i$; its amplitude is one twentieth and the SNR is $1/400$. The information a single view carries about the site's fourth cumulant is, by the paper's Theorem 3, proportional to e^{-8t}, which between these two levels falls by the factor $e^{-8(3-0.6769)} = e^{-18.58} \approx 8.5 \times 10^{-9}$.

Where it is used. The channel is equation (2) of the paper (Section 2, "Data and channel") and Tweedie's identity is the first sentence of "The score is a posterior mean"; the Gaussian corollary is the covariance twin of the same section and of Proposition 1. The twelve-level schedule is the frozen configuration of Appendix C, and the one-view interface is interface C of Section 5. Code: the channel and its likelihood table are `src/markov_score/noising.py`; the LMMSE mean and score and the mean-to-score conversion are in `src/markov_score/theory.py`; the score-to-mean readout is in `src/markov_score/reverse.py`; the schedule is `t_grid` in `experiments/frozen_config.py` and the one-view flag is in `experiments/protocol.py`. Experiments: every registered experiment E01–E21 noises its data through the two lines of Equation (7.3).

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Chapter 8

Score matching

The paper's networks are trained to approximate the score of the noised law, and its structured estimator is compared with them through the score. This chapter explains what "learning a score" means, derives the two objectives that make it possible without ever knowing the score (Hyvärinen's implicit and Vincent's denoising score matching), works out what each becomes under the Ornstein–Uhlenbeck channel of Chapter 7, and then follows the code: the two output parameterisations, the three ways the divergence term is computed, the minibatch estimate of the loss, the estimand the paper reports, and the three information interfaces at which the comparison is run.

8.1 What learning a score means

A **score model** is a family of vector fields $s_\theta(\mathbf{x}, t)\colon \mathbb{R}^L \times (0,\infty) \to \mathbb{R}^L$ indexed by parameters θ (the weights of a network). The target is the score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ of the noised law, and the natural measure of success is the **explicit score-matching** loss at one level,

$$J_{\mathrm{ESM}}(s) = \mathbb{E}_{P_t}\big|s(\mathbf{x}_t) - \nabla \log P_t(\mathbf{x}_t)\big|^2. \tag{8.1}$$

The difficulty is that $\nabla \log P_t$ is unknown: for real data one has samples of P_t (or of P_0, which one can noise) and nothing else. In the paper's laboratory S is known exactly (Section 2), which is what allows J_{ESM} to be measured; but the networks are still trained as they would be in practice, by one of the two surrogates below, which have the same minimiser as Equation (8.1) and can be estimated from samples. In both, t is fixed for the moment; the paper's schedule enters in Section 8.6.

8.2 Hyvärinen's implicit score matching

Theorem 8.1 (Implicit score matching)**.** *Let p be a continuously differentiable density on $\mathbb{R}^L$ and let $s\colon \mathbb{R}^L \to \mathbb{R}^L$ be continuously differentiable, with $\mathbb{E}_p|s|^2 < \infty$, $\mathbb{E}_p|\nabla \log p|^2 < \infty$, $\mathbb{E}_p|\operatorname{div} s| < \infty$, and $p(\mathbf{x})\,|s(\mathbf{x})| \to 0$ as $|\mathbf{x}| \to \infty$. Then*

$$\mathbb{E}_p\big|s - \nabla \log p\big|^2 = 2\,\mathbb{E}_p\Big[\tfrac{1}{2}|s|^2 + \operatorname{div} s\Big] + \mathbb{E}_p|\nabla \log p|^2. \tag{8.2}$$

*The last term does not depend on s, so the **implicit score-matching** objective $J_{\mathrm{ISM}}(s) = \mathbb{E}_p[\tfrac{1}{2}|s|^2 + \operatorname{div} s]$ has the same minimiser as J_{ESM}, and J_{ISM} involves only s and samples of p.*

Proof. Scalar case first. Expand the square:

$$\int p\,(s - (\log p)')^2\,\mathrm{d}x = \int p\,s^2\,\mathrm{d}x - 2\int p\,s\,\frac{p'}{p}\,\mathrm{d}x + \int p\,((\log p)')^2\,\mathrm{d}x.$$

In the middle term $p \cdot p'/p = p'$, so it is $-2\int s\,p'\,\mathrm{d}x$, and integration by parts on $[-R, R]$ gives

$$\int_{-R}^{R} s\,p'\,\mathrm{d}x = [s\,p]_{-R}^{R} - \int_{-R}^{R} s'\,p\,\mathrm{d}x.$$

The boundary term tends to zero as $R \to \infty$ by the decay hypothesis, and the remaining integral converges by the integrability of s' against p. Hence $\int p\,(s - (\log p)')^2 = \int p\,s^2 + 2\int p\,s' + \int p\,((\log p)')^2$, which is Equation (8.2) with $\operatorname{div} s = s'$. In L dimensions the middle term is $-2\int \langle s, \nabla p\rangle\,\mathrm{d}\mathbf{x}$, and the divergence theorem on the ball B_R gives $\int_{B_R}\langle s, \nabla p\rangle = \oint_{\partial B_R} p\,\langle s, \mathbf{n}\rangle\,\mathrm{d}\sigma - \int_{B_R} p \operatorname{div} s$; the surface term is bounded by the area of the sphere times $\sup_{|\mathbf{x}|=R} p|s|$, which tends to zero by hypothesis (the hypothesis is to be read as decay fast enough for this product to vanish). □

Two remarks. First, at the true score $J_{\mathrm{ISM}}(S) = -\frac{1}{2}\mathbb{E}|S|^2$, i.e. $\mathbb{E}_p[\operatorname{div}\nabla \log p] = -\mathbb{E}_p|\nabla \log p|^2$; combined with Proposition 7.11 this reproduces Proposition 7.14. Second, the price of avoiding $\nabla \log p$ is the divergence, a sum of L diagonal entries of the Jacobian of s, which is what Section 8.5 is about.

8.3 Vincent's denoising score matching

Suppose the samples of p arise by corrupting clean samples: $\mathbf{a} \sim P_0$ and then $\mathbf{x} \sim p(\cdot \mid \mathbf{a})$, so that $p(\mathbf{x}) = \int P_0(\,\mathrm{d}\mathbf{a})\,p(\mathbf{x} \mid \mathbf{a})$. The conditional score $\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{a})$ is known in closed form whenever the corruption is.

Theorem 8.2 (Denoising score matching)**.** *Let $p(\mathbf{x} \mid \mathbf{a})$ be differentiable in $\mathbf{x}$ with $\mathbb{E}|\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{a})|^2 < \infty$ under the joint law, and let $\mathbb{E}_p|s|^2 < \infty$. Then*

$$J_{\mathrm{DSM}}(s) := \mathbb{E}_{(\mathbf{a},\mathbf{x})}\big|s(\mathbf{x}) - \nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{a})\big|^2 = J_{\mathrm{ESM}}(s) + \mathbb{E}_{(\mathbf{a},\mathbf{x})}\big|\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{a}) - \nabla_{\mathbf{x}} \log p(\mathbf{x})\big|^2. \tag{8.3}$$

The second term does not depend on s, so J_{DSM} and J_{ESM} have the same minimiser, and J_{DSM} needs only pairs $(\mathbf{a}, \mathbf{x})$.

Proof. The key identity is that the marginal score is the conditional expectation of the conditional score,

$$\mathbb{E}\big[\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{a}) \;\big|\; \mathbf{x}\big] = \nabla_{\mathbf{x}} \log p(\mathbf{x}), \tag{8.4}$$

because, with the posterior $p(\mathbf{a} \mid \mathbf{x}) = P_0(\mathbf{a})p(\mathbf{x} \mid \mathbf{a})/p(\mathbf{x})$,

$$\int p(\mathbf{a} \mid \mathbf{x})\,\nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{a})\,\mathrm{d}\mathbf{a} = \frac{1}{p(\mathbf{x})}\int P_0(\mathbf{a})\,\nabla_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{a})\,\mathrm{d}\mathbf{a} = \frac{\nabla_{\mathbf{x}} p(\mathbf{x})}{p(\mathbf{x})}.$$

Now write $Y = \nabla_{\mathbf{x}} \log p(\mathbf{x} \mid \mathbf{a})$ and $S = \nabla \log p(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{x}]$. For any function s of $\mathbf{x}$, the orthogonality argument of Proposition 7.13 applied to Y gives $\mathbb{E}|s(\mathbf{x}) - Y|^2 = \mathbb{E}|s(\mathbf{x}) - S|^2 + \mathbb{E}|S - Y|^2$, which is Equation (8.3). □

The regression reading of the proof is worth stating on its own: J_{DSM} is a least-squares regression of the random target Y on the input $\mathbf{x}$, and the least-squares regression function is the conditional expectation, which by Equation (8.4) is the score. The irreducible part of the loss is the conditional variance of the target.

Corollary 8.3 (The channel)**.** *Under $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$, the conditional law is $\mathcal{N}(e^{-t}\mathbf{a}, \Delta_t \mathbf{I})$ and the target is*

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}_t \mid \mathbf{a}) = -\frac{\mathbf{x}_t - e^{-t}\mathbf{a}}{\Delta_t} = -\frac{\mathbf{z}}{\sqrt{\Delta_t}}. \tag{8.5}$$

Identity Equation (8.4) *is Tweedie's identity Equation* (7.9) *in another guise, and the irreducible loss is $\mathbb{E}|-\mathbf{z}/\sqrt{\Delta_t} - S|^2 = (e^{-2t}/\Delta_t^2)\,\mathrm{MMSE}_t$ by Proposition 7.14.*

The target's variance is L/Δ_t per example, which diverges as $t \to 0$; a network asked to output the score directly would have to produce values of size $1/\sqrt{\Delta_t}$ at the lowest levels. The parameterisations of the next section are the standard way to avoid that.

8.4 The two output parameterisations

Every network of the paper maps a noised sequence and a level to an output $f_\theta(\mathbf{x}, t) \in \mathbb{R}^L$, and the output is converted to a posterior mean and then to a score by Theorem 7.9. The code supports two conventions; they are implemented by `Denoiser.posterior_mean`, `_score_from_out` and `dsm_loss` in `src/markov_score/nn_torch.py`.

Definition 8.4 (`eps` and `x0`)**.** With $\alpha = e^{-t}$:

- `eps`: the output predicts the noise, target $\mathbf{z}$; the posterior mean is $\widehat{m} = (\mathbf{x} - \sqrt{\Delta_t}\, f)/\alpha$ and the score is $\widehat{s} = -(\mathbf{x} - \alpha\widehat{m})/\Delta_t = -f/\sqrt{\Delta_t}$.
- `x0`: the output predicts the clean sequence, target $\mathbf{a}$; the posterior mean is $\widehat{m} = f$ and the score is $\widehat{s} = -(\mathbf{x} - \alpha f)/\Delta_t$.

The training loss in both cases is the mean squared distance between output and target over the batch, $\mathbb{E}|f - \mathbf{z}|^2$ or $\mathbb{E}|f - \mathbf{a}|^2$.

Proposition 8.5 (Both losses are weighted denoising score matching)**.** *With $s^* = -(\mathbf{x} - \alpha\mathbf{a})/\Delta_t$ the target of Equation* (8.5)*,*

$$|f - \mathbf{z}|^2 = \Delta_t\, |\widehat{s} - s^*|^2 \quad (\texttt{eps}), \qquad |f - \mathbf{a}|^2 = \frac{\Delta_t^2}{\alpha^2}\, |\widehat{s} - s^*|^2 \quad (\texttt{x0}).$$

Hence the `eps` loss is J_{DSM} weighted by Δ_t, and the `x0` loss is J_{DSM} weighted by $\Delta_t^2/\alpha^2 = \Delta_t/\mathrm{SNR}_t$. Both have the score as their minimiser at every fixed t.

Proof. From $\widehat{s} = -(\mathbf{x} - \alpha\widehat{m})/\Delta_t$ and $s^* = -(\mathbf{x} - \alpha\mathbf{a})/\Delta_t$, $\widehat{s} - s^* = \alpha(\widehat{m} - \mathbf{a})/\Delta_t$, so $|\widehat{m} - \mathbf{a}|^2 = (\Delta_t^2/\alpha^2)|\widehat{s} - s^*|^2$; this is the `x0` case since $\widehat{m} = f$. In the `eps` case $\widehat{m} - \mathbf{a} = (\mathbf{x} - \sqrt{\Delta_t} f)/\alpha - (\mathbf{x} - \sqrt{\Delta_t}\mathbf{z})/\alpha = -\sqrt{\Delta_t}(f - \mathbf{z})/\alpha$, so $|f - \mathbf{z}|^2 = (\alpha^2/\Delta_t)|\widehat{m} - \mathbf{a}|^2 = \Delta_t|\widehat{s} - s^*|^2$. □

The weights matter once levels are mixed (Section 8.6): `eps` weights the levels almost equally in score units at high noise ($\Delta_t \approx 1$) and suppresses the lowest levels ($\Delta_t \approx 2t$); `x0` suppresses them more strongly. The paper trains both and lets the screen choose (Appendix C, "both output parameterisations"). The conversion $\widehat{s} = -f/\sqrt{\Delta_t}$ for `eps` is the reason a network output of order one corresponds to a score of order $1/\sqrt{\Delta_t}$, as the target requires.

8.5 Computing the divergence

Implicit score matching needs $\operatorname{div}\widehat{s} = \sum_i \partial\widehat{s}_i/\partial x_i = \operatorname{tr} J$ for every example in a batch, where $J = \partial\widehat{s}/\partial\mathbf{x}$ is the $L \times L$ Jacobian of the network's score. Automatic differentiation gives Jacobian-transpose-vector products cheaply: for any $\mathbf{v}$, one backward pass through $\langle\widehat{s}, \mathbf{v}\rangle$ returns $J^\top\mathbf{v}$, since $\partial_j \sum_i \widehat{s}_i v_i = \sum_i v_i J_{ij}$. The function `divergence` in `src/markov_score/nn_torch.py` offers three ways to turn such products into a trace.

8.5.1 The exact trace by L backward passes

Taking $\mathbf{v} = \mathbf{e}_i$ returns the i-th row of J, whose i-th entry is J_{ii}. Summing over i costs L backward passes per batch (the `"exact"` branch: for each i, the gradient of $\sum_{\text{batch}} \widehat{s}_i$ with respect to $\mathbf{x}$, read at coordinate i). At $L = 32$ this is thirty-two passes for every training step, and it is what the frozen matched arm used for the window head (Appendix C: "computed exactly for the window head").

8.5.2 Hutchinson's estimator

Proposition 8.6 (Hutchinson)**.** *Let* $\mathbf{v}$ *be a random vector with* $\mathbb{E}[\mathbf{v}\mathbf{v}^\top] = \mathbf{I}$*, independent of* J*. Then* $\mathbb{E}[\mathbf{v}^\top J\mathbf{v}] = \operatorname{tr} J$*. If* $\mathbf{v} \sim \mathcal{N}(0, \mathbf{I})$*,*

$$\operatorname{Var}(\mathbf{v}^\top J\mathbf{v}) = \|J\|_F^2 + \operatorname{tr}(J^2) = \|J\|_F^2 + \langle J, J^\top\rangle,$$

which equals $2\|J\|_F^2$ *when* J *is symmetric.*

Proof. $\mathbb{E}[\mathbf{v}^\top J\mathbf{v}] = \sum_{ij} J_{ij}\mathbb{E}[v_i v_j] = \sum_i J_{ii}$. For the variance with Gaussian $\mathbf{v}$, use $\mathbb{E}[v_i v_j v_k v_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$:

$$\mathbb{E}\big[(\mathbf{v}^\top J\mathbf{v})^2\big] = \sum_{ijkl} J_{ij}J_{kl}\mathbb{E}[v_i v_j v_k v_l] = (\operatorname{tr} J)^2 + \sum_{ij} J_{ij}^2 + \sum_{ij} J_{ij}J_{ji},$$

and subtract $(\operatorname{tr} J)^2$. □

The score of a density has a symmetric Jacobian (Proposition 7.11); a network's need not, but is close to symmetric when it is close to a score. With one standard-normal probe per example per step (the `"hutchinson"` branch, used by the frozen matched arm for the transformer), the loss estimate is unbiased but its variance adds to the minibatch noise of Section 8.6. Hutchinson (1990) used Rademacher probes ($v_i = \pm 1$), for which the diagonal contributes no variance and $\operatorname{Var} = 2\sum_{i\neq j} J_{ij}J_{ji}$, at most the Gaussian value; the code uses Gaussian probes. For scale: if $J = \mathbf{I}_{32}$ a Gaussian probe estimates the trace 32 with standard deviation $\sqrt{2 \cdot 32} = 8$.

8.5.3 Coloured probes for a banded Jacobian

The window head of the paper computes its output at site i from the inputs at sites $|j - i| \le r$ only (radius r; `receptive_radius` in the code). Its Jacobian is therefore **banded**: $J_{ij} = 0$ for $|i - j| > r$. For such a map the exact trace costs $2r + 1$ backward passes rather than L.

Proposition 8.7 (Coloured probes)**.** *Let* $J_{ij} = 0$ *whenever* $|i - j| > r$*, put* $p = 2r + 1$*, and for* $k = 0, \dots, p - 1$ *let* $\mathbf{v}_k \in \{0, 1\}^L$ *have entries* $(\mathbf{v}_k)_j = \mathbf{1}\{j \equiv k \pmod p\}$*. Then* $\sum_{k=0}^{p-1} \mathbf{v}_k^\top J\mathbf{v}_k = \operatorname{tr} J$ *exactly.*

Proof. $\mathbf{v}_k^\top J\mathbf{v}_k = \sum_{i\equiv k}\sum_{j\equiv k} J_{ij}$. If $i \equiv j \pmod p$ and $i \neq j$ then $|i - j| \ge p = 2r + 1 > r$, so $J_{ij} = 0$. Only $i = j$ survives: $\mathbf{v}_k^\top J\mathbf{v}_k = \sum_{i\equiv k} J_{ii}$. Every i has exactly one residue class, so the sum over k is $\sum_i J_{ii}$. □

This is the colouring of a banded Jacobian of Curtis, Powell and Reid (1974): two columns may share a probe when no row has a nonzero in both, and the residue classes modulo $2r + 1$ are such a partition. The code (`"coloured"` branch) computes, for each k, one backward pass $J^\top \mathbf{v}_k$ and then the inner product with $\mathbf{v}_k$, exactly as for Hutchinson but with p deterministic probes and no variance. Zero padding at the ends of the chain only removes dependencies, so the boundary sites are covered. The branch refuses a model that declares no receptive radius, since for a dense Jacobian the cross terms would not vanish. The identity is checked against the L-pass trace by the test `test_coloured_trace_equals_exact_trace_for_the_window_head` in `tests/test_nn_torch.py`. For the radius-2 head this is five passes in place of thirty-two.

Example 8.8 (Seven sites, radius one)**.** With $L = 7$, $r = 1$ and $p = 3$ the probes are $\mathbf{v}_0 = \mathbf{e}_1 + \mathbf{e}_4 + \mathbf{e}_7$, $\mathbf{v}_1 = \mathbf{e}_2 + \mathbf{e}_5$ and $\mathbf{v}_2 = \mathbf{e}_3 + \mathbf{e}_6$. For a tridiagonal J, $\mathbf{v}_0^\top J\mathbf{v}_0 = J_{11} + J_{44} + J_{77} + (J_{14} + J_{41} + J_{47} + J_{74} + J_{17} + J_{71})$ and every bracketed entry is zero because its indices differ by at least three. Adding the three probes gives $J_{11} + \cdots + J_{77}$.

8.6 Why the loss is estimated per batch

The population objective is an expectation; training minimises it by stochastic gradient descent on unbiased minibatch estimates. Under the paper's recipe (`TrainConfig` and `train` in `src/markov_score/nn_torch.py`) one step draws a batch of 64 examples, and for each example a level t uniformly from the twelve of Table 7.1 and a fresh corruption $\mathbf{z}$; the loss is the mean of the per-example losses; the gradient is taken; its norm is clipped at 1; and Adam takes a step at learning rate 10^{-3} or 3×10^{-3}. Because t and $\mathbf{z}$ are drawn independently of θ, the batch mean is an unbiased estimate of

$$\bar{J}(\theta) = \frac{1}{12} \sum_{k=1}^{12} w(t_k)\, J_{\mathrm{DSM}}^{(t_k)}(s_\theta(\cdot, t_k)) \tag{8.6}$$

with the weights w of Proposition 8.5, and its gradient is an unbiased estimate of $\nabla \bar{J}$. The minimiser of Equation (8.6) over all vector fields is the score at every level simultaneously, since the levels do not compete (the field may depend on t); with a finite network they do compete, and the weights decide the compromise. Checkpoints are saved on a ladder up to 32,000 steps and one is selected afterwards on a validation bundle; the selection rule is part of the interface (Section 8.8). For implicit score matching the same loop draws 64 examples from the fixed set of noised views, computes $\frac{1}{2}|\widehat{s}|^2 + \operatorname{div} \widehat{s}$ per example with one of the three divergence branches, and averages; with a Hutchinson probe the estimate is still unbiased, since the probe is drawn independently of θ, at the cost of extra variance.

8.7 The estimand: schedule-pooled relative score error

Definition 8.9 (The estimand)**.** The paper's headline number for any estimator $\widehat{s}$ is the **schedule-pooled relative score error** (`experiments/protocol.py`, `estimand`; Appendix C, "Estimand"),

$$\operatorname{err}(\widehat{s}) = \left(\frac{\sum_t \sum_j \sum_i \big(\widehat{s}_t(\mathbf{x}_j)_i - s_t(\mathbf{x}_j)_i\big)^2}{\sum_t \sum_j \sum_i s_t(\mathbf{x}_j)_i^2} \right)^{1/2}, \tag{8.7}$$

where t runs over the twelve levels, j over the sequences of a test bundle noised at that level, i over the sites of a region, and s_t is the tree-exact reference score. Both sums are formed before the square root.

Three features of Equation (8.7) matter for reading the paper. It is *relative*: an estimator that outputs zero scores 1, and the isotropic score $-\mathbf{x}$ and the LMMSE twin give the two reference values of the paper's oracle ladder. It is *pooled* rather than averaged over levels: the numerator and the denominator are energies summed over the schedule, so a level contributes in proportion to its score energy, which by Proposition 7.14 is close to L/Δ_t per sequence at low noise. Taking $1/\Delta_t$ as a proxy for the per-site energy, the twelve levels of Table 7.1 carry the weights $10.5, 7.4, 5.3, 3.8, 2.8, 2.1, 1.65, 1.35, 1.16, 1.06, 1.02, 1.00$, whose sum is 39.1; the lowest four levels carry about 69% of the pooled denominator and the highest four about 11%. A pooled error is therefore a low-noise-dominated summary, which the paper says in words (Section 3, "What this says about score learning") and which explains why the fourth-order Edgeworth correction, exact at high noise, is worse than the twin when pooled (Chapter 10). Finally it is computed on *regions*: all 32 sites (the headline), the bulk, sites 5 to 28 (four trimmed at each end, used for every selection so that boundary effects do not drive the choice of checkpoint), and the centre site (a diagnostic). Ratios between arms are formed within seed and averaged, and intervals are percentile bootstraps over seeds (Appendix C).

8.8 The three information interfaces

The paper's Section 5 declares three interfaces and reports the structured estimator against the networks at each. The definitions below follow `experiments/e03_baselines.py` line by line.

Definition 8.10 (Interface A: the standard denoising recipe)**.** The network sees the M clean training sequences. At every step it draws a batch, a level per example from the twelve, and a fresh corruption per example, and minimises the `eps` or `x0` denoising loss (`_train_select` with `loss="dsm"`; `_draw_batch` in `nn_torch.py`). The checkpoint is selected on a validation bundle of 1024 sequences, scored by Equation (8.7) on the bulk sites against the tree-exact reference under the true kernel.

Definition 8.11 (Interface B: clean targets, one fixed corruption)**.** Each training sequence is corrupted once, at one level, with the sequences split evenly across the twelve levels, by the same function `_one_view_groups` that builds the structured estimator's data. The pair $(\mathbf{a}, \mathbf{x})$ is kept and never redrawn, and the target $\mathbf{z} = (\mathbf{x} - e^{-t}\mathbf{a})/\sqrt{\Delta_t}$ (or $\mathbf{a}$) is regressed by the same denoising loss on batches drawn from the fixed pairs; the training loop is `_train_select_fixed_corruption`. The checkpoint is selected as in interface A, on the validation bundle against the true score.

Definition 8.12 (Interface C: one noised view per sequence, no oracle)**.** The network sees only the noised views $(\mathbf{x}^\mu, t_\mu)$ of interface B, without the clean sequences: the same one-view bundle the structured estimator sees. It is trained by implicit score matching (`loss="ism"`), with the divergence computed exactly for the window head and by Hutchinson's estimator for the transformer. In E03 the exact trace of the window head is the L-pass branch (`hutchinson=(arch != "window")` with no `trace` override); in E14, whose network arms train exactly as E03's, it is the coloured-probe branch of Proposition 8.7 (`trace_by_arch`), which returns the same number at a fifth of the cost. The checkpoint is selected on the implicit objective $\mathbb{E}[\frac{1}{2}|\widehat{s}|^2 + \operatorname{div}\widehat{s}]$ of a held-out noisy bundle, one view per validation sequence, with the exact trace (`_ism_objective`, `select_on="ism"`); the true score enters only in the final scoring on the test bundle.

In the paper's words (Section 5): (A) is "clean sequences with a fresh corruption at every step"; (B) is "clean sequences with one fixed corruption each"; (C) is "one noised view per sequence and nothing else, which is what EM–BP sees", and the matched comparison adds that "since the true score is not available for selection in that setting, the checkpoint [is] selected on the implicit score-matching objective of a held-out noisy bundle". The sigma-algebras are nested: C sees strictly less than B, which sees strictly less than A (A sees unboundedly many corruptions of each sequence as training proceeds). The audit of 18–19 September 2026 (`docs/SCIENCE_AUDIT.md`) is the reason the paper now names the interface of every number: an earlier draft called the interface-B arm "information-matched", which it is not, because its regression target is the clean sequence.

Why it matters here. The paper's headline factor at "matched information" is the ratio of the interface-C network's error to the structured estimator's on the same bundles; it is a statement about what the factorisation and the exact E-step are worth at equal information, and Definition 8.12 is what "equal information" means. Interface B isolates the value of fresh corruptions, and the difference between A and B is that value.

8.9 Worked example

Example 8.13 (The three objectives on a Gaussian)**.** Let $L = 1$, $P_0 = \mathcal{N}(0, 1)$, so that $P_t = \mathcal{N}(0, 1)$ at every t and the score is $S(x) = -x$. Consider the linear family $s_c(x) = -cx$.

- Explicit: $J_{\mathrm{ESM}}(c) = \mathbb{E}(-cx + x)^2 = (1 - c)^2$.

- Implicit: $J_{\mathrm{ISM}}(c) = \mathbb{E}[\frac{1}{2}c^2x^2 + (-c)] = \frac{1}{2}c^2 - c = \frac{1}{2}(c-1)^2 - \frac{1}{2}$. The offset $-\frac{1}{2}$ is $-\frac{1}{2}\mathbb{E}|\nabla \log p|^2$, as Equation (8.2) says, and the minimiser is $c = 1$ in both.
- Denoising: with $x = \alpha a + \sqrt{\Delta_t}z$ and target $-z/\sqrt{\Delta_t}$, $J_{\mathrm{DSM}}(c) = \mathbb{E}(-cx + z/\sqrt{\Delta_t})^2 = c^2\mathbb{E}x^2 - 2c\,\mathbb{E}[xz]/\sqrt{\Delta_t} + \mathbb{E}z^2/\Delta_t = c^2 - 2c + 1/\Delta_t = (c-1)^2 + e^{-2t}/\Delta_t$, since $\mathbb{E}[xz] = \sqrt{\Delta_t}$ and $1/\Delta_t - 1 = e^{-2t}/\Delta_t$. The offset is the irreducible term $(e^{-2t}/\Delta_t^2)\mathrm{MMSE}_t$ of Corollary 8.3 with $\mathrm{MMSE}_t = \mathrm{Var}(a \mid x) = 1 - \alpha^2/(\alpha^2 + \Delta_t) = \Delta_t$. At $t = \log 2$ it equals $(1/4)/(3/4) = \frac{1}{3}$, the SNR.

The `eps` loss for the same family is $\Delta_t J_{\mathrm{DSM}}$ and the `x0` loss is $(\Delta_t^2/\alpha^2)J_{\mathrm{DSM}}$: at $t = \log 2$ these are $\frac{3}{4}J_{\mathrm{DSM}}$ and $\frac{9}{4}J_{\mathrm{DSM}}$, and at level 1 of Table 7.1 they are $0.095\,J_{\mathrm{DSM}}$ and $0.010\,J_{\mathrm{DSM}}$.

Where it is used. Score matching is the training principle of every network in the paper's Section 5 and Appendix C; the interfaces are defined in Section 5 and used in the baselines and matched-comparison paragraphs; the estimand is Appendix C. Experiments: E03 (screen, confirmatory table, matched arm, length extrapolation), E08 (misspecification network arm), E14 (regime baselines, whose network arms train exactly as E03's: window head with the exact trace by coloured probes, transformer with Hutchinson's estimator), and E05 for the network samplers. Code: `src/markov_score/nn_torch.py` (`TrainConfig`, `train`, `dsm_loss`, `ism_loss`, `divergence`, `ism_objective`, `Denoiser`), `experiments/e03_baselines.py` (`_train_select`, `_one_view_groups`, `_train_select_fixed_corruption`, `_ism_objective`, `_risk`, `_sq`), `experiments/protocol.py` (`estimand`, `bulk_margin`, `net_steps`), `tests/test_nn_torch.py`.

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Chapter 9

Reverse-time diffusion and generation

A diffusion model generates by running the channel of Chapter 7 backwards in time, and the only ingredient the backward equation needs that the forward one does not is the score. This chapter derives the reverse-time equation (Anderson's theorem) from the Fokker–Planck equation, explains why an error in the score is exactly what an error in the generated law is made of, writes down the Euler–Maruyama scheme the code uses, and then describes the paper's generation experiment E05: its same-Brownian-path design, its arms, and every metric by which the samples are judged, including the transition-law metric (Hellinger distance) used in the learning section.

9.1 Anderson's reverse-time equation

Consider a forward diffusion on $[0, T]$,

$$\mathrm{d}\mathbf{x}_t = f(\mathbf{x}_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 \sim P_0, \tag{9.1}$$

with smooth positive marginal densities p_t. The evolution of p_t is the **Fokker–Planck equation**

$$\partial_t p_t = -\nabla \cdot (f\, p_t) + \tfrac{1}{2} g^2\, \Delta p_t, \tag{9.2}$$

where $\nabla\cdot$ is the divergence and $\Delta = \nabla \cdot \nabla$ the Laplacian (not to be confused with Δ_t). The reader who has not met Equation (9.2) may read it as a conservation law: the drift transports probability with velocity f, and the noise spreads it by diffusion with coefficient $\frac{1}{2}g^2$.

Theorem 9.1 (Time reversal)**.** *Let $\tau = T - t$ and let $\bar{\mathbf{x}}_\tau$ solve*

$$\mathrm{d}\bar{\mathbf{x}}_\tau = \big[-f(\bar{\mathbf{x}}_\tau, T-\tau) + g(T-\tau)^2\, \nabla \log p_{T-\tau}(\bar{\mathbf{x}}_\tau)\big]\,\mathrm{d}\tau + g(T-\tau)\,\mathrm{d}\bar{\mathbf{W}}_\tau, \qquad \bar{\mathbf{x}}_0 \sim p_T, \tag{9.3}$$

with $\bar{\mathbf{W}}$ a Brownian motion. Then $\bar{\mathbf{x}}_\tau$ has the law $p_{T-\tau}$ for every $\tau \in [0, T]$; in particular $\bar{\mathbf{x}}_T \sim P_0$. Anderson (1982) proved more: the process $t \mapsto \mathbf{x}_{T-t}$ itself is a solution of Equation (9.3)*.*

Proof of the marginal statement. Scalar case, and write $p = p_t(x)$. Since $\frac{1}{2}g^2\partial_x^2 p = \partial_x(g^2\partial_x p) - \frac{1}{2}g^2\partial_x^2 p$ and $\partial_x p = p\,\partial_x \log p$, the Fokker–Planck equation can be rewritten as

$$\partial_t p = -\partial_x\Big[(f - g^2\partial_x \log p)p\Big] - \tfrac{1}{2} g^2\, \partial_x^2 p.$$

Now set $q(x, \tau) = p_{T-\tau}(x)$, so that $\partial_\tau q = -\partial_t p$:

$$\partial_\tau q = -\partial_x\Big[(-f + g^2\partial_x \log q)q\Big] + \tfrac{1}{2} g^2\, \partial_x^2 q.$$

This is the Fokker–Planck equation of the diffusion with drift $-f+g^2\partial_x \log q$ and diffusion coefficient g, run in the time τ, and $q(\cdot,0)=p_T$. The solution of Equation (9.3) has marginals obeying the same equation with the same initial condition, hence the same marginals, by uniqueness for this parabolic equation. In L dimensions replace ∂_x by $\nabla\cdot$ and ∂_x^2 by Δ line by line. □

A useful by-product of the same rewriting is the **probability-flow ODE**: $\frac{1}{2}g^2\Delta p = \nabla\cdot(\frac{1}{2}g^2 p\nabla\log p)$ turns Equation (9.2) into $\partial_t p = -\nabla\cdot[(f-\frac{1}{2}g^2\nabla\log p)p]$, the transport equation of the deterministic flow $\dot{\mathbf{x}} = f-\frac{1}{2}g^2\nabla\log p_t$, which has the same marginals as the SDE and is implemented in `probability_flow_ode` of `src/markov_score/reverse.py`.

Corollary 9.2 (The channel reversed)**.** *For the channel $f(\mathbf{x},t)=-\mathbf{x}$, $g=\sqrt{2}$, the reverse equation in the original time t running from T down to 0 is*

$$\mathrm{d}\mathbf{x} = \big[-\mathbf{x} - 2\,S(\mathbf{x},t)\big]\,\mathrm{d}t + \sqrt{2}\;\mathrm{d}\bar{\mathbf{W}}_t \qquad (\,\mathrm{d}t<0), \tag{9.4}$$

that is, in the forward-running time τ, $\mathrm{d}\bar{\mathbf{x}} = [\bar{\mathbf{x}} + 2S(\bar{\mathbf{x}},T-\tau)]\,\mathrm{d}\tau + \sqrt{2}\;\mathrm{d}\bar{\mathbf{W}}_\tau$.

The drift has two parts. The term $\bar{\mathbf{x}}$ undoes the contraction of the forward drift; the term $2S$ pushes towards regions where P_t is large. At high noise, where $S\approx-\mathbf{x}$, the two nearly cancel and the reverse process is again an Ornstein–Uhlenbeck process (Example 9.7); at low noise the score carries all the structure of P_0.

9.2 Why the score is exactly what drives generation

Everything in Equation (9.4) is known except S: the drift $-\mathbf{x}$ and the coefficient $\sqrt{2}$ are the channel's. A generator that replaces S by an estimate $\widehat{s}$ runs a different diffusion and produces a different law, and the discrepancy is controlled by the score error alone.

Proposition 9.3 (Score error controls the generated law)**.** *Let $\mathbb{P}$ be the law of the paths of Equation (9.4) on $[t_{\min},T]$ started at $\mathbf{x}_T\sim P_T$, and $\mathbb{Q}$ the law of the paths of the same equation with $\widehat{s}$ in place of S and started at $\gamma=\mathcal{N}(0,\mathbf{I})$. Then*

$$D_{\mathrm{KL}}(P_{t_{\min}}\,\|\,\mathbb{Q}_{t_{\min}}) \le D_{\mathrm{KL}}(P_T\,\|\,\gamma) + \int_{t_{\min}}^{T}\mathbb{E}_{P_t}\big|\widehat{s}(\mathbf{x}_t,t)-S(\mathbf{x}_t,t)\big|^2\,\mathrm{d}t, \tag{9.5}$$

where $\mathbb{Q}_{t_{\min}}$ is the law of the generated sample at time $t_{\min}$.

Proof sketch. Two diffusions with the same diffusion coefficient g and drifts μ_1,μ_2 have path laws whose Kullback–Leibler divergence is, by Girsanov's theorem, $\frac{1}{2}\int\mathbb{E}|\mu_1-\mu_2|^2/g^2\,\mathrm{d}t$ plus the divergence of the initial laws. Here $\mu_1-\mu_2 = 2(S-\widehat{s})$ and $g^2=2$, so the integrand is $\frac{1}{2}\cdot 4|S-\widehat{s}|^2/2 = |S-\widehat{s}|^2$, with the expectation under the true path law, whose marginal at time t is P_t. The divergence of the marginals at $t_{\min}$ is at most that of the path laws (data processing). □

This is Theorem 1 of Song, Durkan, Murray and Ermon (2021). It says that the time-integrated mean squared score error under the true noised laws is the quantity a generator pays for, in nats. The paper's estimand (Definition 8.9) is the square root of that integrand summed over the twelve levels and divided by the reference energy; the generation experiment E05 measures whether the ordering of estimators by score error is the ordering by sample quality, which Equation (9.5) predicts but does not guarantee, because it is an upper bound.

9.3 The Euler–Maruyama scheme and its grid

Definition 9.4 (Euler–Maruyama for the reverse channel)**.** Given a decreasing sequence of times $T = t_0 > t_1 > \cdots > t_n = t_{\min}$ with steps $h_k = t_k - t_{k+1} > 0$, the scheme is

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h_k\big[\mathbf{x}_k + 2\,\widehat{s}(\mathbf{x}_k, t_k)\big] + \sqrt{2h_k}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, \mathbf{I}) \text{ i.i.d.}, \qquad \mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})\,. \quad (9.6)$$

This is `reverse_sde` and `reverse_sde_with_increments` in `src/markov_score/reverse.py`: the drift is evaluated at the left endpoint, and the Brownian increment over the step is $\sqrt{h_k}\,\xi_k$, scaled by the $\sqrt{2}$ of the channel. Three choices need explaining.

The grid is geometric. The code uses $t_k = t_{\max}\, r^k$ with $r = (t_{\min}/t_{\max})^{1/n}$ (`time_grid`), so that $h_k = (1 - r)\, t_k$: the step is a fixed fraction of the current time. The reason is that the reverse drift can be stiff at small t: by Equation (7.10), $\nabla S = -\mathbf{I}/\Delta_t + (e^{-2t}/\Delta_t^2)\,\mathrm{Cov}(\mathbf{a} \mid \mathbf{x})$, whose entries can be of size $1/\Delta_t \approx 1/(2t)$ when the posterior is narrower than the likelihood (a kinked or discrete prior), and an explicit scheme is only stable for steps below the reciprocal of the stiffness. A step proportional to t keeps $h_k/\Delta_{t_k} \approx (1 - r)/2$ fixed. With the paper's $t_{\max} = 3$, $t_{\min} = 0.02$ and $n = 400$ steps (`experiments/e05_generation.py`), $\ln(t_{\min}/t_{\max}) = -5.0106$, $r = e^{-0.012527} = 0.98755$, the first step is $h_0 = 3 \times 0.01245 = 0.0374$ and the last is $h_{399} = 0.0203 \times 0.01245 = 0.00025$; the ratio h_k/Δ_{t_k} is about 0.006 throughout.

The start is $\mathcal{N}(0, \mathbf{I})$, not P_T. The true law at $T = 3$ is not available to sample from, and γ is its limit (Proposition 7.6). The cost is the first term of Equation (9.5). For the covariance twin it is computable: $\mathbf{\Sigma}_T = \mathbf{I} + e^{-2T}(\mathbf{\Sigma}_0 - \mathbf{I})$ and, to second order in e^{-2T}, $D_{\mathrm{KL}}(\mathcal{N}(0, \mathbf{\Sigma}_T)\,\|\gamma) = \frac{1}{2}[\operatorname{tr}\mathbf{\Sigma}_T - L - \log\det\mathbf{\Sigma}_T] \approx \frac{1}{4}e^{-4T}\|\mathbf{\Sigma}_0 - \mathbf{I}\|_F^2$. For the frozen chain, $\|\mathbf{\Sigma}_0 - \mathbf{I}\|_F^2 = 2\sum_{k=1}^{31}(32 - k)\rho^{2k} \approx 148$ at $\rho = 0.85$ (a geometric sum), so the start costs about $\frac{1}{4}e^{-12} \cdot 148 \approx 2 \times 10^{-4}$ nats: negligible next to the discretisation, and the reason $t_{\max} = 3$ suffices.

The stop is $t_{\min} = 0.02$, not 0. At $t = 0$ the score of a law with a density that is not smooth (the Laplace chain has a kink) is singular, and for a law without a density it does not exist. The generated sample is therefore a sample at $t_{\min}$ and is compared with a reference sample of the true law at $t_{\min}$, obtained by drawing clean sequences and noising them once (`rng_ref` in `_gen_cell`); comparing with clean sequences would be a category error. The alternative readout $\langle\mathbf{a}\rangle_{x,t} = e^t(\mathbf{x} + \Delta_t S)$ of Theorem 7.9 (`denoising_readout`) returns a posterior mean, which is not a sample.

Order of accuracy. For an additive noise, as here, Euler–Maruyama has strong order one (the pathwise error is $O(h)$) and weak order one (errors of expectations are $O(h)$); the discretisation bias is measured, not assumed, by the step study below.

9.4 The same-Brownian-path design of E05

The experiment (`experiments/e05_generation.py`) is organised in cells (seed, M). In each cell one initial sample $\mathbf{x}_0$ of 2000 sequences and one array of Brownian increments for all 400 steps are drawn once, and every arm is integrated by `reverse_sde_with_increments` on exactly those increments. The arms differ only in the score function $\widehat{s}$:

- *exact*: the tree-exact score under the true kernel (grid message passing; its distance to the true law is the sampler's own discretisation floor);
- *em–bp*: the mixture kernel fitted in E02a for the same seed and M, at its validation-selected checkpoint, driven through the same message passing;

- *lmmse*: the covariance twin, $\widehat{s} = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$ (Corollary 7.12);
- *isotropic*: $\widehat{s} = -\mathbf{x}$, the score of γ;
- *net*: network checkpoints from E03's confirmatory part for the same cell, converted from posterior means by Equation (7.12).

Because the noise is shared, the difference between two arms' samples is due to their scores and not to Monte Carlo scatter, and the metrics can be compared within cell and then paired across seeds, as everywhere else in the paper. The reference sample R of 4000 sequences at $t_{\min}$ is drawn from its own stream. The *step* part of the experiment integrates the exact arm at 100, 200, 400 and 800 steps on **nested Brownian paths**: the increments of the finest grid are drawn once and the coarse grid's increment over an interval is the sum of the fine increments inside it (`nested_brownian_path`, `coarsen_increments`). Two resolutions are then driven by the same Brownian path and the root mean square of their difference is a **strong error**, a pure measure of discretisation. With independent noise at each resolution the difference would be dominated by sampling scatter and the study would show only robustness, which is the mistake the code's docstring records.

9.5 The sample-quality metrics

Let $G \in \mathbb{R}^{n_g \times L}$ be the generated sample and $R \in \mathbb{R}^{n_r \times L}$ the reference sample, both at $t_{\min}$. All definitions are those of `src/markov_score/gen_metrics.py` (`compare`); the paper's generation table prints the first, the second, the fourth and the sliced Wasserstein distance.

Residuals. For a sample Y the **autoregressive residuals** are $r_{j,i} = y_{j,i+1} - \rho\, y_{j,i}$ with the true ρ, $i = 1, \dots, L-1$. At $t_{\min} = 0$ they would be the innovations ε_{i+1}; at $t_{\min} > 0$ they are $e^{-t_{\min}}\varepsilon_{i+1} + \sqrt{\Delta_{t_{\min}}}(z_{i+1} - \rho z_i)$.

Excess kurtosis. For a pooled sample $\{u\}$ with central moments m_2, m_4, $\widehat{\kappa} = m_4/m_2^2 - 3$. Applied to the residuals it measures the non-Gaussian shape of the innovation that the sampler reproduces; a Gaussian-innovation sampler (the twin) gives zero up to sampling error. Its standard error is a bootstrap over sequences (200 resamples).

Proposition 9.5 (Residual kurtosis at $t_{\min}$)**.** *If the innovation has variance q and excess kurtosis κ, then the residual of the true law at level t has*

$$\mathrm{Var}(r) = e^{-2t}q + \Delta_t(1+\rho^2), \qquad \mathrm{cum}_4(r) = e^{-4t}\kappa\, q^2, \qquad \textit{excess kurtosis} = \frac{\kappa\, e^{-4t} q^2}{\left(e^{-2t}q + \Delta_t(1+\rho^2)\right)^2}.$$

Proof. $r = e^{-t}\varepsilon + \sqrt{\Delta_t}(z' - \rho z)$ is a sum of independent terms; variances add, and fourth cumulants add with the Gaussian terms contributing nothing. □

So the reference row of the paper's table does not show the innovation's excess kurtosis 3 but its attenuated and diluted value at $t_{\min}$; the worked example computes it. The same holds for every arm, which is why the table is read as a comparison with the reference row and not with 3.

Lag fourth cumulant. $\mathrm{cum}(r_i, r_i, r_{i+1}, r_{i+1}) = \mathbb{E}[r_i^2 r_{i+1}^2] - \mathbb{E}[r_i^2]\mathbb{E}[r_{i+1}^2] - 2\mathbb{E}[r_i r_{i+1}]^2$ for centred residuals, pooled over sites. For the true law it is exactly zero at every t: r_i and r_{i+1} share only the Gaussian z_{i+1}, joint cumulants are additive over independent components, and a Gaussian has no fourth cumulant. It is therefore a check for spurious dependence between neighbouring residuals, which a sampler with a wrong score can create; it is not a signature of the innovation's shape.

Marginal Wasserstein-1. Pooling all entries of G and of R into two one-dimensional samples with quantile functions F_G^{-1}, F_R^{-1}, $W_1 = \int_0^1 |F_G^{-1}(u) - F_R^{-1}(u)|\,\mathrm{d}u$, approximated by the mean over 1999 equally spaced u. It compares the site marginals only.

Sliced Wasserstein-2. For 256 random unit vectors θ_k (common seed), project both samples, $G\theta_k$ and $R\theta_k$, compute the one-dimensional W_2^2 of the projections by quantile matching, $W_2^2(\mu,\nu) = \int_0^1 (F_\mu^{-1} - F_\nu^{-1})^2\,\mathrm{d}u$, and report $\left(\frac{1}{256}\sum_k W_2^2\right)^{1/2}$. It sees the joint law through its one-dimensional projections, and is dominated by second-order structure: two laws with the same covariance and different fourth cumulants project to nearly the same one-dimensional laws, which is why the twin and the exact arm are close in this column.

Energy distance. $2\mathbb{E}|G-R| - \mathbb{E}|G-G'| - \mathbb{E}|R-R'|$ with independent copies, estimated on subsamples of 1000 (Székely and Rizzo); it is zero if and only if the laws coincide, and is again dominated by low-order structure in practice.

Covariance errors. The sample covariance $\widehat{\mathbf{\Sigma}}$ of G against a target $\mathbf{\Sigma}$: the relative Frobenius error $\|\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_F / \|\mathbf{\Sigma}\|_F$ and the worst absolute error of the mean diagonal at lags 0 to 5. The code passes $\mathbf{\Sigma}_0$ as the target although the sample lives at $t_{\min}$, where the covariance is $\mathbf{\Sigma}_{t_{\min}} = e^{-2t_{\min}}\mathbf{\Sigma}_0 + \Delta_{t_{\min}}\mathbf{I}$; the difference is $\Delta_{t_{\min}}(\mathbf{\Sigma}_0 - \mathbf{I})$, whose relative Frobenius size at $t_{\min} = 0.02$ is about $0.039 \times \|\mathbf{\Sigma}_0 - \mathbf{I}\|_F / \|\mathbf{\Sigma}_0\|_F \approx 0.035$ for the frozen chain, a floor the exact arm cannot go below. The paper's table does not print this column.

Residual autocorrelation. $\rho_k = \mathbb{E}[r_i r_{i+k}]/\mathbb{E}[r_i^2]$ for $k = 1,\dots,5$ on the pooled centred residuals; the reported number is the largest absolute difference between the generated and the reference values. For the true law at $t_{\min}$ the residuals are not white ($\rho_1 = -\rho\Delta_{t_{\min}}/\operatorname{Var}(r)$, from the shared z_{i+1}), which is why the reference values are subtracted rather than zero.

Histogram divergence of the residuals. With 60 bins between the 0.001 and 0.999 quantiles of the reference residuals, p the reference histogram and q the generated one with add-one smoothing, $\sum_{p_b>0} p_b \log(p_b/q_b)$: a Kullback–Leibler divergence of the innovation shape, true against generated.

Log-likelihood under the true law. For every sequence of G and of R the evidence $\log P_{t_{\min}}(\mathbf{x})$ under the true kernel is computed exactly for the discretised model by the same message passing that gives the score (the accumulated normalisation constants of the recursion; the function `log_likelihood_true`), and the reported gap is

$$\Delta\ell = \frac{1}{n_r}\sum_R \log P_{t_{\min}}(\mathbf{x}) - \frac{1}{n_g}\sum_G \log P_{t_{\min}}(\mathbf{x}). \tag{9.7}$$

Writing P for the true law and Q for the generated one, in expectation $\Delta\ell = \mathbb{E}_P \log P - \mathbb{E}_Q \log P = D_{\mathrm{KL}}(Q\|P) + H(Q) - H(P)$ with H the differential entropy: a difference of cross-entropies, not a divergence. It is zero when $Q = P$, positive when Q is misplaced or too spread, and can be negative when Q is more concentrated than P on the high-density region. The exact arm's value is the discretisation floor of the sampler, since its only error is the scheme's; the superseded four-seed run recorded a negative value for that arm, which is consistent with the definition and is the reason the sign is worth stating.

9.6 A transition-law metric: the Hellinger distance

The learning section of the paper reports that the transition law is recovered "to Hellinger distance" a stated value at the lowest level. The metric is the function `transition_hellinger` of `src/markov_score/metrics.py`.

Definition 9.6 (Hellinger distance)**.** For two densities p, q on $\mathbb{R}$,

$$H^2(p,q) = \tfrac{1}{2}\int \left(\sqrt{p} - \sqrt{q}\right)^2 \mathrm{d}u = 1 - \int \sqrt{pq}\ \mathrm{d}u \in [0,1],$$

the second form because $\int p = \int q = 1$. On the quadrature grid with weights w_k the integral is $\frac{1}{2}\sum_k w_k(\sqrt{p_k} - \sqrt{q_k})^2$ after each column has been normalised under the same weights.

For a transition kernel every parent state u_j gives one density $K(\cdot \mid u_j)$, hence one distance H_j between the fitted and the true kernel. The code reports the median, mean, 90th percentile and maximum over parents, the same over the interior parents $|u_j| \le A/2$ (near the edge of the domain the true column has mass outside the grid and the distance picks up truncation), and versions weighted by the parent law, which defaults to the standard normal because $\mathrm{Var}(a_i) = 1$: the weighted mean $\sum_j \pi_j H_j$, the weighted median, and $\sum_j \pi_j H_j^2$. The second form of the definition is the one the code evaluates, because $1 - \int \sqrt{pq}$ subtracts two numbers near one and loses everything below 10^{-15} to cancellation; the first form returns exactly zero for identical columns. Which of these summaries the paper's sentence quotes is not said in the paper; the reader should take the weighted mean as the intended one and check the generator.

9.7 Worked examples

Example 9.7 (The reverse of a Gaussian, and the bias of a step)**.** Let $P_0 = \mathcal{N}(0,1)$ in one dimension, so that $S(x,t) = -x$ at every t. The reverse equation Equation (9.4) in the time τ is $\mathrm{d}\bar{x} = [\bar{x} - 2\bar{x}]\,\mathrm{d}\tau + \sqrt{2}\ \mathrm{d}\bar{W} = -\bar{x}\ \mathrm{d}\tau + \sqrt{2}\ \mathrm{d}\bar{W}$: the forward channel again, for which $\mathcal{N}(0,1)$ is stationary, as it must be. The Euler step Equation (9.6) with constant h is $x \leftarrow (1-h)x + \sqrt{2h}\,\xi$, and the variance v of the iterate obeys $v \leftarrow (1-h)^2 v + 2h$, whose fixed point is

$$v^* = \frac{2h}{1-(1-h)^2} = \frac{2h}{2h-h^2} = \frac{2}{2-h} = 1 + \tfrac{h}{2} + O(h^2).$$

The scheme's stationary variance is biased by $h/2$: with the first geometric step $h_0 = 0.0374$ the bias would be 1.9%, with the last, 0.01%. This is the weak error of order one, and it is what the exact arm's log-likelihood gap measures on the chain.

Example 9.8 (The residual kurtosis of the frozen chain at $t_{\min}$)**.** Take $\rho = 0.85$, $q = 1 - \rho^2 = 0.2775$, a Laplace innovation ($\kappa = 3$) and $t_{\min} = 0.02$, so $e^{-2t_{\min}} = 0.96079$, $e^{-4t_{\min}} = 0.92312$, $\Delta_{t_{\min}} = 0.03921$. By Proposition 9.5,

$$\mathrm{Var}(r) = 0.96079 \times 0.2775 + 0.03921 \times 1.7225 = 0.26662 + 0.06754 = 0.33416,$$

$$\mathrm{cum}_4(r) = 3 \times 0.92312 \times 0.2775^2 = 3 \times 0.92312 \times 0.07701 = 0.21327,$$

and the excess kurtosis is $0.21327/0.33416^2 = 0.21327/0.11166 = 1.91$. A perfect sampler at $t_{\min} = 0.02$ should therefore show a residual excess kurtosis near 1.9, not 3; the paper's reference row is to be read against this value (see the paper, Section 5, the generation table). The Gaussian twin shows 0; a sampler that reproduces only part of the fourth cumulant lands in between.

Example 9.9 (Hellinger distances one can compute by hand)**.** For $p = \mathcal{N}(0,1)$ and $q = \mathcal{N}(0,\sigma^2)$, $\int \sqrt{pq} = \sqrt{2\sigma/(1+\sigma^2)}$, so at $\sigma = 1.1$: $2.2/2.21 = 0.99548$, square root 0.99774, $H^2 = 0.00226$, $H = 0.048$. For $p = \mathcal{N}(0,1)$ and $q = \mathcal{N}(\mu,1)$, $\int \sqrt{pq} = e^{-\mu^2/8}$, so at $\mu = 0.5$: $H^2 = 1 - e^{-0.03125} = 0.0308$, $H = 0.175$. A ten per cent error in the width of an innovation costs less than a half-unit shift of its centre; a column-wise Hellinger summary is sensitive to both.

Where it is used. Generation is Section 5 of the paper ("Generation") with its table, and the reverse SDE is named there with its step count, grid and shared Brownian path; the Hellinger sentence is in Section 4. Experiment E05 (parts `generate` and `steps`); the arms come from E02a (fitted kernels) and E03 (network checkpoints). Code: `src/markov_score/reverse.py` (`time_grid`, `reverse_sde`, `reverse_sde_with_increments`, `nested_brownian_path`, `coarsen_increments`, `probability_flow_ode`, `denoising_readout`), `src/markov_score/gen_metrics.py` (`compare` and the functions it calls), `src/markov_score/metrics.py` (`transition_hellinger`), `experiments/e05_generation.py`.

References. B. D. O. Anderson, 1982, "Reverse-time diffusion equation models", Stochastic Processes and their Applications 12, 313–326. U. G. Haussmann and E. Pardoux, 1986, "Time reversal of diffusions", Annals of Probability 14, 1188–1205. Y. Song, C. Durkan, I. Murray and S. Ermon, 2021, "Maximum likelihood training of score-based diffusion models", Advances in Neural Information Processing Systems 34. P. E. Kloeden and E. Platen, 1992, *Numerical Solution of Stochastic Differential Equations*, Springer. G. J. Székely and M. L. Rizzo, 2013, "Energy statistics: a class of statistics based on distances", Journal of Statistical Planning and Inference 143, 1249–1272. J. Rabin, G. Peyré, J. Delon and M. Bernot, 2012, "Wasserstein barycenter and its application to texture mixing", Scale Space and Variational Methods in Computer Vision, Lecture Notes in Computer Science 6667, 435–446.

Chapter 10

The attenuation theorems, step by step

This chapter is a line-by-line reading of Section 3 of the paper and of the proofs in its Appendix A. It states each result, then derives it: the exact attenuation of Hermite moments, the expansion of the noised log-density in the signal-to-noise ratio, the leading term of the gap between the true score and that of the covariance twin with its constant $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$, the Fisher information constants $\frac{1}{2}\|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$ and $\frac{1}{24}\|\partial_\theta \boldsymbol{\kappa}_4\|_F^2$, the general-order theorem, the sample-complexity corollary and its caveats, the crossover t_*, what happens for heavy tails, and the Edgeworth correction that puts the missing term back. The scalar case is always worked first. The notation is that of Chapter 7: $\alpha = e^{-t}$, $\Delta_t = 1 - \alpha^2$, $\mathbf{x}_t = \alpha \mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$, and γ is the standard Gaussian on $\mathbb{R}^L$.

10.1 The statements

The paper's Section 3 contains, in order: Theorem 2 (Hermite attenuation), which is exact at every t and needs only finite moments; Assumption (E), the exponential moment; Theorem 3 (order-by-order information decay), three asymptotic equalities as $t \to \infty$ at fixed L,

$$\mathbb{E}_{P_t}\|S - S_{\mathrm{G}}\|^2 = \tfrac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}(1+\epsilon_t), \tag{10.1}$$

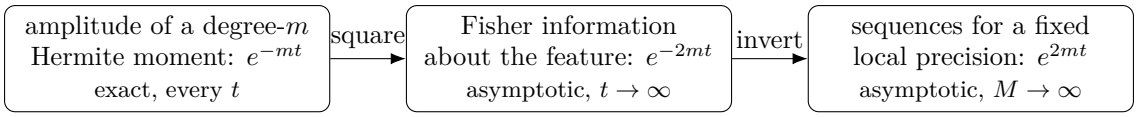
$$I_t(\theta) = \tfrac{1}{2}\|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2 e^{-4t}(1+\epsilon_t) \quad \text{if } \partial_\theta \boldsymbol{\Sigma}_0 \neq 0, \tag{10.2}$$

$$I_t(\theta) = \tfrac{1}{24}\|\partial_\theta \boldsymbol{\kappa}_4\|_F^2 e^{-8t}(1+\epsilon_t) \quad \text{if } \partial_\theta \boldsymbol{\Sigma}_0 = 0,\ \partial_\theta \boldsymbol{\kappa}_4 \neq 0, \tag{10.3}$$

with $\epsilon_t = O(e^{-2t})$; Theorem 4 (general order), which replaces the constants by the Hermite norm of the derivative of the degree-m moment tensor; Corollary 5 (sample complexity by order), a local asymptotic minimax statement; a paragraph on the crossover $t_* = \frac{1}{2}\log \lambda_{\max}(\boldsymbol{\Sigma}_0)$; and a paragraph on heavy tails, stated as an observation and not a theorem. Here S is the score of P_t, $S_{\mathrm{G}} = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$ that of the covariance twin (Corollary 7.12), $\boldsymbol{\kappa}_4$ the fourth-cumulant tensor of P_0, $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ the Fisher information one noised sequence carries about a parameter θ of P_0, and $\|\cdot\|_F$ the Frobenius norm over ordered indices.

10.2 Three clocks

The paper's abstract and Section 3 use three exponentials for one feature, and the audit of 18–19 September 2026 asked that the dictionary be stated once. Figure 10.1 is that dictionary. A **feature** of P_0 is any function of its law; its **Hermite degree** m is the lowest degree of a Hermite moment $\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]$ that the feature moves. Correlation moves $\mathbb{E}[a_i a_j]$, degree two; the shape of the innovation at fixed covariance first moves the fourth cumulant, degree four. The amplitude of the moment decays as e^{-mt} exactly (Theorem 10.2); the Fisher information, being



correlation ($m = 2$): e^{-2t}	e^{-4t}	e^{4t}
shape ($m = 4$): e^{-4t}	e^{-8t}	e^{8t}

Figure 10.1: The three clocks of the paper for a feature first visible at Hermite degree m. The first is a theorem at every t; the second and third are asymptotic statements, in t and in M respectively, and the two limits do not commute (Section 10.11).

the expectation of a squared score, decays as the square, e^{-2mt} (Section 10.9); and the number of sequences needed to hold a local precision fixed is the reciprocal of the information, e^{2mt} (Section 10.11). The squared score gap Equation (10.1) is the square of a degree-four amplitude, hence e^{-8t}.

10.3 Hermite attenuation

The probabilists' **Hermite polynomials** are $\mathrm{He}_0 = 1$, $\mathrm{He}_1(x) = x$, $\mathrm{He}_2(x) = x^2 - 1$, $\mathrm{He}_3(x) = x^3 - 3x$, $\mathrm{He}_4(x) = x^4 - 6x^2 + 3$, and in general $\mathrm{He}_{n+1} = x\mathrm{He}_n - n\mathrm{He}_{n-1}$; they are generated by

$$e^{sx - s^2/2} = \sum_{n \geq 0} \mathrm{He}_n(x) \frac{s^n}{n!} \tag{10.4}$$

and are orthogonal under γ with $\mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m] = n!\, \delta_{nm}$. For a multi-index $\nu \in \mathbb{N}^L$ put $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$, of degree $|\nu| = \sum_i \nu_i$, with $\mathbb{E}_\gamma[H_\nu H_\mu] = \nu!\, \delta_{\nu\mu}$, $\nu! = \prod_i \nu_i!$. The compendium's chapter on Gaussian and Hermite structure develops these; here only Equation (10.4) and the orthogonality are used.

Lemma 10.1 (Mehler)**.** *For $z \sim \mathcal{N}(0,1)$, real a and $\alpha \in [0,1]$ with $\beta = \sqrt{1 - \alpha^2}$, $\mathbb{E}_z\big[\mathrm{He}_n(\alpha a + \beta z)\big] = \alpha^n \mathrm{He}_n(a)$ for every n.*

Proof. Put $x = \alpha a + \beta z$ in Equation (10.4) and average over z. On the left,

$$\mathbb{E}_z e^{s(\alpha a + \beta z) - s^2/2} = e^{s\alpha a - s^2/2}\, \mathbb{E}_z e^{s\beta z} = e^{s\alpha a - s^2/2 + s^2\beta^2/2} = e^{s\alpha a - s^2\alpha^2/2},$$

which by Equation (10.4) again (with $x = a$ and s replaced by αs) is $\sum_n \mathrm{He}_n(a)(\alpha s)^n / n!$. On the right, exchanging $\mathbb{E}_z$ and the sum (the series converges absolutely and uniformly on compact sets of s), $\sum_n \mathbb{E}_z[\mathrm{He}_n(\alpha a + \beta z)] s^n / n!$. Comparing coefficients of s^n gives the claim. □

Theorem 10.2 (Hermite attenuation; the paper's Theorem 2)**.** *For every multi-index ν with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$,*

$$\mathbb{E}_{P_t}\big[H_\nu(\mathbf{x}_t)\big] = e^{-|\nu| t}\, \mathbb{E}_{P_0}\big[H_\nu(\mathbf{a})\big].$$

Proof. Conditionally on $\mathbf{a}$ the coordinates $x_{t,i} = \alpha a_i + \sqrt{\Delta_t} z_i$ are independent, so $\mathbb{E}[H_\nu(\mathbf{x}_t) \mid \mathbf{a}] = \prod_i \mathbb{E}_{z_i}[\mathrm{He}_{\nu_i}(\alpha a_i + \sqrt{\Delta_t} z_i)] = \prod_i \alpha^{\nu_i} \mathrm{He}_{\nu_i}(a_i) = \alpha^{|\nu|} H_\nu(\mathbf{a})$ by Lemma 10.1 with $\beta = \sqrt{\Delta_t}$. Averaging over $\mathbf{a} \sim P_0$ needs $\mathbb{E}|H_\nu(\mathbf{a})| < \infty$, which holds because H_ν is a polynomial of degree $|\nu|$ and $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$. □

Corollary 10.3 (Consequences used by the paper)**.** *1. Cumulants of order $m \geq 3$ scale exactly: $\kappa_m(\mathbf{x}_t) = e^{-mt} \kappa_m(\mathbf{a})$.*

2. *Two laws with the same covariance $\mathbf{\Sigma}_0$ have degree-four Hermite moments that differ exactly by their fourth-cumulant tensors: for $|\nu| = 4$ with index tuple (i, j, k, l), $\mathbb{E}[H_\nu(\mathbf{a})] - \mathbb{E}'[H_\nu(\mathbf{a})] = \boldsymbol{\kappa}_4[ijkl] - \boldsymbol{\kappa}'_4[ijkl]$, attenuated by e^{-4t} under the channel.*

Proof. (i) is independent of Hermite polynomials: $\mathbf{x}_t = \alpha\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ is a sum of independent terms, joint cumulants are additive over independent summands and homogeneous of degree m under scaling, and a Gaussian has no cumulants of order three or more. (ii): H_ν is monic in the monomial $\mathbf{x}^\nu$ plus polynomials of degree at most two, so $\mathbb{E}[H_\nu(\mathbf{a})]$ is the fourth moment $\mathbb{E}[a_i a_j a_k a_l]$ plus a polynomial in second moments; the fourth moment is $\boldsymbol{\kappa}_4[ijkl] + \mathbf{\Sigma}_{ij}\mathbf{\Sigma}_{kl} + \mathbf{\Sigma}_{ik}\mathbf{\Sigma}_{jl} + \mathbf{\Sigma}_{il}\mathbf{\Sigma}_{jk}$, and everything except $\boldsymbol{\kappa}_4$ is shared by the two laws. □

Nothing here requires $P_0/\gamma \in L^2(\gamma)$, which fails for the Laplace chain and for the Gaussian chain at $\rho = 0.85$ (Section 10.12). This is the reason the paper proves Theorem 3 through the function Λ of the next section rather than through the spectral theory of the Ornstein–Uhlenbeck semigroup on $L^2(\gamma)$.

10.4 The exact identities behind the expansion

Definition 10.4 (The tilted cumulant generating function)**.** For $s \in \mathbb{R}^L$ and $b \geq 0$,

$$\Lambda(s, b) = \log \mathbb{E}_{P_0} \exp\Big(s \cdot \mathbf{a} - \frac{b}{2}|\mathbf{a}|^2\Big) \in (-\infty, +\infty]. \tag{10.5}$$

Proposition 10.5 (Exact identities; the paper's Appendix A.3, first paragraph)**.** *With $s = \alpha\mathbf{x}/\Delta_t$ and $b = \alpha^2/\Delta_t$,*

$$\log P_t(\mathbf{x}) = \log \gamma_{\Delta_t}(\mathbf{x}) + \Lambda(s, b), \tag{10.6}$$

$$\langle \mathbf{a} \rangle_{x,t} = \nabla_s \Lambda(s, b), \tag{10.7}$$

$$S(\mathbf{x}, t) = -\frac{\mathbf{x}}{\Delta_t} + \frac{\alpha}{\Delta_t}\nabla_s \Lambda(s, b), \tag{10.8}$$

$$\partial_\theta \log P_t(\mathbf{x}) = \partial_\theta \Lambda(s, b) \quad \textit{for a parameter } \theta \textit{ of } P_0. \tag{10.9}$$

These need only that P_0 be a probability measure with $\mathbb{E}|\mathbf{a}| < \infty$.

Proof. Expand the square in the Gaussian density of Equation (7.6):

$$\gamma_{\Delta_t}(\mathbf{x} - \alpha\mathbf{a}) = (2\pi\Delta_t)^{-L/2} \exp\Big(-\frac{|\mathbf{x}|^2}{2\Delta_t} + \frac{\alpha\,\mathbf{x}\cdot\mathbf{a}}{\Delta_t} - \frac{\alpha^2|\mathbf{a}|^2}{2\Delta_t}\Big) = \gamma_{\Delta_t}(\mathbf{x}) \exp\Big(s \cdot \mathbf{a} - \frac{b}{2}|\mathbf{a}|^2\Big).$$

Averaging over P_0 and taking logarithms gives Equation (10.6). The tilted law $\propto P_0(\,\mathrm{d}\mathbf{a}) \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$ is, by the same display, the posterior of $\mathbf{a}$ given $\mathbf{x}_t = \mathbf{x}$; differentiating Equation (10.5) in s under the integral gives $\nabla_s \Lambda = \mathbb{E}_{\text{tilted}}[\mathbf{a}]$, which is Equation (10.7). Then Equation (10.8) is the gradient of Equation (10.6) by the chain rule, $\nabla_{\mathbf{x}} s = (\alpha/\Delta_t)\mathbf{I}$, and it agrees with Tweedie's identity Equation (7.9). Finally γ_{Δ_t} does not depend on θ, which gives Equation (10.9). □

Example 10.6 (The Gaussian Λ)**.** For $P_0 = \mathcal{N}(0, \mathbf{\Sigma}_0)$, completing the square gives $\Lambda^{\mathrm{G}}(s, b) = -\frac{1}{2}\log\det(\mathbf{I} + b\mathbf{\Sigma}_0) + \frac{1}{2}s^\top(\mathbf{\Sigma}_0^{-1} + b\mathbf{I})^{-1}s$. In one dimension with $\mathbf{\Sigma}_0 = 1$: $\Lambda^{\mathrm{G}} = -\frac{1}{2}\log(1 + b) + s^2/(2(1 + b))$, and inserting $s = \alpha x/\Delta_t$, $b = \alpha^2/\Delta_t$ (so that $\Delta_t(1 + b) = 1$) into Equation (10.6) returns $\log P_t(x) = -\frac{1}{2}x^2 - \frac{1}{2}\log 2\pi$: the noised law of $\mathcal{N}(0, 1)$ is $\mathcal{N}(0, 1)$, as it must be.

10.5 Grading by order in e^{-t}

Both arguments of Λ are small at high noise: $s = \alpha\mathbf{x}/\Delta_t = O(\alpha)$ and $b = \alpha^2/\Delta_t = O(\alpha^2)$ for fixed $\mathbf{x}$. A monomial $s^j b^k$ is therefore of order α^{j+2k}, and the paper calls $j + 2k$ its **grade**. The expansion of Λ at the origin is a cumulant expansion. Write $X = s \cdot \mathbf{a}$ and $W = |\mathbf{a}|^2/2$, so that $\Lambda = \log \mathbb{E} e^{X - bW}$; if the expansion is legitimate (the subject of Section 10.6),

$$\Lambda = \sum_{n \geq 1} \frac{\kappa_n(X - bW)}{n!}, \qquad \kappa_n(X - bW) = \sum_{k=0}^{n} \binom{n}{k} (-b)^k \operatorname{cum}\big(\underbrace{X, \dots, X}_{n-k}, \underbrace{W, \dots, W}_{k}\big), \tag{10.10}$$

by the multilinearity of joint cumulants. The term with $n - k$ copies of X and k copies of W has grade $(n - k) + 2k = n + k$. Under the paper's standing symmetry $\mathbf{a} \mapsto -\mathbf{a}$ of P_0, a joint cumulant with an odd number of odd arguments vanishes, so $n - k$ is even and only even grades occur: $\Lambda = \Lambda_2 + \Lambda_4 + R_6$.

Proposition 10.7 (Grades two and four)**.** *With* cum *the joint cumulant under P_0 and $\mathbf{\Sigma}_0 = \operatorname{Cov}(\mathbf{a})$,*

$$\Lambda_2 = \tfrac{1}{2} s^\top \mathbf{\Sigma}_0 s - \tfrac{b}{2} \operatorname{tr} \mathbf{\Sigma}_0, \tag{10.11}$$

$$\Lambda_4 = \tfrac{1}{24} \boldsymbol{\kappa}_4[s, s, s, s] - \tfrac{b}{4} \sum_i \operatorname{cum}(s \cdot \mathbf{a},\, s \cdot \mathbf{a},\, a_i^2) + \tfrac{b^2}{8} \sum_{ij} \operatorname{cum}(a_i^2,\, a_j^2), \tag{10.12}$$

where $\boldsymbol{\kappa}_4[s, s, s, s] = \sum_{ijkl} \boldsymbol{\kappa}_4[ijkl] s_i s_j s_k s_l$. Moreover

$$\operatorname{cum}(s \cdot \mathbf{a}, s \cdot \mathbf{a}, a_i^2) = \boldsymbol{\kappa}_4[s, s, i, i] + 2\,(\mathbf{\Sigma}_0 s)_i^2, \qquad \operatorname{cum}(a_i^2, a_j^2) = \boldsymbol{\kappa}_4[iijj] + 2\,\mathbf{\Sigma}_{0,ij}^2, \tag{10.13}$$

so that Λ_2 is the same for P_0 and its covariance twin, and

$$\Lambda_4 - \Lambda_4^{\mathrm{G}} = \tfrac{1}{24} \boldsymbol{\kappa}_4[s, s, s, s] - \tfrac{b}{4} \sum_i \boldsymbol{\kappa}_4[s, s, i, i] + \tfrac{b^2}{8} \sum_{ij} \boldsymbol{\kappa}_4[iijj] \tag{10.14}$$

is linear in $\boldsymbol{\kappa}_4$ and vanishes if and only if $\boldsymbol{\kappa}_4 = 0$.

Proof. Grade two collects $(n, k) = (2, 0)$, giving $\kappa_2(X)/2 = \frac{1}{2} \operatorname{Var}(s \cdot \mathbf{a}) = \frac{1}{2} s^\top \mathbf{\Sigma}_0 s$, and $(n, k) = (1, 1)$, giving $-b\,\mathbb{E}W = -\frac{b}{2} \operatorname{tr} \mathbf{\Sigma}_0$. Grade four collects $(4, 0)$: $\kappa_4(X)/24 = \boldsymbol{\kappa}_4[s, s, s, s]/24$, since the fourth cumulant of a linear form is the contraction of the cumulant tensor; $(3, 1)$ and $(2, 2)$:

$$\binom{3}{1}(-b)\frac{\operatorname{cum}(X, X, W)}{3!} = -\frac{b}{2} \operatorname{cum}(X, X, W) = -\frac{b}{4} \sum_i \operatorname{cum}(X, X, a_i^2),$$

$$\binom{2}{2} b^2 \frac{\operatorname{cum}(W, W)}{2!} = \frac{b^2}{2} \operatorname{Var}(W) = \frac{b^2}{8} \sum_{ij} \operatorname{cum}(a_i^2, a_j^2).$$

For Equation (10.13), use the formula for a cumulant with a product argument (Leonov and Shiryaev): $\operatorname{cum}(X, X, a_i^2)$ is the sum over partitions of the four variables X, X, a_i, a_i that are indecomposable with respect to the grouping $\{X\}, \{X\}, \{a_i, a_i\}$ of products of cumulants over blocks. The full block gives $\operatorname{cum}(X, X, a_i, a_i) = \boldsymbol{\kappa}_4[s, s, i, i]$; the two pairings that join each X to one a_i give $2 \operatorname{cum}(X, a_i)^2 = 2(\mathbf{\Sigma}_0 s)_i^2$; the pairing $\{X, X\}, \{a_i, a_i\}$ is decomposable and excluded; means vanish. Likewise

$$\operatorname{cum}(a_i^2, a_j^2) = \operatorname{Cov}(a_i^2, a_j^2) = \mathbb{E}[a_i^2 a_j^2] - \mathbf{\Sigma}_{ii} \mathbf{\Sigma}_{jj} = \boldsymbol{\kappa}_4[iijj] + 2\mathbf{\Sigma}_{ij}^2.$$

The twin has the same $\mathbf{\Sigma}_0$ and $\boldsymbol{\kappa}_4 = 0$, so subtracting gives Equation (10.14). It is a nonzero linear functional of $\boldsymbol{\kappa}_4$ unless $\boldsymbol{\kappa}_4 = 0$, because its first term alone is a nonzero quartic form in s whenever the symmetric tensor is nonzero. □

Remark 10.8 (On the paper's displayed Λ_4). The paper's Appendix A.3 prints the second and third terms of Equation (10.12) with the fourth-order joint cumulants $\operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i, a_i)$ and $\operatorname{cum}(a_i, a_i, a_j, a_j)$ in place of the cumulants of the squares. By Equation (10.13) the two differ by $-\frac{b}{2}|\boldsymbol{\Sigma}_0 s|^2 + \frac{b^2}{4}\|\boldsymbol{\Sigma}_0\|_F^2$, which is the grade-four part of Λ^{G}: the scalar Gaussian of Example 10.6 has $\Lambda_4^{\mathrm{G}} = \frac{b^2}{4} - \frac{b}{2}s^2 \neq 0$, whereas the printed formula gives zero for any Gaussian. The omission is common to P_0 and its twin, so Equation (10.14) and everything that follows from it are unaffected; the displayed Λ_4 itself is not correct as printed.

10.5.1 From the grade-four difference to a Hermite polynomial

Substituting $s = \alpha\mathbf{x}/\Delta_t = \alpha\mathbf{x}(1+\alpha^2+\dots)$ and $b = \alpha^2/\Delta_t = \alpha^2(1+\alpha^2+\dots)$ into Equation (10.14) and keeping the leading order,

$$\Lambda_4 - \Lambda_4^{\mathrm{G}} = \alpha^4\Big[\tfrac{1}{24}\boldsymbol{\kappa}_4[\mathbf{x},\mathbf{x},\mathbf{x},\mathbf{x}] - \tfrac{1}{4}\sum_i \boldsymbol{\kappa}_4[\mathbf{x},\mathbf{x},i,i] + \tfrac{1}{8}\sum_{ij}\boldsymbol{\kappa}_4[iijj]\Big] + O(\alpha^6(1+|\mathbf{x}|)^4).$$

The bracket is a Hermite form. The **Wick product** (multivariate Hermite polynomial) of four coordinates of the standard Gaussian is

$$\begin{aligned} :x_ix_jx_kx_l: = {} & x_ix_jx_kx_l - \big(\delta_{ij}x_kx_l + \delta_{ik}x_jx_l + \delta_{il}x_jx_k + \delta_{jk}x_ix_l + \delta_{jl}x_ix_k + \delta_{kl}x_ix_j\big) \\ & + \big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big), \end{aligned} \tag{10.15}$$

which reduces to H_ν when the indices are grouped into a multi-index (for instance $:x_1^4: = \mathrm{He}_4(x_1)$ and $:x_1^2x_2^2: = \mathrm{He}_2(x_1)\mathrm{He}_2(x_2)$). Contracting Equation (10.15) with a symmetric tensor, the six single contractions are equal and the three double contractions are equal, so $\frac{1}{24}\sum_{ijkl}\boldsymbol{\kappa}_4[ijkl]:x_ix_jx_kx_l:$ is exactly the bracket. Hence

$$\Lambda_4 - \Lambda_4^{\mathrm{G}} = \frac{\alpha^4}{24}\sum_{ijkl}\boldsymbol{\kappa}_4[ijkl]\,:x_ix_jx_kx_l: + O\big(\alpha^6(1+|\mathbf{x}|)^4\big), \tag{10.16}$$

which is the paper's "the three contributions of Λ_4 combine into the Hermite form". In one dimension with unit variance the bracket is $\kappa_4[\frac{1}{24}x^4 - \frac{1}{4}x^2 + \frac{1}{8}] = \frac{\kappa_4}{24}(x^4 - 6x^2 + 3) = \frac{\kappa_4}{24}\mathrm{He}_4(x)$.

10.6 Where the expansion lives: the exponential moment

Assumption 10.9 (Exponential moment; the paper's Assumption (E))**.** $\mathbb{E}_{P_0}e^{\lambda|\mathbf{a}|} < \infty$ for some $\lambda > 0$.

Gaussian, Laplace, uniform, bounded mixtures and the generalised Gaussian with $\beta \geq 1$ satisfy it; Student-t does not.

Lemma 10.10 (Analyticity and uniform tilted moments)**.** *Under Assumption 10.9: (i) $\Lambda(s,b)$ is finite on $\{|s|\leq\lambda\}\times\{b\geq 0\}$ and real-analytic in (s,b) on $\{|s|<\lambda\}\times\{b\geq 0\}$, with Taylor coefficients at the origin equal to the joint cumulants of $X = s\cdot\mathbf{a}$ and $W = |\mathbf{a}|^2/2$, so that Equation* (10.10) *is a convergent expansion there; (ii) the tilted laws $\propto P_0(\,\mathrm{d}\mathbf{a})\,e^{\vartheta(s\cdot\mathbf{a}-b|\mathbf{a}|^2/2)}$, $\vartheta\in[0,1]$, have moments of every order bounded uniformly on $\{|s|\leq\lambda/2,\ 0\leq b\leq 1\}$ by constants that depend only on λ and $\mathbb{E}e^{\lambda|\mathbf{a}|}$.*

Proof. (i) For $|s|\leq\lambda$ and $b\geq 0$, $|e^{s\cdot\mathbf{a}-b|\mathbf{a}|^2/2}| \leq e^{\lambda|\mathbf{a}|}$, integrable. For complex s with $|\operatorname{Re} s| < \lambda$ the same bound holds, the integrand is entire in s and in b, and dominated convergence makes the integral holomorphic; a holomorphic function of the complex variables that is real on the real set is real-analytic there, and the logarithm of a positive real-analytic function is real-analytic. The Taylor coefficients of $\log\mathbb{E}e^{X-bW}$ are the joint cumulants by the definition of cumulants as the coefficients of the log-moment generating function. (ii) For $|s|\leq\lambda/2$ and $0\leq b\leq 1$ the tilting weight is at most $e^{\lambda|\mathbf{a}|/2}$, so $\mathbb{E}_{\mathrm{tilted}}|\mathbf{a}|^k \leq \mathbb{E}[|\mathbf{a}|^k e^{\lambda|\mathbf{a}|/2}]/Z$ with $|\mathbf{a}|^k e^{\lambda|\mathbf{a}|/2} \leq C_k(\lambda)e^{\lambda|\mathbf{a}|}$; the normaliser $Z = \mathbb{E}e^{\vartheta(X-bW)}$ is at least $\exp(\mathbb{E}[\vartheta(X-bW)]) \geq \exp(-\frac{1}{2}\mathbb{E}|\mathbf{a}|^2)$ by Jensen, and $\mathbb{E}|\mathbf{a}|^2$ is bounded by $\mathbb{E}e^{\lambda|\mathbf{a}|}$ and λ. □

Why finite moments of all orders do not suffice. The audit of 18–19 September 2026 (`docs/SCIENCE_AUDIT.md`; `reports/audit-2026-09-18.tex`, Section 3.3) found that an earlier draft stated Theorem 3 under "finite eighth moments" while its proof used analyticity and an exponential tail bound; the theorem is now stated under Assumption 10.9. Three things break without it, and the third is not repairable by adding more finite moments.

1. The expansion is taken at $b = 0$, where $\Lambda(s, 0) = \log \mathbb{E} e^{s \cdot \mathbf{a}}$ is the log-moment generating function. For a law with polynomial tails, such as Student-t, it is $+\infty$ for every $s \neq 0$: the Taylor coefficients (the cumulants) exist up to the order of the finite moments, but there is no function for them to expand. Even a law with *all* moments finite can have $\mathbb{E} e^{\lambda |\mathbf{a}|} = \infty$ for every $\lambda > 0$ (a density with tails $\propto e^{-|a|^{1/2}}$ is an example); its cumulants then grow so fast that the Taylor series at the origin has radius of convergence zero, and $\Lambda(\cdot, 0)$ is smooth but not analytic.

2. The Lagrange remainder R_6 is a sixth derivative of Λ at an intermediate point $(\vartheta s, \vartheta b)$, i.e. a tilted sixth-order joint cumulant. On the bulk $\{|s| \leq \lambda/2\}$ these are uniformly bounded by Lemma 10.10(ii); without Assumption 10.9 there is no bulk of fixed size on which they are, and for $\vartheta \to 0$ they tend to the untilted sixth cumulant, which for Student-t_5 is infinite.

3. The tail of the bulk must have probability below every power of α for the remainder to be negligible at the order α^8 of the theorem. Under Assumption 10.9 it is $\exp(-c\, e^{2t})$ (Section 10.8); under finite moments of order p it is only $O(\alpha^{2p})$ by Markov's inequality, polynomial in α, and for a regularly varying tail of index ν the tail's own contribution to the gap is of order α^{ν}, which leads whenever $\nu < 8$ (Section 10.13).

The audit also observes that the *conclusion* Equation (10.1) may well hold under a moment condition of order strictly greater than eight (the heuristic of Section 10.13 makes finite eighth moments the borderline), but the present proof does not show it, and the paper claims the theorem only under Assumption 10.9.

10.7 The leading term of the score gap and Wick's theorem

10.7.1 The scalar computation in full

Let $L = 1$, $\mathrm{Var}(a) = 1$ and κ_4 the fourth cumulant of P_0. By Equation (10.8) and Equation (10.16),

$$\begin{aligned} S - S_{\mathrm{G}} &= \frac{\alpha}{\Delta_t}\, \partial_s (\Lambda - \Lambda^{\mathrm{G}}) = \partial_x (\Lambda - \Lambda^{\mathrm{G}}) \\ &= \partial_x \Big[\frac{\kappa_4 \alpha^4}{24} \mathrm{He}_4(x) \Big] + O(\alpha^6 (1 + |x|)^5) = \frac{\kappa_4}{6} \mathrm{He}_3(x)\, \alpha^4 + O(\alpha^6 (1 + |x|)^5), \end{aligned}$$

using $\partial_x = (\alpha/\Delta_t)\partial_s$ and $\mathrm{He}_4' = 4\mathrm{He}_3$. This is the paper's "the scalar leading term of (3) is $(\kappa_4/6)\mathrm{He}_3(x) e^{-4t}$". Its γ-second moment is

$$\mathbb{E}_\gamma \Big[\Big(\frac{\kappa_4}{6} \mathrm{He}_3 \Big)^2 \Big] \alpha^8 = \frac{\kappa_4^2}{36}\, 3!\, \alpha^8 = \frac{\kappa_4^2}{6} \alpha^8,$$

which is Equation (10.1) with $\|\kappa_4\|_F^2 = \kappa_4^2$.

10.7.2 Wick's theorem

Theorem 10.11 (Wick)**.** *Under γ, for index tuples $J = (j_1, \ldots, j_p)$ and $K = (k_1, \ldots, k_q)$,*

$$\mathbb{E}_\gamma \big[{:}x_{j_1} \cdots x_{j_p}{:}\, {:}x_{k_1} \cdots x_{k_q}{:} \big] = \delta_{pq} \sum_{\sigma \in S_p} \prod_{r=1}^{p} \delta_{j_r, k_{\sigma(r)}},$$

the number of perfect matchings between the two tuples that pair equal indices.

This is the orthogonality $\mathbb{E}_\gamma[H_\nu H_\mu] = \nu!\delta_{\nu\mu}$ written for ordered tuples: when both tuples are the same multiset ν, the matchings that pair equal indices number $\nu!$.

Proposition 10.12 (The constant $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$)**.** *The m-th component of the leading term of $S - S_{\mathrm{G}}$ is $\alpha^4 \frac{1}{6}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]\,{:}x_j x_k x_l{:}$, and*

$$\mathbb{E}_\gamma\Big[\Big(\tfrac{1}{6}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]{:}x_j x_k x_l{:}\Big)^2\Big] = \frac{3!}{36}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]^2, \qquad \sum_m(\cdots) = \tfrac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2.$$

Proof. Differentiating Equation (10.16) with respect to x_m: each of the four index slots can carry m, and $\partial_{x_m}{:}x_i x_j x_k x_l{:} = \delta_{im}{:}x_j x_k x_l{:} + \dots$ (the derivative of a Wick monomial removes one factor, as for Hermite polynomials, $\mathrm{He}_n' = n\mathrm{He}_{n-1}$); by symmetry of $\boldsymbol{\kappa}_4$ the four contributions are equal, giving $\frac{4}{24} = \frac{1}{6}$. For the second moment, expand the square and apply Theorem 10.11 with $p = q = 3$:

$$\begin{aligned}\sum_{jkl}\sum_{j'k'l'} \boldsymbol{\kappa}_4[mjkl]\boldsymbol{\kappa}_4[mj'k'l'] \sum_{\sigma\in S_3}\prod_r \delta_{(jkl)_r,(j'k'l')_{\sigma(r)}} &= \sum_{\sigma\in S_3}\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]\boldsymbol{\kappa}_4[m\,\sigma(jkl)] \\ &= 3!\sum_{jkl} \boldsymbol{\kappa}_4[mjkl]^2,\end{aligned}$$

since $\boldsymbol{\kappa}_4$ is symmetric under permutation of its slots. Multiply by $\frac{1}{36}$ and sum over m; the sum over all four ordered indices of $\boldsymbol{\kappa}_4[mjkl]^2$ is $\|\boldsymbol{\kappa}_4\|_F^2$. □

10.7.3 From γ to P_t, and the remainder

The theorem is a statement under P_t, not γ. Two corrections are needed on the bulk, both of relative size $O(\alpha^2)$, and both are what the factor $(1 + \epsilon_t)$ absorbs.

- *Change of measure.* By Mehler's kernel (Section 10.10), $P_t/\gamma = 1 + \frac{\alpha^2}{2}\langle \boldsymbol{\Sigma}_0 - \mathbf{I}, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle + O(\alpha^4(1 + |\mathbf{x}|)^4)$. Integrating the square of the degree-three leading term against the degree-two correction gives a term of relative order α^2. The paper says the correction "has zero γ-mean against the square of a degree-3 polynomial up to terms of relative size α^2"; in fact its γ-mean is not zero in general (in one dimension $\mathrm{He}_3^2 = \mathrm{He}_6 + 9\mathrm{He}_4 + 18\mathrm{He}_2 + 6$, so $\mathbb{E}_\gamma[\mathrm{He}_2\mathrm{He}_3^2] = 36$), but it carries the explicit factor α^2, which is all the theorem needs.
- *Remainder.* The cross term between the leading term (order α^4) and $\nabla_s R_6$ (order α^6 times a polynomial in $|\mathbf{x}|$, by Lemma 10.10(ii)) is $O(\alpha^{10})$ by Cauchy–Schwarz, and the square of the remainder is $O(\alpha^{12})$.

The constant in $O(\alpha^2)$ depends on P_0 and on L; for the Gaussian chain it is $\lambda_{\max}(\boldsymbol{\Sigma}_0)$ (Section 10.12).

10.8 The tail

The expansion is used only on the **bulk** $B = \{|s| \le \lambda/2\} = \{|\mathbf{x}| \le \lambda\Delta_t/(2\alpha)\}$, a ball whose radius grows like e^t.

Lemma 10.13 (Tail probability)**.** *Under Assumption 10.9, with $c = \lambda\Delta_t/4$,*

$$P_t(B^c) \le e^{-\lambda c/\alpha^2}\,\mathbb{E}e^{\lambda|\mathbf{a}|} + \mathbb{P}(\sqrt{\Delta_t}|\mathbf{z}| > c/\alpha),$$

and the second term is at most $C_L e^{-c^2/(4\alpha^2)}$; both are $\exp(-c'e^{2t})$.

Proof. On B^c, $|\alpha\mathbf{x}| > \lambda\Delta_t/2 = 2c$. Since $\alpha\mathbf{x} = \alpha^2\mathbf{a} + \alpha\sqrt{\Delta_t}\mathbf{z}$, either $\alpha^2|\mathbf{a}| > c$ or $\alpha\sqrt{\Delta_t}|\mathbf{z}| > c$. For the first, $\mathbb{P}(|\mathbf{a}| > c/\alpha^2) \le e^{-\lambda c/\alpha^2}\mathbb{E}e^{\lambda|\mathbf{a}|}$ by Markov's inequality applied to $e^{\lambda|\mathbf{a}|}$. For the second, $|\mathbf{z}|^2$ is χ^2 with L degrees of freedom and $\mathbb{E}e^{|\mathbf{z}|^2/4} = 2^{L/2}$, so $\mathbb{P}(|\mathbf{z}| > R) \le 2^{L/2}e^{-R^2/4}$ with $R = c/(\alpha\sqrt{\Delta_t})$. □

Proposition 10.14 (The tail contributes below every power of α)**.** $\mathbb{E}_{P_t}[\|S - S_{\mathrm{G}}\|^2 \mathbf{1}_{B^c}] = O(e^{-c'' e^{2t}})$ *for some* $c'' > 0$.

Proof. By Equation (7.9) and Corollary 7.12, $\|S - S_{\mathrm{G}}\|^2 \le C(|\mathbf{x}|^2 + \alpha^2 \|\langle \mathbf{a} \rangle_{x,t}\|^2 + \alpha^2 \|\alpha \boldsymbol{\Sigma}_0 \boldsymbol{\Sigma}_t^{-1} \mathbf{x}\|^2)/\Delta_t^2$. Cauchy–Schwarz gives $\mathbb{E}[\|\cdot\|^2 \mathbf{1}_{B^c}] \le (\mathbb{E}\|\cdot\|^4)^{1/2} P_t(B^c)^{1/2}$, and the fourth moment is finite: $\mathbb{E}|\mathbf{x}|^4 < \infty$, and $\mathbb{E}\|\langle \mathbf{a} \rangle_{x,t}\|^4 \le \mathbb{E}|\mathbf{a}|^4$ by Jensen's inequality for the conditional expectation. The tail probability is $\exp(-c' e^{2t})$ by Lemma 10.13, and its square root still is. □

Putting Proposition 10.12, the bulk corrections and Proposition 10.14 together gives Equation (10.1): the paper's Theorem 3, first line. The paper notes what changes without Assumption 10.9: the tail probability becomes polynomial in α, of order $\alpha^{2\nu}$ for a tail index ν, and the tail's own contribution is of order α^{ν}; this is Section 10.13.

10.9 Fisher information: degree two and degree four

By Equation (10.9), $\partial_\theta \log P_t = \partial_\theta \Lambda$, and the graded expansion of Λ gives the graded expansion of the score with respect to θ.

Proposition 10.15 (Degree-two constant)**.** *If* $\partial_\theta \boldsymbol{\Sigma}_0 \neq 0$*, the leading term of* $\partial_\theta \log P_t$ *is*

$$\tfrac{\alpha^2}{2} \langle \partial_\theta \boldsymbol{\Sigma}_0, :\mathbf{x}\mathbf{x}^\top: \rangle = \tfrac{\alpha^2}{2} \sum_{ij} \partial_\theta \boldsymbol{\Sigma}_{0,ij} \, :x_i x_j:,$$

and its γ*-second moment is* $\frac{\alpha^4}{2} \|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$.

Proof. From Equation (10.11) with $s \approx \alpha \mathbf{x}$, $b \approx \alpha^2$,

$$\partial_\theta \Lambda_2 = \tfrac{\alpha^2}{2} (\mathbf{x}^\top \partial_\theta \boldsymbol{\Sigma}_0 \mathbf{x} - \operatorname{tr} \partial_\theta \boldsymbol{\Sigma}_0) = \tfrac{\alpha^2}{2} \sum_{ij} \partial_\theta \boldsymbol{\Sigma}_{0,ij} \, (x_i x_j - \delta_{ij}),$$

and $:x_i x_j: = x_i x_j - \delta_{ij}$. By Theorem 10.11 with $p = q = 2$, $\mathbb{E}_\gamma[:x_i x_j: :x_{i'} x_{j'}:] = \delta_{ii'}\delta_{jj'} + \delta_{ij'}\delta_{ji'}$, so the second moment is

$$\tfrac{\alpha^4}{4} \sum_{ij} (\partial \boldsymbol{\Sigma}_{ij}^2 + \partial \boldsymbol{\Sigma}_{ij} \partial \boldsymbol{\Sigma}_{ji}) = \tfrac{\alpha^4}{4} \cdot 2 \|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$$

by symmetry. □

Proposition 10.16 (Degree-four constant)**.** *If* $\partial_\theta \boldsymbol{\Sigma}_0 = 0$ *and* $\partial_\theta \boldsymbol{\kappa}_4 \neq 0$*, the leading term is* $\frac{\alpha^4}{24} \sum_{ijkl} \partial_\theta \boldsymbol{\kappa}_4[ijkl] :x_i x_j x_k x_l:$*, with* γ*-second moment* $\frac{4!}{24^2} \|\partial_\theta \boldsymbol{\kappa}_4\|_F^2 \alpha^8 = \frac{1}{24} \|\partial_\theta \boldsymbol{\kappa}_4\|_F^2 \alpha^8$.

Proof. The grade-two term has vanishing θ-derivative. Differentiating Equation (10.16) in θ (only $\boldsymbol{\kappa}_4$ depends on θ) gives the leading term; Theorem 10.11 with $p = q = 4$ and the symmetry of $\partial_\theta \boldsymbol{\kappa}_4$ give $4! \, \|\partial_\theta \boldsymbol{\kappa}_4\|_F^2$ for the sum over matchings, exactly as in Proposition 10.12. □

The bulk corrections and the tail are handled as in Sections 10.7 and 10.8, which gives Equations (10.2) and (10.3).

Proposition 10.17 (The Gaussian chain, exactly)**.** *If* $P_0 = \mathcal{N}(0, \boldsymbol{\Sigma}_0(\theta))$ *then* $\mathbf{x}_t \sim \mathcal{N}(0, \boldsymbol{\Sigma}_t(\theta))$ *with* $\boldsymbol{\Sigma}_t = \alpha^2 \boldsymbol{\Sigma}_0 + \Delta_t \mathbf{I}$*, and at every* t

$$I_t(\theta) = \tfrac{1}{2} \operatorname{tr}(\boldsymbol{\Sigma}_t^{-1} \partial_\theta \boldsymbol{\Sigma}_t \, \boldsymbol{\Sigma}_t^{-1} \partial_\theta \boldsymbol{\Sigma}_t), \qquad \partial_\theta \boldsymbol{\Sigma}_t = \alpha^2 \partial_\theta \boldsymbol{\Sigma}_0,$$

so that $e^{4t} I_t(\theta) \to \frac{1}{2} \|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2$ *with relative correction* $O(\alpha^2 \lambda_{\max}(\boldsymbol{\Sigma}_0))$.

Proof. The Fisher information of a centred Gaussian family about a parameter of its covariance is $\frac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial\boldsymbol{\Sigma}\,\boldsymbol{\Sigma}^{-1}\partial\boldsymbol{\Sigma})$ (differentiate $\log\det\boldsymbol{\Sigma}$ and $\mathbf{x}^\top\boldsymbol{\Sigma}^{-1}\mathbf{x}$ twice and take expectations). With $\boldsymbol{\Sigma}_t = \mathbf{I} + \alpha^2(\boldsymbol{\Sigma}_0 - \mathbf{I})$ the Neumann series gives $\boldsymbol{\Sigma}_t^{-1} = \mathbf{I} - \alpha^2(\boldsymbol{\Sigma}_0 - \mathbf{I}) + O(\alpha^4)$ in operator norm once $\alpha^2\|\boldsymbol{\Sigma}_0 - \mathbf{I}\| < 1$, hence $\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\partial\boldsymbol{\Sigma}_0) = \|\partial\boldsymbol{\Sigma}_0\|_F^2(1 + O(\alpha^2\|\boldsymbol{\Sigma}_0 - \mathbf{I}\|))$, and $\|\boldsymbol{\Sigma}_0 - \mathbf{I}\| \le \lambda_{\max}(\boldsymbol{\Sigma}_0)$. □

The code computes this closed form (`gaussian_chain_fisher` in `src/markov_score/theory.py`), and the paper's Figure (theory), panel (a), plots $I_t e^{4t}$ divided by the constant for five correlations: every curve reaches 1, and how fast is governed by t_*.

Efficient information and nuisance parameters. When θ is estimated together with other coordinates η, the relevant quantity is the **efficient information** $I_t^{\mathrm{eff}}(\theta) = I_{\theta\theta} - I_{\theta\eta}I_{\eta\eta}^{-1}I_{\eta\theta}$, the reciprocal of the θ-entry of the inverse information matrix. For a degree-four θ and degree-two nuisances η the cross information $I_{\theta\eta} = \mathbb{E}[\partial_\theta \log P_t\, \partial_\eta \log P_t]$ has no α^6 term, because the leading terms are a degree-four and a degree-two Hermite polynomial, which are γ-orthogonal (Theorem 10.11 with $p \neq q$); so $I_{\theta\eta} = O(\alpha^8)$, $I_{\eta\eta} = \Theta(\alpha^4)$, and the correction is $O(\alpha^{12})$, negligible against $I_{\theta\theta} = \Theta(\alpha^8)$. This is the paper's "the leading orders of a degree-2 and a degree-4 coordinate do not couple". A nuisance of the *same* degree (another shape coordinate, as in the eight-component mixture the paper fits) has a cross term of the same order and reduces the efficient information by a factor of order one; that is why the paper's Section 4 calls its one-parameter Cramér–Rao reference a yardstick and not an efficiency bound. The code's `efficient_information` in `src/markov_score/information.py` inverts the outer-product estimate of the full matrix.

10.10 The general order

Theorem 10.18 (General order; the paper's Theorem 4)**.** *Let P_0^θ satisfy Assumption 10.9 uniformly near θ_0, and let m be the smallest degree at which $\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})] \neq 0$ for some $|\nu| = m$. Then, at fixed L,*

$$I_t(\theta) = e^{-2mt} \sum_{|\nu|=m} \frac{\big(\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})]\big)^2}{\nu!}\,(1 + O(e^{-2t})).$$

The proof rests on a second expansion of the same function, in Hermite polynomials rather than in cumulants.

Lemma 10.19 (Mehler's kernel)**.** *Formally, and in $L^2(\gamma)$ whenever $\sum_\nu \alpha^{2|\nu|}(\mathbb{E}_{P_0}H_\nu)^2/\nu! < \infty$,*

$$\frac{P_t(\mathbf{x})}{\gamma(\mathbf{x})} = \sum_\nu \alpha^{|\nu|}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]\,\frac{H_\nu(\mathbf{x})}{\nu!} = 1 + \frac{\alpha^2}{2}\langle \boldsymbol{\Sigma}_0 - \mathbf{I}, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle + \dots \tag{10.17}$$

Proof. In one dimension the transition density of the channel divided by γ is $\gamma_{\Delta_t}(x - \alpha a)/\gamma(x) = (1-\alpha^2)^{-1/2}\exp(-\frac{(x-\alpha a)^2}{2(1-\alpha^2)} + \frac{x^2}{2}) = \sum_n \alpha^n \mathrm{He}_n(a)\mathrm{He}_n(x)/n!$, Mehler's formula, which follows from Equation (10.4) by integrating $e^{sa - s^2/2}e^{ux - u^2/2}$ against the joint Gaussian of (a, x) with correlation α. Multiply over coordinates, average over $\mathbf{a} \sim P_0$, and read off the $|\nu| \le 2$ terms: $\mathbb{E}[a_i] = 0$, $\mathbb{E}[\mathrm{He}_2(a_i)] = \boldsymbol{\Sigma}_{0,ii} - 1$, $\mathbb{E}[a_i a_j] = \boldsymbol{\Sigma}_{0,ij}$. □

Proof of Theorem 10.18. Under Assumption 10.9 the graded expansion of Λ is a convergent series in α on the bulk, and $\log(P_t/\gamma_{\Delta_t}) = \Lambda$. On the other hand

$$\log(P_t/\gamma) = \log\Big(1 + \sum_{|\nu|\ge 2} \alpha^{|\nu|}\,\mathbb{E}[H_\nu]\,\frac{H_\nu(\mathbf{x})}{\nu!}\Big)$$

is a second formal series in α for the same function up to the explicit $\log(\gamma_{\Delta_t}/\gamma)$. At each grade both coefficients are polynomials in the moments of P_0 up to that grade; on compactly supported P_0 both series converge and represent the function, so the two polynomials agree on the class of moment sequences of compactly supported laws, which determines a polynomial identity; hence they agree identically. Now differentiate in θ. The grade-m coefficient of $\partial_\theta \log(P_t/\gamma)$ is

$$\alpha^m \sum_{|\nu|=m} \partial_\theta \mathbb{E}[H_\nu] \, \frac{H_\nu(\mathbf{x})}{\nu!}$$

plus ∂_θ of products of lower-grade terms of the logarithm's series; every such product contains a factor $\mathbb{E}[H_\mu]$ with $|\mu| < m$, whose θ-derivative vanishes by the minimality of m, and all terms of grade below m vanish for the same reason. The γ-second moment of the leading term is $\alpha^{2m} \sum_{|\nu|=m} (\partial_\theta \mathbb{E} H_\nu)^2/\nu!$ by orthogonality, and the passage from γ to P_t on the bulk and the tail are as in Sections 10.7 and 10.8. □

Consistency with the two constants. For $m = 2$: the multi-indices are $\nu = 2\mathbf{e}_i$, with $H_\nu = \mathrm{He}_2(x_i)$, $\mathbb{E}H_\nu = \mathbf{\Sigma}_{0,ii} - 1$ and $\nu! = 2$, and $\nu = \mathbf{e}_i + \mathbf{e}_j$ $(i < j)$, with $H_\nu = x_i x_j$, $\mathbb{E}H_\nu = \mathbf{\Sigma}_{0,ij}$ and $\nu! = 1$; the sum is $\frac{1}{2}\sum_i (\partial \mathbf{\Sigma}_{ii})^2 + \sum_{i<j} (\partial \mathbf{\Sigma}_{ij})^2 = \frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$. For $m = 4$ at fixed covariance: by Corollary 10.3(ii), $\partial_\theta \mathbb{E}[H_\nu] = \partial_\theta \boldsymbol{\kappa}_4[\nu]$ with ν read as an index multiset; the Frobenius norm over *ordered* tuples counts each multiset ν exactly $4!/\nu!$ times, so $\|\partial \boldsymbol{\kappa}_4\|_F^2 = \sum_\nu (4!/\nu!)(\partial \boldsymbol{\kappa}_4[\nu])^2$ and $\sum_\nu (\partial \boldsymbol{\kappa}_4[\nu])^2/\nu! = \|\partial \boldsymbol{\kappa}_4\|_F^2/24$. This is the paper's sentence "since the multiset ν collects $4!/\nu!$ ordered index tuples". A direction with $\partial_\theta \mathbb{E}[H_\nu] = 0$ for every ν has no information at any level; under Assumption 10.9 the law is determined by its moments (the moment problem is determinate when the moment generating function is finite near zero), so such a direction does not move P_0 at all.

10.11 The sample-complexity corollary and its three caveats

Corollary 10.20 (Local asymptotic minimax; the paper's Corollary 5)**.** *Fix $t > 0$ and a coordinate θ of lowest degree m with efficient information $I_t^{\mathrm{eff}}(\theta) > 0$. For M i.i.d. sequences observed at level t and any estimator sequence $\hat{\theta}_M$,*

$$\lim_{c\to\infty} \liminf_{M\to\infty} \sup_{|\theta'-\theta|\le c/\sqrt{M}} M\, \mathbb{E}_{\theta'}(\hat{\theta}_M - \theta')^2 \;\ge\; \frac{1}{I_t^{\mathrm{eff}}(\theta)} = \Theta(e^{2mt}),$$

and the maximum-likelihood estimator attains the bound.

This is the local asymptotic minimax theorem of Hájek and Le Cam (van der Vaart 1998, Theorem 8.11) for a family that is **differentiable in quadratic mean** at θ with non-singular information; at fixed $t > 0$ the family $\{P_t^\theta\}$ is smooth in θ (through Λ, under Assumption 10.9), which gives the differentiability, and the efficiency of maximum likelihood is the standard consequence of local asymptotic normality. The statement has the limit $c \to \infty$ outside: for a fixed c the local minimax risk is strictly smaller than $1/I$ (a bounded normal mean can be estimated better than the unrestricted one), and the paper's display, which fixes c, should be read with that limit understood. Three things the corollary does not say, all listed by the paper and all visible in its Section 4:

1. Nothing uniform in t. The regime is $M \to \infty$ at fixed t; local asymptotic normality needs $MI_t \gg 1$, i.e. $M \gg e^{2mt}$, so the limits in M and in t do not commute, and the $t \to \infty$ form of the constant is a second limit taken afterwards.

2. Nothing about absolute precision for a bounded coordinate. Once $1/\sqrt{MI_t}$ exceeds the range of the coordinate (the excess kurtosis of the paper's family lives on a bounded interval) the local statement is empty; this is the level at which the fitted shape in the paper's learning experiment collapses.

3. Nothing about an optimiser at a finite budget: EM at a validation-selected iteration count is not the maximum-likelihood estimator, and the paper reports where its iteration ladder binds.

The ratio statement of the paper, "a shape coordinate needs e^{4t} times more sequences than a covariance coordinate", is the ratio of two such bounds and inherits the same regime.

10.12 The crossover

For the Gaussian chain the expansion parameter is not α but $\alpha^2\lambda_{\max}(\mathbf{\Sigma}_0)$.

Proposition 10.21 (Three faces of t_*)**.** *Let $\mathbf{\Sigma}_0$ have largest eigenvalue $\lambda_{\max}$ and put $t_* = \frac{1}{2}\log\lambda_{\max}$.*

1. *The Neumann series $\mathbf{\Sigma}_t^{-1} = \sum_{n\geq 0}(-\alpha^2)^n(\mathbf{\Sigma}_0 - \mathbf{I})^n$ converges in operator norm if and only if $\alpha^2\|\mathbf{\Sigma}_0 - \mathbf{I}\| < 1$; when $\lambda_{\max} - 1 \geq 1 - \lambda_{\min}(\mathbf{\Sigma}_0)$ this is $e^{2t} > \lambda_{\max} - 1$.*
2. *$P_t/\gamma \in L^2(\gamma)$ if and only if $\lambda_{\max}(\mathbf{\Sigma}_t) < 2$, i.e. $e^{2t} > \lambda_{\max} - 1$; above that time the Hermite series Equation (10.17) converges in $L^2(\gamma)$, and below it it does not.*
3. *For the paper's chain, $\mathbf{\Sigma}_0 = (\rho^{|i-j|})_{ij}$ and $\lambda_{\max} \leq \sum_{k\in\mathbb{Z}}\rho^{|k|} = \frac{1+\rho}{1-\rho}$, so $t_* \leq \frac{1}{2}\log\frac{1+\rho}{1-\rho}$.*

Proof. (i) $\mathbf{\Sigma}_t = \mathbf{I} + \alpha^2(\mathbf{\Sigma}_0 - \mathbf{I})$ and the geometric series for $(\mathbf{I} + E)^{-1}$ converges iff $\|E\| < 1$; $\|\mathbf{\Sigma}_0 - \mathbf{I}\| = \max(\lambda_{\max} - 1, 1 - \lambda_{\min})$. (ii) $\int (P_t/\gamma)^2\gamma\,\mathrm{d}\mathbf{x} = \int P_t^2/\gamma\,\mathrm{d}\mathbf{x}$, and $P_t^2/\gamma \propto \exp(-\mathbf{x}^\top\mathbf{\Sigma}_t^{-1}\mathbf{x} + \frac{1}{2}|\mathbf{x}|^2) = \exp(-\frac{1}{2}\mathbf{x}^\top(2\mathbf{\Sigma}_t^{-1} - \mathbf{I})\mathbf{x})$, integrable iff $2\mathbf{\Sigma}_t^{-1} - \mathbf{I} \succ 0$ iff $\mathbf{\Sigma}_t \prec 2\mathbf{I}$ iff $\alpha^2\lambda_{\max} + \Delta_t < 2$ iff $\alpha^2(\lambda_{\max} - 1) < 1$. The $L^2(\gamma)$ norm of Equation (10.17) is $\sum_\nu \alpha^{2|\nu|}(\mathbb{E}H_\nu)^2/\nu!$, which is $\|P_t/\gamma\|^2_{L^2(\gamma)}$ by orthogonality, finite exactly then. (iii) $\mathbf{\Sigma}_0$ is a principal submatrix of the infinite Toeplitz matrix with symbol $g(\omega) = \sum_k \rho^{|k|}e^{ik\omega} = \frac{1-\rho^2}{1-2\rho\cos\omega+\rho^2}$, whose eigenvalues lie in $[\min g, \max g] = [\frac{1-\rho}{1+\rho}, \frac{1+\rho}{1-\rho}]$ (Szegő), and the eigenvalues of a principal submatrix interlace. □

At $\rho = 0.85$ the bound is $\frac{1}{2}\log(1.85/0.15) = \frac{1}{2}\log 12.33 = 1.26$; at $\rho = 0.5$ it is $\frac{1}{2}\log 3 = 0.55$; at $\rho = 0.95$ it is $\frac{1}{2}\log 39 = 1.83$. The exact $\lambda_{\max}$ at $L = 32$ is a little below the bound and is the value the paper's macro reports (Section 3, "Where the asymptote begins"; computed by `crossover_time` in `src/markov_score/theory.py`). Since $\lambda_{\max} > 2$ at the frozen correlation, $P_0/\gamma \notin L^2(\gamma)$ for the Gaussian chain itself, and P_t/γ enters $L^2(\gamma)$ only above $\frac{1}{2}\log(\lambda_{\max} - 1)$. For the Laplace chain P_t/γ is never in $L^2(\gamma)$: in one dimension a density with tails $\propto e^{-c|a|}$ convolved with a Gaussian has tails at least $\propto e^{-c|x|/\alpha}$ (restrict the convolution integral to $|a - x/\alpha| \leq 1$), so $P_t^2/\gamma \gtrsim e^{-2c|x|/\alpha + x^2/2}$ is not integrable; for the chain the site marginal has the same tail, and $\|P_t/\gamma\|_{L^2(\gamma)}$ dominates the marginal's norm by Jensen. This is the sentence "for the Laplace chain it never does, which is why the proof goes through Λ". Below t_* the long-wavelength modes of the covariance are still of order one and every rate is shallower than its asymptote; the paper uses this to explain the slopes of about -3 and -6 that earlier measurements on the model obtained over $t \leq 1.6$, below t_*, as pre-asymptotic values rather than as a failure of the law. For non-Gaussian laws the same scale is a heuristic that the measurements support.

10.13 Heavy tails

A density is **regularly varying** with index $\nu > 0$ if $\mathbb{P}(|\varepsilon| > u) = u^{-\nu}\ell(u)$ with ℓ slowly varying; the Student-t_ν density $\varphi(\varepsilon) \propto (1 + \varepsilon^2/((\nu - 2)q))^{-(\nu+1)/2}$, scaled to variance q, has $\varphi(\varepsilon) \sim c|\varepsilon|^{-\nu-1}$ and is the paper's example with $\nu = 5$. Its absolute moment of order k is finite if and only if $k < \nu$: t_5 has a fourth cumulant (excess kurtosis $6/(\nu - 4) = 6$) and an infinite sixth moment. Its moment generating function is infinite for every $s \neq 0$, so Assumption 10.9 fails and $\Lambda(s, 0) = +\infty$.

The $e^{-\nu t}$ term. The following heuristic, from the audit (Section 3.3) and the paper's "Heavy tails" paragraph, explains what replaces Equation (10.1). Take one site with unit variance. For $|x|$ large, $P_t(x) = \int P_0(a)\gamma_{\Delta_t}(x - \alpha a)\,\mathrm{d}a$ receives two contributions: from $a \approx x/\alpha$, where the prior's tail is slowly varying on the scale of the likelihood, giving $P_t(x) \approx P_0(x/\alpha)/\alpha \approx c\,\alpha^\nu |x|^{-\nu-1}$; and from $a \approx 0$, giving the Gaussian tail $\approx \gamma_{\Delta_t}(x)$. The first dominates once $|x| > K_t$ with $K_t^2 \approx 2\nu t$ (solve $\alpha^\nu |x|^{-\nu-1} = e^{-x^2/2}$ in the leading order). On that region the posterior of a concentrates near x/α, $\mathbb{E}[a \mid x] \approx x/\alpha - \Delta_t(\nu+1)/(\alpha x)$ (the prior's log-slope shifts the mean by $(\Delta_t/\alpha^2)\partial_a \log P_0(x/\alpha)$), so by Tweedie's identity $S(x) \approx -(\nu+1)/x$, while $S_{\mathrm{G}}(x) = -x$. The squared gap is $\approx x^2$ there, and its expectation over the region is

$$\int_{|x|>K_t} x^2\, c\,\alpha^\nu |x|^{-\nu-1}\,\mathrm{d}x = \frac{2c}{\nu-2}\,\alpha^\nu K_t^{2-\nu}\,:$$

a term of order $e^{-\nu t}$ up to a power of t. The probability of the region is $\approx \frac{2c}{\nu}\alpha^\nu K_t^{-\nu}$, also of order $e^{-\nu t}$, and it is carried by sequences with a site of size $|a| \approx |x|/\alpha \gtrsim K_t e^t$. In expectation the tail term leads once it exceeds $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$, which for $\nu < 8$ always happens eventually, late if $\|\boldsymbol{\kappa}_4\|_F^2$ is large; for $\nu > 8$ the Hermite term leads for ever, and finite eighth moments are the borderline of the conclusion, as remarked in Section 10.6. The paper states this as an observation supported by an untruncated scalar quadrature (E18) and not as a theorem.

Remark 10.22 (A discrepancy in the paper's wording). The paper writes that the tail term is "carried by sequences with $|\mathbf{a}| \gtrsim e^{2t}$, whose probability is itself of order $e^{-\nu t}$". For a tail of index ν the probability that a site exceeds e^{2t} is of order $e^{-2\nu t}$, not $e^{-\nu t}$; the events of probability $e^{-\nu t}$ that carry the term have $|a|$ of order e^t (times the slowly growing K_t), as derived above. The threshold e^{2t} is the boundary $|\mathbf{a}| > c/\alpha^2$ of the proof's bulk, which is a different object. One of the two numbers in the sentence should change.

What "never drawn" means. A sample of M sequences has ML sites, and the number of them beyond a threshold u is approximately Poisson with mean $ML\mathbb{P}(|a| > u)$. At $t = 3$ the carriers have $|a| \gtrsim K_t e^3 \approx \sqrt{30} \times 20 \approx 110$ in units of the site's standard deviation. For a variance-one t_5 variable, whose density is $\approx 47|a|^{-6}$ in standardised units for large $|a|$ (the constant is $\Gamma(3)5^{5/2}/(\sqrt{5\pi}\Gamma(5/2))$ times the scale factor $\sqrt{3/5}$ to the fifth power), $\mathbb{P}(|a| > 110) \approx 2 \times 9.5 \times (110/0.775)^{-5} \approx 3 \times 10^{-10}$, so a sample of 8192×32 sites contains, in expectation, about 10^{-4} carriers. A Monte Carlo estimate of the squared gap therefore reports the Hermite law $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$ at every level a finite sample can reach, and the $e^{-\nu t}$ term lives only in the expectation, which an untruncated quadrature can evaluate and a sample cannot. The paper's Figure (theory), panel (c), shows exactly this: the sampled Student-t_5 chain, on a domain wide enough to hold its sample, reaches the constant. What the earlier draft attributed to the moment condition on the narrow grid was the truncated prior in the reference (E18; the audit's Section 6.2).

10.14 The Edgeworth correction: adding the fourth-order term back

Equation (10.1) says what the twin lacks to leading order, and E16 (the script `experiments/e16_edgeworth_oracle.py`) adds it back exactly, with the true $\boldsymbol{\Sigma}_0$ and $\boldsymbol{\kappa}_4$, to test whether the gap is made of that term and nothing else.

Proposition 10.23 (First Edgeworth correction around the twin)**.** *Let $\mathbf{x}_t$ have covariance $\boldsymbol{\Sigma}_t$ and fourth-cumulant tensor $K = \alpha^4\boldsymbol{\kappa}_4$ (Corollary 10.3(i)), with vanishing odd cumulants. Then,*

formally, $P_t(\mathbf{x}) = P_t^{\mathrm{G}}(\mathbf{x})(1+\epsilon(\mathbf{x})) + \dots$ *with*

$$\epsilon(\mathbf{x}) = \frac{1}{24}\Big[K[y,y,y,y] - 6\,y^\top A y + 3\langle A, \mathbf{\Sigma}_t^{-1}\rangle\Big], \qquad y = \mathbf{\Sigma}_t^{-1}\mathbf{x}, \quad A_{ml} = \sum_{jk} K_{mjkl}(\mathbf{\Sigma}_t^{-1})_{jk}, \tag{10.18}$$

and the corrected score is

$$S^{\mathrm{E4}} = S_{\mathrm{G}} + \frac{\nabla\epsilon}{1+\epsilon}, \qquad \nabla\epsilon = \tfrac{1}{6}\mathbf{\Sigma}_t^{-1}K[\cdot,y,y,y] - \tfrac{1}{2}\mathbf{\Sigma}_t^{-1}Ay. \tag{10.19}$$

Proof. The characteristic function is $\varphi(\xi) = \exp(-\frac{1}{2}\xi^\top\mathbf{\Sigma}_t\xi + \frac{1}{24}K[\xi,\xi,\xi,\xi] + \dots) = \varphi^{\mathrm{G}}(\xi)(1 + \frac{1}{24}K[\xi,\xi,\xi,\xi] + \dots)$. Fourier inversion turns multiplication by $\xi_i\xi_j\xi_k\xi_l$ into $\partial_i\partial_j\partial_k\partial_l$ (the four factors of $-i$ give $(-i)^4 = 1$), so the correction to the density is $\frac{1}{24}\sum K_{ijkl}\partial_{ijkl}P_t^{\mathrm{G}}$. With $P_t^{\mathrm{G}} \propto \exp(-\frac{1}{2}\mathbf{x}^\top\mathbf{\Sigma}_t^{-1}\mathbf{x})$ and $y = \mathbf{\Sigma}_t^{-1}\mathbf{x}$, $P = \mathbf{\Sigma}_t^{-1}$, repeated differentiation gives $\partial_{ijkl}P_t^{\mathrm{G}} = \big[y_iy_jy_ky_l - (P_{ij}y_ky_l + 5\text{ terms}) + (P_{ij}P_{kl} + 2\text{ terms})\big]P_t^{\mathrm{G}}$, the Hermite tensor of $\mathcal{N}(0,\mathbf{\Sigma}_t)$. Contracting with the symmetric K gives the bracket of Equation (10.18). The score of $P_t^{\mathrm{G}}(1+\epsilon)$ is $S_{\mathrm{G}} + \nabla\log(1+\epsilon)$; and $\nabla_{\mathbf{x}}y = P$, so $\nabla K[y,y,y,y] = 4PK[\cdot,y,y,y]$ and $\nabla(y^\top Ay) = 2PAy$ (A is symmetric because K is), giving Equation (10.19). □

At high noise $\mathbf{\Sigma}_t \to \mathbf{I}$, $y \to \mathbf{x}$ and $\epsilon \to \frac{\alpha^4}{24}\sum \boldsymbol{\kappa}_4[ijkl]{:}x_ix_jx_kx_l{:}$, which is exactly $\Lambda_4 - \Lambda_4^{\mathrm{G}}$ of Equation (10.16); the corrected score then differs from the true one at order α^6 instead of α^4, so the fraction of the squared gap it closes tends to one. At low noise $y = \mathbf{\Sigma}_t^{-1}\mathbf{x}$ is amplified by up to $1/\lambda_{\min}(\mathbf{\Sigma}_0)$, ϵ is a quartic of order one or larger, $1+\epsilon$ can approach or cross zero (the code clips it at 0.1), and the correction hurts: the series in α does not converge there. E16 measures the fraction $1 - \mathbb{E}\|S^{\mathrm{E4}} - S\|^2/\mathbb{E}\|S_{\mathrm{G}} - S\|^2$ level by level on the Laplace, bimodal, uniform and Student-t_5 chains, with the "lin" form $S_{\mathrm{G}} + \nabla\epsilon$ as a control; the paper reports the fractions (Section 3, "The gap is the fourth-order term"; Appendix E) and the level below which the expansion diverges. Pooled over the twelve levels the corrected score is worse than the twin, because the pooled estimand is dominated by the lowest levels (Section 8.7); the correction is a statement about what the gap is made of at each level, not a competitor on the schedule. On Student-t_5 the correction can close the Hermite part only, and the levels where it fails are the levels where the tail term of Section 10.13 leads.

10.15 Reading the theory figure

Figure 10.2 is the paper's Figure in Section 3. Panel (a) is Proposition 10.17: no Monte Carlo, a closed form on a fine grid of t (`part_gauss` of `experiments/e01_information_decay.py`); the curves reach one, and the tick marks show that a strongly correlated chain approaches its asymptote later, as $\alpha^2\lambda_{\max}$ requires. Panel (b) is Propositions 10.15 and 10.16 measured on the chain: the per-sequence scores of the marginal likelihood are obtained by Fisher's identity through the tree-exact recursion (`per_chain_scores` in `src/markov_score/information.py`), the outer-product matrix is inverted for the efficient information, and the slopes above t_* are compared with -4 and -8; the constant of the shape rate is verified to high precision only in the scalar case (E21), because the outer-product estimate on the chain is a small residual of order-one quantities and reaches a quadrature floor. Panel (c) is Proposition 10.12: the gap is measured by tree-exact message passing against the LMMSE twin, divided by the predicted constant, on a domain wide enough for each family's sample (E18), with the bootstrap intervals from many sequences because the per-sequence gap is heavy-tailed (a degree-six polynomial in nearly Gaussian variables). The uniform family, with the smallest constant, reaches the double-precision floor of the posterior means first. All numbers are in the paper; none is repeated here.

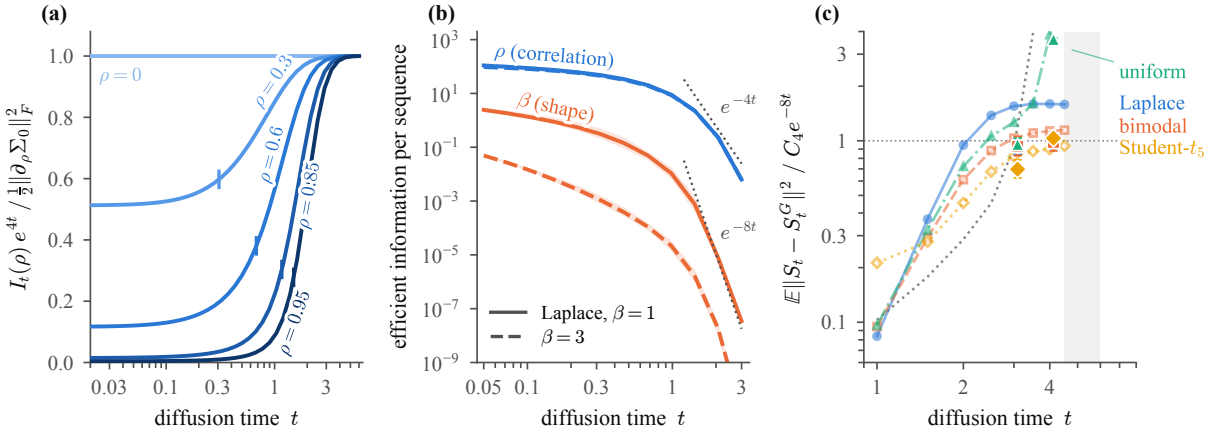


Figure 10.2: The paper's theory figure (Section 3), reproduced for reference. (a) Exact Fisher information of the Gaussian chain about ρ divided by its asymptote $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 e^{-4t}$, five correlations, ticks at $t_*(\rho)$ (Propositions 10.17 and 10.21). (b) Efficient information per sequence about ρ (degree two) and about the shape exponent β (degree four) of the generalised-Gaussian chain, from Fisher-identity scores, against the guides e^{-4t} and e^{-8t} (Section 10.9). (c) The closure gap $\mathbb{E}\|S - S_{\mathrm{G}}\|^2$ divided by $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$ for four innovation families (Proposition 10.12), with the large-sample constants and the double-precision floor.

10.16 Worked examples

Example 10.24 (The scalar Laplace law)**.** A unit-variance Laplace variable has excess kurtosis 3, so $\kappa_4 = 3$ and the constant of Equation (10.1) is $\kappa_4^2/6 = 1.5$. The leading term of the score gap is $\frac{3}{6}\mathrm{He}_3(x)e^{-4t} = \frac{1}{2}(x^3 - 3x)e^{-4t}$; at $x = 2$ and $t = 1$ it is $\frac{1}{2} \cdot 2 \cdot e^{-4} = 0.0183$, against a twin score of -2. The expected squared gap at $t = 2$ is $1.5\, e^{-16} = 1.5 \times 1.13 \times 10^{-7} = 1.7 \times 10^{-7}$, and since the score energy at that level is close to one (Proposition 7.14), the relative score error of the twin is about $\sqrt{1.7 \times 10^{-7}} = 4 \times 10^{-4}$.

Example 10.25 (The shape constant at the Gaussian member)**.** The generalised Gaussian $\varphi_\beta \propto e^{-|\varepsilon/s_\beta|^\beta}$ at unit variance has excess kurtosis $\kappa(\beta) = \Gamma(5/\beta)\Gamma(1/\beta)/\Gamma(3/\beta)^2 - 3$. Its derivative at the Gaussian member $\beta = 2$ can be done by hand with the digamma recursion $\psi(x+1) = \psi(x) + 1/x$: $\frac{\mathrm{d}}{\mathrm{d}\beta}\log(\kappa+3) = \beta^{-2}[-5\psi(5/\beta) - \psi(1/\beta) + 6\psi(3/\beta)]$, and at $\beta = 2$, with $\psi(5/2) = \psi(1/2) + 2 + \frac{2}{3}$ and $\psi(3/2) = \psi(1/2) + 2$, the bracket is $(-5-1+6)\psi(\frac{1}{2}) - \frac{40}{3} + 12 = -\frac{4}{3}$, so the derivative of the logarithm is $-\frac{1}{3}$ and $\partial_\beta \kappa|_{\beta=2} = 3 \times (-\frac{1}{3}) = -1$ exactly. In the scalar case Equation (10.3) therefore predicts $I_t(\beta) = \frac{1}{24}e^{-8t}(1 + O(e^{-2t}))$ at the Gaussian member: an information of one part in twenty-four times e^{-8t} per observation. The paper's E21 checks this constant by untruncated quadrature (Appendix E).

Example 10.26 (The two-site Gaussian chain, exactly)**.** For $L = 2$, $\mathbf{\Sigma}_0 = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ and $\partial_\rho \mathbf{\Sigma}_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 = 1$. With $\mathbf{\Sigma}_t = \left(\begin{smallmatrix} 1 & \alpha^2\rho \\ \alpha^2\rho & 1 \end{smallmatrix}\right)$, $\mathbf{\Sigma}_t^{-1}\partial_\rho \mathbf{\Sigma}_t = \frac{\alpha^2}{1-\alpha^4\rho^2}\left(\begin{smallmatrix} -\alpha^2\rho & 1 \\ 1 & -\alpha^2\rho \end{smallmatrix}\right)$, whose square has trace $2\alpha^4(1+\alpha^4\rho^2)/(1-\alpha^4\rho^2)^2$, so by Proposition 10.17

$$I_t(\rho) = \frac{\alpha^4(1+\alpha^4\rho^2)}{(1-\alpha^4\rho^2)^2} = e^{-4t}(1 + 3\alpha^4\rho^2 + O(\alpha^8)).$$

At $\rho = 0.85$ and $t = 1$: $\alpha^4 = e^{-4} = 0.01832$ and $\alpha^4\rho^2 = 0.01323$, so

$$I_1 = \frac{0.01832 \times 1.01323}{0.98677^2} = 0.01906,$$

four per cent above the asymptote. Here the correction is $O(\alpha^4)$ rather than $O(\alpha^2)$ because $\mathrm{tr}(\mathbf{\Sigma}_0 - \mathbf{I}) = 0$ and the first-order term of the Neumann series drops out of the trace; for the 32-site chain it does not, and the correction is $O(\alpha^2\lambda_{\max})$.

Example 10.27 (The fourth-cumulant tensor of a three-site chain)**.** Let $a_1 \sim \mathcal{N}(0,1)$, $a_2 = \rho a_1 + \varepsilon_2$, $a_3 = \rho a_2 + \varepsilon_3 = \rho^2 a_1 + \rho\varepsilon_2 + \varepsilon_3$, with i.i.d. innovations of variance $q = 1 - \rho^2$ and fourth cumulant $\kappa_4(\varepsilon) = 3q^2$ (Laplace). Joint cumulants are additive over the independent summands and a_1 is Gaussian, so $\boldsymbol{\kappa}_4[ijkl] = \kappa_4(\varepsilon) \sum_{m=2}^{\min(i,j,k,l)} \rho^{i+j+k+l-4m}$ (`chain_kappa4` in `src/markov_score/theory.py`). The nonzero entries up to permutation are $\boldsymbol{\kappa}_4[2222] = 3q^2$, $\boldsymbol{\kappa}_4[2223] = 3q^2\rho$, $\boldsymbol{\kappa}_4[2233] = 3q^2\rho^2$, $\boldsymbol{\kappa}_4[2333] = 3q^2\rho^3$ and $\boldsymbol{\kappa}_4[3333] = 3q^2(\rho^4 + 1)$. Over ordered tuples the Frobenius norm is

$$\|\boldsymbol{\kappa}_4\|_F^2 = 9q^4\,[1 + 4\rho^2 + 6\rho^4 + 4\rho^6 + (1 + \rho^4)^2],$$

the multiplicities being $1, 4, 6, 4, 1$. At $\rho = 0.85$: $q = 0.2775$, $q^4 = 0.00593$, $\rho^2 = 0.7225$, $\rho^4 = 0.5220$, $\rho^6 = 0.3772$; the bracket is $1 + 2.890 + 3.132 + 1.509 + 2.317 = 10.85$, so $\|\boldsymbol{\kappa}_4\|_F^2 = 9 \times 0.00593 \times 10.85 = 0.579$ and $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 = 0.0965$. The same recipe at $L = 32$ gives the constant the paper quotes for the Laplace chain (Section 3, "Verification with the exact score"), which grows linearly in L away from the first site.

Where it is used. Section 3 of the paper (Theorems 2–4, Assumption (E), Corollary 5, the crossover and heavy-tail paragraphs, the Edgeworth paragraph) and Appendix A (the proofs). Experiments: E01 (Gaussian closed form, shape information on the chain, closure gap), E16 (Edgeworth correction), E18 (closure constant and heavy tails on wide domains, untruncated scalar quadrature), E21 (information curve at $\rho = 0.5$ and the scalar shape constant). Code: `src/markov_score/theory.py` (`he`, `chain_covariance`, `chain_kappa4`, `gaussian_chain_fisher`, `score_gap_constant`, `shape_information_constant`, `correlation_information_constant`, `crossover_time`), `src/markov_score/information.py` (`per_chain_scores`, `outer_product_information`, `efficient_information`), `experiments/e01_information_decay.py`, `experiments/e16_edgeworth_oracle.py`, `experiments/e18_closure_wide.py`, `experiments/e21_information_curve.py`; the Lean targets for the constants are listed in `lean/README.md`. Audit: `docs/SCIENCE_AUDIT.md`, `reports/audit-2026-09-18.tex`, Section 3.

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Part III

Exact inference

Chapter 11

Factor graphs and belief propagation

The paper's central computational claim is that the score of a non-Gaussian, correlated law in L dimensions is available exactly, by "one forward–backward sweep of functional message passing". This chapter builds the objects in that sentence from scratch: what a factor graph is, what the sum-product algorithm does on it, why the algorithm is exact when the graph is a tree, and why conditioning the chain prior on coordinatewise noised observations leaves a tree. It then specialises the general algorithm to the two message sweeps of the paper's equation for $\overrightarrow{m}_i$ and $\overleftarrow{m}_i$ (Appendix B, the equation labelled *bp*), reads off the site marginals, the pairwise marginals and the evidence, and places the recursion next to its two classical relatives, the forward–backward algorithm for hidden Markov models and the Kalman smoother.

11.1 Graphical models and factor graphs

11.1.1 Factorised functions

Let $a = (a_1, \dots, a_L)$ be a vector of variables, each taking values in a set $\mathcal{A}$ (a finite alphabet, or $\mathbb{R}$). A **factorisation** of a non-negative function f of a is a representation

$$f(a) = \prod_{\alpha \in \mathcal{F}} \psi_\alpha(a_{\partial\alpha}), \tag{11.1}$$

where $\mathcal{F}$ is a finite index set of **factors**, each factor α has a **scope** $\partial\alpha \subseteq \{1, \dots, L\}$, and $a_{\partial\alpha}$ is the sub-vector of the variables in that scope. The **factor graph** of the factorisation is the bipartite graph with one **variable node** per index i, one **factor node** per factor α, and an edge between i and α whenever $i \in \partial\alpha$. A factor with a scope of size one is called **unary**; a scope of size two is **pairwise**.

When f is a probability density up to a constant, $p(a) = f(a)/Z$ with $Z = \int f(a)\,\mathrm{d}a$, the factor graph is a **graphical model** of p. The constant Z is the **normaliser**, also called the **partition function** in physics and, when f is a joint density of data and latent variables evaluated at the observed data, the **evidence** or **marginal likelihood**.

Example 11.1 (The chain prior)**.** The paper's clean law, its Section 2, equation *prior*, is $P_0(\mathbf{a}) = \mu(a_1) \prod_{i=2}^{L} K(a_i \mid a_{i-1})$. It is a factorisation with one unary factor μ of scope $\{1\}$ and $L-1$ pairwise factors $K(\cdot \mid \cdot)$ of scopes $\{i-1, i\}$. Its factor graph is a path: variable nodes $a_1, a_2, \dots, a_L$ with a factor node between each consecutive pair, and a dangling unary factor at a_1. Here $Z = 1$ because each $K(\cdot \mid a)$ integrates to one.

11.1.2 Conditional independence

The factor graph encodes which variables are conditionally independent of which. The statement we need is the following. If removing the variable node a_i (together with its edges) disconnects the

graph into components $\mathcal{C}_1, \dots, \mathcal{C}_r$, then under p the groups of variables in different components are conditionally independent given a_i. The proof is one line: every factor lies inside one component (its scope, minus i, is connected through the factor node), so $f(a) = \prod_j g_j(a_{\mathcal{C}_j}, a_i)$, and a joint density that is a product of functions of the separate groups (each also depending on a_i) is, given a_i, a product density.

Applied to the path of Example 11.1, removing a_i separates the past $\{a_1, \dots, a_{i-1}\}$ from the future $\{a_{i+1}, \dots, a_L\}$: this is the **Markov property**, and it is the only use we make of the word "Markov".

11.1.3 Trees, and why marginalisation is hard in general

A factor graph is a **tree** when it is connected and has no cycles. A path is a tree. The **marginal** of p at a variable i is $p_i(a_i) = \frac{1}{Z} \int f(a)\, \mathrm{d}a_{\setminus i}$, integrating over every variable but the i-th. Written out on a finite alphabet of size N_g, the integral is a sum over N_g^{L-1} configurations, which for the paper's $N_\mathrm{g} = 401$ and $L = 32$ is not a computation. Every algorithm in this chapter is a way of organising that sum so that it costs $O(LN_\mathrm{g}^2)$ instead, and the organisation is possible exactly because the graph is a tree.

11.2 The sum-product algorithm

11.2.1 Messages

The **sum-product algorithm**, also called **belief propagation**, passes two kinds of functions along the edges of the factor graph. For an edge between variable i and factor α:

$$\text{variable to factor:} \qquad m_{i\to\alpha}(a_i) = \prod_{\beta \in \partial i \setminus \{\alpha\}} m_{\beta\to i}(a_i), \tag{11.2}$$

$$\text{factor to variable:} \qquad m_{\alpha\to i}(a_i) = \int \psi_\alpha(a_{\partial\alpha}) \prod_{j \in \partial\alpha \setminus \{i\}} m_{j\to\alpha}(a_j)\, \mathrm{d}a_{\partial\alpha\setminus i}, \tag{11.3}$$

where ∂i is the set of factors whose scope contains i. An empty product is 1, so a leaf variable (one attached to a single factor) sends the constant message 1, and a leaf factor (a unary factor) sends its own function $m_{\alpha\to i}(a_i) = \psi_\alpha(a_i)$. The names are literal: a message is a function of the variable at the edge's variable end, and Equation (11.3) sums (integrates) a product.

On a tree, every message can be computed once and in a definite order: root the tree anywhere, compute messages from the leaves towards the root, then from the root back towards the leaves. Each edge carries one message in each direction, and each message is computed after all the messages it depends on. The **belief** at a variable and at a factor are then

$$b_i(a_i) \propto \prod_{\alpha \in \partial i} m_{\alpha\to i}(a_i), \qquad b_\alpha(a_{\partial\alpha}) \propto \psi_\alpha(a_{\partial\alpha}) \prod_{j \in \partial\alpha} m_{j\to\alpha}(a_j). \tag{11.4}$$

11.2.2 Exactness on a tree

Theorem 11.2 (Sum-product is exact on a tree)**.** *Let the factor graph of Equation* (11.1) *be a tree. Then the beliefs of Equation* (11.4) *are the exact marginals: $b_i = p_i$ and $b_\alpha = p_{\partial\alpha}$, where $p_{\partial\alpha}$ is the marginal of p on the scope of α. Moreover, for every variable i, $Z = \int \prod_{\alpha\in\partial i} m_{\alpha\to i}(a_i)\, \mathrm{d}a_i$.*

Proof. The proof is an induction on subtrees, and it rests on one fact about integrals of products: the **distributive law**, $\int g(u)h(v)\, \mathrm{d}u\, \mathrm{d}v = \int g\, \mathrm{d}u \int h\, \mathrm{d}v$ when the two factors depend on disjoint sets of variables.

Fix an edge (i, α) and delete it. Because the graph is a tree, this splits it into two subtrees; call $T_{\alpha\to i}$ the one containing α and $T_{i\to\alpha}$ the one containing i. Write $F_{\alpha\to i}(a_i)$ for the integral,

over every variable of $T_{\alpha\to i}$ other than a_i, of the product of all factors in $T_{\alpha\to i}$; and $F_{i\to\alpha}(a_i)$ likewise for $T_{i\to\alpha}$ (the factors on the i side other than α). We claim

$$m_{\alpha\to i} = F_{\alpha\to i}, \qquad m_{i\to\alpha} = F_{i\to\alpha}. \tag{11.5}$$

The claim is proved by induction on the number of nodes in the subtree behind the message. If $T_{\alpha\to i}$ consists of α alone, then α is unary and both sides equal $\psi_\alpha(a_i)$. If $T_{i\to\alpha}$ consists of i alone, both sides are 1. Otherwise, suppose the claim holds for all smaller subtrees. The subtree $T_{\alpha\to i}$ is α together with, for each $j \in \partial\alpha \setminus \{i\}$, the subtree $T_{j\to\alpha}$; these subtrees are disjoint (no cycles) and share no variables with one another (a shared variable would be a cycle through α). Hence the product of all factors in $T_{\alpha\to i}$ is $\psi_\alpha \prod_j (\text{factors of } T_{j\to\alpha})$, and integrating out the variables of each $T_{j\to\alpha}$ other than a_j can be done separately by the distributive law, giving $\psi_\alpha \prod_j F_{j\to\alpha}(a_j)$; the remaining integral over $a_{\partial\alpha\setminus i}$ is exactly Equation (11.3) once $F_{j\to\alpha} = m_{j\to\alpha}$ by the induction hypothesis. The variable-to-factor case is the same argument with the subtrees $T_{\beta\to i}$, $\beta \in \partial i \setminus \{\alpha\}$, which share only the variable a_i; no integral is needed and Equation (11.2) follows.

Now take a variable i. The whole graph is the union of the subtrees $T_{\alpha\to i}$, $\alpha \in \partial i$, which pairwise share only a_i. By the distributive law again, $\int f(a)\,\mathrm{d}a_{\setminus i} = \prod_{\alpha\in\partial i} F_{\alpha\to i}(a_i)$, and by Equation (11.5) this is the unnormalised belief. Integrating once more over a_i gives Z. For a factor α, the whole graph is α together with the subtrees $T_{j\to\alpha}$, $j \in \partial\alpha$, sharing no variables, and the same reasoning gives $b_\alpha = p_{\partial\alpha}$. □

Three remarks on the theorem. First, nothing in the proof depends on the variables being discrete: on $\mathbb{R}$ the messages are functions and the sums are integrals, which is what the paper means by *functional* message passing. Second, the proof shows that the belief at a factor integrates, over the variables of the factor other than i, to the belief at i: beliefs are **locally consistent**. Third, on a graph with cycles the messages of Equations (11.2) and (11.3) still make sense and can be iterated (**loopy belief propagation**), but the induction fails at the first cycle and the beliefs are approximations. The paper's claim is that its inference is *tree-exact*: the posterior graph is a tree, so no such approximation is made.

11.3 The chain posterior of the paper

11.3.1 Conditioning adds only unary factors

The observation model is the coordinatewise channel of the paper's equation *channel*: given the clean chain $\mathbf{a}$, the coordinates of $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ are independent, each with density $\mathcal{N}(e^{-t}a_i, \Delta_t)$ in x_i. By Bayes' rule,

$$p_t(\mathbf{a} \mid \mathbf{x}) = \frac{P_0(\mathbf{a}) \prod_{i=1}^{L} \ell_t(x_i \mid a_i)}{P_t(\mathbf{x})}, \qquad \ell_t(x_i \mid a_i) = \frac{1}{\sqrt{2\pi\Delta_t}} \exp\Big[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\Big]. \tag{11.6}$$

Compared with the prior, the numerator has one extra factor per site, each of scope $\{i\}$: a **unary likelihood factor**. No factor couples two latent variables that were not already coupled. The factor graph of the posterior is therefore the prior's path with a unary factor hanging from every variable node; it is still a tree (Figure 11.1). This is the whole content of the sentence "conditioning leaves the posterior factor graph a tree". It would fail for a channel that mixed coordinates (a non-diagonal noise covariance would add pairwise factors among the a_i, closing cycles), and it holds for any prior whose graph is a tree, not only for a path.

11.3.2 From the general rules to the two sweeps

Specialise Equations (11.2) and (11.3) to Figure 11.1. Each latent variable a_i has at most three neighbouring factors: the transition factor on its left, the transition factor on its right, and

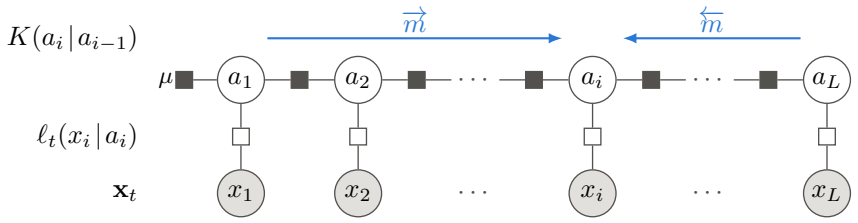


Figure 11.1: The factor graph of the chain posterior, drawn as in the paper's Figure 1. Latent sites (white circles) are joined by the transition factors K (filled squares) and, at the left end, by the initial factor μ. Conditioning attaches one unary likelihood factor (open square) per site, joining a_i to its own observation x_i (shaded circle), and adds no edge among the latent variables. The graph is a tree, so the forward sweep $\overrightarrow{m}$ and the backward sweep $\overleftarrow{m}$ of the sum-product algorithm are exact; they meet at the query site i.

its unary likelihood. Write K_i for the factor between a_{i-1} and a_i (with $K_1 = \mu$ the unary initial factor). Going left to right, the message that K_i sends into a_i is, by Equation (11.3), an integral over a_{i-1} of $K(a_i \mid a_{i-1})$ times the message a_{i-1} sends into K_i; and that message is, by Equation (11.2), the product of what a_{i-1} receives from its other two neighbours, its own unary factor $\ell_t(x_{i-1} \mid \cdot)$ and the transition factor on its left. Naming $\overrightarrow{m}_i(a) = m_{K_i \to a_i}(a)$, this reads

$$\overrightarrow{m}_1(a) = \mu(a), \qquad \overrightarrow{m}_i(a) = \int K(a \mid a')\, \ell_t(x_{i-1} \mid a')\, \overrightarrow{m}_{i-1}(a')\, \mathrm{d}a', \quad i \geq 2. \tag{11.7}$$

Going right to left, name $\overleftarrow{m}_i(a) = m_{K_{i+1} \to a_i}(a)$ the message that the transition factor on the right of a_i sends into a_i; for the last site there is no such factor and $\overleftarrow{m}_L \equiv 1$ (an empty product). The same two rules give

$$\overleftarrow{m}_L(a) = 1, \qquad \overleftarrow{m}_i(a) = \int K(a'' \mid a)\, \ell_t(x_{i+1} \mid a'')\, \overleftarrow{m}_{i+1}(a'')\, \mathrm{d}a'', \quad i \leq L-1. \tag{11.8}$$

These are, up to the proportionality signs discussed in Section 11.3.5, the two lines of the paper's equation *bp*. The paper's convention for $\overrightarrow{m}_1$ differs from Equation (11.7) in one cosmetic respect: it writes $\overrightarrow{m}_1 \equiv 1$ and treats μ as if it were part of the first site's unary factor. The code follows Equation (11.7), with `L[0] = exp(log_mu)` and `R[-1] = 1` in `src/markov_score/bp_grid.py`. Note the asymmetry of the kernel's arguments: forward, the integration variable is the *conditioning* argument of K; backward, it is the *conditioned* one. The transition matrix on a grid, $K[k,l] = K(u_k \mid u_l)$ with column index the parent, enters the forward sweep as K and the backward sweep as $K^\top$, which is exactly what the code does.

Two conventions are worth fixing, because both appear in the literature. The messages Equations (11.7) and (11.8) *exclude* the unary factor of the site they arrive at; they are the factor-to-variable messages of the general algorithm. The alternative convention (the "filtered" forward variable of the HMM literature) multiplies the local likelihood in and would be $\overrightarrow{m}_i \ell_t$. The paper and the code use the excluded form, so that the local factor appears once, explicitly, in every formula below.

11.3.3 Site marginals, pairwise marginals and the evidence

By Equation (11.4), the belief at site i is the product of what it receives from its three neighbours,

$$p_t(a_i \mid \mathbf{x}) = \frac{\overrightarrow{m}_i(a_i)\, \ell_t(x_i \mid a_i)\, \overleftarrow{m}_i(a_i)}{Z_i}, \qquad Z_i = \int \overrightarrow{m}_i \ell_t(x_i \mid \cdot) \overleftarrow{m}_i \, \mathrm{d}a_i, \tag{11.9}$$

and by Theorem 11.2 it is the exact posterior marginal. The paper's phrase "site marginals $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$" is this line. The belief at the transition factor between a_{i-1} and a_i is

$$p_t(a_{i-1}, a_i \mid \mathbf{x}) = \frac{\overrightarrow{m}_{i-1}(a_{i-1})\, \ell_t(x_{i-1} \mid a_{i-1})\, K(a_i \mid a_{i-1})\, \ell_t(x_i \mid a_i)\, \overleftarrow{m}_i(a_i)}{Z_{i-1,i}}, \tag{11.10}$$

the exact **pairwise marginal**, whose two single-site margins are Equation (11.9) at $i-1$ and at i (local consistency). The pairwise marginals are what the expectation–maximisation chapter (Chapter 15) sums into the expected transition statistic Ξ; they come from the same two sweeps at no extra cost beyond the outer products.

The normaliser is available at any site, $Z_i = P_t(\mathbf{x})$ for every i, because every Z_i equals $\int P_0(\mathbf{a}) \prod_j \ell_t(x_j \mid a_j)\,\mathrm{d}\mathbf{a}$ by the proof of Theorem 11.2; the same holds for every $Z_{i-1,i}$. In particular the forward sweep alone gives it, as $Z_L = \int \overrightarrow{m}_L \ell_t(x_L \mid \cdot)\,\mathrm{d}a$ once $\overleftarrow{m}_L \equiv 1$. This number, $\log P_t(\mathbf{x})$, is the **log-evidence**: the marginal log-likelihood of one noised sequence under the chain prior, which the paper's Section 4 maximises over the parameters of the kernel and which its Appendix E uses, summed over a held-out bundle, as an oracle-free model-selection criterion.

11.3.4 From the marginal to the score

The paper needs one number per site, the posterior mean $\langle a_i \rangle_{x,t} = \int a_i\, p_t(a_i \mid \mathbf{x})\,\mathrm{d}a_i$, because Tweedie's identity (the chapter on the Ornstein–Uhlenbeck channel) converts it into the score, $S(\mathbf{x}, t)_i = -(x_i - e^{-t}\langle a_i \rangle_{x,t})/\Delta_t$. Thus the whole of "the exact score" is: two sweeps, L one-dimensional integrals for the means, and one subtraction per site. The word "exact" refers to Theorem 11.2; what is not exact is the numerical representation of the messages, which is the subject of Chapter 12.

11.3.5 Normalisation and the log-normaliser

Messages are defined up to a positive constant: multiplying $\overrightarrow{m}_i$ by c multiplies every later forward message and every belief involving it by c, and the normalised beliefs Equations (11.9) and (11.10) are unchanged. This is why the paper writes $\propto$ in its recursion. Numerically, unnormalised messages on a long chain underflow or overflow (each step multiplies by a likelihood of order $1/\sqrt{2\pi\Delta_t}$ and by a kernel of order $1/\sqrt{q}$), so an implementation rescales each message to unit mass and keeps the discarded constants:

$$\widetilde{\overrightarrow{m}}_i = \frac{\overrightarrow{m}_i}{c_i}, \qquad c_i = \int \overrightarrow{m}_i\,\mathrm{d}a, \qquad \log P_t(\mathbf{x}) = \sum_{i=2}^{L} \log c_i + \log \int \widetilde{\overrightarrow{m}}_L \ell_t(x_L \mid \cdot)\,\mathrm{d}a, \tag{11.11}$$

where each c_i is computed from the *already rescaled* previous message, so that the product of the c_i is the total constant removed. The log-evidence is thus assembled from a sum of logarithms and never as a product of small numbers. Chapter 12 gives the grid version, including the constants of the likelihood factors that the implementation removes for stability and restores at the end.

11.3.6 Cost

On a grid of N_g points, a message is a vector of length N_g and a transition integral is a matrix–vector product with the $N_\mathrm{g} \times N_\mathrm{g}$ matrix $K[k, l]$, costing $O(N_\mathrm{g}^2)$. The forward sweep performs $L-1$ of them, the backward sweep the same, and the beliefs cost $O(LN_\mathrm{g})$. The total is $O(LN_\mathrm{g}^2)$ per sequence and per noise level, as the paper states in its conclusion, and it is linear in the chain length: this is why the length-extrapolation study of the paper's Section 5 costs nothing extra at $L = 256$. For a batch of B sequences the matrix–vector products become matrix–matrix products $(N_\mathrm{g} \times N_\mathrm{g})(N_\mathrm{g} \times B)$, which is how `grid_bp_batch` in `src/markov_score/bp_grid.py` organises the computation.

11.4 Three names for one recursion

Forward–backward for hidden Markov models. When the state space is a finite set of N_{g} symbols, Equations (11.7) and (11.8) are the forward and backward recursions of Baum and colleagues (1970) and Rabiner's tutorial (1989), usually written for the filtered variable $\alpha_i(k) = P(a_i = k, x_1, \dots, x_i)$ and the backward variable $\beta_i(k) = P(x_{i+1}, \dots, x_L \mid a_i = k)$. In our notation $\alpha_i = \overrightarrow{m}_i \ell_t(x_i \mid \cdot)$ and $\beta_i = \overleftarrow{m}_i$; the posterior at a site is $\alpha_i \beta_i / P_t(\mathbf{x})$ and the pairwise posterior is $\alpha_{i-1} K \ell_t \beta_i / P_t(\mathbf{x})$, both as above. The pairwise posteriors summed over sites are the "expected transition counts" of the Baum–Welch algorithm, and their continuous analogue on the grid is exactly the paper's $\boldsymbol{\Xi}$. The paper's recursion is therefore not a new algorithm; the paper says so in its related work, citing Baum and Rabiner.

The Kalman smoother. When μ, K and ℓ_t are all Gaussian, every message is Gaussian (a product of Gaussians is Gaussian, and the convolution of a Gaussian with a Gaussian kernel is Gaussian), so each message is carried by two numbers, a mean and a variance, or equivalently a precision and an information vector. The forward sweep is then the Kalman filter and the combination of forward and backward messages is the Rauch–Tung–Striebel smoother. Chapter 13 writes out the two moment maps and proves the paper's Proposition 1, which says that projecting every message onto a Gaussian, for *any* innovation law with the same variance, reproduces this smoother. The implementation of the Gaussian case is `gaussian_chain_bp` in `src/markov_score/bp_gaussian.py`.

Functional message passing and Kitagawa's grid smoother. When the state is continuous and non-Gaussian, the messages are genuine functions on $\mathbb{R}$ and have no finite parametrisation. The paper calls the recursion "functional message passing" to stress that the algorithm is defined at the level of functions, before any discretisation, and "tree-exact inference" to stress that the only approximation is in representing those functions. Representing them by their values on a grid and replacing integrals by quadrature is Kitagawa's non-Gaussian smoother (1987), which the paper also cites. The phrase "the score" throughout the paper means: the posterior mean obtained by this recursion, evaluated on a grid whose adequacy is measured (Chapter 12), inserted into Tweedie's identity.

11.5 A worked example: three sites, three states

Take $L = 3$ sites and a three-point alphabet $\{u_1, u_2, u_3\} = \{-1, 0, 1\}$ with no quadrature weights (a finite-state model, so that every integral is a sum). Let the initial law, the transition matrix and the three likelihood rows be

$$\mu = \begin{pmatrix} 0.25 \\ 0.5 \\ 0.25 \end{pmatrix}, \qquad K = \begin{pmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.6 & 0.3 \\ 0.1 & 0.2 & 0.6 \end{pmatrix}, \qquad \ell_1 = \begin{pmatrix} 0.2 \\ 0.5 \\ 0.3 \end{pmatrix},\ \ell_2 = \begin{pmatrix} 0.1 \\ 0.3 \\ 0.6 \end{pmatrix},\ \ell_3 = \begin{pmatrix} 0.4 \\ 0.4 \\ 0.2 \end{pmatrix},$$

where $K[k, l]$ is the probability of moving from state l (column) to state k (row), so the columns sum to one, and $\ell_i[k]$ (the unary factor $\ell_t(x_i \mid u_k)$ of the chapter, written as a plain vector here) is the likelihood of the i-th observation given $a_i = u_k$ (any positive numbers; they are not normalised over k and need not be).

Forward sweep. $\overrightarrow{m}_1 = \mu$. The message into site 2 is $\overrightarrow{m}_2 = K(\overrightarrow{m}_1 \circ \ell_1)$, with $\circ$ the elementwise product: $\overrightarrow{m}_1 \circ \ell_1 = (0.05,\ 0.25,\ 0.075)$, and

$$\overrightarrow{m}_2 = \begin{pmatrix} 0.6 \cdot 0.05 + 0.2 \cdot 0.25 + 0.1 \cdot 0.075 \\ 0.3 \cdot 0.05 + 0.6 \cdot 0.25 + 0.3 \cdot 0.075 \\ 0.1 \cdot 0.05 + 0.2 \cdot 0.25 + 0.6 \cdot 0.075 \end{pmatrix} = \begin{pmatrix} 0.0875 \\ 0.1875 \\ 0.1 \end{pmatrix}.$$

Then $\overrightarrow{m}_2 \circ \ell_2 = (0.00875, 0.05625, 0.06)$ and $\overrightarrow{m}_3 = K(\overrightarrow{m}_2 \circ \ell_2) = (0.0225, 0.054375, 0.048125)$.

Backward sweep. $\overleftarrow{m}_3 = (1, 1, 1)$. Then $\overleftarrow{m}_2 = K^\top(\overleftarrow{m}_3 \circ \ell_3) = K^\top(0.4, 0.4, 0.2)$: the first entry is $0.6 \cdot 0.4 + 0.3 \cdot 0.4 + 0.1 \cdot 0.2 = 0.38$, and altogether $\overleftarrow{m}_2 = (0.38, 0.36, 0.28)$. Next $\overleftarrow{m}_2 \circ \ell_2 = (0.038, 0.108, 0.168)$ and $\overleftarrow{m}_1 = K^\top(\overleftarrow{m}_2 \circ \ell_2) = (0.072, 0.106, 0.137)$.

Evidence. From the end of the forward sweep, $Z = \sum_k \overrightarrow{m}_3[k]\ell_3[k] = 0.0225 \cdot 0.4 + 0.054375 \cdot 0.4 + 0.048125 \cdot 0.2 = 0.040375$. The same number is obtained at site 1, $\sum_k \mu[k]\ell_1[k]\overleftarrow{m}_1[k] = 0.25 \cdot 0.2 \cdot 0.072 + 0.5 \cdot 0.5 \cdot 0.106 + 0.25 \cdot 0.3 \cdot 0.137 = 0.040375$, and at site 2, as Theorem 11.2 requires.

Site marginals. At site 1 the unnormalised belief is $\mu \circ \ell_1 \circ \overleftarrow{m}_1 = (0.0036, 0.0265, 0.010275)$; dividing by Z gives $p(a_1 \mid \mathbf{x}) = (0.0892, 0.6564, 0.2545)$ and posterior mean $-0.0892 + 0.2545 = 0.1653$. At site 2 the unnormalised belief is

$$\overrightarrow{m}_2 \circ \ell_2 \circ \overleftarrow{m}_2 = (0.003325, 0.02025, 0.0168),$$

so that the normalised marginal is $p(a_2 \mid \mathbf{x}) = (0.0824, 0.5015, 0.4161)$, with mean 0.3337. At site 3 the same steps give $p(a_3 \mid \mathbf{x}) = (0.2229, 0.5387, 0.2384)$ and the mean 0.0155. Enumerating all 27 configurations of (a_1, a_2, a_3) and summing gives the same Z and the same three marginals, which is the finite version of Theorem 11.2.

Pairwise marginal. On the edge $(1, 2)$, Equation (11.10) gives the matrix with entries

$$p(a_2 = u_k, a_1 = u_l \mid \mathbf{x}) = \overleftarrow{m}_2[k]\, \ell_2[k]\, K[k, l]\, \overrightarrow{m}_1[l]\, \ell_1[l]/Z;$$

for instance the entry $(k, l) = (2, 2)$, staying at 0, is $0.108 \cdot 0.6 \cdot 0.25/0.040375 = 0.4012$, and the full matrix is

$$p(a_2 = u_k, a_1 = u_l \mid \mathbf{x}) = \begin{pmatrix} 0.0282 & 0.0471 & 0.0071 \\ 0.0401 & 0.4012 & 0.0602 \\ 0.0208 & 0.2081 & 0.1872 \end{pmatrix},$$

whose column sums are the site-1 marginal and whose row sums are the site-2 marginal, to the rounding shown. This is the object that Chapter 15 accumulates into $\boldsymbol{\Xi}$: summing the two edge matrices of this chain gives a 3×3 matrix of total mass 2, the number of transitions.

Where it is used. The paper, Section 2 ("the score is a posterior mean, and the posterior is a tree", Figure 1) and Appendix B (the recursion *bp*); Section 4 (the pairwise marginals behind $\boldsymbol{\Xi}$); Appendix E (the log-evidence as a selection criterion, E20). Code: `src/markov_score/bp_grid.py` (`grid_bp` and `grid_bp_batch`), `src/markov_score/em.py` (`e_step`, which adds the pairwise accumulation to the same sweeps), `src/markov_score/bp_gaussian.py` (the Gaussian special case), and `paper/figures/fig_mechanism.tex` (the figure mirrored above). Tests: the brute-force enumeration test in `tests/test_em_bp.py`; `tests/test_fisher_bruteforce.py`, which checks the evidence and the pairwise marginals against enumeration on a five-state, four-site chain; and `tests/test_gaussian_bp_equivalence.py`. Experiments: every registered experiment, E01 to E21, scores against this recursion.

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Chapter 12

Messages on a grid: quadrature, truncation and their diagnostics

The recursion of Chapter 11 is exact for functions. A computer holds a function by finitely many numbers, and the paper's choice is the simplest one: the values of each message at N_{g} equally spaced points of an interval $[-A, A]$, with integrals replaced by the composite trapezoidal rule. The paper summarises the consequence in one clause, "exact for the discretised factor model and controlled for the continuous one", and devotes Appendix B to measuring the control. This chapter explains the clause word by word: what the discretised model is and why forward–backward is exact for it; where the trapezoidal rule's error comes from, why it is far smaller than its textbook order for smooth rapidly decaying integrands and why it is not for a discontinuous kernel; what happens when the domain is too narrow for the data; and what each of the paper's diagnostics (column-mass defect, boundary belief mass, drift, exceedance count) measures. It ends with the normalisation bookkeeping of the implementation.

12.1 Functions as vectors

12.1.1 The grid and its weights

Fix a **half-width** $A > 0$ and a number of points $N_{\mathrm{g}} \geq 2$. The **grid** is $u_k = -A + (k-1)h$ for $k = 1, \ldots, N_{\mathrm{g}}$ with **spacing** $h = 2A/(N_{\mathrm{g}} - 1)$; the paper's operating grid has $N_{\mathrm{g}} = 401$, $A = 8$ and hence $h = 0.04$. A function g on $\mathbb{R}$ is represented by the vector $(g(u_1), \ldots, g(u_{N_{\mathrm{g}}}))$, and an integral by the **composite trapezoidal rule**

$$\int_{-A}^{A} g(a)\,\mathrm{d}a \;\approx\; \sum_{k=1}^{N_{\mathrm{g}}} w_k\, g(u_k), \qquad w_1 = w_{N_{\mathrm{g}}} = \tfrac{h}{2}, \quad w_k = h \text{ otherwise.} \tag{12.1}$$

The weights come from integrating the piecewise linear interpolant of g through the grid values: each interior cell $[u_k, u_{k+1}]$ contributes $h\,(g(u_k) + g(u_{k+1}))/2$, and summing over cells gives every interior point the weight h and the two end points half of it. The code is `make_grid` in `src/markov_score/bp_grid.py`.

12.1.2 The discretised factor model

Replacing the integrals of the chain posterior by Equation (12.1) amounts to replacing the continuous model by a finite-state one. Define the **discretised factor model** as the probability mass function on $\{1, \ldots, N_{\mathrm{g}}\}^L$

$$\hat{p}(k_1, \ldots, k_L \mid \mathbf{x}) \propto \mu(u_{k_1}) w_{k_1} \prod_{i=2}^{L} K(u_{k_i} \mid u_{k_{i-1}}) w_{k_i} \prod_{i=1}^{L} \ell_t(x_i \mid u_{k_i}). \tag{12.2}$$

It is a hidden Markov model with N_{g} states whose transition matrix is $\hat{K}[k,l] = K(u_k \mid u_l)\, w_k$, that is, the continuous kernel sampled on the grid and multiplied by the quadrature weight of the destination state. Two remarks. First, $\hat{K}$ is not exactly stochastic: its column sums $\sum_k \hat{K}[k,l]$ are trapezoidal estimates of $\int K(a \mid u_l)\,\mathrm{d}a = 1$ and miss one by the quadrature and truncation error of that integral. The recursion normalises every message, so this deficit never accumulates, but it is the natural scalar diagnostic of the discretisation at its source and is measured as the **column-mass defect** $\max_l \left|\sum_k K(u_k \mid u_l) w_k - 1\right|$ (Section 12.4). Second, running the two sweeps of Chapter 11 with the matrix $\hat{K}$ is the forward–backward algorithm on this finite-state model, and by the exactness theorem of that chapter it returns the marginals and the normaliser of Equation (12.2) *exactly*, up to floating point. The returned evidence is $\log \hat{Z}(\mathbf{x})$, the normaliser of the discretised model: exact bookkeeping of an approximate object. This is the first half of the paper's clause. The tests `test_xi_matches_brute_force_enumeration` in `tests/test_em_bp.py` and `tests/test_fisher_bruteforce.py` check exactly this statement, by enumerating Equation (12.2) on a tiny grid.

The second half, "controlled for the continuous one", concerns the distance between the marginals of Equation (12.2), read as functions on the grid, and those of the continuous posterior. That distance has two sources, treated in turn: **quadrature error**, from replacing integrals over $[-A, A]$ by sums, and **truncation error**, from ignoring the part of $\mathbb{R}$ outside the domain.

12.2 The trapezoidal rule and its error

12.2.1 Euler–Maclaurin

Let g be $2m$ times continuously differentiable on $[-A, A]$ and write $T_h(g)$ for the right-hand side of Equation (12.1). The **Euler–Maclaurin formula** states

$$\begin{aligned} T_h(g) - \int_{-A}^{A} g\,\mathrm{d}a &= \sum_{j=1}^{m-1} \frac{B_{2j}}{(2j)!} h^{2j} \Big[g^{(2j-1)}(A) - g^{(2j-1)}(-A) \Big] + R_m, \\ |R_m| &\le C_m\, h^{2m} \int_{-A}^{A} |g^{(2m)}|\,\mathrm{d}a, \end{aligned} \tag{12.3}$$

where $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, … are the Bernoulli numbers and C_m is an absolute constant. The derivation for one cell is worth seeing once. On $[0, h]$, integrate by parts with the polynomial $P_1(a) = a - h/2$ (whose derivative is 1 and which is odd about the midpoint): $\int_0^h g\,\mathrm{d}a = [P_1 g]_0^h - \int_0^h P_1 g'\,\mathrm{d}a = \frac{h}{2}(g(0) + g(h)) - \int_0^h P_1 g'\,\mathrm{d}a$. The first term is the trapezoidal rule on the cell; the second is the error. Integrating by parts again with P_2 chosen so that $P_2' = P_1$ and $P_2(0) = P_2(h) = h^2/12$ turns it into $-\frac{h^2}{12}(g'(h) - g'(0)) + \int_0^h P_2 g''\,\mathrm{d}a$, and summing over the cells telescopes the boundary terms to the two ends of $[-A, A]$, giving the $j = 1$ term of Equation (12.3) with $B_2/2! = 1/12$. Continuing the integrations by parts generates the higher Bernoulli terms. The textbook conclusion is that the trapezoidal rule is **second order**: its error is $O(h^2)$, with constant $\frac{1}{12}|g'(A) - g'(-A)|$. The paper's Appendix B calls the quadrature error "second order in the spacing" for this reason.

12.2.2 Why the rule is much better than second order on a Gaussian

Look again at Equation (12.3). Every term of the sum involves derivatives of g at the end points of the domain. If g and all its derivatives are negligible there, as they are for a Gaussian-like integrand whose domain is several standard deviations wide, every power of h in the expansion has a vanishing coefficient, and the error is smaller than any power of h. This property is called **spectral accuracy** (or super-algebraic convergence). It can be made exact for an integrand on

the whole line: for g smooth and rapidly decaying, the **Poisson summation formula** gives

$$h\sum_{k\in\mathbb{Z}} g(kh) - \int_{\mathbb{R}} g\,\mathrm{d}a = \sum_{n\neq 0} \hat{g}\Big(\frac{2\pi n}{h}\Big), \qquad \hat{g}(\omega) = \int_{\mathbb{R}} g(a)e^{-i\omega a}\,\mathrm{d}a, \tag{12.4}$$

so the error of the infinite trapezoidal sum is the sum of the Fourier transform of g at the frequencies $2\pi n/h$: it decays as fast as $\hat{g}$ does. For the standard Gaussian density, $\hat{g}(\omega) = e^{-\omega^2/2}$ and the error is $2\sum_{n\geq 1} e^{-2\pi^2 n^2/h^2}$, dominated by $2e^{-2\pi^2/h^2}$. At $h = 1$ this is $2e^{-19.74} \approx 5\times 10^{-9}$; at $h = 0.5$ it is $2e^{-79} \approx 10^{-34}$, below double precision; at the operating $h = 0.04$ it is zero for every practical purpose. The worked example of Section 12.6 checks the $h = 1$ figure on a finite domain.

This is why the quadrature is not the binding error in the paper's numerics for smooth families, and why the paper writes that "the domain, not the spacing, is what a heavy-tailed family tests". Two situations degrade the picture. If the integrand is narrow compared with h, which happens to the unary factor $\ell_t(x_i \mid \cdot)$ at small t (its standard deviation as a function of a is $\sqrt{\Delta_t}/e^{-t}$, about 0.32 at $t = 0.05$, eight cells at $h = 0.04$), the Fourier transform decays slowly and the terms of Equation (12.4) are no longer negligible; the code exposes `likelihood_resolution_ok` in `src/markov_score/noising.py` for this check, and the paper's E04 refines the grid on the reported cells to show the effect is small. And if g is not smooth the expansion Equation (12.3) fails at the first derivative that does not exist: the Laplace kernel has a kink at $a' = \rho a$ (its first derivative jumps), which costs the Bernoulli terms beyond $j = 1$ but still leaves an $O(h^2)$ error; the uniform kernel is discontinuous, and that is the case of the next subsection.

12.2.3 A discontinuous kernel is first order

The uniform innovation has kernel $K(a' \mid a) = 1/(2h_u)$ for $|a' - \rho a| \leq h_u$ and zero otherwise, with $h_u = \sqrt{3q}$ so that the variance is q. As a function of the integration variable it is a step, and the trapezoidal rule on a step is only **first order**. The mechanism is elementary. If a jump of height J falls at a distance $\delta \in [0, h)$ from the nearest grid point, the piecewise linear interpolant misplaces the jump by up to h and the integral by an amount between $-Jh/2$ and $+Jh/2$, depending on δ; the error oscillates with the position of the jump and its envelope shrinks only like h, not h^2. Section 12.6 exhibits the error exactly in the aligned case. In the chain the jumps sit at $a' = \rho a \pm h_u$ and move with a, so every column of $\hat{K}$ has its own misplacement, and the paper's Appendix B records the signature: for the uniform family the worst drift is largest on the *coarsest* spacing rather than on the narrowest domain, the opposite of the heavy-tailed families. The column-mass defect of the uniform kernel is likewise the largest among the families at moderate correlation, and it does fall with N_{g}.

12.3 Truncation

12.3.1 A site beyond the domain

The second error is the one the paper's sweep found to be binding. The grid represents only $[-A, A]$, so every message is implicitly zero outside. For a family whose marginal has light tails (Gaussian, Laplace, the bimodal mixture, the generalised Gaussian) the probability that any site of any sequence lies beyond $A = 8$ is negligible: for a standard Gaussian site it is about 10^{-15}, and for the Laplace innovation at the frozen correlation the probability that a single innovation exceeds 8 in absolute value is of order 10^{-10}. For a Student-t innovation the tail is polynomial and the probability is not negligible, which is the case treated next.

Suppose a clean site has $|a_i| > A$. Its observation is $x_i = e^{-t}a_i + \sqrt{\Delta_t}z_i$, and the unary factor $\ell_t(x_i \mid a)$, as a function of the latent a, is a Gaussian centred at $x_i/e^{-t} \approx a_i$, outside the grid. Restricted to the grid the factor is a monotone ramp, largest at the boundary point nearest to a_i; the transition factors from the neighbouring sites add little against it. The belief

at site i is therefore concentrated in the last few cells at the edge: the paper's phrase is that the posterior is **pinned to the boundary**. The posterior mean is then close to $\pm A$ instead of close to a_i, and by Tweedie's identity the score error at that site is about $(e^{-t}/\Delta_t)\,(|a_i| - A)$, which at the lowest level of the schedule is a factor $e^{-t}/\Delta_t \approx 10$ times the excess. One such site in a bundle of thousands can dominate a pooled relative error, and, crucially, it is pinned on *every* grid with the same half-width, so refining N_{g} at fixed A does not move it. That is the signature the paper's Appendix B reports for the binding cell: drift flat in N_{g}, falling only when the domain widens.

12.3.2 The exceedance count and its Poisson law

The right summary of truncation for a polynomial tail is not a maximum over sites but a count. Let N_A be the number of sites among the ML of a bundle with $|a_i| > A$. Its expectation is

$$\mathbb{E}N_A = ML\,\mathbb{P}(|a| > A) \tag{12.5}$$

when every site has the same marginal law (at $\rho = 0$ every site after the first is an independent draw of the innovation). Because the events are rare and, along a chain, only weakly dependent, the count is close to a Poisson variable with that mean: this is the law of rare events (Poisson's limit theorem, or Le Cam's inequality for dependent indicators with small total variation). No finite A makes Equation (12.5) zero for a polynomial tail; widening the domain only lowers the mean, and a Poisson variable with mean λ is zero with probability $e^{-\lambda}$ and at least one with probability $1 - e^{-\lambda}$. For the Student-t innovation with ν degrees of freedom, variance-matched so that $a = s\,T_\nu$ with $s^2 = q(\nu - 2)/\nu$, the tail probability is $\mathbb{P}(|a| > A) = 2\,\bar{F}_\nu(A/s)$ with $\bar{F}_\nu$ the survival function of T_ν; it decays like $A^{-\nu}$. Section 12.6 computes the mean count for the paper's grids. The paper's Appendix B states this law and gives the expected counts at $A = 12$ and 16; the point is that the Student-t cell of the family ladder is affected by a handful of sites, whose number is predicted, and not by "the family" in some diffuse sense.

12.4 The diagnostics

The paper measures four quantities, all produced by `experiments/e11_grid_validation.py` and the diagnostic branch of `grid_bp_batch`. For each the definition, what it detects, and what it does not.

Column-mass defect. $\max_l \left|\sum_k K(u_k \mid u_l) w_k - 1\right|$: the worst deviation of a column of $\hat{K}$ from unit mass. It combines two effects. In the interior of the grid it is the quadrature error of a one-dimensional integral of the kernel, spectrally small for a smooth kernel (machine precision for the Gaussian and mixture families at the operating grid) and $O(h)$ for the uniform one. In the columns near the boundary it is a *truncation* effect: the column with parent $u_l = \pm A$ is a density centred at $\pm\rho A$, and the part of it beyond $\pm A$ is cut off. As ρ rises, ρA moves towards the edge and the cut-off mass grows, from nothing at $\rho = 0$ to a sizeable fraction of a column at $\rho = 0.95$; this part is flat in N_{g} and falls only with A. A reader checking the paper's Appendix B should therefore read the growth of the defect with correlation as a statement about the outermost columns, not about resolution in the interior, where the smooth families are resolved to machine precision at every correlation. In either case the defect is not an error bar for the posterior: the recursion renormalises every message, so a defect that is the same across a column's neighbours cancels in the normalised beliefs, and the paper reports it as a diagnostic only.

Boundary belief mass. For each site of each chain, the mass of the normalised belief (or of the forward message) in the outermost 2% of the grid on each side, $\sum_{k \le e} b_i(u_k) w_k + \sum_{k > N_{\mathrm{g}} - e} b_i(u_k) w_k$ with $e = \lfloor 0.02\, N_{\mathrm{g}} \rfloor$, maximised over sites and chains. It is computed inside

the recursion, on the very objects the recursion propagates, which is the reason the diagnostic lives in `grid_bp_batch` rather than in a separate tool: an earlier version computed a "boundary mass" from the likelihood table alone, which involved no message and measured nothing the algorithm used. A value near one means that some site's posterior is pinned to the boundary, and the count of such sites is the exceedance count above. A value of a few per cent at $\rho = 0.85$ with no site beyond the domain is a different, benign situation: a site near $|a| \approx 6$ whose posterior at the lowest level has a tail reaching the edge. The paper quotes the frozen configuration's value and the binding cell's value in Appendix B.

Drift. The relative root-mean-square difference between the posterior-mean field computed on a candidate grid and on the finest grid of the ladder, on the same chains and noise draws, pooled over the twelve levels (and separately at the worst level and at the lowest level). Drift is what a reported number would move by under refinement, and it is the quantity a claim is judged against; the other diagnostics say why it is small. Its reference is the finest grid, not the truth, so it is a self-convergence diagnostic: a site pinned on both grids contributes no drift, which is why the binding cell's reference grid in E11 could itself miss the largest site and why E18 and the wide-domain reruns E07b and E07c widen the reference further.

Exceedance count. Not computed by the sweep but by Equation (12.5) from the family's tail; it is the interpretation of the previous three for heavy tails, and it was the audit's re-reading of the E11 cell that turned "all the belief mass outside the domain" into "one or two sites of eight thousand".

12.4.1 How the experiments use them

E11 (the envelope). Five families, four correlations, twelve levels, four seeds, seven candidate grids against a reference at a finer and wider grid. Its findings are the paper's Appendix B: at the frozen configuration the drift is far below every reported effect; across the sweep the worst drift is flat in N_{g} at fixed A and falls only when A grows (truncation, in the Student-t cell at $\rho = 0$); the column-mass defect grows with correlation but does not predict the drift; the uniform family is the one case whose worst drift sits on the coarsest spacing. *E11b* repeats the sweep at chain lengths 64 and 128 to show that the envelope measured at $L = 32$ covers the longer chains of the length study; its prediction is that per-site errors compound at most linearly in L. *E07b and E07c* recompute the family ladder of the paper's Appendix F on wider domains ($A = 12$ and 16), which by Equation (12.5) lowers the expected exceedance count by an order of magnitude and then by another. *E18* recomputes the closure-gap panel of the paper's Figure 2 at $(N_{\mathrm{g}}, A) = (1601, 16)$ and $(3201, 32)$ and shows that the Student-t departure from the e^{-8t} law on the operating grid is the truncated prior's tail in the reference, not the moment condition; on the wide domain the family reaches the predicted constant. *E04* re-scores the validation-selected fitted kernels of the capacity experiment at $N_{\mathrm{g}} = 801$, 1601 and at $A = 10$, with the reference recomputed at each resolution, to bound the movement of the reported estimands under refinement; the paper's Appendix B quotes the bound and notes it is orders of magnitude below the effects.

12.5 Normalisation and the evidence on the grid

The implementation in `src/markov_score/bp_grid.py` keeps messages in the linear domain and rescales them after every step, accumulating the logarithms of the scale factors. The details matter for reading the code and for trusting the returned log-evidence.

The likelihood table. The unary factors are tabulated with the row maximum removed,

$$\log \ell_t(x_i \mid u_k) \text{ is stored as } -\frac{(x_i - e^{-t}u_k)^2}{2\Delta_t} + c_i, \qquad c_i = \max_k \frac{(x_i - e^{-t}u_k)^2}{2\Delta_t},$$

so that the largest entry of every row is exactly 1 after exponentiation and no row can underflow to zero (`log_likelihood_matrix` in `src/markov_score/noising.py`; `grid_bp_batch` recomputes the same table internally). The Gaussian constant $-\frac{1}{2}\log(2\pi\Delta_t)$ is dropped at the same time. Both removals are harmless for the beliefs, which are normalised, and both are recorded: the sum of the row shifts and L copies of the constant are added back to the evidence at the end.

The sweeps. With $\hat{K}$ the matrix $K(u_k \mid u_l)$ (weights applied on the incoming side) the forward step is $\overrightarrow{m}_{i+1} = \hat{K}\,(\overrightarrow{m}_i \circ \ell_i \circ w)/c_{i+1}$, where $c_{i+1} = \sum_k w_k[\hat{K}(\overrightarrow{m}_i \circ \ell_i \circ w)]_k$ is the trapezoidal mass of the unnormalised result; the backward step is the same with $\hat{K}^\top$ and its own constants, which are not needed for the evidence. Each step is a matrix–vector product, or, for a batch of B chains, a matrix–matrix product; the batch is processed in chunks of at most 256 chains so that the $(L \times B \times N_\mathrm{g})$ temporaries fit in memory, with results identical to round-off across chunk sizes (`tests/test_bp_grid_chunk.py`). A guard raises an error if any normaliser is non-finite or non-positive: a message that has lost all its mass is a failure to report, not to hide.

The evidence. Putting the pieces together,

$$\log \hat{Z}(\mathbf{x}) = \sum_{i=2}^{L} \log c_i + \log \sum_k w_k\, \overrightarrow{m}_L(u_k)\, \ell_L(u_k) + \sum_{i=1}^{L} (-c_i^{\mathrm{shift}}) - \frac{L}{2}\log(2\pi\Delta_t), \tag{12.6}$$

which is the log-normaliser of Equation (12.2) to floating point. Its correctness is checked in three independent ways: against the closed-form Gaussian marginal likelihood; against an enumeration of the discretised posterior; and against a Monte Carlo estimate on the Laplace chain, where no closed form exists. The first and the third are tests in `tests/test_em_bp.py`; the second is the file `tests/test_fisher_bruteforce.py`. Its role in the paper is twofold: it is the objective that expectation–maximisation increases (Chapter 15), and, summed over a held-out one-view bundle and divided by the number of transitions, it is the oracle-free selection rule of E20.

Diagnostic outputs. When called with `boundary_frac`, `grid_bp_batch` also returns a dictionary with the maximum and mean boundary mass of the forward messages and of the beliefs, and the number of edge cells used. These are the numbers behind the paper's Appendix B and behind the frozen configuration's docstring in `experiments/frozen_config.py`, which is careful to say that a maximum over sampled sites is a statistic and not a bound.

12.6 Worked examples

The Gaussian, spectral accuracy. Integrate the standard Gaussian density on $[-A, A]$ with the trapezoidal rule. At $A = 8$ and $N_\mathrm{g} = 17$, so $h = 1$, the nine non-negligible grid values are $\phi(0) = 0.398942$, $\phi(\pm1) = 0.241971$, $\phi(\pm2) = 0.053991$, $\phi(\pm3) = 0.004432$, $\phi(\pm4) = 0.000134$, and the sum $h\sum_k \phi(k)$ (the end weights are irrelevant at this precision) is 1.0000000054: an error of 5×10^{-9}, which is $2e^{-2\pi^2}$ to two digits, as Equation (12.4) predicts. Halving the spacing to $h = 0.5$ gives an error below 10^{-15}. By contrast, at $h = 2$ the sum is 1.0144: the Poisson term $2e^{-2\pi^2/4} = 0.0144$ again. The convergence is not h^2; it is exponential in $1/h^2$.

The step, first order. Integrate the uniform density $\frac{1}{2}\mathbf{1}\{|a| \le 1\}$ on $[-2, 2]$ with $N_g = 9$ points, $h = 0.5$. The grid values at $a = \pm 1$ are $\frac{1}{2}$ (the jumps sit on grid points), at $|a| < 1$ they are $\frac{1}{2}$, elsewhere zero; the trapezoidal sum is $h \cdot \frac{1}{2} \cdot 5 = 1.25$, an error of $0.25 = h/2$. At $h = 0.25$ the error is $0.125 = h/2$; at $h = 0.125$ it is 0.0625. The error halves when h halves: first order, with constant $J/2$ per jump, exactly as Section 12.2.3 says. Shifting the grid so that the jumps fall between grid points changes the sign and the size of the error but not its order.

The column-mass defect on a coarse grid. Take the Gaussian kernel $K(a' \mid a) = \mathcal{N}(\rho a, q)$ with $\rho = 0.5$, $q = 0.75$ on the five-point grid $\{-2, -1, 0, 1, 2\}$, $h = 1$. The column with parent $u_l = 0$ is the density $\mathcal{N}(0, 0.75)$ sampled at the five points, (0.0320, 0.2365, 0.4607, 0.2365, 0.0320), and its trapezoidal mass with end weights $\frac{1}{2}$ is 0.9657: a defect of 0.034, almost all of it the tail of a $\mathcal{N}(0, 0.75)$ beyond ± 2, which is 0.021, plus the quadrature error of $h = 1$ on a Gaussian of standard deviation 0.87. The column with parent $u_l = 2$ is centred at 1 and loses the whole of its tail beyond 2; its mass is 0.848, a defect of 0.152 that no refinement of h removes and that a wider domain does. Widening to $[-4, 4]$ at the same spacing brings the worst column to 0.983. This is the mechanism behind the growth of the paper's defect with correlation: the extreme column's centre ρA approaches the edge.

The exceedance count. For the variance-matched Student-t_5 innovation, $s^2 = 0.6$ and $\mathbb{P}(|a| > A) = 2\bar{F}_5(A/\sqrt{0.6})$. At $A = 8$ the argument is 10.33 and the two-sided tail is 1.46×10^{-4}; over $ML = 256 \times 32 = 8192$ sites the expected count is 1.2, so the probability of no exceedance in a bundle is $e^{-1.2} \approx 0.30$ and of two or more is 0.34. At $A = 12$ the expected count is 0.17 and at $A = 16$ it is 0.04. These are the analytic figures the paper's Appendix B quotes next to the observed counts of the E11 cell; a reader can reproduce them from the t survival function alone.

Where it is used. The paper, Section 2 ("messages are functions on $\mathbb{R}$; we carry them on a grid ... exact for the discretised factor model and controlled for the continuous one") and Appendix B (the three levels of exactness, the validation at the frozen configuration, the envelope, the refinement, the floors); Appendix F (the Student-t cell excluded pending a wider domain); Appendix E (E18, the wide-domain closure panel). Experiments E11, E11b, E07b, E07c, E18 and E04. Code: the grid and the two sweeps live in `src/markov_score/bp_grid.py` (`make_grid`, `grid_bp`, and `grid_bp_batch` with its `chunk` and `boundary_frac` arguments); the row-shifted likelihood table and `likelihood_resolution_ok` in `src/markov_score/noising.py`; the drivers are `experiments/e11_grid_validation.py`, `experiments/e18_closure_wide.py`, `experiments/e04_refinement.py` and `experiments/e07_phase_diagram.py`, with the macros emitted by `tools/make_grid_envelope.py`. Tests: for the grid itself `tests/test_grid_convergence.py` and `tests/test_bp_grid_chunk.py`; for the evidence, `tests/test_em_bp.py` and `tests/test_fisher_bruteforce.py`.

References. P. J. Davis and P. Rabinowitz (1984), *Methods of Numerical Integration*, 2nd ed., Academic Press, Chapters 2 and 4. L. N. Trefethen and J. A. C. Weideman (2014), "The exponentially convergent trapezoidal rule", *SIAM Review* 56(3). G. Kitagawa (1987), "Non-Gaussian state-space modeling of nonstationary time series", *Journal of the American Statistical Association* 82(400). L. Le Cam (1960), "An approximation theorem for the Poisson binomial distribution", *Pacific Journal of Mathematics* 10(4). N. J. Higham (2002), *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, Chapter 2 (floating-point arithmetic and rescaling).

Chapter 13

Gaussian closure and the LMMSE floor

The paper's Proposition 1 is short: run the sum-product recursion of Chapter 11, but after every transition integral replace the message by the Gaussian with the same mean and variance; then, whatever the innovation law, the posterior means you obtain are those of the *Gaussian* chain with the same correlation and innovation variance, and those are the linear minimum-mean-square estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. The proposition turns an approximation that practitioners make for convenience (Gaussian message passing, assumed-density filtering, the Kalman smoother applied to a non-Gaussian model) into an object with a name and a closed form, and it gives the paper an apparatus check that is used in every experiment: a Gaussian-innovation kernel fitted by maximum likelihood must land on that closed form. This chapter defines the projection, derives the linear estimator from first principles, shows that the Gaussian chain's forward–backward is the Kalman smoother, proves the proposition in full, derives the two consequences the paper draws from it, and explains what the gap between the true score and the Gaussian one measures.

13.1 The Gaussian moment projection

Definition 13.1 (Gaussian moment projection)**.** For a non-negative integrable function g on $\mathbb{R}$ with finite second moment, let $\bar{g} = g/\int g$ be its normalisation, $m = \int a\,\bar{g}(a)\,\mathrm{d}a$ its mean and $v = \int (a-m)^2\bar{g}(a)\,\mathrm{d}a$ its variance. The **Gaussian moment projection** of g is $\Pi_{\mathrm{G}}g = \mathcal{N}(m, v)$, the Gaussian density with the same first two moments as $\bar{g}$.

The operator is defined on shapes: the normalisation of g is discarded, which is harmless because messages are defined up to a constant. Two facts make Π_{G} the natural projection rather than one choice among many. First, among all Gaussians, $\Pi_{\mathrm{G}}g$ is the one closest to $\bar{g}$ in Kullback–Leibler divergence $D_{\mathrm{KL}}(\bar{g}\,\|\,\cdot)$: writing $\mathcal{N}(m', v')$ for a candidate, $-\int \bar{g}\log\mathcal{N}(m', v') = \frac{1}{2}\log(2\pi v') + (v + (m-m')^2)/2v'$ is minimised at $m' = m$ and $v' = v$, since the term is a function of the first two moments of $\bar{g}$ only. Second, and this is the property the proof of Proposition 1 uses, a Gaussian is determined by its first two moments, so Π_{G} forgets everything about g except (m, v).

Moment-projected sum-product is the recursion of Chapter 11 with Π_{G} applied to each message after every transition integral: in the notation of that chapter,

$$\begin{aligned}\overrightarrow{m}_i^{\mathrm{p}} &= \Pi_{\mathrm{G}}\Big[\int K(\cdot \mid a')\ell_t(x_{i-1} \mid a')\overrightarrow{m}_{i-1}^{\mathrm{p}}(a')\,\mathrm{d}a'\Big],\\ \overleftarrow{m}_i^{\mathrm{p}} &= \Pi_{\mathrm{G}}\Big[\int K(a'' \mid \cdot)\ell_t(x_{i+1} \mid a'')\overleftarrow{m}_{i+1}^{\mathrm{p}}(a'')\,\mathrm{d}a''\Big],\end{aligned} \tag{13.1}$$

with the same starting values as before. In the literature this is Gaussian assumed-density filtering, or expectation propagation with a single Gaussian per message and one forward and

one backward pass. The paper's phrase is "single-Gaussian message closure". Note where the projection is applied: after the transition integral, and not after the multiplication by the unary factor, which would be a different algorithm. For the Gaussian likelihood of the channel the difference is immaterial, because a Gaussian message times a Gaussian likelihood is already Gaussian; the paper fixes the order once so that the statement is unambiguous.

13.2 The linear minimum-mean-square estimator

13.2.1 Derivation

Let $(\mathbf{a}, \mathbf{x})$ be a pair of random vectors with finite second moments, means zero, and second-moment matrices $\mathbf{\Sigma}_{aa} = \mathbb{E}[\mathbf{a}\mathbf{a}^\top]$, $\mathbf{\Sigma}_{ax} = \mathbb{E}[\mathbf{a}\mathbf{x}^\top]$, $\mathbf{\Sigma}_{xx} = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$, the last invertible. A **linear estimator** of $\mathbf{a}$ from $\mathbf{x}$ is $\hat{\mathbf{a}} = G\mathbf{x}$ for a fixed matrix G, and its **mean-square error** is $R(G) = \mathbb{E}\|\mathbf{a} - G\mathbf{x}\|^2 = \operatorname{tr}\mathbf{\Sigma}_{aa} - 2\operatorname{tr}(G\mathbf{\Sigma}_{ax}^\top) + \operatorname{tr}(G\mathbf{\Sigma}_{xx}G^\top)$, by expanding the square and using linearity of expectation. The function R is a quadratic in the entries of G with positive semidefinite Hessian (it is $\operatorname{tr}(G\mathbf{\Sigma}_{xx}G^\top)$ plus lower-order terms, and $\mathbf{\Sigma}_{xx} \succ 0$), so its stationary point is its minimum. Differentiating, $\partial R/\partial G = -2\mathbf{\Sigma}_{ax} + 2G\mathbf{\Sigma}_{xx}$, which vanishes at

$$G^* = \mathbf{\Sigma}_{ax}\mathbf{\Sigma}_{xx}^{-1}, \qquad \hat{\mathbf{a}}_{\mathrm{LMMSE}} = \mathbf{\Sigma}_{ax}\mathbf{\Sigma}_{xx}^{-1}\mathbf{x}, \qquad R(G^*) = \operatorname{tr}(\mathbf{\Sigma}_{aa} - \mathbf{\Sigma}_{ax}\mathbf{\Sigma}_{xx}^{-1}\mathbf{\Sigma}_{ax}^\top). \tag{13.2}$$

This is the **linear minimum-mean-square estimator** (LMMSE), also called the Wiener or normal-equation estimator; the matrix G^* solves the **normal equations** $G\mathbf{\Sigma}_{xx} = \mathbf{\Sigma}_{ax}$. The error $\mathbf{a} - G^*\mathbf{x}$ is uncorrelated with $\mathbf{x}$ (the **orthogonality principle**: $\mathbb{E}[(\mathbf{a} - G^*\mathbf{x})\mathbf{x}^\top] = \mathbf{\Sigma}_{ax} - G^*\mathbf{\Sigma}_{xx} = 0$), which is another way of saying that no linear correction can lower the error.

Apply this to the channel $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ with $\mathbf{a} \sim P_0$ of covariance $\mathbf{\Sigma}_0$ and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ independent of $\mathbf{a}$. Then $\mathbf{\Sigma}_{ax} = \mathbb{E}[\mathbf{a}(e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z})^\top] = e^{-t}\mathbf{\Sigma}_0$ and $\mathbf{\Sigma}_{xx} = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I} = \mathbf{\Sigma}_t$, so

$$\hat{\mathbf{a}}_{\mathrm{LMMSE}} = e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}, \qquad \mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I}. \tag{13.3}$$

The derivation used only the first two moments of P_0: every law with covariance $\mathbf{\Sigma}_0$ has the same LMMSE, which is why the paper calls the Gaussian with covariance $\mathbf{\Sigma}_0$ the **covariance twin** of P_0 and Equation (13.3) "the LMMSE of the covariance twin".

13.2.2 When it is the MMSE, and when it is only the best linear map

The **minimum-mean-square estimator** without the word "linear" is the conditional mean $\mathbb{E}[\mathbf{a} \mid \mathbf{x}]$, the minimiser of $\mathbb{E}\|\mathbf{a} - f(\mathbf{x})\|^2$ over all measurable f (the standard argument: for any f, $\mathbb{E}\|\mathbf{a} - f\|^2 = \mathbb{E}\|\mathbf{a} - \mathbb{E}[\mathbf{a} \mid \mathbf{x}]\|^2 + \mathbb{E}\|\mathbb{E}[\mathbf{a} \mid \mathbf{x}] - f\|^2$, the cross term vanishing by the tower property). If $(\mathbf{a}, \mathbf{x})$ is jointly Gaussian, the conditional mean is linear, $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] = \mathbf{\Sigma}_{ax}\mathbf{\Sigma}_{xx}^{-1}\mathbf{x}$, so the LMMSE *is* the MMSE. This is the case of the Gaussian chain: $\mathbf{a}$ Gaussian and $\mathbf{z}$ Gaussian independent make $(\mathbf{a}, \mathbf{x})$ jointly Gaussian. For any other P_0 the conditional mean is a nonlinear function of $\mathbf{x}$, the LMMSE is only the best among linear maps, and the difference $\mathbb{E}\|\mathbb{E}[\mathbf{a} \mid \mathbf{x}] - \hat{\mathbf{a}}_{\mathrm{LMMSE}}\|^2 = \mathbb{E}\|\mathbf{a} - \hat{\mathbf{a}}_{\mathrm{LMMSE}}\|^2 - \mathbb{E}\|\mathbf{a} - \mathbb{E}[\mathbf{a} \mid \mathbf{x}]\|^2$ is the price of linearity. Through Tweedie's identity, the score of the covariance twin is

$$S_{\mathrm{G}}(\mathbf{x}, t) = -\frac{\mathbf{x} - e^{-t}\hat{\mathbf{a}}_{\mathrm{LMMSE}}}{\Delta_t} = -\frac{1}{\Delta_t}(\mathbf{I} - e^{-2t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1})\mathbf{x} = -\mathbf{\Sigma}_t^{-1}\mathbf{x}, \tag{13.4}$$

where the last step uses $\mathbf{I} - e^{-2t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1} = (\mathbf{\Sigma}_t - e^{-2t}\mathbf{\Sigma}_0)\mathbf{\Sigma}_t^{-1} = \Delta_t\mathbf{\Sigma}_t^{-1}$. So $S_{\mathrm{G}} = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$ is at once the score of $\mathcal{N}(0, \mathbf{\Sigma}_t)$, which is the noised law of the Gaussian twin, and the Tweedie image of the LMMSE. The paper's Section 2 states both forms.

13.3 The Gaussian chain and the Kalman smoother

For the Gaussian chain, $\mu = \mathcal{N}(0,1)$, $K(a' \mid a) = \mathcal{N}(\rho a, q)$, $\ell_t(x \mid a) \propto \mathcal{N}(x, \Delta_t)$ evaluated at $e^{-t}a$, every message of Chapter 11 is a Gaussian in a and is carried by two numbers. It is convenient to use the **information form**: a Gaussian shape $\propto \exp(-\lambda a^2/2 + ha)$ has precision λ, information h, mean h/λ and variance $1/\lambda$; products of shapes add (λ, h); a flat message is $\lambda = 0$, $h = 0$. The unary factor, as a function of a, has $\lambda_{\ell_t} = e^{-2t}/\Delta_t$ and $h_{\ell_t} = e^{-t}x_i/\Delta_t$.

Forward step. Let $\overrightarrow{m}_{i-1} = \mathcal{N}(m, v)$. Multiplying by $\ell_t(x_{i-1} \mid \cdot)$ gives the **tilted** Gaussian with $\lambda' = 1/v + \lambda_{\ell_t}$, $h' = m/v + h_{\ell_t}$, mean $m' = h'/\lambda'$ and variance $v' = 1/\lambda'$. The transition integral is $\int \mathcal{N}(\rho a', q)\,(a)\,\mathcal{N}(m', v')\,(a')\,\mathrm{d}a'$, the density of $\rho a' + \varepsilon$ with $a' \sim \mathcal{N}(m', v')$ and $\varepsilon \sim \mathcal{N}(0, q)$ independent, which is $\mathcal{N}(\rho m', \rho^2 v' + q)$. Hence

$$(m, v) \xrightarrow{\text{tilt}} (m', v') \xrightarrow{\text{transition}} (\rho m',\ \rho^2 v' + q). \tag{13.5}$$

This is the Kalman filter: the tilt is the measurement update and the transition is the prediction step.

Backward step. Let $\overleftarrow{m}_{i+1} = \mathcal{N}(m, v)$ and tilt it by $\ell_t(x_{i+1} \mid \cdot)$ to $\mathcal{N}(m', v')$. The backward integral is $\int \mathcal{N}(\rho a, q)\,(a'')\,\mathcal{N}(m', v')\,(a'')\,\mathrm{d}a''$, a function of a: it is $h(\rho a)$ with $h(u) = \int \mathcal{N}(u, q)\,(a'')\mathcal{N}(m', v')\,(a'')\,\mathrm{d}a''$ the density of $a'' - \varepsilon$ at u, which is $\mathcal{N}(m', v' + q)\,(u)$. As a function of a, $\mathcal{N}(m', v' + q)\,(\rho a) \propto \exp[-(\rho a - m')^2/2(v' + q)]$ is a Gaussian shape in a with mean m'/ρ and variance $(v' + q)/\rho^2$, provided $\rho \neq 0$:

$$(m, v) \xrightarrow{\text{tilt}} (m', v') \xrightarrow{\text{transition}^{-1}} (m'/\rho,\ (v' + q)/\rho^2). \tag{13.6}$$

At $\rho = 0$ the backward message is constant in a, has no normalised moments, and the map is undefined; the recursion is still fine (a flat message is $\lambda = 0$), but the moment description is not. This is the origin of the condition $\rho \neq 0$ in the proposition. The terminal message $\overleftarrow{m}_L \equiv 1$ is flat and is never projected; the backward induction starts from the tilt by $\ell_t(x_L \mid \cdot)$.

Beliefs. At site i the belief has precision $\lambda_i^{\overrightarrow{m}} + \lambda_{\ell_t} + \lambda_i^{\overleftarrow{m}}$ and information $h_i^{\overrightarrow{m}} + h_{\ell_t} + h_i^{\overleftarrow{m}}$; its mean is the posterior mean and its variance the posterior variance. Combining a forward filter with a backward pass in this way is the Rauch–Tung–Striebel smoother written in information form, and the code is `gaussian_chain_bp` in `src/markov_score/bp_gaussian.py`, which is tested against the closed form $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$ to machine precision (`tests/test_gaussian_bp_equivalence.py`).

13.4 Proposition 1 and its proof

Proposition 13.2 (Gaussian closure computes the LMMSE of the covariance twin)**.** *Let the chain have $a_1 \sim \mathcal{N}(0,1)$, $a_i = \rho a_{i-1} + \varepsilon_i$ with $\rho \neq 0$ and innovations ε_i i.i.d. from any law φ of mean zero and variance q. Then moment-projected sum-product Equation* (13.1) *on the chain posterior returns, at every site, the belief of the covariance-matched Gaussian chain, and hence the posterior means $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$, the same for every such φ.*

The proof needs one lemma about convolutions.

Lemma 13.3 (First two moments of a convolution)**.** *Let U and V be independent with finite variances, means m_U, m_V and variances v_U, v_V. Then $\rho U + V$ has mean $\rho m_U + m_V$ and variance $\rho^2 v_U + v_V$, and $U - V$ has mean $m_U - m_V$ and variance $v_U + v_V$.*

Proof. Linearity of expectation gives the means. For the variances, $\operatorname{Var}(\rho U + V) = \rho^2 \operatorname{Var} U + \operatorname{Var} V + 2\rho \operatorname{Cov}(U, V)$ and the covariance vanishes by independence; likewise for $U - V$. No other property of the laws of U and V enters. □

Proof of Proposition 13.2. Write $\overrightarrow{m}_i^{\mathrm{p}}$ and $\overleftarrow{m}_i^{\mathrm{p}}$ for the projected messages and $\overrightarrow{m}_i^{\mathrm{G}}$, $\overleftarrow{m}_i^{\mathrm{G}}$ for the exact messages of the Gaussian chain with the same (ρ, q), which are Gaussians with the moments given by Equations (13.5) and (13.6). We prove by induction along the chain that $\overrightarrow{m}_i^{\mathrm{p}} = \overrightarrow{m}_i^{\mathrm{G}}$ and $\overleftarrow{m}_i^{\mathrm{p}} = \overleftarrow{m}_i^{\mathrm{G}}$ for every i.

Forward, base case. $\overrightarrow{m}_1 = \mu = \mathcal{N}(0, 1)$ in both chains; it is Gaussian, so projecting it changes nothing.

Forward, inductive step. Suppose $\overrightarrow{m}_{i-1}^{\mathrm{p}} = \overrightarrow{m}_{i-1}^{\mathrm{G}} = \mathcal{N}(m, v)$. Multiplying by the Gaussian unary factor gives the tilted Gaussian $\mathcal{N}(m', v')$, identical in the two chains. The transition integral in the non-Gaussian chain, $\int \varphi(a - \rho a')\mathcal{N}(m', v')\,(a')\,\mathrm{d}a'$, is the density of $\rho U + V$ with $U \sim \mathcal{N}(m', v')$ and $V \sim \varphi$ independent; by Lemma 13.3 its mean is $\rho m'$ and its variance $\rho^2 v' + q$, whatever φ is beyond its first two moments. The projection therefore returns $\mathcal{N}(\rho m', \rho^2 v' + q)$, which by Equation (13.5) is $\overrightarrow{m}_i^{\mathrm{G}}$. The non-Gaussian shape of the integral before projection (a mixture, a Laplace-smoothed Gaussian, whatever it may be) is discarded by Π_{G} and never influences a later step.

Backward, base case. $\overleftarrow{m}_L \equiv 1$ in both chains; it is not projected. The first backward integral starts from the tilt of the flat message by $\ell_t(x_L \mid \cdot)$, which is the Gaussian $\mathcal{N}(x_L/e^{-t}, \Delta_t/e^{-2t})$ in both chains.

Backward, inductive step. Suppose $\overleftarrow{m}_{i+1}^{\mathrm{p}} = \overleftarrow{m}_{i+1}^{\mathrm{G}}$, and let the tilt by $\ell_t(x_{i+1} \mid \cdot)$ be $\mathcal{N}(m', v')$. The backward integral in the non-Gaussian chain is $\int \varphi(a'' - \rho a)\mathcal{N}(m', v')\,(a'')\,\mathrm{d}a'' = h(\rho a)$ with $h(u) = \int \varphi(a'' - u)\mathcal{N}(m', v')\,(a'')\,\mathrm{d}a''$, the density at u of $U - V$ with $U \sim \mathcal{N}(m', v')$, $V \sim \varphi$; by Lemma 13.3 h integrates to one and has mean m' and variance $v' + q$. As a function of a, $h(\rho a)$ has mass $1/|\rho|$, normalised mean m'/ρ and normalised variance $(v' + q)/\rho^2$ (substitute $u = \rho a$ in the moment integrals). Here $\rho \neq 0$ is used: at $\rho = 0$ the function $h(0)$ is constant in a and has no normalised moments. The projection returns $\mathcal{N}(m'/\rho, (v' + q)/\rho^2)$, which by Equation (13.6) is $\overleftarrow{m}_i^{\mathrm{G}}$.

Beliefs. At every site the projected belief is the product of three Gaussians, $\overrightarrow{m}_i^{\mathrm{G}} \ell_t \overleftarrow{m}_i^{\mathrm{G}}$, the belief of the Gaussian chain; its mean is the Gaussian chain's posterior mean, which is the conditional mean of a jointly Gaussian pair and hence, by Equation (13.3), the LMMSE $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. The recursion read φ only through $(0, q)$, so the result is the same for every such law. Nothing assumed stationarity: the induction carries site-dependent (m, v), and $\mathbf{\Sigma}_0$ is whatever covariance the chain has (with $a_1 \sim \mathcal{N}(0, 1)$ and $q = 1 - \rho^2$ it is $\rho^{|i-j|}$, but the proposition does not need that). □

Two remarks on what is and is not checked by machine. The paper reports that the two moment maps as integrals, the covariance identities $\mathbf{\Sigma}_{ax} = e^{-t}\mathbf{\Sigma}_0$, $\mathbf{\Sigma}_{xx} = \mathbf{\Sigma}_t$, and the optimality of the normal-equation gain in arbitrary dimension are verified in Lean 4 (`lean/README.md`), for a *Gaussian* innovation; the projection step and the moments-only clause of Lemma 13.3, which are the content of the proposition, are prose. That is an accurate description, and the reader who wants to check the load-bearing step should check the two applications of Lemma 13.3 above. Numerically, the proposition is checked in `tests/test_gaussian_bp_equivalence.py` by running a grid-mediated projection on a Laplace chain and comparing with the information-form Gaussian recursion; the two agree to the grid's accuracy at moderate t. The same file records an instability of the grid-mediated projection at weakly informative t (a nearly flat outgoing message whose maximiser lies outside the domain is dragged to the boundary by the truncation before projection), which is why the closed-form information recursion, and not the grid projection, is what the paper uses for the LMMSE arm.

13.5 Two consequences

The paper draws two corollaries from the proposition "as checks". They are different facts and each needs its own line.

Corollary 13.4 (Every Gaussian-kernel sum-product is a linear map)**.** *Sum-product with a Gaussian transition kernel $\mathcal{N}(\rho' a, q')$ (any ρ', q', not necessarily the truth) on the chain posterior returns posterior means that are a fixed linear map $G(\rho', q')\mathbf{x}$. Consequently its mean-square error is at least that of the LMMSE with the true covariance, and its score error is at least the LMMSE floor $\mathbb{E}\|S - S_{\mathrm{G}}\|^2$.*

Proof. With a Gaussian kernel every message is Gaussian without any projection (Section 13.3), the recursion is the Kalman smoother of the Gaussian chain with parameters (ρ', q'), and its posterior means are $\mathbf{\Sigma}'_{ax}\mathbf{\Sigma}'^{-1}_{xx}\mathbf{x}$ for the covariances of that chain: a linear map. By Equation (13.2) the LMMSE with the true second moments minimises the mean-square error over all linear maps, so it is a floor for this one. For the score, Tweedie's identity gives $\hat{S} - S = (e^{-t}/\Delta_t)(\hat{\mathbf{a}} - \langle \mathbf{a} \rangle_{x,t})$ for any estimator $\hat{\mathbf{a}}$ of the posterior mean, so the squared score error is (e^{-2t}/Δ_t^2) times the excess mean-square error over the MMSE, and the ordering of estimators is the same. □

This is the sense in which the LMMSE is a **floor for every second-order estimator**: a Gaussian-innovation kernel however fitted, a linear denoiser however trained, and the Kalman smoother with any parameters all compute linear maps of $\mathbf{x}$, and none can beat $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. The paper uses the floor as a reference line in every figure and table.

Corollary 13.5 (The Gaussian quasi-likelihood sees only the covariance)**.** *Fit the Gaussian chain with parameters (ρ', q') to noised data by maximising the Gaussian marginal log-likelihood. That objective depends on the data only through the sample covariance $\hat{\mathbf{\Sigma}} = \frac{1}{M}\sum_\mu \mathbf{x}^\mu \mathbf{x}^{\mu\top}$ (per level), whose expectation is $e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$ whatever the innovation law; the population maximiser is the true (ρ, q), so the fitted Gaussian kernel is consistent for them and its sum-product lands on the LMMSE floor.*

Proof. The Gaussian marginal log-likelihood of one sequence at level t is

$$-\tfrac{1}{2}\log\det(2\pi\mathbf{\Sigma}'_t) - \tfrac{1}{2}\mathbf{x}^\top\mathbf{\Sigma}'^{-1}_t\mathbf{x}, \qquad \mathbf{\Sigma}'_t = e^{-2t}\mathbf{\Sigma}'_0 + \Delta_t\mathbf{I};$$

summed over sequences the data enter only through $\sum_\mu \mathbf{x}^{\mu\top}\mathbf{\Sigma}'^{-1}_t\mathbf{x}^\mu = M\,\mathrm{tr}(\mathbf{\Sigma}'^{-1}_t\hat{\mathbf{\Sigma}})$. The expectation of the per-sequence log-likelihood is $-\frac{1}{2}\log\det(2\pi\mathbf{\Sigma}'_t) - \frac{1}{2}\,\mathrm{tr}(\mathbf{\Sigma}'^{-1}_t\mathbf{\Sigma}_t)$, which as a function of $\mathbf{\Sigma}'_t$ is maximised uniquely at $\mathbf{\Sigma}'_t = \mathbf{\Sigma}_t$ (the Gaussian cross-entropy is minimised at the matching covariance); in the two-parameter family $\mathbf{\Sigma}_t$ determines (ρ, q), since $\mathbf{\Sigma}_0 = (\mathbf{\Sigma}_t - \Delta_t\mathbf{I})e^{2t}$ has entries $\rho^{|i-j|}$ scaled by the variances determined by q. Consistency of the maximiser follows from the law of large numbers for $\hat{\mathbf{\Sigma}}$ and the continuity of the objective. □

The distinction between the two corollaries is the distinction between the proposition's two uses in the paper. Corollary 13.4 says that the $C = 1$ arm *cannot go below* the floor; Corollary 13.5 says that it *lands on* it, which is what the capacity table and the regime sweep verify at every cell, and what makes those cells an apparatus check of the whole pipeline (grid, sweeps, EM, scoring).

13.6 The price of second-order closure

Define the **closure gap** at level t as $\mathbb{E}_{P_t}\|S - S_{\mathrm{G}}\|^2$, the mean-square distance between the true score and that of the covariance twin, and the **relative score error** of any estimator $\hat{S}$ as $\left(\sum_t\sum_{\mu,i}(\hat{S} - S)_i^2 / \sum_t\sum_{\mu,i} S_i^2\right)^{1/2}$, pooled over the schedule (the paper's estimand, Appendix C). By Proposition 13.2 and Corollary 13.4 the closure gap is three things at once: the error of Gaussian message passing on the non-Gaussian chain; the error of the best linear denoiser; and the error of a Gaussian-innovation kernel fitted with infinite data. That is why the paper calls it "the price of second-order closure" and "what any estimator that carries only second-order structure must leave on the table". Its size is the subject of the paper's order-by-order decay

theorem (its Section 3): to leading order $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$, the degree-four Hermite term the twin lacks, and the compendium's chapter on the attenuation theorems derives it. Pooled over the twelve-level schedule and expressed as a relative error it is the number the paper's Appendix D lists in the LMMSE column, per innovation family.

The other reference point in that table is the **isotropic score** $-\mathbf{x}$, the score of $\mathcal{N}(0, \mathbf{I})$, which is the noised law of independent unit-variance sites. It is the estimator that knows the marginal variance and nothing else. Through Tweedie, $-\mathbf{x} = -(\mathbf{x} - e^{-t}\hat{\mathbf{a}})/\Delta_t$ gives $\hat{\mathbf{a}} = (1 - \Delta_t)\mathbf{x}/e^{-t} = e^{-t}\mathbf{x}$: the isotropic denoiser shrinks the observation by e^{-t}, which is the LMMSE for $\boldsymbol{\Sigma}_0 = \mathbf{I}$, that is, for a chain with $\rho = 0$. Its relative error rises with ρ because the covariance itself is then most of what there is to learn, which is the paper's reading of the figure in its Appendix F on the second-order floor across correlation. The code for both references is `lmmse_posterior_mean`, `lmmse_score` and `isotropic_score` in `src/markov_score/theory.py`; the isotropic posterior mean is written there as $(\mathbf{x} - \Delta_t\mathbf{x})/e^{-t}$, the same thing.

13.7 A worked example: two sites

Take $L = 2$, $\rho = \frac{1}{2}$, $q = \frac{3}{4}$ (so that $\operatorname{Var} a_2 = \rho^2 + q = 1$), and a level with $e^{-t} = 0.8$, $\Delta_t = 0.36$. Observe $\mathbf{x} = (1, 0)$.

Closed form. $\boldsymbol{\Sigma}_0 = \left(\begin{smallmatrix} 1 & 0.5 \\ 0.5 & 1 \end{smallmatrix}\right)$, $\boldsymbol{\Sigma}_t = 0.64\boldsymbol{\Sigma}_0 + 0.36\mathbf{I} = \left(\begin{smallmatrix} 1 & 0.32 \\ 0.32 & 1 \end{smallmatrix}\right)$ with determinant $1 - 0.32^2 = 0.8976$, so $\boldsymbol{\Sigma}_t^{-1} = \frac{1}{0.8976}\left(\begin{smallmatrix} 1 & -0.32 \\ -0.32 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1.1141 & -0.3565 \\ -0.3565 & 1.1141 \end{smallmatrix}\right)$. Then $\boldsymbol{\Sigma}_t^{-1}\mathbf{x} = (1.1141, -0.3565)$, $\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\mathbf{x} = (1.1141 - 0.1783, 0.5570 - 0.3565) = (0.9358, 0.2005)$, and the LMMSE is $0.8 \times (0.9358, 0.2005) = (0.7487, 0.1604)$. The twin's score is $-\boldsymbol{\Sigma}_t^{-1}\mathbf{x} = (-1.1141, 0.3565)$, and Tweedie's identity from the LMMSE gives the same: $-(1 - 0.8 \cdot 0.7487)/0.36 = -1.1141$ and $-(0 - 0.8 \cdot 0.1604)/0.36 = 0.3565$.

The two sweeps. The unary factor has $\lambda_{\ell_t} = 0.64/0.36 = 1.7778$ and $h_{\ell_t} = 0.8x_i/0.36$, that is 2.2222 at site 1 and 0 at site 2. Forward: the prior at site 1 is $(\lambda, h) = (1, 0)$; tilting gives $\lambda' = 2.7778$, $m' = 2.2222/2.7778 = 0.8$, $v' = 0.36$; the transition maps this to $m = 0.5 \cdot 0.8 = 0.4$, $v = 0.25 \cdot 0.36 + 0.75 = 0.84$: $\overrightarrow{m}_2 = \mathcal{N}(0.4, 0.84)$. Backward: the terminal message is flat; tilting it at site 2 gives $m' = 0$, $v' = 1/1.7778 = 0.5625$; the inverse transition maps this to $m = 0/0.5 = 0$, $v = (0.5625 + 0.75)/0.25 = 5.25$: $\overleftarrow{m}_1 = \mathcal{N}(0, 5.25)$. Belief at site 1: precision $1 + 1.7778 + 1/5.25 = 2.9683$, information $0 + 2.2222 + 0 = 2.2222$, mean $2.2222/2.9683 = 0.7487$. Belief at site 2: precision $1/0.84 + 1.7778 = 2.9683$, information $0.4/0.84 + 0 = 0.4762$, mean 0.1604. Both agree with the closed form, and the common posterior variance $1/2.9683 = 0.3369$ is the diagonal of $\boldsymbol{\Sigma}_0 - e^{-2t}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\boldsymbol{\Sigma}_0$.

A non-Gaussian innovation. Replace φ by the two-point law $\varepsilon = \pm\sqrt{0.75}$ with probability $\frac{1}{2}$ each, which has the same mean and variance. The exact forward message at site 2 is now the mixture $\frac{1}{2}\mathcal{N}(0.4 + 0.866, 0.09) + \frac{1}{2}\mathcal{N}(0.4 - 0.866, 0.09)$, a bimodal shape whose mean is 0.4 and whose variance is $0.09 + 0.75 = 0.84$ (Lemma 13.3). Its projection is $\mathcal{N}(0.4, 0.84)$, the Gaussian chain's message, and every later step is identical: the projected recursion returns (0.7487, 0.1604) for this law too. The exact posterior mean for the two-point law is different, and the difference is the closure gap of this tiny example.

Where it is used. The paper, Section 2 ("the covariance twin and the price of second-order closure", Proposition 1 and the two consequences), Appendix A (the proof), Section 3 (the size of the gap, the decay theorem), Section 5 and Appendices D and F (the LMMSE and isotropic reference points, the $C = 1$ arm landing on the floor in E02a, E02b, E13), Appendix E (the Gaussian twin's windowed LMMSE in E15). Code: `src/markov_score/bp_gaussian.py`, which holds `gaussian_chain_bp` and the legacy grid-projected variant kept to document its instability; `src/markov_score/theory.py`, which holds `chain_covariance`, `lmmse_posterior_mean`, `lmmse_score` and `isotropic_score`; `src/markov_score/exact_scores.py`; and `lean/README.md` for what is machine-checked. Tests: `tests/test_gaussian_bp_equivalence.py` and the Gaussian-recovery test of `tests/test_em_bp.py`.

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Part IV

Learning

Chapter 14

Maximum likelihood and expectation–maximisation

The paper's learning method is expectation–maximisation (EM) for the marginal likelihood of noised sequences, with the exact E-step of Chapter 11. This chapter supplies the statistical background in the order a reader needs it: what a likelihood and a maximum-likelihood estimator are and what consistency and efficiency mean; what changes when the model has latent variables; Jensen's inequality; the EM algorithm derived twice, as coordinate ascent on a free-energy bound (Neal and Hinton) and in the original form of Dempster, Laird and Rubin, with the monotonicity proof; its generalisations (generalised EM, expectation–conditional maximisation) and why they keep the monotone ascent; its convergence rate and the fraction of missing information; stopping rules and the unit of the paper's tolerance; and why EM slows when the data carry little information about a coordinate, which is the mechanism the paper's Section 4 invokes when it says the iteration budget binds at high noise.

14.1 Likelihood, maximum likelihood, consistency, efficiency

14.1.1 Definitions

A **statistical model** is a family of densities $\{p_\theta : \theta \in \Theta\}$ on a sample space, indexed by a parameter in an open set $\Theta \subseteq \mathbb{R}^P$. Given observations $x^1, \dots, x^M$ drawn independently from p_{θ_0} for some unknown **true parameter** θ_0, the **likelihood** is the joint density as a function of the parameter, and its logarithm is the **log-likelihood**

$$\mathcal{L}(\theta) = \sum_{\mu=1}^{M} \log p_\theta(x^\mu). \tag{14.1}$$

The **maximum-likelihood estimator** (MLE) is any maximiser $\hat\theta = \arg\max_\theta \mathcal{L}(\theta)$. Its gradient $\nabla_\theta \log p_\theta(x)$ is the **score** of the model (not to be confused with the score of a density in the diffusion sense, which is a gradient in x; the paper uses "score" for the latter and "Fisher's identity" for the former); the covariance of the score under p_θ is the **Fisher information** $I(\theta)$, treated in the compendium's chapter on Fisher information and the Cramér–Rao bound.

14.1.2 Consistency and efficiency in one page

Two facts justify maximum likelihood. Both are stated for a **regular** model: the densities have a common support, are twice continuously differentiable in θ, differentiation under the integral is allowed, the information matrix is finite and non-singular, and the parameter is **identifiable**, $p_\theta = p_{\theta'}$ only if $\theta = \theta'$.

Consistency. By the law of large numbers, $\mathcal{L}(\theta)/M \to \mathbb{E}_{\theta_0} \log p_\theta(x)$ for every θ, and the limit equals $\mathbb{E}_{\theta_0} \log p_{\theta_0}(x) - D_{\mathrm{KL}}(p_{\theta_0} \| p_\theta)$. The Kullback–Leibler divergence is non-negative and zero only at $p_\theta = p_{\theta_0}$ (this is Gibbs' inequality, itself a consequence of Jensen's inequality below), so the limiting objective has a unique maximum at θ_0 under identifiability. With a uniform law of large numbers on compact sets, the maximiser of the sample objective converges to the maximiser of the limit: $\hat{\theta} \to \theta_0$ in probability. This is Wald's argument.

Asymptotic normality and efficiency. Expand the score equation $\nabla \mathcal{L}(\hat{\theta}) = 0$ around θ_0: $0 = \nabla \mathcal{L}(\theta_0) + \nabla^2 \mathcal{L}(\bar{\theta})(\hat{\theta} - \theta_0)$ for some intermediate $\bar{\theta}$. The first term is a sum of M independent zero-mean vectors (the score has mean zero: $\mathbb{E}_\theta \nabla \log p_\theta = \nabla \int p_\theta = 0$), so by the central limit theorem $\nabla \mathcal{L}(\theta_0)/\sqrt{M} \to \mathcal{N}(0, I(\theta_0))$; the second factor divided by M converges to $-I(\theta_0)$ (the information identity $\mathbb{E}[-\nabla^2 \log p] = \mathbb{E}[\nabla \log p \, \nabla \log p^\top]$, from differentiating $\int \nabla p = 0$ once more). Hence

$$\sqrt{M}\,(\hat{\theta} - \theta_0) \;\to\; \mathcal{N}\Big(0, I(\theta_0)^{-1}\Big). \tag{14.2}$$

The Cramér–Rao bound says no unbiased estimator has covariance below $I(\theta_0)^{-1}/M$; an estimator attaining Equation (14.2) is **asymptotically efficient**, and the MLE is one. The local asymptotic minimax theorem of Hájek and Le Cam, which the paper's sample-complexity corollary invokes, is the sharp form: over shrinking neighbourhoods $|\theta' - \theta_0| \le c/\sqrt{M}$ no estimator sequence does better than $1/I(\theta_0)$ in the limit, and the MLE attains it. Three caveats travel with these statements and the paper repeats them: they are limits in M at fixed model, they say nothing about a finite optimisation budget, and for a bounded coordinate the local statement is empty once $1/\sqrt{MI}$ exceeds the coordinate's range.

14.2 Latent-variable models

A **latent-variable model** specifies a joint density $p_\theta(x, z)$ of an observed variable x and an unobserved (**latent**, **hidden**, **missing**) variable z, and the likelihood of the data is the **marginal likelihood**

$$p_\theta(x) = \int p_\theta(x, z)\, \mathrm{d}z. \tag{14.3}$$

The joint is usually simple, often in an exponential family whose log-likelihood is linear in a few sufficient statistics of (x, z); the marginal is usually not, because the integral destroys the structure. In the paper x is a noised sequence, z is the clean chain, the joint is $P_0(\mathbf{a}) \prod_i \ell_t(x_i \mid a_i)$, and the integral over the chain is what the two sweeps of Chapter 11 compute. The **complete-data log-likelihood** $\log p_\theta(x, z)$ is the object EM works with: were z observed, maximising it would be easy; it is not, so EM replaces it by its conditional expectation given the data.

14.3 Jensen's inequality

Lemma 14.1 (Jensen)**.** *Let ϕ be a concave function on an interval J and Y a random variable with values in J and finite expectation. Then $\phi(\mathbb{E}Y) \ge \mathbb{E}\phi(Y)$, with equality if ϕ is strictly concave only when Y is almost surely constant.*

Proof. A concave function lies below every one of its supporting lines: at the point $m = \mathbb{E}Y$ there is a slope s with $\phi(y) \le \phi(m) + s(y - m)$ for all $y \in J$ (for a differentiable ϕ, $s = \phi'(m)$; in general any element of the superdifferential). Take expectations: $\mathbb{E}\phi(Y) \le \phi(m) + s(\mathbb{E}Y - m) = \phi(m)$. For strict concavity the inequality $\phi(y) < \phi(m) + s(y - m)$ is strict for $y \ne m$, so equality of expectations forces $Y = m$ almost surely. □

With $\phi = \log$, which is strictly concave on $(0, \infty)$, two consequences follow. **Gibbs' inequality**: for densities p, q with $q \ll p$, $\int p \log(q/p) \le \log \int p\,(q/p) = \log \int q \le 0$, so $D_{\mathrm{KL}}(p \| q) = \int p \log(p/q) \ge 0$ with equality only at $p = q$. And the **evidence lower bound** of the next section.

14.4 Expectation–maximisation

14.4.1 The free-energy view

Fix the data x and consider an arbitrary density $q(z)$ over the latent variable. Write the log marginal likelihood as an expectation under q and apply Lemma 14.1:

$$\log p_\theta(x) = \log \int q(z) \frac{p_\theta(x,z)}{q(z)} \,\mathrm{d}z \;\geq\; \int q(z) \log \frac{p_\theta(x,z)}{q(z)} \,\mathrm{d}z = \underbrace{\mathbb{E}_q \log p_\theta(x,z)}_{\text{energy}} + \underbrace{H(q)}_{\text{entropy}} =: F(q,\theta). \tag{14.4}$$

The bound F is the **free energy** (negative, in the physicists' sign) or **evidence lower bound**; it is exact in the sense that

$$\log p_\theta(x) - F(q,\theta) = D_{\mathrm{KL}}\big(q \,\|\, p_\theta(\cdot \mid x)\big) \;\geq 0, \tag{14.5}$$

which one checks by writing $p_\theta(x,z) = p_\theta(x) p_\theta(z \mid x)$ inside the integral. So $F(q,\theta) \leq \log p_\theta(x)$ for every q, with equality exactly when q is the posterior $p_\theta(\cdot \mid x)$.

The EM algorithm of Neal and Hinton (1998) is **coordinate ascent** on F:

- **E-step**: with $\theta^{(k)}$ fixed, maximise F over q. By Equation (14.5) the maximiser is the posterior $q^{(k+1)} = p_{\theta^{(k)}}(\cdot \mid x)$, and at it $F = \log p_{\theta^{(k)}}(x)$.
- **M-step**: with $q^{(k+1)}$ fixed, maximise F over θ. The entropy does not depend on θ, so this is maximising the expected complete-data log-likelihood $\mathbb{E}_{q^{(k+1)}} \log p_\theta(x,z)$; call the maximiser $\theta^{(k+1)}$.

Each step does not decrease F, and after every E-step F equals the log-likelihood. Hence $\log p_{\theta^{(k+1)}}(x) = F(q^{(k+2)},\theta^{(k+1)}) \geq F(q^{(k+1)},\theta^{(k+1)}) \geq F(q^{(k+1)},\theta^{(k)}) = \log p_{\theta^{(k)}}(x)$: the marginal likelihood never decreases. The view also explains partial versions of either step: any q that increases F and any θ that increases F preserve the monotonicity, because F is the single objective both steps ascend.

14.4.2 The Dempster–Laird–Rubin view and the Q-function

Define the **Q-function**

$$\mathcal{Q}(\theta \mid \theta') = \mathbb{E}_{p_{\theta'}(z \mid x)}\big[\log p_\theta(x,z)\big] = \int p_{\theta'}(z \mid x) \log p_\theta(x,z) \,\mathrm{d}z, \tag{14.6}$$

the expected complete-data log-likelihood at θ under the posterior computed at θ'. The E-step computes whatever expectations $\mathcal{Q}(\cdot \mid \theta^{(k)})$ needs; the M-step sets $\theta^{(k+1)} = \arg\max_\theta \mathcal{Q}(\theta \mid \theta^{(k)})$. The original monotonicity proof (Dempster, Laird and Rubin, 1977) is the following.

Theorem 14.2 (Monotone ascent of EM)**.** *If $\mathcal{Q}(\theta^{(k+1)} \mid \theta^{(k)}) \geq \mathcal{Q}(\theta^{(k)} \mid \theta^{(k)})$, then $\log p_{\theta^{(k+1)}}(x) \geq \log p_{\theta^{(k)}}(x)$, with equality only if the posteriors at the two parameters coincide and the $\mathcal{Q}$ values are equal.*

Proof. Write $\log p_\theta(x) = \log p_\theta(x,z) - \log p_\theta(z \mid x)$ for any z and take the expectation under $p_{\theta'}(z \mid x)$, which leaves the left side unchanged:

$$\log p_\theta(x) = \mathcal{Q}(\theta \mid \theta') - \mathcal{H}(\theta \mid \theta'), \qquad \mathcal{H}(\theta \mid \theta') = \mathbb{E}_{p_{\theta'}(z \mid x)} \log p_\theta(z \mid x). \tag{14.7}$$

By Gibbs' inequality, $\mathcal{H}(\theta \mid \theta') \leq \mathcal{H}(\theta' \mid \theta')$ for every θ (the cross-entropy of the posterior at θ' against the posterior at θ is largest when the two agree). Subtracting Equation (14.7) at $\theta = \theta^{(k+1)}$ from itself at $\theta = \theta^{(k)}$, both with $\theta' = \theta^{(k)}$,

$$\log p_{\theta^{(k+1)}}(x) - \log p_{\theta^{(k)}}(x) = \big[\mathcal{Q}(\theta^{(k+1)} \mid \theta^{(k)}) - \mathcal{Q}(\theta^{(k)} \mid \theta^{(k)})\big] + \big[\mathcal{H}(\theta^{(k)} \mid \theta^{(k)}) - \mathcal{H}(\theta^{(k+1)} \mid \theta^{(k)})\big],$$

and both brackets are non-negative. □

The theorem asks only that $\mathcal{Q}$ *increase*, not that it be maximised. This is the definition of **generalised EM** (GEM): any M-step that ascends $\mathcal{Q}$ preserves the monotone increase of the likelihood. A useful special case is **expectation–conditional maximisation** (ECM, Meng and Rubin, 1993): partition the parameter into blocks $\theta = (\theta_1, \dots, \theta_B)$ and replace the M-step by a sweep of **conditional maximisations**, each maximising $\mathcal{Q}$ over one block with the others held at their most recent values. Each conditional maximisation cannot decrease $\mathcal{Q}$ (it maximises it over a subset containing the current point), so the sweep ascends $\mathcal{Q}$ and Theorem 14.2 applies; several sweeps, or a different order, are equally admissible. Meng and Rubin also show that if every conditional maximiser is unique and the blocks together span the parameter space in a suitable sense, the fixed points of ECM are the stationary points of the likelihood, as for EM. The paper's mixture-innovation M-step is an ECM sweep, block by block, which is why the paper calls it "a generalised EM whose ascent is monotone" and cites Meng and Rubin.

14.4.3 Fixed points are stationary points

At a fixed point θ^* of EM, θ^* maximises $\mathcal{Q}(\cdot \mid \theta^*)$, so $\nabla_\theta \mathcal{Q}(\theta \mid \theta^*)|_{\theta=\theta^*} = 0$. Differentiating Equation (14.7) in θ at $\theta = \theta'$ shows $\nabla_\theta \mathcal{H}(\theta \mid \theta')|_{\theta=\theta'} = \mathbb{E}_{p_{\theta'}(z|x)} \nabla \log p_{\theta'}(z \mid x) = 0$ (the posterior score has mean zero under the posterior), hence

$$\nabla_\theta \log p_\theta(x) = \nabla_\theta \mathcal{Q}(\theta \mid \theta')\big|_{\theta'=\theta} = \mathbb{E}_{p_\theta(z|x)}\big[\nabla_\theta \log p_\theta(x, z)\big]. \tag{14.8}$$

This is **Fisher's identity**: the gradient of the marginal log-likelihood is the posterior expectation of the complete-data gradient. It says that fixed points of EM are stationary points of the likelihood, and, more usefully for the paper, that the E-step delivers the exact gradient of the marginal likelihood as a by-product. Chapter 15 derives it again directly, in the form the paper uses.

14.5 Convergence rate and the fraction of missing information

EM defines a map $\theta^{(k+1)} = \mathcal{M}(\theta^{(k)})$. Near a fixed point θ^* with a non-singular Hessian, $\theta^{(k+1)} - \theta^* \approx J(\theta^{(k)} - \theta^*)$ with $J = \nabla \mathcal{M}(\theta^*)$ the Jacobian of the map, so the error contracts linearly with rate equal to the largest eigenvalue of J. Dempster, Laird and Rubin identified J. Define the **complete information** $I_{\text{com}} = -\nabla^2_\theta \mathcal{Q}(\theta \mid \theta')|_{\theta=\theta'=\theta^*}$, the expected negative Hessian of the complete-data log-likelihood under the posterior, and the **observed information** $I_{\text{obs}} = -\nabla^2 \log p_{\theta^*}(x)$. Then

$$J = I_{\text{mis}}\, I_{\text{com}}^{-1}, \qquad I_{\text{mis}} = I_{\text{com}} - I_{\text{obs}}, \qquad \text{so} \quad J = \mathbf{I} - I_{\text{obs}} I_{\text{com}}^{-1}. \tag{14.9}$$

The matrix I_{mis} is the **missing information**, the expected information that the latent variable would have added, and J is the **fraction of missing information**. The derivation is one line from the M-step's stationarity condition: $\nabla_\theta \mathcal{Q}(\mathcal{M}(\theta') \mid \theta') = 0$ for all θ'; differentiate in θ' at the fixed point to get $-I_{\text{com}} J + \nabla_{\theta'} \nabla_\theta \mathcal{Q} = 0$, and the mixed derivative equals $I_{\text{com}} - I_{\text{obs}}$ by differentiating Equation (14.7) twice (the second derivative of $\mathcal{H}$ in its first argument at the fixed point is minus the posterior information, and the identity rearranges to Equation (14.9)).

The reading is the important part. When the data determine the parameter almost as well as the complete data would, $I_{\text{obs}} \approx I_{\text{com}}$, $J \approx 0$ and EM converges in a few iterations. When the observed information is a small fraction of the complete information, the eigenvalues of J approach one and EM crawls: each iteration removes only the fraction $1 - \lambda_{\max}(J)$ of the remaining error, and the number of iterations to reduce the error by a factor e is about $1/(1 - \lambda_{\max}(J))$. In a multi-parameter problem the rate is direction-dependent: a coordinate about which the data are nearly uninformative has its own eigenvalue near one and converges last, however fast the others settle. This is the observation the next section, and the paper, build on.

14.6 Stopping rules

An EM run is stopped by a **tolerance** on the increase of the log-likelihood, by an **iteration cap**, or by an external **selection rule**. The paper uses all three and the compendium should say what each measures.

Units. The total log-likelihood Equation (14.1) of M sequences of L sites is a sum over $M(L-1)$ transitions (plus the first sites), so it grows with the dataset, and so does its increment per iteration. A tolerance in absolute nats means something different at every sample size. A relative tolerance $|\Delta\mathcal{L}| \le \tau|\mathcal{L}|$ is scale-free, because the increment and the total are sums over the same transitions; the docstring of `fit_em` in `src/markov_score/em.py` records that this was measured to be invariant to the sample size to every digit, so the relative rule was not wrong. What the paper's code does instead is to express the increment in **nats per transition**, $\Delta\mathcal{L}/(M(L-1))$, and to stop when it lies in $[0, \tau]$ with $\tau = 10^{-9}$. The unit buys interpretability: a constant threshold in nats per transition means the same thing at every sample size and every noise level, whereas a relative threshold tightens as $|\mathcal{L}|$ falls along the run and changes its meaning with the level (at high noise $|\mathcal{L}|$ per transition is close to the entropy of a standard Gaussian and the increments are tiny in comparison).

Sign. An exact EM cannot decrease the likelihood. A stopping test on $|\Delta\mathcal{L}|$ accepts a small *decrease* as convergence and thereby hides a defect (a wrong M-step, or a generalised step that overshot). The code's test is signed: convergence requires $0 \le \Delta\mathcal{L} \le \tau$ per transition; any decrease is recorded as a `monotone_violation` and is a bug report, not a stop. A run that exhausts its iteration cap without meeting the signed test is labelled **censored**, and the label travels with the fitted kernel so that a comparison of estimators can tell whether it compares converged runs. The returned kernel is always the iterate whose likelihood is the last trace entry: the loop scores each proposal after the M-step that produced it. An earlier version returned a kernel one update beyond anything evaluated, which made the reported likelihood belong to the previous iterate; the tests in `tests/test_em_bp.py` and `tests/test_gradient_trace_alignment.py` pin this alignment.

Selection. At the budgets of the paper's experiments, the per-transition increments EM takes are far above 10^{-9} for hundreds of iterations, so the tolerance rarely fires and what ends a run is the cap of the **checkpoint ladder** (Chapter 15). Which rung is reported is then decided by a validation bundle, not by the likelihood: the rung with the smallest score error on held-out sequences. This is **early stopping**, and it is a form of regularisation, because successive EM iterates move from the initialisation towards the maximiser and an early rung is a shrunken estimate. The paper's Section 4 reads its own fits this way: at high noise the validation-selected iterate is a shrinkage estimator whose shape collapses towards a Gaussian innovation. The paper's Appendix E (E20) checks that selecting the rung by held-out log-evidence instead of by the validation score error picks the same rung in most seeds, so the selection needs no oracle.

14.7 Why EM slows when information is scarce

Combine Equation (14.9) with the paper's order-by-order decay theorem. In the noised-sequence problem the complete data are the clean chains, and the complete information about the kernel parameters is of order $M(L-1)$ times a constant that does not depend on the noise level, since it is the information the clean transitions carry. The observed information is M times the per-sequence Fisher information $I_t(\theta)$, which decays as e^{-4t} for a covariance coordinate and e^{-8t} for

a shape coordinate. Hence the fraction of missing information along each coordinate,

$$\lambda(\theta_j) \approx 1 - \frac{I_t(\theta_j)}{I_{\mathrm{com}}(\theta_j)}, \tag{14.10}$$

approaches one as t grows, and does so faster for the shape coordinate. Two consequences follow, both visible in the paper. First, the number of iterations to converge grows like $1/(1-\lambda)$, so at high noise the ladder's cap is reached before the tolerance fires: the paper reports that at the two highest levels of its learning experiment the validation-selected rung is the last of the ladder in most cells, "whether the excess over the reference at those two levels is information or budget the run cannot say", and registers a control with a sixfold longer ladder (E06c). Second, at any fixed noise level the correlation coordinate (degree two) converges in far fewer iterations than the shape coordinate (degree four), because its missing fraction is smaller by roughly e^{-4t} in the information ratio; the paper's "convergence asymmetry" (correlation settles in tens of updates, shape in hundreds) is Equation (14.10) read coordinate by coordinate. Chapter 15 adds the second mechanism specific to the mixture kernel, the inner latent label, which slows the shape further.

14.8 A worked example: one mixing weight

Let $y_1, y_2, y_3 = -1.5,\ 0.2,\ 1.0$ be three observations from the two-component mixture $p_\pi(y) = \pi\,\phi(y-1) + (1-\pi)\,\phi(y+1)$ with ϕ the standard Gaussian density and the mixing weight π unknown. The latent variable is the component label $z_i \in \{0, 1\}$; the complete-data log-likelihood is $\sum_i [z_i \log \pi + (1 - z_i) \log(1 - \pi)] + \text{const}$, linear in the z_i, so the E-step needs only the **responsibilities** $r_i = \mathbb{E}[z_i \mid y_i] = \pi\phi(y_i - 1)/p_\pi(y_i)$, and the M-step maximises $\sum_i [r_i \log \pi + (1 - r_i) \log(1 - \pi)]$, giving $\pi^{\mathrm{new}} = \frac{1}{3}\sum_i r_i$.

The densities are

$$\phi(y_i - 1) = (0.01753,\ 0.28969,\ 0.39894), \qquad \phi(y_i + 1) = (0.35207,\ 0.19419,\ 0.05399).$$

Start at $\pi^{(0)} = 0.5$. The log-likelihood is $\sum_i \log[0.5\phi(y_i - 1) + 0.5\phi(y_i + 1)] = -4.59273$. The responsibilities are $r = (0.04743,\ 0.59869,\ 0.88080)$ and the M-step gives $\pi^{(1)} = 0.50897$. The log-likelihood rises to -4.59199; the new responsibilities are $(0.04907,\ 0.60728,\ 0.88451)$ and $\pi^{(2)} = 0.51362$, with log-likelihood -4.59180; then $\pi^{(3)} = 0.51602$, $\pi^{(4)} = 0.51725$, $\pi^{(5)} = 0.51789$, with log-likelihoods -4.59174, -4.59173, -4.59173: monotone, as Theorem 14.2 requires, and slowing.

The maximiser is $\pi^* = 0.51855$. The successive errors $|\pi^{(k)} - \pi^*|$ shrink by the factor $0.514,\ 0.514,\ 0.513,\ \ldots$ per iteration. Equation (14.9) predicts this number: at π^* the complete information is $I_{\mathrm{com}} = \sum_i [r_i/\pi^2 + (1-r_i)/(1-\pi)^2] = 12.02$, the observed information is minus the second derivative of the log-likelihood, $I_{\mathrm{obs}} = 5.86$ (a finite difference suffices), and $1 - I_{\mathrm{obs}}/I_{\mathrm{com}} = 0.513$. Half the information about π is missing, and EM removes half of the remaining error per iteration. Had the two components been closer (means ± 0.2 instead of ± 1), the responsibilities would be near $\frac{1}{2}$ for every observation, the observed information a small fraction of the complete one, and the rate near one: the situation of Section 14.7.

Where it is used. The paper, Section 4 ("expectation–maximisation around the same exact recursion", generalised EM with conditional-maximisation sweeps, the convergence asymmetry, the iteration budget at high noise), the sample-complexity corollary and its caveats (local asymptotic minimax, the MLE attains it), Appendix C (the checkpoint ladder and validation selection), Appendix E (E20). Code: `src/markov_score/em.py`, whose `fit_em` carries the docstring on the stopping rule and whose `EMTrace` records `monotone_violation`, `converged` and `stop_reason`, with `q_value` and `q_gradient` beside them; and the ECM sweeps of the mixture kernel's `m_step` in `src/markov_score/kernels.py`. Tests: monotonicity, the signed stopping test, censored against converged and the stationarity of the fixed point in `tests/test_em_bp.py`; the iterate-to-trace alignment in `tests/test_gradient_trace_alignment.py`.

References. A. P. Dempster, N. M. Laird and D. B. Rubin (1977), "Maximum likelihood from incomplete data via the EM algorithm", *Journal of the Royal Statistical Society B* 39(1). R. M. Neal and G. E. Hinton (1998), "A view of the EM algorithm that justifies incremental, sparse, and other variants", in *Learning in Graphical Models*, ed. M. I. Jordan, Kluwer. X.-L. Meng and D. B. Rubin (1993), "Maximum likelihood estimation via the ECM algorithm: a general framework", *Biometrika* 80(2). C. F. J. Wu (1983), "On the convergence properties of the EM algorithm", *Annals of Statistics* 11(1). A. W. van der Vaart (1998), *Asymptotic Statistics*, Cambridge University Press, Chapters 5, 7 and 8. O. Cappé, E. Moulines and T. Rydén (2005), *Inference in Hidden Markov Models*, Springer, Chapter 10.

Chapter 15

EM for the transition law from noised sequences

This chapter assembles the paper's estimator from the pieces of the previous four. The data are M noised sequences, each observed once at its own noise level; the unknown is the transition kernel K_θ of the clean chain; the objective is the marginal likelihood; the algorithm is expectation–maximisation with the exact E-step of Chapter 11 on the grid of Chapter 12. Everything the M-step needs is one $N_{\mathrm{g}} \times N_{\mathrm{g}}$ matrix, the expected transition statistic $\boldsymbol{\Xi}$, and Fisher's identity turns the same matrix into the exact gradient of the marginal likelihood. The chapter derives Fisher's identity in full, defines $\boldsymbol{\Xi}$ and its properties, derives the Gaussian M-step (the belief-weighted Yule–Walker equations), describes the mixture-innovation kernel and its conditional-maximisation sweeps, explains the checkpoint ladder, the validation selection and the `at_cap` flag, the clean-data limit, and how E06, E06c and E02a use all of it. It ends with the information-theoretic reason for the convergence asymmetry the paper reports.

15.1 The marginal likelihood of noised sequences

The paper's Section 4 drops the assumption that K is known and keeps everything else: $a_1 \sim \mathcal{N}(0,1)$, $a_i = \rho a_{i-1} + \varepsilon_i$ with a parameterised transition kernel $K_\theta(a' \mid a)$, and the channel $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$. The observations are M sequences $\mathbf{x}^\mu$, the μ-th at level t_μ. Under the **one-view protocol** each clean sequence is noised once, at one level, and never seen clean; in the headline experiments the sequences are split evenly across the twelve levels of the schedule, and in the learning experiment E06 all of them sit at one level. The **marginal log-likelihood** is

$$\mathcal{L}(\theta) = \sum_{\mu=1}^{M} \log P_t[t_\mu, \theta](\mathbf{x}^\mu), \qquad P_t[t,\theta](\mathbf{x}) = \int \mu(a_1) \prod_{i=2}^{L} K_\theta(a_i \mid a_{i-1}) \prod_{i=1}^{L} \ell_t(x_i \mid a_i)\, \mathrm{d}\mathbf{a}, \quad (15.1)$$

where the unary factors ℓ_t depend on t_μ but not on θ, and only the transition factors carry θ. Each term of Equation (15.1) is the evidence of Chapter 11, returned by the forward sweep on the grid. Two structural facts drive everything below: the parameter enters only through the $L-1$ transition factors of each sequence, and the noise level enters only through the unary factors, so sequences at different levels contribute additively and identically to every statistic (`e_step_multi` in `src/markov_score/em.py` simply adds them).

15.2 Fisher's identity

15.2.1 Derivation

Let $p_\theta(x, z)$ be a joint density of data and latent variables, with $p_\theta(x) = \int p_\theta(x, z)\,\mathrm{d}z$, and assume differentiation under the integral sign is permitted (the integrand and its θ-derivative are dominated by an integrable function, which holds for the kernels used here on the truncated grid, where the integral is a finite sum). Then

$$\begin{aligned}\nabla_\theta \log p_\theta(x) &= \frac{\nabla_\theta p_\theta(x)}{p_\theta(x)} = \frac{1}{p_\theta(x)} \int \nabla_\theta p_\theta(x, z)\,\mathrm{d}z \\ &= \int \frac{p_\theta(x, z)}{p_\theta(x)}\, \nabla_\theta \log p_\theta(x, z)\,\mathrm{d}z = \mathbb{E}_{p_\theta(z \mid x)}\big[\nabla_\theta \log p_\theta(x, z)\big],\end{aligned} \tag{15.2}$$

using $\nabla p = p \nabla \log p$ in the third step and $p_\theta(x, z)/p_\theta(x) = p_\theta(z \mid x)$ in the last. This is **Fisher's identity**: the gradient of the marginal log-likelihood equals the posterior expectation of the gradient of the complete-data log-likelihood. It was met in Chapter 14 as the statement that EM's fixed points are stationary points; here it is a computational device.

Apply it to one sequence of Equation (15.1). The complete-data log-likelihood is $\log \mu(a_1) + \sum_{i \geq 2} \log K_\theta(a_i \mid a_{i-1}) + \sum_i \log \ell_t(x_i \mid a_i)$, and only the middle sum depends on θ, so

$$\begin{aligned}\nabla_\theta \log P_t[t, \theta](\mathbf{x}) &= \sum_{i=2}^{L} \mathbb{E}\big[\nabla_\theta \log K_\theta(a_i \mid a_{i-1}) \,\big|\, \mathbf{x}\big] \\ &= \sum_{i=2}^{L} \int p_t(a_{i-1}, a_i \mid \mathbf{x})\, \nabla_\theta \log K_\theta(a_i \mid a_{i-1})\,\mathrm{d}a_{i-1}\,\mathrm{d}a_i.\end{aligned} \tag{15.3}$$

Each term needs only the *pairwise* posterior marginal on one edge, which the two sweeps return exactly (equation *pair* of Chapter 11). No derivative of the recursion is taken: the sweeps supply an expectation, and the gradient is an integral against it. This is the sense of the paper's "no differentiation through the recursion is needed".

15.2.2 On the grid: the expected transition statistic

Replace the integrals by the discretised model of Chapter 12. The pairwise posterior on edge $(i-1, i)$ of sequence μ becomes a matrix of probabilities $P(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu)$ over grid pairs, and $\nabla_\theta \log K_\theta$ becomes the array $\nabla_\theta \log K_\theta(u_k \mid u_l)$. Summing over edges and sequences,

$$\begin{aligned}\nabla_\theta \mathcal{L}(\theta) &= \sum_{k,l} \Xi_{kl}(\theta)\, \nabla_\theta \log K_\theta(u_k \mid u_l) = \big\langle \Xi(\theta), \nabla_\theta \log K_\theta \big\rangle, \\ \Xi_{kl} &= \sum_{\mu} \sum_{i \geq 2} P(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu),\end{aligned} \tag{15.4}$$

which is the paper's equation *fisher*. The matrix Ξ is the **expected transition statistic**: entry (k, l) is the expected posterior mass on the transition $u_l \to u_k$ (column the parent, row the child, the same orientation as the transition matrix), summed over every edge of every sequence. It is the continuous-state analogue of the expected transition counts of Baum–Welch. Its computation from the sweeps is a rank-one accumulation: with $f_i = w \circ \overrightarrow{m}_i \circ \ell_i$ and $g_i = w \circ \overleftarrow{m}_i \circ \ell_i$ (messages with the local factor and the quadrature weight folded in), the unnormalised pairwise belief on edge $(i, i+1)$ is $f_i(l)\, K[k, l]\, g_{i+1}(k)$ with normaliser $Z_i = g_{i+1}^\top K f_i$, so

$$\Xi = C \circ K, \qquad C = \sum_{\mu} \sum_{i} \frac{g_{i+1}^{\mu}\, (f_i^{\mu})^\top}{Z_i^{\mu}}, \tag{15.5}$$

with $\circ$ the elementwise product. The code (`_e_step_chunk` in `src/markov_score/em.py`) forms C as one matrix product of the stacked g and f vectors over all edges of a chunk, then multiplies by K.

Three properties of $\boldsymbol{\Xi}$ are used constantly.

1. *Mass.* $\sum_{kl} \boldsymbol{\Xi}_{kl} = M(L-1)$, the number of transitions, because each pairwise posterior sums to one. This is checked exactly in `tests/test_em_bp.py` and is what lets the clean-data histogram of Section 15.6 be a drop-in replacement.

2. *Additivity.* Sequences at different levels contribute additively; one $\boldsymbol{\Xi}$ summarises a dataset spread over the whole schedule.

3. *What it is not.* $\boldsymbol{\Xi}$ is a complete summary for the current M-step (the maximisation never revisits the data), but it is not a sufficient statistic of the observations: it is computed at the current θ and depends on it, on the grid, on the fixed initial law and on the homogeneity of the transition. The code's docstring insists on this distinction, and so should the reader.

15.2.3 The same matrix is the whole E-step

With $\boldsymbol{\Xi}$ in hand, the Q-function of Chapter 14 for the discretised model is

$$\mathcal{Q}(\theta \mid \theta') = \langle \boldsymbol{\Xi}(\theta'), \log K_\theta \rangle + (\text{site-1 term}) + \text{const}, \tag{15.6}$$

a finite bilinear form; the site-1 term involves the initial law, which the paper holds fixed at $\mathcal{N}(0,1)$, and the constant collects the likelihood terms and the posterior entropy, none of which depend on θ. Its gradient at $\theta = \theta'$ is Equation (15.4): the Q-function's gradient and the marginal likelihood's gradient coincide where the E-step was taken, which is Fisher's identity once more. The E-step is therefore "one forward–backward pass" plus the accumulation Equation (15.5), at cost $O(MLN_{\mathrm{g}}^2)$ independent of the number of parameters, and the M-step touches only $\boldsymbol{\Xi}$, at cost $O(PN_{\mathrm{g}}^2)$ for P parameters.

15.2.4 Verification

The paper states that the identity is verified against finite differences to 10^{-9}. Three tests do the work, and it matters which checks what. `test_fisher_identity_matches_finite_differences` in `tests/test_em_bp.py` compares $\langle \boldsymbol{\Xi}, \nabla \log K_\theta \rangle$ with a central difference of the grid evidence for the Gaussian kernel: a real check, but not an independent one, since the evidence and $\boldsymbol{\Xi}$ come from the same recursion and the same discretised kernel, so a shared error (a transposed index, a misplaced factor) could satisfy both sides. `tests/test_fisher_bruteforce.py` removes that coupling by enumerating every latent path of a five-state, four-site chain from the model definition, with no recursion, and checking the evidence, $\boldsymbol{\Xi}$ *including its orientation* (the matrix must match and must not match its transpose, on a grid asymmetric enough that the two differ), and the gradient against a finite difference of the *enumerated* evidence. The Laplace kernel is included there deliberately: its ρ-derivative carries $\operatorname{sign}(a' - \rho a)$, a discontinuity along the diagonal, and trapezoidal quadrature of a discontinuous integrand is first order (Chapter 12); the test in `tests/test_em_bp.py` named for the Laplace scale gradient pins the fact that the smooth b-derivative matches a finite difference to high accuracy while the ρ-derivative is off by a grid-dependent percentage. That is a discretisation artefact of differentiating a non-smooth density on a mesh, not a failure of the identity; the Laplace M-step avoids it by never using the gradient (it is a weighted median). Finally, `tests/test_gradient_trace_alignment.py` guards a different invariant of the gradient-ascent route: the returned parameter is the one at which the last recorded likelihood was evaluated, and the monotonicity record covers the final update.

15.3 The M-step for the Gaussian kernel

Take $K_\theta(a' \mid a) = \mathcal{N}(\rho a, q)$, $\theta = (\rho, q)$. The edge part of Equation (15.6) is

$$\mathcal{Q}(\rho, q) = \sum_{k,l} \boldsymbol{\Xi}_{kl} \Big[-\frac{(u_k - \rho u_l)^2}{2q} - \frac{1}{2}\log(2\pi q) \Big] = -\frac{S_{11} - 2\rho S_{01} + \rho^2 S_{00}}{2q} - \frac{W}{2}\log(2\pi q), \quad (15.7)$$

where $W = \sum_{kl} \boldsymbol{\Xi}_{kl}$ and the three belief-weighted moments are

$$S_{00} = \sum_{kl} \boldsymbol{\Xi}_{kl}\, u_l^2, \qquad S_{01} = \sum_{kl} \boldsymbol{\Xi}_{kl}\, u_k u_l, \qquad S_{11} = \sum_{kl} \boldsymbol{\Xi}_{kl}\, u_k^2,$$

the second moment of the parent, the cross moment, and the second moment of the child. The subscripts index the two variables of a transition (0 for the parent u_l, 1 for the child u_k) and say which pair is multiplied; they are not powers. The paper's Section 4 writes the same three quantities as $S_{ab} = \sum_{kl} \boldsymbol{\Xi}_{kl} u_k^a u_l^b$, which read literally as powers would make $S_{00} = W$ and $S_{01} = \sum \boldsymbol{\Xi}_{kl} u_l$ and would not give the formulas that follow; the index reading above is the one the formulas and the code use. Setting the derivatives to zero,

$$\begin{aligned} \frac{\partial \mathcal{Q}}{\partial \rho} &= \frac{S_{01} - \rho S_{00}}{q} = 0 && \Rightarrow \quad \rho = \frac{S_{01}}{S_{00}}, \\ \frac{\partial \mathcal{Q}}{\partial q} &= \frac{S_{11} - 2\rho S_{01} + \rho^2 S_{00}}{2q^2} - \frac{W}{2q} = 0 && \Rightarrow \quad q = \frac{S_{11} - 2\rho S_{01} + \rho^2 S_{00}}{W} = \frac{S_{11} - \rho S_{01}}{W}, \end{aligned} \quad (15.8)$$

the last equality because $\rho^2 S_{00} = \rho S_{01}$ at the optimum. These are the **belief-weighted Yule–Walker equations**: the least-squares regression of the child on the parent through the origin, with the pairwise posterior mass as weights, and the residual variance as the innovation variance. The Hessian of Equation (15.7) is negative definite at this point (it is a weighted least-squares problem in ρ and a concave profile in q), so this is the maximiser and the M-step is exact. In the clean-data limit, where $\boldsymbol{\Xi}$ is a histogram of observed transitions, Equation (15.8) reduces to the ordinary AR(1) least-squares estimator, which the clean-statistics test of `tests/test_em_bp.py` confirms. The Gaussian kernel's `m_step` in `src/markov_score/kernels.py` does not clamp $|\rho| < 1$: the update is the exact maximiser, a finite chain with $|\rho| \geq 1$ is a valid joint law, and what fails there is only the stationary interpretation $q = 1 - \rho^2$, which the code flags with a warning rather than silently biasing the step.

15.4 The mixture-innovation kernel

15.4.1 Definition and moments

The paper's headline estimator keeps the linear autoregression and frees the innovation law:

$$K_\theta(a' \mid a) = \sum_{c=1}^{C} \pi_c \mathcal{N}\Big(a' - \rho a - \nu_c, s_c^2\Big), \qquad \theta = (\rho, \pi_1, \ldots, \pi_C, \nu_1, \ldots, \nu_C, s_1^2, \ldots, s_C^2), \quad (15.9)$$

a C-component Gaussian mixture as the density of the innovation $\varepsilon = a' - \rho a$. Heavy tails (Laplace, Student-t), bimodality and skew are all approximable; the estimator never sees a Laplace density and must reproduce one from noised data. The component means are unconstrained: recentering them after each update to enforce $\sum_c \pi_c \nu_c = 0$ is not a maximiser of $\mathcal{Q}$ and was found to break monotone ascent, so the fitted innovation mean is reported as a diagnostic instead (on zero-mean data it lands near zero by itself). The fitted innovation's moments are read from the components in closed form,

$$\begin{aligned} m_1 &= \sum_c \pi_c \nu_c, \qquad q_{\text{fit}} = \sum_c \pi_c (s_c^2 + (\nu_c - m_1)^2), \\ m_4 &= \sum_c \pi_c (3 s_c^4 + 6 s_c^2 (\nu_c - m_1)^2 + (\nu_c - m_1)^4), \qquad \hat{\kappa}_4^{\text{ex}} = \frac{m_4}{q_{\text{fit}}^2} - 3, \end{aligned} \quad (15.10)$$

using the Gaussian moments $\mathbb{E}(\nu + sZ - m_1)^4$ expanded in powers of sZ (the odd terms vanish). The last quantity is the fitted excess kurtosis, the scalar "non-Gaussianity knob" the paper compares with the truth (+3 for Laplace); the formula is `innovation_moments` in `src/markov_score/kernels.py` and is tested against draws from the same mixture.

15.4.2 The complete-data problem has a second latent variable

For the Gaussian kernel the M-step was a closed-form maximisation because $\log K_\theta$ was a quadratic in θ. For Equation (15.9) it is the logarithm of a sum, and the maximiser of $\langle \Xi, \log K_\theta \rangle$ has no closed form. The remedy is the classical one: the mixture has its own latent variable, the component label, and given Ξ the M-step is itself an EM problem over that label. Define the **responsibilities** on the grid,

$$r_c(k,l) = \frac{\pi_c \, \mathcal{N}\big(u_k - \rho u_l - \nu_c, s_c^2\big)}{\sum_{c'} \pi_{c'} \, \mathcal{N}\big(u_k - \rho u_l - \nu_{c'}, s_{c'}^2\big)}, \tag{15.11}$$

the posterior probability of component c given the transition $u_l \to u_k$. A sweep of **conditional maximisations** (Chapter 14) then alternates two blocks, each a closed-form maximiser of the inner Q-function with the other block fixed, with $e_{kl} = u_k - \rho u_l$ the residual:

$$\text{mixture block:} \qquad \pi_c = \frac{\langle \Xi, r_c \rangle}{W}, \quad \nu_c = \frac{\langle \Xi, r_c \circ e \rangle}{\langle \Xi, r_c \rangle}, \quad s_c^2 = \frac{\langle \Xi, r_c \circ (e - \nu_c)^2 \rangle}{\langle \Xi, r_c \rangle}; \tag{15.12}$$

$$\text{autoregression block:} \qquad \rho = \frac{\sum_{c,k,l} \Xi_{kl} \, r_c(k,l) \, u_l \, (u_k - \nu_c)/s_c^2}{\sum_{c,k,l} \Xi_{kl} \, r_c(k,l) \, u_l^2/s_c^2}, \tag{15.13}$$

the first being the usual Gaussian-mixture M-step with the weights $\Xi_{kl} r_c(k,l)$ and the second a weighted least-squares fit of the child on the parent with component-specific offsets ν_c and precisions $1/s_c^2$ (differentiate $\sum \Xi \, r_c[-(u_k - \rho u_l - \nu_c)^2/2s_c^2]$ in ρ). The responsibilities are recomputed between the blocks. Each block increases the inner Q-function, which by Jensen's inequality (the standard mixture-EM argument applied with Ξ as the "data weights") increases $\langle \Xi, \log K_\theta \rangle$; so the sweep ascends $\mathcal{Q}$, and by the monotonicity theorem of Chapter 14 the outer EM's marginal likelihood cannot decrease. This is the sense of the paper's "a generalised EM whose ascent is monotone" and of its citation of Meng and Rubin.

The implementation, the mixture kernel's `m_step` in `src/markov_score/kernels.py`, sweeps until the per-transition gain in $\langle \Xi, \log K_\theta \rangle / W$ falls below 10^{-8} or sixteen sweeps are spent, and records which happened on the returned kernel; a negative gain beyond round-off raises an error, since a conditional maximiser cannot decrease its objective. The docstring records why sixteen: with four sweeps the innovation *shape* was materially under-converged on short outer runs while ρ barely moved, the same asymmetry as the outer loop's (Section 15.8), and sixteen removes the interaction at a quarter of the cost of sixty-four. Because Ξ is fixed during the sweeps, each costs $O(CN_g^2)$ on a matrix that never grows with the dataset, which is why the M-step's cost per iteration is independent of M and why, at the operating grid, it is the M-step rather than the E-step that dominates a fit below a few thousand sequences.

15.4.3 Resolution

A mixture component can be narrower than a grid cell: the variance floor in the code is 10^{-6}, a standard deviation of 10^{-3}, against a spacing of 0.04. A spike narrower than the mesh can raise the quadrature likelihood without describing any feature of the innovation law. The diagnostic `scale_diagnostics` reports the narrowest fitted standard deviation in units of h and a flag `resolved` equal to $s_{\min} \geq 2h$; every reported cell carries it, the paper's Appendix C says such cells are flagged rather than dropped, and its capacity table marks them.

15.5 The checkpoint ladder and validation selection

15.5.1 The ladder

The protocol (`experiments/protocol.py`) fixes an iteration ladder

$$\{10, 20, 40, 60, 80, 120, 160, 220, 300, 400, 600, 800, 1200\}.$$

`fit_em` runs EM to the top rung and snapshots the kernel at each rung. The snapshot semantics are precise: at iteration k the kernel about to be scored is saved *before* its E-step, so the saved kernel at rung k is the one whose likelihood is the k-th trace entry; the final iterate is always available under its own index, so a caller asking for a rung past an early convergence break still receives the best kernel rather than an error. The likelihood tolerance (10^{-9} nats per transition, signed) can end a run before the top; at the paper's sizes it rarely does, and the run's `stop_reason` records which happened.

15.5.2 Selection and the `at_cap` flag

Which rung is reported is chosen on a **validation bundle**: sequences drawn from a random stream disjoint from the training stream, noised, and scored by the relative score error on the bulk sites (the four sites at each end trimmed) against the reference posterior means computed with the *true* kernel. The rung with the smallest validation error is selected and then scored, once, on a separate test bundle. The selected rung is compared with the top of the ladder, and the boolean `at_cap` records whether they coincide (`best == max(saved)` in `experiments/e02_capacity.py` and `experiments/e06_learning_through_noise.py`). A cell with `at_cap` true is one whose validation curve was still falling at the last rung, so its error is an upper bound on what a longer ladder would give; the paper reports the fraction of such cells (as the macro that its Section 4 calls the at-cap rate) and refuses to read a difference between two arms as final when either is at cap. E20 shows that the oracle in this rule is dispensable: selecting the rung by the held-out one-view log-evidence per transition, which needs neither the clean chain nor the true kernel, picks the same rung in most seeds and the same test error within noise.

15.6 The clean-data limit

As $t \to 0$ the unary factors sharpen to delta functions at the observations, the posterior collapses onto the clean chain, the pairwise beliefs become products of deltas, and Ξ degenerates to a histogram of the observed transitions. The code makes this limit an explicit input: `clean_statistics` in `src/markov_score/em.py` places each observed pair (a_{i-1}, a_i) on the grid by bilinear interpolation (a value a between u_k and u_{k+1} is split into weights $1-\theta$ and θ with $\theta = (a - u_k)/h$, which preserves the first moment exactly), and accumulates unit mass per transition. No quadrature weights enter: the clean Ξ is a counting measure, whereas the BP Ξ is belief mass that has already absorbed the weights; both sum to the number of transitions, which is the only property the M-steps rely on. `fit_clean` then applies the same M-step code, so the clean-data maximum-likelihood fit and the noisy-data EM differ only in how Ξ is built. The test `test_small_noise_em_approaches_clean_mle` pins the continuity between the two. E06 includes a clean arm as the "optimiser's own floor"; for one code path it is implemented as the channel at a tiny level, which the paper's Figure 3 shows as the dotted reference.

15.7 How the experiments use this

E02a (capacity). Laplace chain at the frozen configuration; for each seed and size M the same hashed bundles; mixture kernels with $C \in \{1, 2, 4, 8, 16\}$ from the same initialisation stream (ρ

initialised at 0.3, components spread over the innovation range), one view per sequence split over the twelve levels, the ladder above, the rung selected on validation identically for every capacity, then the test error in every region, the fitted ρ, innovation variance and excess kurtosis, the resolution flag and `at_cap`. The $C = 1$ arm must land on the LMMSE floor by the corollaries of Chapter 13; the paper's capacity table (Section 5) reports that it does, and that the mixture goes far below it. The earlier reading that "capacity does not help" came from a sweep in which only the single-Gaussian arm stopped at its optimum (the tolerance fired for it) while the richer arms ran to a fixed cap past theirs: the stopping rule, not the model, produced the verdict (the paper's Appendix F).

E06 (learning through noise). The same estimator with $C = 8$ and $M = 512$, but with every training sequence noised at *one* level t, over a ladder of levels and sixteen replicates, the validation bundle at the same level, the test bundle on the twelve-level schedule. Recorded per cell: the selected rung, `at_cap`, the stop reason, $\hat{\rho}$, the fitted innovation variance and excess kurtosis, the Hellinger distance of the fitted kernel to the truth, the resolution flag and the schedule-pooled test error. The paper's Figure 3 and Section 4 read these against the Cramér–Rao reference $1/\sqrt{MI_t}$ level by level: unimodal and at the reference at low levels; a split into a shrunk cluster and Gaussian-collapsed fits at an intermediate level; collapse where the reference crosses the coordinate's range; $\hat{\rho}$ recovered until past the crossover. At the two highest levels the selected rung is the top of the ladder in most cells, which is the budget question.

E06c. The control that answers it: the same cells with the ladder extended sixfold. Its result is what decides whether the excess of $\hat{\rho}$'s error over the reference at those levels is information or optimisation; the paper leaves the question open pending it, and Section 15.8 says why the budget binds exactly there.

15.8 The convergence asymmetry

The paper reports, as a fact about this model, that the correlation coordinate settles in tens of EM updates and the innovation shape in hundreds. Two mechanisms produce it, and both can be read off the information at each Hermite degree.

The outer loop. By the rate formula of Chapter 14, EM contracts the error along a coordinate by the factor $1 - I_{\text{obs}}/I_{\text{com}}$ per iteration, where I_{obs} is the information the noised data carry about the coordinate and I_{com} the information the clean chains would carry. The complete information is of the same order for both coordinates: clean transitions determine ρ and the innovation shape at the same rate in M. The observed information differs by the Hermite degree at which each coordinate first appears: by the paper's decay theorem, per sequence it is $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 e^{-4t}$ for the correlation (degree two) and $\frac{1}{24}\|\partial_{\boldsymbol{\theta}} \boldsymbol{\kappa}_4\|_F^2 e^{-8t}$ for a shape coordinate (degree four), up to corrections. So the missing fraction is closer to one for the shape at every level, by roughly a factor e^{-4t} in the retained fraction (0.37 at $t = 0.25$, 0.14 at $t = 0.5$, 0.018 at $t = 1$), and the number of iterations the shape needs grows correspondingly. In the clean limit both informations are complete and the asymmetry disappears, which the clean arm of E06 shows.

The inner loop. For the mixture kernel there is a second latent variable, the component label, and the shape parameters are learned through it: the M-step's inner sweeps are themselves an EM whose rate is governed by how well the responsibilities Equation (15.11) separate the components. A nearly Gaussian innovation gives nearly equal responsibilities, a missing fraction near one, and slow inner convergence, which the code's sixteen-sweep budget was introduced to absorb; ρ is not learned through the label at all (its block is a single least-squares solve) and does not suffer this. The two mechanisms compound, and both act on the same coordinate.

The consequence for the one-view protocol is the paper's arithmetic: of the twelve levels, spread over $[0.05, 3]$, only the lowest few carry appreciable information about the shape, since e^{-8t} has fallen by a factor of several hundred by $t = 0.7$, so for the shape the twelve-level schedule is effectively a four-level one, and the estimator's shape is learned from a quarter of the data at a rate set by the smallest of its information ratios. That is why the ladder is long, why the shape is what the inner budget affects, and why the validation-selected rung sits at the cap precisely at the levels where the information is smallest.

15.9 A worked example: the Gaussian M-step from a small Ξ

Return to the three-site, three-state chain of Chapter 11, grid $\{-1, 0, 1\}$. Its two pairwise posteriors summed over the two edges give, for that one sequence,

$$\Xi = \begin{pmatrix} 0.0802 & 0.1585 & 0.0665 \\ 0.0661 & 0.7356 & 0.2385 \\ 0.0251 & 0.2638 & 0.3656 \end{pmatrix}, \qquad W = \sum_{kl} \Xi_{kl} = 2,$$

rows indexed by the child u_k, columns by the parent u_l (the first edge matrix was written out in that chapter; the second is obtained the same way). The belief-weighted moments are

$$S_{00} = \sum_{kl} \Xi_{kl} u_l^2 = (0.0802 + 0.0661 + 0.0251) + (0.0665 + 0.2385 + 0.3656) = 0.8421,$$
$$S_{11} = \sum_{kl} \Xi_{kl} u_k^2 = (0.0802 + 0.1585 + 0.0665) + (0.0251 + 0.2638 + 0.3656) = 0.9598,$$
$$S_{01} = \sum_{kl} \Xi_{kl} u_k u_l = 0.0802 - 0.0665 - 0.0251 + 0.3656 = 0.3542,$$

where S_{01} collects the four corner entries with the sign of $u_k u_l$. Then Equation (15.8) gives $\rho = 0.3542/0.8421 = 0.4206$ and $q = (0.9598 - 0.4206 \cdot 0.3542)/2 = 0.4054$. Evaluating $\mathcal{Q}(\rho, q)$ of Equation (15.7) at this point gives -1.9350, and at the perturbed points (0.4706, 0.4054), (0.4206, 0.4459) and (0.3706, 0.3649) it gives -1.9376, -1.9394 and -1.9436: the closed form is the maximiser, as the test of the Gaussian M-step in `tests/test_em_bp.py` checks by random perturbation. A two-component mixture started at $\pi = (\frac{1}{2}, \frac{1}{2})$, $\nu = (-0.3, 0.3)$, $s^2 = (0.5, 0.5)$, $\rho = 0.2$ has $\langle \Xi, \log K_\theta \rangle / W = -1.0155$; one mixture block Equation (15.12) moves it to -0.9700 and the following autoregression block Equation (15.13) to -0.9592, with $\rho = 0.328$ after one sweep: each block increases the objective, as the conditional-maximisation argument requires. (The example has a single sequence and a three-point grid, so the numbers illustrate the algebra, not the statistics.)

Where it is used. The paper, Section 4 (the marginal likelihood, Fisher's identity and equation *fisher*, the Yule–Walker M-step, the mixture kernel and its conditional-maximisation sweeps, the one-view protocol and validation selection, the convergence asymmetry and the twelve-level arithmetic), Appendix C (the ladder and its selection rule, the resolution flag), Appendix E (E19's moment estimators as the competitor, E20's oracle-free selection), Appendix F (the stopping-rule artefact of the earlier capacity reading). Code: the E-step, its multi-level sum, the clean-data statistics, the two fitting loops and the Q-function helpers are all in `src/markov_score/em.py`; the three kernels, with the mixture's `innovation_moments` and `scale_diagnostics`, are in `src/markov_score/kernels.py`; the ladder `em_checkpoints` is in `experiments/protocol.py`; the drivers are `experiments/e02_capacity.py`, `experiments/e06_learning_through_noise.py` and `experiments/e20_evidence_selection.py`. Tests: `tests/test_em_bp.py`, `tests/test_fisher_bruteforce.py` and `tests/test_gradient_trace_alignment.py`.

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Chapter 16

Quasi-likelihood and moment estimators

The paper's headline estimator maximises the exact marginal likelihood of the noised data by expectation–maximisation around the tree-exact recursion (Chapters 14 and 15). Two simpler estimators appear beside it. The first is the *Gaussian quasi-likelihood*: the Gaussian log-likelihood written down for data that are not Gaussian. The paper needs it in one sentence of Section 2 (a Gaussian-innovation kernel fitted by maximum likelihood is consistent for (ρ, q) and lands on the LMMSE floor) and uses that sentence as an apparatus check in every capacity table and in the fifteen cells of the regime sweep. The second is the *method of moments*: two closed-form statistics, one for the correlation and one for the fourth cumulant, which the paper places against EM as the rate-correct, inefficient competitor at high noise (Section 4 of the paper, experiment E19). This chapter builds both from scratch, derives the two properties the paper asserts (consistency of the quasi-MLE; unbiasedness of the moment estimators), computes their variances and their efficiency relative to maximum likelihood, and records how E19 measured them.

16.1 The Gaussian quasi-likelihood of noised data

16.1.1 Likelihood of the wrong model

Fix one noise level t and write $\alpha = e^{-t}$, $\Delta_t = 1 - \alpha^2$. A noised sequence is $\mathbf{x} = \alpha\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$ with $\mathbf{a} \sim P_0$ and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$. Whatever the innovation law, the chain of the paper has $\mathbb{E}\mathbf{a} = 0$ and $\mathrm{Cov}(\mathbf{a}) = \mathbf{\Sigma}_0$ with $(\mathbf{\Sigma}_0)_{ij} = \rho^{|i-j|}$, so

$$\mathbb{E}\mathbf{x} = 0, \qquad \mathrm{Cov}(\mathbf{x}) = \alpha^2\mathbf{\Sigma}_0 + \Delta_t\mathbf{I} =: \mathbf{\Sigma}_t. \tag{16.1}$$

Only the first two moments of P_0 enter Equation (16.1); this is the whole reason the Gaussian quasi-likelihood works.

Definition 16.1 (Gaussian quasi-likelihood)**.** Let $\mathbf{\Sigma}_t(\theta) = e^{-2t}\mathbf{\Sigma}_0(\theta) + \Delta_t\mathbf{I}$ be the covariance that the Gaussian chain with parameters $\theta = (\rho, q)$ would give to a noised sequence at level t. The **Gaussian quasi-likelihood** of M noised sequences $\mathbf{x}^1, \dots, \mathbf{x}^M$ observed at level t is the Gaussian log-likelihood

$$\ell_{\mathrm{G}}(\theta) = -\frac{1}{2}\sum_{\mu=1}^{M}\Big[\log\det\mathbf{\Sigma}_t(\theta) + \mathbf{x}^{\mu\top}\mathbf{\Sigma}_t(\theta)^{-1}\mathbf{x}^{\mu}\Big] - \frac{ML}{2}\log 2\pi, \tag{16.2}$$

evaluated on data whose true law is not Gaussian. Its maximiser is the **quasi-maximum-likelihood estimator** (quasi-MLE) $\hat{\theta}_{\mathrm{G}}$.

The word *quasi* records that Equation (16.2) is not the log-density of the data. The true log-density of $\mathbf{x}$ is $\log P_t(\mathbf{x})$, a non-Gaussian function whose evaluation needs the forward–backward sweep of Chapter 11. The Gaussian quasi-likelihood is what one obtains by pretending that the innovation law is Gaussian, and Equation (16.2) is then exactly the marginal likelihood of the Gaussian chain, since the Gaussian chain noised by a Gaussian channel is Gaussian with covariance $\mathbf{\Sigma}_t(\theta)$.

16.1.2 It depends on the data only through the sample covariance

Let $\mathbf{S} = M^{-1}\sum_\mu \mathbf{x}^\mu \mathbf{x}^{\mu\top}$ be the **sample covariance** (about zero, because the mean is known to be zero). Using $\mathbf{x}^\top \mathbf{A}\mathbf{x} = \operatorname{tr}(\mathbf{A}\mathbf{x}\mathbf{x}^\top)$ and the linearity of the trace,

$$\sum_\mu \mathbf{x}^{\mu\top}\mathbf{\Sigma}_t^{-1}\mathbf{x}^\mu = \operatorname{tr}\Big(\mathbf{\Sigma}_t^{-1}\sum_\mu \mathbf{x}^\mu\mathbf{x}^{\mu\top}\Big) = M\ \operatorname{tr}(\mathbf{\Sigma}_t^{-1}\mathbf{S}), \tag{16.3}$$

so

$$\ell_{\mathrm{G}}(\theta) = -\frac{M}{2}\Big[\log\det\mathbf{\Sigma}_t(\theta) + \operatorname{tr}\big(\mathbf{\Sigma}_t(\theta)^{-1}\mathbf{S}\big)\Big] + \text{const.} \tag{16.4}$$

This is the statement in the paper's Section 2: *the Gaussian quasi-likelihood of noised data depends on the data only through their covariance.* Two remarks make it precise. First, the dependence is through the *sample* covariance $\mathbf{S}$, a random matrix; the population statement is about its expectation $\mathbb{E}\mathbf{S} = \mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I}$, which is Equation (16.1). Second, nothing about the shape of P_0 beyond its covariance can influence $\hat\theta_{\mathrm{G}}$: two data sets with the same $\mathbf{S}$ give the same quasi-MLE, whether their innovations were Laplace, uniform or Gaussian.

16.1.3 The one-view protocol: a sum over levels

Under the paper's one-view protocol each sequence is observed at one level t_μ and the M sequences are split evenly across the twelve levels of the schedule. The quasi-likelihood is then a sum of twelve terms of the form Equation (16.4), one per level, each with its own sample covariance $\mathbf{S}_t$ over the $M/12$ sequences at that level:

$$\ell_{\mathrm{G}}(\theta) = -\frac{1}{2}\sum_t M_t\Big[\log\det\mathbf{\Sigma}_t(\theta) + \operatorname{tr}\big(\mathbf{\Sigma}_t(\theta)^{-1}\mathbf{S}_t\big)\Big] + \text{const}, \qquad M_t = M/12. \tag{16.5}$$

Every level identifies the same θ, so the sum only adds information; the argument below is given for one level and applies term by term.

16.2 Consistency of the quasi-MLE for (ρ, q)

16.2.1 The population objective is a Kullback–Leibler divergence

Divide Equation (16.4) by M and let $M \to \infty$. By the law of large numbers $\mathbf{S} \to \mathbf{\Sigma}_t^\star := e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I}$, the true covariance of the noised data, so the normalised quasi-likelihood converges, uniformly on compact parameter sets, to the **population quasi-likelihood**

$$\bar\ell(\theta) = -\frac{1}{2}\Big[\log\det\mathbf{\Sigma}_t(\theta) + \operatorname{tr}\big(\mathbf{\Sigma}_t(\theta)^{-1}\mathbf{\Sigma}_t^\star\big)\Big]. \tag{16.6}$$

Compare this with the Kullback–Leibler divergence between two centred Gaussians (Chapter 1 for the definition of D_{KL}):

$$D_{\mathrm{KL}}\big(\mathcal{N}(0,\mathbf{\Sigma}^\star)\,\|\,\mathcal{N}(0,\mathbf{\Sigma})\big) = \frac{1}{2}\Big[\operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{\Sigma}^\star) - L + \log\det\mathbf{\Sigma} - \log\det\mathbf{\Sigma}^\star\Big]. \tag{16.7}$$

The derivation of Equation (16.7) is short. With $p = \mathcal{N}(0, \boldsymbol{\Sigma}^\star)$ and $q = \mathcal{N}(0, \boldsymbol{\Sigma})$, $\log(p/q) = \frac{1}{2}[\log\det\boldsymbol{\Sigma} - \log\det\boldsymbol{\Sigma}^\star] - \frac{1}{2}\mathbf{x}^\top(\boldsymbol{\Sigma}^{\star-1} - \boldsymbol{\Sigma}^{-1})\mathbf{x}$, and $\mathbb{E}_p[\mathbf{x}^\top\mathbf{A}\mathbf{x}] = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^\star)$ gives $\mathbb{E}_p[\mathbf{x}^\top\boldsymbol{\Sigma}^{\star-1}\mathbf{x}] = L$ and $\mathbb{E}_p[\mathbf{x}^\top\boldsymbol{\Sigma}^{-1}\mathbf{x}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}^\star)$. Hence

$$\bar{\ell}(\theta) = -D_{\mathrm{KL}}\big(\mathcal{N}(0, \boldsymbol{\Sigma}_t^\star) \,\big\|\, \mathcal{N}(0, \boldsymbol{\Sigma}_t(\theta))\big) + \text{const}, \tag{16.8}$$

where the constant does not depend on θ. Maximising the population quasi-likelihood is minimising a Kullback–Leibler divergence between Gaussians, and that divergence is zero if and only if $\boldsymbol{\Sigma}_t(\theta) = \boldsymbol{\Sigma}_t^\star$, because $D_{\mathrm{KL}}(p \,\|\, q) \geq 0$ with equality only at $p = q$ (Chapter 1).

16.2.2 Identifiability of (ρ, q) from the noised covariance

It remains to show that $\boldsymbol{\Sigma}_t(\theta) = \boldsymbol{\Sigma}_t^\star$ forces $\theta = \theta^\star$. Since $\boldsymbol{\Sigma}_t(\theta) - \Delta_t\mathbf{I} = e^{-2t}\boldsymbol{\Sigma}_0(\theta)$ and $e^{-2t} > 0$, equality of the two noised covariances is equality of the two clean covariances, $\boldsymbol{\Sigma}_0(\theta) = \boldsymbol{\Sigma}_0(\theta^\star)$. For the chain with $a_1 \sim \mathcal{N}(0, 1)$ and general (ρ, q) (the two-parameter family the EM fits), the code's `theory.chain_covariance` gives

$$(\boldsymbol{\Sigma}_0)_{ij} = \rho^{|i-j|}\, v_{\min(i,j)}, \qquad v_i = \rho^{2(i-1)} + q\,\frac{1 - \rho^{2(i-1)}}{1 - \rho^2}, \tag{16.9}$$

so $(\boldsymbol{\Sigma}_0)_{11} = 1$, $(\boldsymbol{\Sigma}_0)_{12} = \rho$ and $(\boldsymbol{\Sigma}_0)_{22} = \rho^2 + q$: the $(1,2)$ entry reads off ρ and the $(2,2)$ entry then reads off q. Two different parameter pairs give two different covariances, so θ is identifiable from $\boldsymbol{\Sigma}_0$, hence from $\boldsymbol{\Sigma}_t^\star$.

Proposition 16.2 (Consistency of the Gaussian quasi-MLE)**.** *Let the noised data at level t have a law with mean zero and covariance $\boldsymbol{\Sigma}_t^\star = e^{-2t}\boldsymbol{\Sigma}_0(\theta^\star) + \Delta_t\mathbf{I}$, and let $\hat{\theta}_{\mathrm{G}}$ maximise Equation* (16.2) *over a compact set containing $\theta^\star$ in its interior. Then $\hat{\theta}_{\mathrm{G}} \to \theta^\star$ in probability as $M \to \infty$, whatever the innovation law.*

Proof. The normalised objective converges uniformly to Equation (16.6), which is continuous and has the unique maximiser $\theta^\star$ by the two facts above; a standard argument (the argmax of a uniformly convergent sequence of continuous functions converges to the unique argmax of the limit) finishes the proof. □

This is the special case of White's theory of misspecified maximum likelihood (Chapter 18) in which the pseudo-true parameter coincides with the true one: the model is wrong about the shape of the law but right about its first two moments, and the Gaussian likelihood only ever asks about those.

16.2.3 The quasi-score equations are moment equations

Differentiate Equation (16.4) with respect to a parameter θ using $\partial\log\det\boldsymbol{\Sigma} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial\boldsymbol{\Sigma})$ and $\partial\boldsymbol{\Sigma}^{-1} = -\boldsymbol{\Sigma}^{-1}(\partial\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}$:

$$\frac{\partial\ell_{\mathrm{G}}}{\partial\theta} = \frac{M}{2}\,\operatorname{tr}\Big(\boldsymbol{\Sigma}_t^{-1}\,\partial_\theta\boldsymbol{\Sigma}_t\,\boldsymbol{\Sigma}_t^{-1}\,(\mathbf{S} - \boldsymbol{\Sigma}_t)\Big). \tag{16.10}$$

The **quasi-score** vanishes when a weighted trace of $\mathbf{S} - \boldsymbol{\Sigma}_t(\theta)$ vanishes: the quasi-MLE solves *moment equations*, one per parameter, that ask the sample covariance to agree with the model covariance in the directions $\mathbf{A}_\theta = \boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_t\boldsymbol{\Sigma}_t^{-1}$. That is the sense in which the quasi-MLE belongs in this chapter beside the method of moments: it is a moment estimator with a particular, likelihood-derived choice of weights.

16.2.4 Variance of the quasi-MLE: the sandwich

The quasi-MLE is consistent for every innovation law, but its *variance* is not the Gaussian one when the data are not Gaussian. The standard result for an estimator that solves $\sum_\mu \psi(\mathbf{x}^\mu;\theta) = 0$ (an **M-estimator**) is

$$\sqrt{M}\,(\hat\theta - \theta^\star) \;\to\; \mathcal{N}\Big(0, \mathbf{J}^{-1}\mathbf{V}\mathbf{J}^{-1}\Big), \qquad \mathbf{J} = -\mathbb{E}\,\partial_\theta\psi, \quad \mathbf{V} = \mathrm{Var}\;\psi, \tag{16.11}$$

the **sandwich covariance**. For the quasi-score $\psi_\theta = \frac{1}{2}\,\mathrm{tr}(\mathbf{A}_\theta(\mathbf{x}\mathbf{x}^\top - \mathbf{\Sigma}_t))$ the outer matrices $\mathbf{J}$ are the Gaussian Fisher information $\frac{1}{2}\,\mathrm{tr}(\mathbf{\Sigma}_t^{-1}\partial_i\mathbf{\Sigma}_t\mathbf{\Sigma}_t^{-1}\partial_j\mathbf{\Sigma}_t)$ of the paper's Theorem 3, and $\mathbf{V}$ needs the variance of a quadratic form under the true law.

Lemma 16.3 (Variance of a quadratic form)**.** *Let* $\mathbf{x}$ *have mean zero, covariance* $\mathbf{\Sigma}$ *and fourth-cumulant tensor* $\boldsymbol{\kappa}_4[\mathbf{x}]$ *(Chapter 2), and let* $\mathbf{A}$ *be symmetric. Then*

$$\mathrm{Var}(\mathbf{x}^\top\mathbf{A}\mathbf{x}) = 2\,\mathrm{tr}(\mathbf{A}\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma}) + \sum_{ijkl} A_{ij}A_{kl}\,(\boldsymbol{\kappa}_4[\mathbf{x}])_{ijkl}. \tag{16.12}$$

Proof. $\mathrm{Var}(\mathbf{x}^\top\mathbf{A}\mathbf{x}) = \sum_{ijkl} A_{ij}A_{kl}(\mathbb{E}[x_ix_jx_kx_l] - \mathbf{\Sigma}_{ij}\mathbf{\Sigma}_{kl})$. The fourth moment of a zero-mean law is its fourth cumulant plus the three pairings of second moments (Chapter 2): $\mathbb{E}[x_ix_jx_kx_l] = \boldsymbol{\kappa}_4[ijkl] + \mathbf{\Sigma}_{ij}\mathbf{\Sigma}_{kl} + \mathbf{\Sigma}_{ik}\mathbf{\Sigma}_{jl} + \mathbf{\Sigma}_{il}\mathbf{\Sigma}_{jk}$. The first pairing cancels against $\mathbf{\Sigma}_{ij}\mathbf{\Sigma}_{kl}$; the other two each contract to $\mathrm{tr}(\mathbf{A}\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma})$ by symmetry of $\mathbf{A}$. □

For the noised chain, $\boldsymbol{\kappa}_4[\mathbf{x}] = e^{-4t}\boldsymbol{\kappa}_4[\mathbf{a}]$ exactly (the cumulant consequence of the paper's Theorem 2, Chapter 10), so the non-Gaussian part of $\mathbf{V}$ is e^{-4t} times a fixed contraction of $\boldsymbol{\kappa}_4[\mathbf{a}]$, while the Gaussian part $2\,\mathrm{tr}(\mathbf{A}\mathbf{\Sigma}_t\mathbf{A}\mathbf{\Sigma}_t)$ is of order e^{-4t} as well through $\mathbf{A}_\theta \propto \partial_\theta\mathbf{\Sigma}_t = e^{-2t}\partial_\theta\mathbf{\Sigma}_0$. The ratio of the two is therefore of order one at every level, not decaying: the quasi-MLE of a Laplace chain is consistent but has a different (typically larger) variance than the same estimator on a Gaussian chain, by an amount proportional to the excess kurtosis. This matters only for error bars; the paper never quotes a variance of the $C = 1$ arm, so the consistency statement is the one it uses.

16.2.5 Why this is an apparatus check

Put together with Proposition 1 of the paper (Chapter 13), the chain of implications is:

1. the $C = 1$ arm of EM–BP fits a Gaussian-innovation kernel by maximum likelihood of the noised data; with a Gaussian kernel the exact marginal likelihood that EM maximises *is* the Gaussian quasi-likelihood Equation (16.5) (up to the grid discretisation), so EM's fixed point is the quasi-MLE;

2. by Proposition 16.2 the quasi-MLE converges to the true (ρ, q);

3. by Proposition 1, sum-product with any Gaussian-innovation kernel returns the LMMSE estimator of that kernel's covariance; at the true (ρ, q) this is the LMMSE of the true covariance, the second-order floor.

So the $C = 1$ row of every capacity table must sit on the LMMSE floor, up to the $O(M^{-1/2})$ sampling error of step 2 and the checkpoint selection. The paper reports that it does, at every sample size, at every member of the generalised-Gaussian family and in all fifteen (ρ, L) cells of the regime sweep (see the paper, Section 5 and Table 1, and the regime paragraph; E02a, E02b, E13). A failure of that identity anywhere would indict the grid, the message passing, the M-step or the scoring, which is why the audit calls it the most reassuring fact about the pipeline.

16.3 The method of moments in general

Definition 16.4 (Method of moments)**.** Let $\mathbf{x}^1, \dots, \mathbf{x}^M$ be i.i.d. with law P_θ, and let $g(\mathbf{x}) \in \mathbb{R}^d$ be a vector of statistics whose expectation $m(\theta) = \mathbb{E}_\theta\, g(\mathbf{x})$ is known in closed form and injective in θ. The **method-of-moments estimator** is the solution of

$$\frac{1}{M}\sum_{\mu=1}^{M} g(\mathbf{x}^\mu) = m(\hat\theta), \qquad \text{i.e.} \qquad \hat\theta = m^{-1}(\bar g). \tag{16.13}$$

Three facts follow from the definition. First, $\bar g$ is an unbiased estimator of $m(\theta)$ for every M, so when m is linear the estimator $\hat\theta$ is exactly unbiased; when m is nonlinear the plug-in $m^{-1}(\bar g)$ is unbiased only to order M^{-1} (the delta method, Chapter 6). Second, by the central limit theorem and the delta method,

$$\operatorname{Var}(\hat\theta) \approx \frac{1}{M}\,(\partial m)^{-1}\,\operatorname{Var}(g)\,(\partial m)^{-\top}, \tag{16.14}$$

which is Equation (16.11) with $\psi = g - m(\theta)$ and $\mathbf{J} = \partial m$. Third, the estimator is asymptotically efficient, that is, attains the Cramér–Rao bound (Chapter 5), exactly when the statistic g is an affine function of the score $\partial_\theta \log P_\theta$: otherwise the ratio of the Cramér–Rao variance to Equation (16.14), the **asymptotic relative efficiency**, is strictly below one.

Moment estimators through the channel. The paper's Theorem 2 (Hermite attenuation, Chapter 10) is what makes the method of moments available for noised data without any inference: it says that for every multi-index ν,

$$\mathbb{E}_{P_t}[H_\nu(\mathbf{x})] = e^{-|\nu| t}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})], \tag{16.15}$$

so any Hermite moment of the clean law has a known image under the channel, and the sample Hermite moment of the noised data, divided by $e^{-|\nu|t}$, is an unbiased estimator of the clean one. The two estimators below are the simplest instances, at degree two and at degree four.

16.4 The paper's two moment estimators

Throughout this section the data are M sequences observed at one common level t, as in E06 and E19 (not the twelve-level split), and $\alpha = e^{-t}$.

16.4.1 The lag-one product for the correlation

Definition 16.5 (Lag-one moment estimator)**.**

$$\hat\rho_{\mathrm{mom}} = \frac{1}{\alpha^2}\cdot\frac{1}{M(L-1)}\sum_{\mu=1}^{M}\sum_{i=1}^{L-1} x_i^\mu\, x_{i+1}^\mu. \tag{16.16}$$

Proposition 16.6 (Unbiasedness)**.** *For the chain of the paper, $\mathbb{E}\,\hat\rho_{\mathrm{mom}} = \rho$ exactly, at every t, every M and every innovation law of mean zero and variance $q = 1 - \rho^2$.*

Proof. $x_i x_{i+1} = (\alpha a_i + \sqrt{\Delta_t} z_i)(\alpha a_{i+1} + \sqrt{\Delta_t} z_{i+1})$. Expanding, the cross terms have a factor $\mathbb{E}z_i = 0$ or $\mathbb{E}z_{i+1} = 0$, and $\mathbb{E}[z_i z_{i+1}] = 0$ because the channel noise is independent across sites; so $\mathbb{E}[x_i x_{i+1}] = \alpha^2 \mathbb{E}[a_i a_{i+1}]$. For the chain, $a_{i+1} = \rho a_i + \varepsilon_{i+1}$ with ε_{i+1} independent of a_i and of mean zero, so $\mathbb{E}[a_i a_{i+1}] = \rho \mathbb{E}[a_i^2] = \rho \operatorname{Var}(a_i)$. The variance choice $q = 1 - \rho^2$ with $a_1 \sim \mathcal{N}(0,1)$ gives $\operatorname{Var}(a_i) = 1$ at every site (induction: $\operatorname{Var}(a_{i+1}) = \rho^2 \operatorname{Var}(a_i) + q = \rho^2 + 1 - \rho^2 = 1$). Hence $\mathbb{E}[x_i x_{i+1}] = \alpha^2 \rho$ for every i, and dividing by α^2 and averaging gives ρ. □

The statistic $x_i x_{i+1}$ is, up to the constant α^2, the degree-two Hermite moment with multi-index $\nu = e_i + e_{i+1}$, that is, the expectation of $\mathrm{He}_1(x_i)\,\mathrm{He}_1(x_{i+1})$; Proposition 16.6 is Theorem 2 at that multi-index. The estimator is a linear function of the data products, so it is unbiased exactly, not only asymptotically.

Variance and the e^{2t} growth. Write $T = \frac{1}{M(L-1)}\sum_{\mu,i} x_i^\mu x_{i+1}^\mu$, so $\hat\rho_{\mathrm{mom}} = \alpha^{-2}T$ and $\mathrm{Var}(\hat\rho_{\mathrm{mom}}) = e^{4t}\,\mathrm{Var}(T)$. Sequences are independent, so $\mathrm{Var}(T) = M^{-1}\,\mathrm{Var}(\bar p)$ with $\bar p = (L-1)^{-1}\sum_i x_i x_{i+1}$ the within-sequence average of the products. For a Gaussian chain Isserlis' theorem (Chapter 3) gives, with $c_k = \mathrm{Cov}(x_i, x_{i+k}) = \alpha^2\rho^k$ for $k \geq 1$ and $c_0 = 1$,

$$\mathrm{Var}(x_i x_{i+1}) = \mathbb{E}[x_i^2 x_{i+1}^2] - c_1^2 = (1+2c_1^2) - c_1^2 = 1 + c_1^2, \qquad \mathrm{Cov}(x_i x_{i+1}, x_{i+k} x_{i+k+1}) = c_k^2 + c_{k-1}c_{k+1}, \tag{16.17}$$

and for a non-Gaussian chain each of these acquires a fourth-cumulant term of order $e^{-4t}\boldsymbol{\kappa}_4$ (Lemma 16.3). All the covariance terms are $O(\alpha^4)$ or $O(\alpha^2)$ while the leading 1 is not, so as t grows

$$\mathrm{Var}(\hat\rho_{\mathrm{mom}}) = \frac{e^{4t}}{M(L-1)}\big(1 + O(e^{-2t})\big), \qquad \mathrm{RMSE}(\hat\rho_{\mathrm{mom}}) = \frac{e^{2t}}{\sqrt{M(L-1)}}\big(1 + O(e^{-2t})\big). \tag{16.18}$$

The error of the degree-two moment estimator grows as e^{2t}: the amplitude of the degree-two signal shrinks as e^{-2t} while the noise in the product stays of order one. This is the first of the two slopes E19 fits (see the paper, Section 4 and Appendix A5).

16.4.2 The per-site fourth cumulant for the shape

The shape coordinate the paper tracks is the excess kurtosis of the innovation, $\boldsymbol{\kappa}_4(\varepsilon)/q^2$, which is $+3$ for the Laplace law. To estimate it from noised data one needs the fourth cumulant of a site of the chain and its image under the channel.

Lemma 16.7 (Fourth cumulant of a chain site)**.** *For the chain $a_1 \sim \mathcal{N}(0,1)$, $a_i = \rho a_{i-1} + \varepsilon_i$ ($i \geq 2$) with i.i.d. innovations of fourth cumulant $\boldsymbol{\kappa}_4(\varepsilon)$,*

$$\boldsymbol{\kappa}_4(a_i) = \boldsymbol{\kappa}_4(\varepsilon)\sum_{m=2}^{i}\rho^{4(i-m)} = \boldsymbol{\kappa}_4(\varepsilon)\,\frac{1-\rho^{4(i-1)}}{1-\rho^4} =: \boldsymbol{\kappa}_4(\varepsilon)\,c_i(\rho). \tag{16.19}$$

Proof. Unrolling the recursion, $a_i = \rho^{i-1}a_1 + \sum_{m=2}^{i}\rho^{i-m}\varepsilon_m$, a sum of independent terms. Cumulants are additive over independent summands and homogeneous of degree four under scaling (Chapter 2): $\boldsymbol{\kappa}_4(\lambda Y) = \lambda^4\boldsymbol{\kappa}_4(Y)$. The Gaussian a_1 has $\boldsymbol{\kappa}_4(a_1) = 0$, so $\boldsymbol{\kappa}_4(a_i) = \sum_{m=2}^{i}\rho^{4(i-m)}\boldsymbol{\kappa}_4(\varepsilon)$, a geometric sum with ratio ρ^4 and $i-1$ terms. □

Under the channel, $x_i = \alpha a_i + \sqrt{\Delta_t}\,z_i$ with z_i Gaussian and independent of a_i, so by the same two properties $\boldsymbol{\kappa}_4(x_i) = \alpha^4\boldsymbol{\kappa}_4(a_i)$: this is the cumulant form of Theorem 2 at degree four. Since x_i has mean zero and variance one at every t, $\boldsymbol{\kappa}_4(x_i) = \mathbb{E}x_i^4 - 3(\mathbb{E}x_i^2)^2 = \mathbb{E}x_i^4 - 3$.

Definition 16.8 (Per-site fourth-cumulant estimator)**.** With $\hat m_{2,i} = M^{-1}\sum_\mu (x_i^\mu)^2$ and $\hat m_{4,i} = M^{-1}\sum_\mu (x_i^\mu)^4$, let $\hat k_{4,i} = \hat m_{4,i} - 3\hat m_{2,i}^2$ be the sample fourth cumulant at site i. The estimator of the innovation's fourth cumulant, and of its excess kurtosis, is

$$\hat{\boldsymbol{\kappa}}_4(\varepsilon) = \frac{\sum_{i=1}^{L}\hat k_{4,i}}{\alpha^4\sum_{i=1}^{L}c_i(\hat\rho)}, \qquad \widehat{\text{exc. kurt.}} = \frac{\hat{\boldsymbol{\kappa}}_4(\varepsilon)}{\hat q^{\,2}}, \quad \hat q = 1 - \hat\rho^2, \tag{16.20}$$

with $\hat\rho = \hat\rho_{\mathrm{mom}}$ from Equation (16.16) (E19 also records the version with the true ρ substituted, to separate the two sources of error).

Proposition 16.9 (Unbiasedness at the population level)**.** *Replace the sample cumulants in Equation* (16.20) *by their population values and $\hat\rho$ by ρ. Then the right-hand side equals $\boldsymbol{\kappa}_4(\varepsilon)$ exactly, at every t.*

Proof. $\sum_i \boldsymbol{\kappa}_4(x_i) = \alpha^4\sum_i \boldsymbol{\kappa}_4(a_i) = \alpha^4\boldsymbol{\kappa}_4(\varepsilon)\sum_i c_i(\rho)$ by Lemma 16.7 and the cumulant scaling; divide. □

This is the sense in which the paper's Section 4 says both estimators are "unbiased by Theorem 2". It is a statement about the population identity, and it is exact. The finite-sample estimator, however, is a nonlinear function of sample moments, and its bias deserves its own paragraph, because the channel amplifies it.

Finite-sample bias of the plug-in cumulant. The sample fourth cumulant $\hat{k}_4 = \hat{m}_4 - 3\hat{m}_2^2$ is not unbiased: $\hat{m}_4$ is unbiased for m_4, but $\mathbb{E}[\hat{m}_2^2] = (\mathbb{E}\hat{m}_2)^2 + \mathrm{Var}(\hat{m}_2) = m_2^2 + (m_4 - m_2^2)/M$, so

$$\mathbb{E}\hat{k}_4 = m_4 - 3m_2^2 - \frac{3(m_4 - m_2^2)}{M} = \boldsymbol{\kappa}_4\Big(1 - \frac{3}{M}\Big) - \frac{6m_2^2}{M}. \tag{16.21}$$

At unit variance the bias is $-(3\boldsymbol{\kappa}_4(x_i) + 6)/M$, of which the $-6/M$ term does not decay with t. Summed over L sites and divided by $\alpha^4 \sum_i c_i$ and q^2, the bias in excess-kurtosis units is about

$$-\frac{6L}{M\,q^2\,e^{-4t}\sum_i c_i(\rho)} \approx -\frac{6\,(1-\rho^4)}{M\,q^2}\,e^{4t} \qquad (\text{large } L), \tag{16.22}$$

which grows as e^{4t}: the $O(1/M)$ bias of a moment estimator is multiplied by the same e^{4t} that multiplies its standard deviation. At $M = 512$, $\rho = 0.85$ and $q = 1 - \rho^2 = 0.2775$ this is about -0.4 at $t = 0.4$ and about -4 at $t = 1$, that is, of the same order as the coordinate's whole range (3) at the level where the paper's own reference bound crosses that range. The remedy is classical: Fisher's **k-statistic** $k_4 = M^2[(M+1)\hat{m}_4 - 3(M-1)\hat{m}_2^2]/[(M-1)(M-2)(M-3)]$ is exactly unbiased for $\boldsymbol{\kappa}_4$ at every M. E19 uses the plug-in form, so its $\hat{\boldsymbol{\kappa}}_4$ is unbiased only up to $O(e^{4t}/M)$; the script's docstring says "unbiased up to $O(1/M)$", and the paper's Appendix A5 reports the measured bias at $t = 0.4$ (see the paper's macro for that level). A second, smaller $O(1/M)$ bias comes from dividing by $c_i(\hat{\rho})$ and $\hat{q}^2$, nonlinear functions of the estimated $\hat{\rho}$.

Variance and the e^{4t} growth. Each $\hat{k}_{4,i}$ has variance of order $M^{-1}\,\mathrm{Var}(x_i^4) = O(M^{-1})$ with a leading constant that does not decay in t (for a standard Gaussian $\mathrm{Var}(x^4) = 96$), while the signal $\boldsymbol{\kappa}_4(x_i) = \alpha^4\boldsymbol{\kappa}_4(a_i)$ decays as e^{-4t}. Dividing by α^4 therefore multiplies the standard deviation by e^{4t}:

$$\mathrm{RMSE}\big(\hat{\boldsymbol{\kappa}}_4(\varepsilon)\big) \propto \frac{e^{4t}}{\sqrt{M}} \qquad (t \to \infty). \tag{16.23}$$

The two slopes, 2 for the degree-two coordinate and 4 for the degree-four one, are the "three clocks" of the paper's Section 3 read in the sample-size clock, and they are what the paper's Appendix A5 reports E19 measuring on $t \geq 1$ (the log-slopes are quoted there as macros).

No collapse. EM at high noise collapses the shape estimate to a Gaussian innovation (fitted excess kurtosis near zero), because the likelihood surface is flat in the shape direction and the validation-selected early stopping shrinks toward the initialisation (Chapter 15; the paper's Section 4). The moment estimator cannot do this: it is a polynomial in the data with no boundary, so its error grows without saturating and its sign is unconstrained. The paper uses this contrast in one sentence ("never collapse"), and it is the reason the asymptotic slopes are visible in E19 where they are not in E06.

16.5 Relative efficiency against maximum likelihood

16.5.1 The reference

Chapter 5 defines the Fisher information and the Cramér–Rao bound; Chapter 10 gives the paper's Theorem 3, which for one noised sequence at level t reads

$$I_t(\rho) = \tfrac{1}{2}\|\partial_\rho \boldsymbol{\Sigma}_0\|_F^2 e^{-4t}(1 + O(e^{-2t})), \qquad I_t(\beta) = \tfrac{1}{24}\|\partial_\beta \boldsymbol{\kappa}_4\|_F^2 e^{-8t}(1 + O(e^{-2t})), \tag{16.24}$$

for a covariance coordinate and a shape coordinate respectively. The reference RMSE for an unbiased estimator from M sequences is $1/\sqrt{MI_t}$, which grows as e^{2t} and e^{4t}: the moment estimators have the right *rates* by construction. What separates them from maximum likelihood is the *constant*.

16.5.2 The lag-one estimator against the covariance information

For the Gaussian chain the information about ρ is exact at every t (the paper's Theorem 3, last sentence), and its $t \to \infty$ form is $e^{-4t} \cdot \frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2$. With $(\mathbf{\Sigma}_0)_{ij} = \rho^{|i-j|}$, the derivative has entries $|i-j|\rho^{|i-j|-1}$ off the diagonal and zero on it, so

$$\tfrac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 = \tfrac{1}{2}\sum_{i \neq j}(|i-j|)^2 \rho^{2(|i-j|-1)} = \sum_{k=1}^{L-1}(L-k)\, k^2 \rho^{2(k-1)}. \tag{16.25}$$

Comparing with Equation (16.18), the asymptotic relative efficiency of the lag-one estimator is

$$\mathrm{ARE}(\hat{\rho}_{\mathrm{mom}}) = \lim_{t\to\infty} \frac{1/(MI_t(\rho))}{\mathrm{Var}(\hat{\rho}_{\mathrm{mom}})} = \frac{L-1}{\sum_{k=1}^{L-1}(L-k)k^2\rho^{2(k-1)}} \;\leq\; 1, \tag{16.26}$$

with equality if and only if $\rho = 0$. For a long chain the per-site sum converges, $\sum_{k\geq 1} k^2 r^{k-1} = (1+r)/(1-r)^3$ with $r = \rho^2$, so

$$\mathrm{ARE}(\hat{\rho}_{\mathrm{mom}}) \;\xrightarrow[L\to\infty]{}\; \frac{(1-\rho^2)^3}{1+\rho^2}. \tag{16.27}$$

The reason is visible in the score. At high noise the likelihood score of ρ is, to leading order, the degree-two Hermite statistic $\frac{1}{2}\alpha^2\langle \partial_\rho \mathbf{\Sigma}_0, :\mathbf{x}\mathbf{x}^\top:\rangle = \frac{1}{2}\alpha^2 \sum_{i\neq j}|i-j|\rho^{|i-j|-1} x_i x_j$ (the paper's Appendix A1, "information and the general order"): it uses *every* lag k, weighted by $k\rho^{k-1}$, because a change of ρ moves the whole covariance. The lag-one product uses lag one only. When $\rho = 0$ the weights vanish for $k \geq 2$ and the two statistics coincide; when ρ is large the longer lags carry most of the information and the lag-one estimator discards it.

Example 16.10 (Efficiency of the lag-one estimator, by hand)**.** *Independent sites.* At $\rho = 0$ the chain is i.i.d. $\mathcal{N}(0,1)$, so $\mathbf{\Sigma}_0 = \mathbf{I}$, $\mathbf{\Sigma}_t = \mathbf{I}$ at every t, and the noised sites x_i are i.i.d. $\mathcal{N}(0,1)$. The exact Gaussian information is $I_t(\rho) = \frac{1}{2}\,\mathrm{tr}(\mathbf{\Sigma}_t^{-1}\partial\mathbf{\Sigma}_t\mathbf{\Sigma}_t^{-1}\partial\mathbf{\Sigma}_t) = \frac{1}{2}e^{-4t}\|\partial_\rho\mathbf{\Sigma}_0\|_F^2 = e^{-4t}(L-1)$, since $\partial_\rho\mathbf{\Sigma}_0$ at $\rho = 0$ has ones on the two first off-diagonals and $2(L-1)$ nonzero entries. On the estimator side, the products $x_i x_{i+1}$ and $x_{i+1}x_{i+2}$ are uncorrelated ($\mathbb{E}[x_i x_{i+1}^2 x_{i+2}] = \mathbb{E}x_i \cdot \mathbb{E}x_{i+1}^2 \cdot \mathbb{E}x_{i+2} = 0$) and each has variance $\mathbb{E}x_i^2\,\mathbb{E}x_{i+1}^2 = 1$, so $\mathrm{Var}(\hat{\rho}_{\mathrm{mom}}) = e^{4t}/(M(L-1)) = 1/(MI_t)$ exactly, at every t: the lag-one estimator is efficient. *The frozen correlation.* At $\rho = 0.85$, $\rho^2 = 0.7225$, $1-\rho^2 = 0.2775$, $(1-\rho^2)^3 = 0.02137$ and $1+\rho^2 = 1.7225$, so by Equation (16.27) the long-chain efficiency is $0.02137/1.7225 \approx 0.0124$: the lag-one product uses about one eightieth of the covariance information, and its RMSE at high noise is about $\sqrt{80.6} \approx 9$ times the Cramér–Rao reference. (At $L = 32$ the finite sum in Equation (16.26) is a little smaller than the infinite one, so the factor is a little below nine.) The exact Gaussian information includes the effect of the nuisance q only through the efficient information, which changes the constant by an $O(1)$ factor (Chapter 10); the order of magnitude is the point.

16.5.3 The per-site cumulant against the shape information

The same argument applies at degree four. The efficient shape score at high noise is

$$\frac{\alpha^4}{24}\sum_{ijkl}\partial_\beta \boldsymbol{\kappa}_4[ijkl]\; :x_i x_j x_k x_l:\,, \tag{16.28}$$

a contraction of the *whole* fourth-cumulant tensor with the degree-four Hermite polynomials, including every off-diagonal entry $\boldsymbol{\kappa}_4[ijkl]$ with i, j, k, l not all equal. The per-site estimator uses only the diagonal entries $\boldsymbol{\kappa}_4[iiii]$. For the chain the off-diagonal entries are

$$\boldsymbol{\kappa}_4[ijkl] = \boldsymbol{\kappa}_4(\varepsilon) \sum_{m=2}^{\min(i,j,k,l)} \rho^{\,i+j+k+l-4m} \tag{16.29}$$

(Chapter 2; `theory.chain_kappa4`), which are comparable to the diagonal ones when ρ is close to one, so again the moment estimator discards most of the information at the frozen correlation. Its constant is therefore far from the Cramér–Rao one, as the paper's Section 4 reports (the ratios are quoted there as macros); its rate is right.

Remark 16.11 (On "the two estimators coincide at high noise"). The paper's Section 4 says that at high noise the moment and likelihood estimators coincide, "since the likelihood score of a degree-m coordinate is itself the degree-m Hermite statistic". The second clause is the general-order theorem of Chapter 10 and is correct. The first clause is correct for the moment estimator built from that *full* degree-m statistic (all lags at degree two; the full tensor at degree four). The paper's two estimators use the lag-one product and the per-site cumulant, which are proper sub-statistics of it; their efficiency relative to maximum likelihood tends to the constant Equation (16.26), not to one, unless $\rho = 0$. What does coincide at high noise is the *rate*, and in a sample of fixed size both estimators are useless there anyway, which is the practical content of the sentence.

16.6 How E19 measured them

E19 (`experiments/e19_moment_estimators.py`) is a deterministic re-analysis of E06's data. It regenerates E06's chains and noise draws from their random-stream tags (`e06-train`, `e06-noise`; Chapter 20 explains the tag mechanism), so every row is paired by seed and level with the EM–BP row of `outputs/E06/levels.csv`. The design is: the Laplace chain at the frozen $\rho = 0.85$ and $L = 32$; $M = 512$ sequences per cell, all noised at one level $t \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1.0, 1.4, 2.0\}$; sixteen seeds. Per cell the script computes $\hat{\rho}_{\mathrm{mom}}$ of Equation (16.16), the per-site sample cumulants, and $\hat{\boldsymbol{\kappa}}_4(\varepsilon)/\hat{q}^2$ of Equation (16.20) with $\hat{\rho}$ and, separately, with the true ρ (column `fit_kurt_oracle_rho`). The predeclared analysis (registry `experiments/registry/E19.json`) is the RMSE, dispersion and bias over the sixteen seeds per level, paired with E06's EM–BP fits, and the ordinary-least-squares log-slope of the RMSE against t on the levels $t \geq 1$, compared with the predicted 2 and 4. The numbers reach the manuscript through `tools/make_audit_numbers.py` as the macros `audMom...` (RMSE per level, bias at each level, the two slopes, and the ratio of the moment RMSE to the EM–BP RMSE over the informative levels $t \leq 0.7$); the paper quotes them in Section 4 (the learning paragraph) and in Appendix A5 (the E19 paragraph). Nothing about the moment estimators is selected: there is no ladder, no validation bundle and no checkpoint, which is what makes them the natural control for the selection-dependent EM fits.

Two readings the paper draws from E19, in the order it draws them: below the crossover the moment estimators are several times worse than EM–BP for the correlation and for the shape, which is the value of exact inference (Example 16.10 gives the analytic reason for the correlation); above it, where the EM shape estimate has collapsed, the moment estimators keep growing at the asymptotic rates, which is the corollary's prediction made visible on an estimator that cannot saturate.

Where it is used. The paper, Section 2 (the quasi-likelihood consequence of Proposition 1), Section 4 (the moment estimators against EM–BP) and Appendix A5 (E19); the $C = 1$ rows of Table 1, of the family figure and of the regime sweep as the apparatus check (E02a, E02b, E13). Code: `experiments/e19_moment_estimators.py`, `experiments/e06_learning_through_noise.py`, `tools/make_audit_numbers.py`, `src/markov_score/theory.py` (`chain_covariance`, `chain_kappa4`,

`correlation_information_constant`), `src/markov_score/kernels.py` (the Gaussian and mixture M-steps), `experiments/e13_regime_sweep.py`.

References. H. White, 1982, Maximum likelihood estimation of misspecified models, *Econometrica* 50, 1–25. P. J. Huber, 1967, The behavior of maximum likelihood estimates under nonstandard conditions, *Proceedings of the Fifth Berkeley Symposium*, 221–233. A. W. van der Vaart, 1998, *Asymptotic Statistics*, Cambridge University Press (chapters 4 and 5: method of moments and M-estimators). R. A. Fisher, 1930, Moments and product moments of sampling distributions, *Proceedings of the London Mathematical Society* 30, 199–238 (the k-statistics). A. Balanov, T. Bendory and D. Edidin, 2026, The generalized method of moments is (almost) statistically efficient in low-SNR Gaussian latent-variable models, arXiv:2605.30095. D. Hsu, S. M. Kakade and T. Zhang, 2012, A spectral algorithm for learning hidden Markov models, *Journal of Computer and System Sciences* 78, 1460–1480.

Chapter 17

Model selection without the truth

Every estimator in the paper has a knob that is not a parameter of the model: the number of EM iterations, or the number of gradient steps, at which the fit is read off. The error of a fit is a function of that knob, usually falling and then rising, and the value reported for an arm is its error at the knob's chosen position. How the position is chosen is therefore part of the estimator, and the paper spends a surprising amount of its protocol on it: validation and test bundles, checkpoint ladders, a selection rule per arm, a screening stage on seeds that never enter a reported number, and a control experiment (E20) whose only purpose is to show that one arm's selection does not depend on the truth. This chapter defines each of these objects, explains why they are there, and gives the three information interfaces A, B and C that the paper uses to say what each learner was allowed to see.

17.1 Bundles: what a fit sees and what it is scored on

17.1.1 Three disjoint samples

A **bundle** is a fixed set of sequences together with everything derived from them that an experiment needs. The paper keeps three per cell:

- the **training bundle**, M clean chains $\mathbf{a}^1, \dots, \mathbf{a}^M$ drawn from the prior, from which each arm builds whatever it is allowed to see (Section 17.7);
- the **validation bundle**, 1024 chains, used only to choose the knob;
- the **test bundle**, 2048 chains, used only to score the chosen fit.

For the validation and test bundles the code draws the chains once and then, for every level t of the twelve-level schedule, one noised view $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ of every chain and the reference posterior mean $\langle \mathbf{a} \rangle_{x,t}$ of that view under the *true* kernel, computed by the tree-exact recursion of Chapter 11 on the frozen grid (`_bundle` in `experiments/e03_baselines.py`; the same function shape in E02, E08, E14). A bundle is therefore a dictionary $t \mapsto (\mathbf{X}_t, \mathbf{M}_t)$ of the noised views and their reference means at each level, and its SHA-256 hash over those two arrays is stored with every row of every CSV, so that two arms claiming to be scored on the same bundle can be checked to have been (Chapter 20).

17.1.2 Keyed random streams

The three bundles are drawn from three random streams keyed by a tag, the seed and, for the training bundle, the size. In the code the training stream is `rng_for("exp31-train", seed, M)`, the validation stream is `rng_for("exp31-val", seed)` and the test stream is `rng_for("exp31-test", seed)`. Two consequences matter for model selection. First, the streams are disjoint by construction (different tags), so no validation or test chain is ever a training chain. Second, they are the

same across experiments: E02a's EM–BP fit and E03's network fit at (seed, M) see the identical training chains and are scored on the identical test bundle, which is what makes their ratio a **paired** comparison rather than a difference of two averages (Chapter 6). The registry calls the pair of tags and sizes the *bundle protocol*; Chapter 20 explains the tag mechanism.

17.1.3 The one-view groups

EM–BP never sees the clean training chains. From them the code derives the **one-view groups**: a random permutation of the M chains is split into twelve consecutive blocks of (nearly) equal size, block k is assigned to level t_k, and each chain in it receives *one* noised view at that level, drawn once from the stream `rng_for("exp31-em-noise", seed)` and never redrawn (`_one_view_groups`). The result is a list of twelve pairs $(\mathbf{X}_{t_k}, t_k)$ with about $M/12$ rows each. This is the data of interface C. A held-out one-view bundle for selection is built the same way from the validation chains with a different noise stream.

17.2 Checkpoint ladders

Definition 17.1 (Checkpoint ladder)**.** A **checkpoint ladder** is a fixed, increasing list of budgets at which a fit is saved. For EM–BP it is the list of iteration counts

$$\{10,\ 20,\ 40,\ 60,\ 80,\ 120,\ 160,\ 220,\ 300,\ 400,\ 600,\ 800,\ 1200\}, \tag{17.1}$$

and for the networks the list of gradient-step counts

$$\{500,\ 1000,\ 2000,\ 4000,\ 8000,\ 16000,\ 32000\}. \tag{17.2}$$

A **rung** is one entry; the **cap** is the last rung. The **selected checkpoint** is the rung the selection rule picks, and a cell is **at cap** when the selected rung is the last one.

Two things the definition does not say deserve saying. The ladder is coarse on purpose: its rungs are spaced roughly geometrically, so the selection can tell a factor of two in budget apart but not ten per cent. And the ladder is not a stopping rule: the fit is always run to the cap (unless EM's own likelihood tolerance ends it first, Chapter 14), the kernel or network state at every rung is saved, and the choice is made afterwards, with every rung available. This is what fixed the stopping-rule artefact of the earlier capacity study, where one arm stopped at its tolerance near its optimum and the others ran to a fixed cap past theirs (see the paper, Appendix on the previous capacity reading).

The at-cap fraction of a set of cells is a diagnostic the paper reports whenever it bears on a claim: if the selected rung is the cap in most cells, validation was still improving when the ladder ran out, and the reported error is an upper bound on what a longer ladder would give. The paper says so for the largest networks at $M = 2048$ and for EM–BP at the two highest levels of the learning experiment; the budget controls E02a_v1 and E06c exist to settle whether the cap binds.

17.3 Oracle selection: validation risk against the true kernel

Definition 17.2 (Oracle selection rule)**.** Let $\hat{m}^{(r)}(\mathbf{x}, t)$ be the posterior-mean map of the fit at rung r. Its **validation risk** is the schedule-pooled relative score error of Chapter 6,

$$R_{\mathrm{val}}(r) = \left(\frac{\sum_t \sum_j \sum_{i \in B} \big(\hat{s}_t^{(r)}(\mathbf{x}_j)_i - s_t(\mathbf{x}_j)_i\big)^2}{\sum_t \sum_j \sum_{i \in B} s_t(\mathbf{x}_j)_i^2} \right)^{1/2}, \tag{17.3}$$

where $s_t(\mathbf{x}) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$ and $\hat{s}_t^{(r)}$ is the same expression with $\hat{m}^{(r)}$ in place of $\langle \mathbf{a} \rangle_{x,t}$, all computed on the validation bundle over the **bulk** region B of sites 5 to $L-4$, with s_t the reference score built from the true kernel's posterior means stored in the bundle. The **oracle rule** selects $r^* = \arg\min_r R_{\mathrm{val}}(r)$.

The rule is called an oracle because s_t is the score of the true law, which a practitioner does not have. Three remarks.

Why the bulk. Every arm is scored on all sites for the headline, but selected on the interior. The boundary sites are where the arms differ for reasons unrelated to what is being selected (a zero-padded convolution behaves differently at the ends; the chain's first site has no left neighbour), and a selection statistic that weights them would let a boundary artefact choose the rung. The paper's protocol appendix records the three regions (all, bulk, centre) and which one is used for what.

Why it is legitimate for a comparison. The oracle rule is applied identically to every arm: EM–BP at every capacity, the six architectures under interface A, the fixed-corruption arm. A comparison between arms that all selected on the same oracle statistic is fair as a comparison; what it cannot support is a claim that any single arm could be deployed at its reported error without the truth. The paper draws exactly this line: the only arm selected without an oracle is the implicit-score-matching arm, and E20 checks EM–BP.

Selection is a shrinkage. Choosing the rung on validation is early stopping, and early stopping is a regulariser: an EM run stopped before convergence sits between the initialisation and the maximum-likelihood estimate. The audit's per-level reading of the learning experiment (Chapter 15) turns on this: at high noise the selected fits are shrunk toward the Gaussian initialisation, which is why they can sit *below* an unbiased-estimator bound.

17.4 Oracle-free selection for EM–BP: held-out log-evidence

17.4.1 The statistic

Definition 17.3 (Held-out one-view log-evidence per edge)**.** Let $\{(\mathbf{X}_{t_k}, t_k)\}$ be a held-out one-view bundle of n chains and let $K_\theta^{(r)}$ be the kernel at rung r. The **held-out log-evidence per edge** is

$$E(r) = \frac{1}{n(L-1)} \sum_k \sum_{\mathbf{x} \in \mathbf{X}_{t_k}} \log P_{t_k, \theta^{(r)}}(\mathbf{x}), \tag{17.4}$$

the average over held-out views of the marginal log-likelihood of the view under the fitted kernel, divided by the number of transitions per chain. The **evidence rule** selects $r^* = \arg\max_r E(r)$.

The marginal log-likelihood $\log P_{t,\theta}(\mathbf{x})$ is exactly what the forward sweep of the discretised model returns when its normalising constants are accumulated (Chapter 12; the field `log_evidence` returned by `e_step_multi` in `src/markov_score/em.py`). No clean chain and no true kernel enters: the statistic is a function of the fitted kernel and of noised data only. Dividing by the number of edges makes it comparable across chain lengths and sizes (E14 uses it at $L = 128$).

17.4.2 Why maximising held-out likelihood is a valid rule

For a held-out sample from the true noised law P_t, the average log-likelihood of a candidate law Q is an unbiased estimate of $\mathbb{E}_{P_t} \log Q(\mathbf{x})$, and

$$\mathbb{E}_{P_t} \log Q(\mathbf{x}) = \mathbb{E}_{P_t} \log P_t(\mathbf{x}) - D_{\mathrm{KL}}(P_t \,\|\, Q). \tag{17.5}$$

The first term does not depend on Q. So the rung with the largest held-out evidence is, up to the sampling error of the held-out set, the rung whose fitted law is closest to the truth in Kullback–Leibler divergence. It is the same principle as cross-validation of a likelihood, and it is oracle-free because Equation (17.5) never requires P_t to be known, only sampled.

Two caveats keep the rule honest. The criterion is Kullback–Leibler divergence of the noised law, not the relative score error the paper reports; the two are different functionals of Q and can rank two rungs differently. And the held-out set is finite, so $E(r)$ carries sampling noise of order $n^{-1/2}$ per edge, which can blur the choice between neighbouring rungs whose evidence differs by less than that. Both are visible in the control.

17.4.3 E20

E20 (`experiments/e20_evidence_selection.py`) replicates E02a cells exactly, at $C = 8$ with the same bundle tags, the same initialisation stream and the same ladder, and records at every rung three numbers: the oracle validation risk Equation (17.3), the held-out one-view evidence Equation (17.4) computed on the validation chains with a fresh noise stream (tags `e20-split`, `e20-noise`), and the test risk at the rungs the two rules and the cap pick. The comparison is the test risk under the evidence rule against the test risk under the oracle rule, paired by seed, and the fraction of seeds in which the two rules pick the same rung. The design concentrates on $M = 32$, the size at which the choice of rung matters most (at $M = 2048$ the frozen validation curves are monotone through the ladder, so every rule picks the cap and the control is empty); the $M = 512$ arm was dropped for cost, as the registry records. The paper quotes the result in Section 5 and Appendix A5: the two rules pick the same rung in most seeds, give the same test error within seed noise, and both are far below the error at the cap. The sentence it licenses is that EM–BP's reported numbers do not depend on the oracle, not that the oracle rule was never used.

17.5 Oracle-free selection for networks: the implicit objective

17.5.1 Hyvärinen's identity

A network trained by implicit score matching (Chapters 8 and 19) minimises, over noisy samples $\mathbf{x} \sim P_t$ only,

$$J(\theta) = \mathbb{E}_{P_t}\Big[\tfrac{1}{2}\|s_\theta(\mathbf{x})\|^2 + \nabla\cdot s_\theta(\mathbf{x})\Big]. \tag{17.6}$$

The reason this is a score-matching objective is an integration by parts. For a smooth density p and a smooth vector field s_θ whose product with p vanishes at infinity,

$$\mathbb{E}_p[\nabla\cdot s_\theta] = \int (\nabla\cdot s_\theta)\, p\,\mathrm{d}\mathbf{x} = -\int s_\theta\cdot\nabla p\,\mathrm{d}\mathbf{x} = -\mathbb{E}_p[s_\theta\cdot\nabla\log p], \tag{17.7}$$

so

$$J(\theta) = \tfrac{1}{2}\mathbb{E}_p\|s_\theta\|^2 - \mathbb{E}_p[s_\theta\cdot s_p] = \tfrac{1}{2}\mathbb{E}_p\|s_\theta - s_p\|^2 - \tfrac{1}{2}\mathbb{E}_p\|s_p\|^2, \tag{17.8}$$

with $s_p = \nabla\log p$ the true score. The last term does not depend on θ. Minimising J is minimising the **Fisher divergence** $\frac{1}{2}\mathbb{E}_p\|s_\theta - s_p\|^2$ without ever evaluating s_p.

17.5.2 As a selection rule

The same identity makes J a model-selection criterion. Evaluate Equation (17.6) on a held-out one-view bundle: the empirical average is an unbiased estimate of J at that rung, hence of the Fisher divergence up to a constant common to all rungs. Selecting $r^* = \arg\min_r J_{\mathrm{held}}(r)$ picks the rung closest to the true score in Fisher divergence, with no clean chain and no true score anywhere (`ism_objective` in `src/markov_score/nn_torch.py`). Three details of the paper's implementation:

- the divergence in the selection statistic is computed *exactly*: by the coloured probes for the window head and by L backward passes for a dense map (Chapter 19); a stochastic estimate would add noise to a comparison between rungs that may differ by little;
- the held-out bundle carries one view per chain split across the twelve levels, exactly as the training data, so the criterion weights the levels as the data do: equal numbers of sequences per level, each contributing its own $\frac{1}{2}\mathbb{E}\|s_\theta - s_p\|^2$; the paper's estimand instead pools squared errors and divides by the pooled reference energy, which weights the low levels more (Chapter 6), so the two can rank rungs differently;
- the whole held-out curve $r \mapsto J_{\text{held}}(r)$ is stored in the CSV, not only its minimum, so that a run whose curve never turns over can be recognised as a divergence of the optimisation rather than reported as the network's error; E03 saw this at $M = 32$, where implicit score matching on a tiny sample is known to be unstable, and the paper says so.

17.6 Screening and confirmation

17.6.1 Two kinds of seeds

The paper's protocol names two disjoint sets of seeds. The **development seeds** 10000 to 10003 (E03's screen used the first two after a pre-data amendment) are for every choice a designer makes: architecture family, hyperparameters, learning rate, output parameterisation. The **confirmatory seeds** 0 to 15 (eight of them in the regime sweeps) are for the reported numbers. The rule is stated twice in the contract and once in the paper: development seeds never enter a reported number; confirmatory seeds never enter a selection.

17.6.2 Why: the winner's curse

Suppose K configurations are tried on the same data and the best one's error is reported. Even if all K have the same true error, the reported minimum is biased downward, because it is the minimum of K noisy measurements. The paper avoids this by choosing on one set of seeds and reporting on another: the confirmatory error of the chosen configuration is then an ordinary, unbiased estimate of that configuration's error, because the choice did not look at the confirmatory data.

Example 17.4 (The winner's curse for two candidates, by hand)**.** Two configurations have the same true risk R, and their measured risks on the development seeds are $R + \sigma Z_1$ and $R+\sigma Z_2$ with Z_1, Z_2 independent standard normal. The screen reports $\min(R+\sigma Z_1, R+\sigma Z_2) = R - \sigma \max(-Z_1, -Z_2)$. For two independent standard normals, $\max(Z_1, Z_2)$ has expectation $\mathbb{E}|Z_1 - Z_2|/2 = \frac{1}{2} \cdot 2/\sqrt{\pi} \cdot \frac{1}{\sqrt{2}} \cdot \sqrt{2} = 1/\sqrt{\pi}$: indeed $\max(a, b) = \frac{1}{2}(a + b) + \frac{1}{2}|a - b|$, the first term has mean zero, and $Z_1 - Z_2$ is normal with variance 2, so $\mathbb{E}|Z_1 - Z_2| = \sqrt{2} \cdot \sqrt{2/\pi} = 2/\sqrt{\pi}$. Hence the screened minimum has expectation $R - \sigma/\sqrt{\pi} \approx R - 0.56\sigma$: with a screen noise of, say, five per cent of the risk, reporting the screen's own number would understate the risk by nearly three per cent for two candidates, and by more for the sixty-eight configurations of E03's screen. Reporting the winner's risk on fresh seeds removes the bias entirely; the fresh measurement has its own noise but no preference for the small side.

17.6.3 What a fair screen is

Definition 17.5 (Fair screen)**.** A screen is **fair** when every candidate is trained under the same recipe (loss, optimiser, batch, gradient clipping, level sampling, corruption), for the same budget (steps and restarts), on the same bundles, with the same checkpoint ladder and the same selection statistic, and the winner of each family is chosen by the same rule on development seeds disjoint from the confirmatory ones.

The definition is a list because each item names a way the earlier screen of this model was unfair, as the audit found: the long-range architectures had been trained with one noise level per gradient step where the window head had twelve, and with one restart where the head had eight, so a difference between families was partly a difference in training. E03's screen (`part_screen` in `experiments/e03_baselines.py`) gives every family the recipe of Chapter 19: per-example level from the twelve, batch 64, Adam, clipping at 1, both parameterisations, learning rates $\{10^{-3}, 3 \cdot 10^{-3}\}$, a family-specific hyperparameter grid (radius and width for the window head; channels for the TCN; hidden size for the GRU; width, heads and depth for the transformer; channels for the U-Net; hidden widths for the MLP), one run of 16,000 steps per configuration at the two sizes $M \in \{32, 2048\}$ on two development seeds, the checkpoint selected by the oracle rule on the bulk, and the family's winner chosen by the pooled bulk risk over development seeds and sizes (`winners_from_screening`). The winners are written to `outputs/E03/winners.json` and are the configurations of every confirmatory row. The paper reports the screen in its protocol appendix (the screen table) and says that the screen's numbers are never a headline.

17.6.4 Confirmation

The **confirmatory run** (`part_confirm`) trains each family's winner on the sixteen confirmatory seeds at the four sizes $\{32, 128, 512, 2048\}$ for the full 32,000 steps, selects the rung by the oracle rule, and scores on the test bundle in all three regions. Its rows are what the baseline table shows and what the paired ratios against EM–BP are computed from, joined on the seed, the size and the test-bundle hash.

17.7 The three information interfaces

The paper's comparisons are organised by what a learner is allowed to see. The definitions below are the code's, made explicit.

Definition 17.6 (Interfaces A, B and C)**.** Let $\{\mathbf{a}^\mu\}_{\mu \le M}$ be the training chains and let each chain be assigned one level t_μ by the even split of Section 17.1.

- **Interface A** (the standard denoising recipe). The learner may query the clean chains at every gradient step, together with a freshly drawn level t (uniform on the twelve) and a freshly drawn corruption $\mathbf{z}$; its training pairs are $(\mathbf{a}^\mu,\ e^{-t}\mathbf{a}^\mu + \sqrt{\Delta_t}\mathbf{z})$ with new $(t, \mathbf{z})$ every time. The checkpoint is selected by the oracle rule.
- **Interface B** (one fixed corruption). The learner sees the pairs $(\mathbf{a}^\mu, \mathbf{x}^\mu)$ with $\mathbf{x}^\mu = e^{-t_\mu}\mathbf{a}^\mu + \sqrt{\Delta_{t_\mu}}\mathbf{z}^\mu$, one corruption per chain, drawn once. It still regresses on the clean chain. The checkpoint is selected by the oracle rule.
- **Interface C** (one noised view per sequence and nothing else). The learner sees $\{(\mathbf{x}^\mu, t_\mu)\}$ only: the one-view groups. It never sees a clean chain, and its checkpoint is selected without the true score, by the held-out implicit objective (networks) or the held-out evidence (EM–BP, E20 and E14's second row).

The three are nested in what they reveal: C sees a function of what B sees, and B sees one draw of what A can draw indefinitely. The paper's Section 5 uses the nesting in one sentence: the gap between the interface-A rows and the interface-C row is the value of clean data with fresh corruptions; the factor that remains at C is what the factorisation and the exact E-step are worth at equal information. Interface B is the control that separates the two ingredients of A: it gives the clean target but not the augmentation, so a network at B that is still far behind EM–BP shows that one corruption per sequence, not the absence of the clean chain, is what it cannot recover from. The audit's correction of 18 September was precisely a relabelling: an arm

that sees $\mathbf{a}^\mu$ is at B, not at C, whatever its loss is called, and the headline "matched" number has to be the interface-C arm.

EM–BP is at interface C by construction: its E-step reads $\{(\mathbf{x}^\mu, t_\mu)\}$ and nothing else (Chapter 15). Its selection rule in the capacity and baseline experiments is the oracle one, which is why E20 exists; in E14 both rules are run side by side.

17.8 The regime sweep of the matched comparison: E14

E14 (`experiments/e14_regime_baselines.py`) repeats the interface-C comparison away from the frozen point: correlations $\rho \in \{0.5, 0.85, 0.95\}$, chain lengths $L \in \{32, 128\}$, Laplace innovations, $M = 2048$, eight confirmatory seeds, with its own bundles (tags `e14-train`, `e14-val`, `e14-test`, `e14-valchains`, keyed by seed and length). Each cell scores six arms on a common test bundle:

`em_bp` the eight-component mixture kernel fitted by EM–BP on the one-view groups, rung selected by the held-out BP risk of the validation bundle against the true kernel's posterior means (the oracle rule of Definition 17.2, on all sites here);

`em_bp_evidence` the same EM run and the same ladder, rung selected by the held-out one-view log-evidence per edge on the noised views of the validation chains (the evidence rule of Definition 17.3): the oracle-free EM–BP;

`ism_one_view` a network trained by implicit score matching on the same one-view groups, rung selected on the held-out implicit objective with the exact divergence; one row per architecture family (window head, transformer), each with the hyperparameters that won a screen *inside the regime* on two development seeds over E03's candidate grid, 8,000 steps for the screen and 32,000 for the confirmation; the window head trains with the coloured-probe divergence and learning rate 10^{-3}, the transformer with Hutchinson's estimator and $3 \cdot 10^{-3}$, as E03's winners did;

`dsm_one_corruption` the same architectures at interface B: denoising score matching on clean targets paired with the very same one noised view per chain (the index arrays of the split are returned so the pairing is exact), rung selected by the validation BP risk;

`lmmse_true` the LMMSE estimator with the true covariance, the second-order floor (Chapter 13);

`isotropic` the score $-\mathbf{x}$ of the standard Gaussian, the reference of a model that learned nothing beyond the unit variance.

The predeclared analysis is, per cell, the mean pooled relative score error of every arm and the paired ratio of the best network to EM–BP over seeds with a percentile bootstrap, with the trend in ρ reported against the marginal's excess kurtosis $3(1-\rho^2)/(1+\rho^2)$ as a direction, not a fitted law. The registry entry carries a pre-data amendment worth reading as a specimen of the protocol: the first screen had trained the window head with Hutchinson's one-probe divergence and obtained risks far above the same network's E03 value; the cause was identified as the estimator, not the regime, the screen was kept as a negative result, and the design was re-run with the coloured probes before any confirmatory cell had run (Chapter 20). At the time of writing E14 is queued on the cluster and no number from it is in the paper; the chapter describes the design, not a result.

Where it is used. The paper, Section 5 (the three interfaces; the oracle-free selection sentence), Appendix A3 (arms and interfaces; the screen table) and Appendix A5 (E20); Chapter 19 for the losses and Chapter 15 for the EM side. Code: `experiments/protocol.py` (bundle tags, ladders, regions, dev and confirmatory seeds), `experiments/e03_baselines.py` (screen, confirm, matched, length), `experiments/e20_evidence_selection.py`, `experiments/e14_regime_baselines.py`, `experiments/e02_capacity.py`, `src/markov_score/nn_torch.py` (`ism_objective`),

`src/markov_score/em.py` (`e_step_multi`, `fit_em`), `tools/make_baseline_numbers.py` (the matched macros by arm).

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Chapter 18

Misspecification and the Edgeworth correction

Two questions in the paper have the same mathematical shape: what does a maximum likelihood estimator converge to when the model it fits is wrong, and what does the Gaussian covariance twin lack when it stands in for the true score? Both are answered by a projection. The first is the Kullback–Leibler projection of the true law onto the model family, whose image is the *pseudo-true parameter* of White; the paper turns it into a closed-form asymptotic bias $b(\varepsilon)$ for each of its four misspecification mechanisms (Section 5 of the paper, experiments E08 and E17). The second is the expansion of the true noised law around its Gaussian twin in Hermite polynomials, whose first non-trivial term is the fourth-cumulant correction of Edgeworth; the paper adds that term back and measures how much of the closure gap it closes (Section 3 of the paper, experiment E16). This chapter builds both, derives the formulas the paper quotes, and records how the experiments used them.

18.1 The Kullback–Leibler projection and the pseudo-true parameter

18.1.1 What a misspecified likelihood converges to

Let the data $\mathbf{x}^1, \dots, \mathbf{x}^M$ be i.i.d. from a law $P^\star$ and let $\{Q_\theta\}$ be a family of densities that may not contain $P^\star$. The **misspecified maximum-likelihood estimator** is $\hat\theta = \arg\max_\theta \sum_\mu \log Q_\theta(\mathbf{x}^\mu)$. Dividing by M, the law of large numbers gives

$$\frac{1}{M}\sum_{\mu} \log Q_\theta(\mathbf{x}^\mu) \;\to\; \mathbb{E}_{P^\star} \log Q_\theta(\mathbf{x}) = \mathbb{E}_{P^\star} \log P^\star(\mathbf{x}) - D_{\mathrm{KL}}\big(P^\star \,\|\, Q_\theta\big), \tag{18.1}$$

since $D_{\mathrm{KL}}(P^\star \| Q) = \mathbb{E}_{P^\star}[\log P^\star - \log Q]$ (Chapter 1). The first term does not depend on θ.

Definition 18.1 (Pseudo-true parameter; Kullback–Leibler projection)**.** The **pseudo-true parameter** is

$$\theta^* = \arg\min_\theta D_{\mathrm{KL}}\big(P^\star \,\|\, Q_\theta\big), \tag{18.2}$$

and Q_{θ^*} is the **Kullback–Leibler projection** of $P^\star$ onto the family. When $P^\star = Q_{\theta_0}$ belongs to the family, $\theta^* = \theta_0$.

Proposition 18.2 (White)**.** *If the minimiser Equation* (18.2) *is unique and the usual regularity holds, then $\hat\theta \to \theta^*$ in probability, and $\sqrt{M}(\hat\theta - \theta^*)$ is asymptotically normal with the sandwich covariance $\mathbf{J}^{-1}\mathbf{V}\mathbf{J}^{-1}$, $\mathbf{J} = -\mathbb{E}_{P^\star}\partial_\theta^2 \log Q_{\theta^*}$, $\mathbf{V} = \mathrm{Var}_{P^\star}(\partial_\theta \log Q_{\theta^*})$.*

The proof is the M-estimator argument of Chapter 16: the objective converges uniformly to Equation (18.1), whose unique maximiser is θ^*. The content of the proposition is the reading: a wrong model does not make the estimator wander; it makes it converge, to a definite point, and more data only sharpens the convergence. The error that remains at θ^* is a **bias** in the strict sense that no sample size removes it. This is the sentence the paper's misspecification section ends on.

18.1.2 Why EM–BP's limit is a Gaussian projection here

The paper's misspecification experiment fits the mixture-innovation AR(1) kernel of Chapter 15 to data from a Gaussian truth that is not an AR(1) chain. The fitted innovation turns out Gaussian to three decimals (the paper reports the fitted excess kurtosis; E08 records it per cell), which is what one expects: with a Gaussian truth there is no fourth-order structure for the mixture to fit, and the extra components only add variance. The paper therefore takes the $M \to \infty$ limit of EM–BP to be the projection onto the *Gaussian* AR(1) family, a two-parameter family (ρ, q), and computes it in closed form. Strictly the projection onto the mixture family could be non-Gaussian even for a Gaussian truth, since the truth is outside both families; the paper's justification is empirical (the fitted kurtosis) and the closed form is checked against the measured $M = 2048$ errors, which is the right order of argument.

18.2 The paper's closed-form bias

18.2.1 The projection under the one-view protocol

Let the truth be Gaussian with clean covariance $\mathbf{\Sigma}(\varepsilon)$, so its noised law at level t is $\mathcal{N}(0, \mathbf{\Sigma}_t(\varepsilon))$ with $\mathbf{\Sigma}_t(\varepsilon) = e^{-2t}\mathbf{\Sigma}(\varepsilon) + \Delta_t \mathbf{I}$. The model is the Gaussian chain with $\hat{\mathbf{\Sigma}}_t(\rho, q) = e^{-2t}\mathbf{\Sigma}_0(\rho, q) + \Delta_t \mathbf{I}$, where $\mathbf{\Sigma}_0(\rho, q)$ is the general chain covariance of Chapter 16. Under the one-view protocol the M sequences are split evenly over the twelve levels, so the population log-likelihood is an average over levels with equal weights, and the pseudo-true parameter is

$$\theta^*(\varepsilon) = \arg\min_{(\rho,q)} \sum_{t \in \mathcal{T}} D_{\mathrm{KL}}\Big(\mathcal{N}(0, \mathbf{\Sigma}_t(\varepsilon)) \,\Big\|\, \mathcal{N}\Big(0, \hat{\mathbf{\Sigma}}_t(\rho, q)\Big)\Big), \tag{18.3}$$

which is the formula of the paper's Appendix A5. The Gaussian divergence is Equation (16.7) of Chapter 16, so up to terms without θ the objective is

$$F(\rho, q) = \frac{1}{2} \sum_t \Big[\log\det \hat{\mathbf{\Sigma}}_t + \mathrm{tr}\big(\hat{\mathbf{\Sigma}}_t^{-1} \mathbf{\Sigma}_t(\varepsilon)\big)\Big], \tag{18.4}$$

exactly what `kl_project` in `experiments/e17_misspec_bias.py` minimises by Nelder–Mead over $(\tanh^{-1}\rho, \log q)$, the reparameterisation keeping $|\rho| < 1$ and $q > 0$. Setting the gradient of F to zero gives the two **projection equations**

$$\sum_t \mathrm{tr}\Big(\hat{\mathbf{\Sigma}}_t^{-1}\, \partial_\theta \hat{\mathbf{\Sigma}}_t\, \hat{\mathbf{\Sigma}}_t^{-1} (\mathbf{\Sigma}_t(\varepsilon) - \hat{\mathbf{\Sigma}}_t)\Big) = 0, \qquad \theta \in \{\rho, q\}, \tag{18.5}$$

the population version of the quasi-score equations of Chapter 16: the projected covariance agrees with the truth in the two directions $\hat{\mathbf{\Sigma}}_t^{-1} \partial_\theta \hat{\mathbf{\Sigma}}_t \hat{\mathbf{\Sigma}}_t^{-1}$, summed over the levels.

18.2.2 The relative score error of the projected model

At the projection the model's score is $-\hat{\mathbf{\Sigma}}_t^{-1}\mathbf{x}$ and the truth's is $-\mathbf{\Sigma}_t^{-1}\mathbf{x}$ (write $\mathbf{\Sigma}_t$ for $\mathbf{\Sigma}_t(\varepsilon)$). The paper's estimand pools squared errors and reference energies over the schedule (Chapter 6); its population version is

$$b(\varepsilon)^2 = \frac{\sum_t \mathbb{E}\|(\hat{\mathbf{\Sigma}}_t^{-1} - \mathbf{\Sigma}_t^{-1})\mathbf{x}\|^2}{\sum_t \mathbb{E}\|\mathbf{\Sigma}_t^{-1}\mathbf{x}\|^2} \qquad (\mathbf{x} \sim \mathcal{N}(0, \mathbf{\Sigma}_t) \text{ at level } t), \tag{18.6}$$

with equal numbers of sequences at every level, as in the test bundles. Both expectations are of quadratic forms: for $\mathbf{x} \sim \mathcal{N}(0, \boldsymbol{\Sigma})$ and any matrix $\mathbf{D}$, $\mathbb{E}\|\mathbf{D}\mathbf{x}\|^2 = \mathbb{E}[\mathbf{x}^\top \mathbf{D}^\top \mathbf{D}\mathbf{x}] = \operatorname{tr}(\mathbf{D}^\top \mathbf{D}\boldsymbol{\Sigma}) = \operatorname{tr}(\mathbf{D}\boldsymbol{\Sigma}\mathbf{D}^\top)$. With $\mathbf{D} = \hat{\boldsymbol{\Sigma}}_t^{-1} - \boldsymbol{\Sigma}_t^{-1}$ (symmetric) in the numerator and $\mathbf{D} = \boldsymbol{\Sigma}_t^{-1}$ in the denominator,

$$b(\varepsilon)^2 = \frac{\sum_t \operatorname{tr}\left[(\hat{\boldsymbol{\Sigma}}_t^{-1} - \boldsymbol{\Sigma}_t^{-1})\, \boldsymbol{\Sigma}_t\, (\hat{\boldsymbol{\Sigma}}_t^{-1} - \boldsymbol{\Sigma}_t^{-1})\right]}{\sum_t \operatorname{tr}\, \boldsymbol{\Sigma}_t^{-1}}, \qquad \hat{\boldsymbol{\Sigma}}_t = \hat{\boldsymbol{\Sigma}}_t(\theta^*(\varepsilon)), \tag{18.7}$$

which is the trace formula of the paper's Appendix A5 and the function `pooled_err` of E17. No Monte Carlo enters: $b(\varepsilon)$ is a deterministic function of the mechanism, its strength, the chain length and the schedule.

E17 computes a second projection beside the Kullback–Leibler one: the **lag-one plug-in**, the AR(1) covariance $\rho_1^{|i-j|}$ whose correlation ρ_1 is the mean lag-one correlation of the truth. It is the population limit of the moment estimator of Chapter 16 and of E08's `lmmse_ar1` arm, and the paper quotes both, noting that the measured EM–BP fit sits between the two because the mixture kernel is not exactly Gaussian and a finite-sample fit undershoots the Kullback–Leibler optimum.

18.2.3 The finite-sample prediction $\sqrt{b^2 + v/M}$

At finite M the fitted parameter is $\hat{\theta} = \theta^* + \delta$ with $\delta = O(M^{-1/2})$ by Proposition 18.2. Expanding the model score to first order in δ, $\hat{s} = s_{\theta^*} + \delta \cdot \partial_\theta s_{\theta^*}$, and the squared error against the truth is

$$\|\hat{s} - s\|^2 = \|s_{\theta^*} - s\|^2 + 2\,\delta \cdot \langle \partial_\theta s_{\theta^*},\, s_{\theta^*} - s \rangle + \|\delta \cdot \partial_\theta s_{\theta^*}\|^2. \tag{18.8}$$

Take expectations over the test data and over the training sample that produced δ. The first term is b^2 times the reference energy. The cross term vanishes to leading order because $\mathbb{E}\delta = o(M^{-1/2})$ and δ is independent of the test sequences. The last term is a quadratic form in δ whose expectation is $\operatorname{tr}(\operatorname{Var}(\delta)\, \mathbf{G})$ for a fixed matrix $\mathbf{G}$, hence v/M for a constant v that depends on the sandwich covariance and the mechanism. Therefore

$$\operatorname{risk}(M, \varepsilon)^2 \approx b(\varepsilon)^2 + \frac{v(\varepsilon)}{M}. \tag{18.9}$$

The paper reads v off the well-specified cells: at $\varepsilon = 0$ the bias vanishes, so $v = M \cdot \operatorname{risk}(M, 0)^2$, and it takes v to be the same at every strength (E17's setting `v_const`), a simplification that the agreement at both sizes in every mechanism justifies after the fact. The two sizes of E08, 512 and 2048, separate the two terms: where the risk falls by a factor of two between them the variance term dominates, and where it does not move the bias does. The paper's Figure 5 (bottom row) draws $b(\varepsilon)$ and Equation (18.9) through the measured points (Chapter 21).

18.3 The order of the bias in the strength

18.3.1 First-order perturbation of a projection

Write $\boldsymbol{\Sigma}(\varepsilon) = \boldsymbol{\Sigma}_0 + \varepsilon \mathbf{D} + O(\varepsilon^2)$ with $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_0(\theta_0)$ in the family, and consider one level for simplicity (the sum over levels adds the same terms). Linearising the projection equations Equation (18.5) around θ_0 gives $\theta^*(\varepsilon) = \theta_0 + \varepsilon\, \delta\theta + O(\varepsilon^2)$ with

$$\delta\theta = \mathbf{I}(\theta_0)^{-1} \left\langle \boldsymbol{\Sigma}_0^{-1} \partial_\theta \boldsymbol{\Sigma}_0 \boldsymbol{\Sigma}_0^{-1}, \mathbf{D} \right\rangle_\theta, \qquad \mathbf{I}_{\theta\theta'} = \langle \boldsymbol{\Sigma}_0^{-1} \partial_\theta \boldsymbol{\Sigma}_0 \boldsymbol{\Sigma}_0^{-1}, \partial_{\theta'} \boldsymbol{\Sigma}_0 \rangle, \tag{18.10}$$

where $\langle \mathbf{A}, \mathbf{B} \rangle = \operatorname{tr}(\mathbf{A}\mathbf{B})$: the projection moves the parameter by the *Fisher-metric projection* of the perturbation $\mathbf{D}$ onto the tangent space of the family, spanned by $\partial_\rho \boldsymbol{\Sigma}_0$ and $\partial_q \boldsymbol{\Sigma}_0$. The projected covariance is then $\hat{\boldsymbol{\Sigma}}(\varepsilon) = \boldsymbol{\Sigma}_0 + \varepsilon\, \delta\theta \cdot \partial_\theta \boldsymbol{\Sigma}_0 + O(\varepsilon^2)$, and the mismatch $\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma} =$

$\varepsilon(\delta\theta \cdot \partial_\theta \boldsymbol{\Sigma}_0 - \mathbf{D}) + O(\varepsilon^2)$ is ε times the component of $\mathbf{D}$ *orthogonal* to the tangent space in the Fisher metric. Inserting into Equation (18.7), with $\hat{\boldsymbol{\Sigma}}^{-1} - \boldsymbol{\Sigma}^{-1} = -\boldsymbol{\Sigma}^{-1}(\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1} + O(\varepsilon^2)$,

$$b(\varepsilon) = \varepsilon \, \|\mathbf{D}^{\perp}\|_{\boldsymbol{\Sigma}_0^{-1}} / \sqrt{\operatorname{tr} \boldsymbol{\Sigma}_0^{-1}} + O(\varepsilon^2), \qquad \|\mathbf{A}\|^2_{\boldsymbol{\Sigma}^{-1}} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{A}\boldsymbol{\Sigma}^{-1}\mathbf{A}\boldsymbol{\Sigma}^{-1}\boldsymbol{\Sigma}) = \operatorname{tr}((\boldsymbol{\Sigma}^{-1}\mathbf{A})^2). \tag{18.11}$$

Proposition 18.3 (Order of the bias)**.** *$b(\varepsilon) = \Theta(\varepsilon)$ if the first-order perturbation $\mathbf{D}$ of the covariance has a nonzero component outside the tangent space of the family at θ_0, and $b(\varepsilon) = \Theta(\varepsilon^2)$ if $\mathbf{D} = 0$, that is, if the covariance depends on the strength only through ε^2.*

The second case is the **rank-one latent** of E08: $a = (y + \beta g)/\sqrt{1+\beta^2}$ with y an AR(1) chain and g a single standard normal shared by all sites, so $\operatorname{Cov}(a_i, a_j) = (\rho^{|i-j|} + \beta^2)/(1+\beta^2)$ for $i \neq j$ (`GaussianAR1PlusGlobal` in `src/markov_score/priors.py`). The covariance is even in β, its first-order term in β vanishes, and the bias is $\Theta(\beta^2)$. The audit adds a second reason the latent is weak: the perturbation $\beta^2 \mathbf{1}\mathbf{1}^\top$ has rank one, so it moves one eigendirection of $\boldsymbol{\Sigma}_0$ among L, and its share of the score energy is $O(1/L)$ as well as $O(\beta^2)$. The other three mechanisms perturb of order L directions at first order and have $b = \Theta(\varepsilon)$, which is what E17's small-ε local exponents report (the paper quotes them as macros).

Example 18.4 (A projection and its bias, by hand)**.** Take $L = 2$ and a single level with $t \to 0$, so that $\boldsymbol{\Sigma}_t \to \boldsymbol{\Sigma}$ and the model covariance is $\hat{\boldsymbol{\Sigma}}(\rho, q) = \left(\begin{smallmatrix} 1 & \rho \\ \rho & \rho^2+q \end{smallmatrix}\right)$, with $\det \hat{\boldsymbol{\Sigma}} = q$ and $\hat{\boldsymbol{\Sigma}}^{-1} = q^{-1}\left(\begin{smallmatrix} \rho^2+q & -\rho \\ -\rho & 1 \end{smallmatrix}\right)$. Let the truth be $\boldsymbol{\Sigma}(\varepsilon) = \left(\begin{smallmatrix} 1+\varepsilon & c \\ c & 1-\varepsilon \end{smallmatrix}\right)$, a chain with lag-one correlation c whose two variances have been tilted by $\pm\varepsilon$; this perturbation lies outside the family, whose first variance is fixed at 1. The objective Equation (18.4) is $F = q^{-1}[(\rho^2+q)(1+\varepsilon) - 2\rho c + 1 - \varepsilon] + \log q$. Setting $\partial_\rho F = 0$ gives $\rho^* = c/(1+\varepsilon)$; setting $\partial_q F = 0$ gives $q^* = \rho^{*2}(1+\varepsilon) - 2\rho^* c + 1 - \varepsilon = 1 - \varepsilon - c^2/(1+\varepsilon)$. The projected covariance is

$$\hat{\boldsymbol{\Sigma}}^* = \begin{pmatrix} 1 & c/(1+\varepsilon) \\ c/(1+\varepsilon) & 1 - \varepsilon - c^2\varepsilon/(1+\varepsilon)^2 \end{pmatrix} = \boldsymbol{\Sigma}(\varepsilon) - \varepsilon \, \mathbf{v}\mathbf{v}^\top + O(\varepsilon^2), \qquad \mathbf{v} = \begin{pmatrix} 1 \\ c \end{pmatrix}. \tag{18.12}$$

So the residual mismatch is rank one, along $\mathbf{v}$: the family absorbs the perturbation of the second variance and of the lag-one entry, and what it cannot absorb is the first variance, which it pins at one. Then $\hat{\boldsymbol{\Sigma}}^{*-1} - \boldsymbol{\Sigma}^{-1} = \varepsilon \, \boldsymbol{\Sigma}^{-1}\mathbf{v}\mathbf{v}^\top\boldsymbol{\Sigma}^{-1} + O(\varepsilon^2)$ and the numerator of Equation (18.7) is $\varepsilon^2(\mathbf{v}^\top\boldsymbol{\Sigma}^{-1}\mathbf{v})^2$. At $\varepsilon = 0$, $\boldsymbol{\Sigma}_0^{-1} = (1-c^2)^{-1}\left(\begin{smallmatrix} 1 & -c \\ -c & 1 \end{smallmatrix}\right)$ and $\mathbf{v}^\top\boldsymbol{\Sigma}_0^{-1}\mathbf{v} = (1 - 2c^2 + c^2)/(1-c^2) = 1$, while the denominator is $\operatorname{tr} \boldsymbol{\Sigma}_0^{-1} = 2/(1-c^2)$. Hence

$$b(\varepsilon) = \varepsilon \sqrt{\frac{1-c^2}{2}} + O(\varepsilon^2), \tag{18.13}$$

linear in the strength as Proposition 18.3 says, with a coefficient that is smaller for a more correlated chain: at $c = 0.85$, $\sqrt{0.2775/2} \approx 0.37$, so a ten per cent tilt of the variances costs a relative score error of about 0.04. Had the truth been $\left(\begin{smallmatrix} 1 & c+\varepsilon \\ c+\varepsilon & 1 \end{smallmatrix}\right)$ instead, the perturbation would lie in the family (it is the chain at correlation $c + \varepsilon$ with $q = 1 - (c+\varepsilon)^2$) and $b \equiv 0$.

18.4 The misspecification mechanisms of E08

E08 (`experiments/e08_misspecification.py`) keeps every truth Gaussian, so that its score is $-\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$ by linear algebra and no grid enters the reference. The four mechanisms, each with a strength whose zero is the well-specified chain at $\rho = 0.85$, $L = 32$:

`global`, rank-one latent β**.** $\operatorname{Cov}(a_i, a_j) = (\rho^{|i-j|} + \beta^2)/(1+\beta^2)$ off the diagonal, unit variances; Gaussian and variance-preserving but not Markov. Strengths $\beta \in \{0, 0.1, 0.25, 0.5, 1\}$.

`longrange`, **long-range precision coupling** γ**.** The precision matrix is the tridiagonal AR(1) precision plus γ times a matrix with entries $-e^{-|i-j|/8}$ for $|i-j| \geq 2$ and a diagonal compensation that keeps it diagonally dominant; the covariance is its inverse rescaled to unit diagonal (`GaussianLongRange`). Strengths $\gamma \in \{0, 0.05, 0.1, 0.2, 0.4\}$.

`ar2`, **a second autoregressive coefficient** ρ_2**.** A stationary AR(2) $a_i = \rho_1 a_{i-1} + \rho_2 a_{i-2} + \varepsilon_i$ with $\rho_1 = \rho(1-\rho_2)$, which keeps the lag-one correlation at ρ (from the first Yule–Walker equation $\gamma_1 = \rho_1/(1-\rho_2)$); the covariance follows from the Yule–Walker recursion $\gamma_k = \rho_1\gamma_{k-1} + \rho_2\gamma_{k-2}$ (`_ar2_cov`). Strengths $\rho_2 \in \{0, 0.1, 0.2, 0.3, 0.4\}$.

`sitevar`, **site-dependent correlation** δ**.** $a_i = \rho_i a_{i-1} + \varepsilon_i$ with $\rho_i = \rho + \delta\cos(2\pi i/L)$ and $\operatorname{Var}\varepsilon_i = 1-\rho_i^2$, so every site keeps unit variance and $\operatorname{Cov}(a_i, a_j) = \prod_{k=i+1}^{j}\rho_k$ (`_sitevar_cov`). The construction needs $|\rho_i| < 1$ at every site, which at $\rho = 0.85$ bounds the amplitude by 0.15; the registered strength 0.2 was invalid by construction and was replaced by 0.14 in a recorded amendment before any number was read (Chapter 20). Strengths $\delta \in \{0, 0.05, 0.1, 0.14\}$.

Each truth is fitted by four arms on identical bundles: `em_bp` (the eight-component mixture kernel, one noised view per sequence, checkpoint on the validation bulk risk); `window` (a radius-2, width-64 window head trained under interface A for 16,000 steps with the ladder $\{500, \ldots, 16000\}$, checkpoint on the validation bulk risk; Chapter 19); `lmmse_ar1` (the LMMSE of the AR(1) covariance with the truth's mean lag-one correlation: the best a Gaussian AR(1) with that correlation can do); and `isotropic`. Two sizes, $M \in \{512, 2048\}$, so that the estimation term and the bias term of Equation (18.9) can be told apart. The predeclared analysis (`tools/make_misspec_numbers.py`) is the mean and standard error of each arm per cell, the paired per-seed ratio window/EM–BP, and the **break-even** strength: the paper reports the two grid points that bracket the sign change of the mean log-ratio and never a fitted crossing, because with the available seeds the crossing is resolved only to the grid. The registry entry declares eight seeds; the script's default is the protocol's seed list, and the paper's macro for the seed count records what was actually run.

18.5 The Edgeworth correction around the covariance twin

18.5.1 Expanding a law around a Gaussian with the same covariance

Let $\mathbf{x} \sim P_t$ have mean zero, covariance $\mathbf{\Sigma}_t$ and fourth-cumulant tensor $\mathbf{K} = \boldsymbol{\kappa}_4[\mathbf{x}] = e^{-4t}\boldsymbol{\kappa}_4[\mathbf{a}]$ (Chapter 10); its third cumulants vanish by the symmetry of the chain. The characteristic function of P_t and of its twin $\mathcal{N}(0, \mathbf{\Sigma}_t)$ have log-characteristic functions that agree to second order and differ at fourth:

$$\log \mathbb{E}e^{i\mathbf{s}\cdot\mathbf{x}} = -\tfrac{1}{2}\mathbf{s}^\top\mathbf{\Sigma}_t\mathbf{s} + \tfrac{1}{24}\mathbf{K}[\mathbf{s},\mathbf{s},\mathbf{s},\mathbf{s}] + \ldots \qquad \text{(cumulant expansion, Chapter 2).} \tag{18.14}$$

Exponentiating and keeping the first correction, $\mathbb{E}e^{i\mathbf{s}\cdot\mathbf{x}} \approx e^{-\frac{1}{2}\mathbf{s}^\top\mathbf{\Sigma}_t\mathbf{s}}(1 + \frac{1}{24}\mathbf{K}[\mathbf{s},\mathbf{s},\mathbf{s},\mathbf{s}])$. Under the inverse Fourier transform a factor (is_j) becomes $-\partial_j$, so $s_i s_j s_k s_l$ becomes $\partial_i\partial_j\partial_k\partial_l$ acting on the Gaussian density $\phi_{\mathbf{\Sigma}_t}$:

$$P_t(\mathbf{x}) \approx \phi_{\mathbf{\Sigma}_t}(\mathbf{x})\Big[1 + \frac{1}{24}\sum_{ijkl} K_{ijkl}\, H^{\mathbf{\Sigma}_t}_{ijkl}(\mathbf{x})\Big], \qquad H^{\mathbf{\Sigma}}_{ijkl}(\mathbf{x}) = \frac{\partial_i\partial_j\partial_k\partial_l\phi_{\mathbf{\Sigma}}(\mathbf{x})}{\phi_{\mathbf{\Sigma}}(\mathbf{x})}. \tag{18.15}$$

This is the **first Edgeworth correction** (also called the Gram–Charlier correction at this order; the two series differ in how they group higher terms, not in this term). The tensors $H^{\mathbf{\Sigma}}_{ijkl}$ are the **multivariate Hermite polynomials** of the Gaussian $\mathcal{N}(0, \mathbf{\Sigma})$. With $\mathbf{y} = \mathbf{\Sigma}^{-1}\mathbf{x}$ one has $\partial_i\phi_{\mathbf{\Sigma}} = -y_i\phi_{\mathbf{\Sigma}}$ and $\partial_i y_j = (\mathbf{\Sigma}^{-1})_{ij}$, and four differentiations give

$$H^{\mathbf{\Sigma}}_{ijkl} = y_i y_j y_k y_l - [y_i y_j(\mathbf{\Sigma}^{-1})_{kl} + \text{five more pairings}] + [(\mathbf{\Sigma}^{-1})_{ij}(\mathbf{\Sigma}^{-1})_{kl} + \text{two more}], \tag{18.16}$$

the exact analogue of $\mathrm{He}_4(x) = x^4 - 6x^2 + 3$ (Chapter 3). Contracting with the symmetric tensor $\mathbf{K}$ collapses the six pairings into one term and the three into one:

$$\epsilon(\mathbf{x}) := \frac{1}{24} \sum_{ijkl} K_{ijkl} H^{\mathbf{\Sigma}_t}_{ijkl}(\mathbf{x}) = \frac{1}{24} \Big[\mathbf{K}[\mathbf{y},\mathbf{y},\mathbf{y},\mathbf{y}] - 6\, \mathbf{y}^\top \mathbf{A} \mathbf{y} + 3\, \langle \mathbf{A}, \mathbf{\Sigma}_t^{-1} \rangle \Big], \tag{18.17}$$

with $A_{ml} = \sum_{jk} K_{mjkl} (\mathbf{\Sigma}_t^{-1})_{jk}$ a symmetric matrix (the partial contraction of the tensor with the precision), which is the formula of the paper's Appendix A5 and of E16.

18.5.2 The corrected score

The score of the corrected density $\phi_{\mathbf{\Sigma}_t}(1+\epsilon)$ is

$$S^{\mathrm{E4}}(\mathbf{x}) = \nabla \log \phi_{\mathbf{\Sigma}_t} + \nabla \log(1+\epsilon) = S_{\mathrm{G}}(\mathbf{x}) + \frac{\nabla \epsilon(\mathbf{x})}{1+\epsilon(\mathbf{x})}, \tag{18.18}$$

with $S_{\mathrm{G}} = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$ the twin's score, and, differentiating Equation (18.17) with $\partial_m y_j = (\mathbf{\Sigma}_t^{-1})_{mj}$,

$$\nabla \epsilon = \frac{1}{6}\, \mathbf{\Sigma}_t^{-1} \mathbf{K}[\,\cdot\,, \mathbf{y}, \mathbf{y}, \mathbf{y}] - \frac{1}{2}\, \mathbf{\Sigma}_t^{-1} \mathbf{A} \mathbf{y} \tag{18.19}$$

(the quartic gives four equal terms, the quadratic two, and the constant none). E16 evaluates both the **log form** Equation (18.18) and the **linear form** $S_{\mathrm{G}} + \nabla\epsilon$, which drops the denominator; the two agree where ϵ is small and differ where it is not.

18.5.3 When the expansion diverges

Equation (18.15) is an asymptotic expansion in the size of the fourth cumulant relative to the covariance. Its correction $\epsilon(\mathbf{x})$ is a quartic in $\mathbf{y}$ with coefficient of order $e^{-4t}\|\boldsymbol{\kappa}_4[\mathbf{a}]\|_F$, so three things go wrong at low noise. The correction is not small on the bulk of the distribution, so the truncated series is not close to P_t; $1+\epsilon$ can be negative, so the corrected "density" is not one and the log form's denominator can change sign (E16 clips it at 0.1); and the quartic always wins in the tail, so far out the correction is large at every t. The expansion is useful exactly where the paper's Theorem 3 is an asymptotic statement, above the crossover, and the paper says so: below a level among the lowest rungs of the schedule (the paper quotes it as a macro) the correction hurts, and pooled over the twelve-level schedule the corrected score is worse than the twin because the lowest levels carry most of the reference energy.

Example 18.5 (The scalar correction, by hand)**.** For $L = 1$ and unit variance, $\mathbf{\Sigma}_t = 1$, $\mathbf{y} = x$, $\mathbf{K} = \mathbf{A} = K$, and Equation (18.17) reads $\epsilon(x) = \frac{K}{24}(x^4 - 6x^2 + 3) = \frac{K}{24}\mathrm{He}_4(x)$, with $\nabla\epsilon = \frac{K}{6}(x^3 - 3x) = \frac{K}{6}\mathrm{He}_3(x)$: the scalar leading term of the paper's Theorem 3, whose γ-second moment $\frac{K^2}{36}\mathbb{E}\mathrm{He}_3^2 = \frac{K^2}{36} \cdot 3! = K^2/6$ is the closure constant. For a unit-variance Laplace law $\boldsymbol{\kappa}_4 = 3$, so $K = 3e^{-4t}$. At $t = 1$, $K = 3e^{-4} \approx 0.0549$ and $\epsilon(x) \approx 0.00229\,\mathrm{He}_4(x)$: at $x = 0$ it is 0.0069, at $x = 2$ ($\mathrm{He}_4 = -5$) it is -0.011, at $x = 3$ ($\mathrm{He}_4 = 30$) it is 0.069, all small, and the correction is a genuine improvement of the twin over the bulk; at $x = 5$ ($\mathrm{He}_4 = 478$) it is 1.09, larger than one, but $x = 5$ is five standard deviations out. At $t = 0.1$, $K = 3e^{-0.4} \approx 2.01$ and $\epsilon(x) \approx 0.0838\,\mathrm{He}_4(x)$: 0.25 at $x = 0$, -0.42 at $x = 2$, 2.5 at $x = 3$. The correction is of order one on the bulk and the series is not an expansion at all there. The dividing line, where $\frac{K}{24}\max_{|x|\le 2}|\mathrm{He}_4| = \frac{5K}{24}$ reaches, say, one tenth, is $K \approx 0.5$, that is $e^{-4t} \approx 1/6$, $t \approx 0.45$; on the chain the relevant size is $e^{-4t}\|\boldsymbol{\kappa}_4[\mathbf{a}]\|_F$ against $\lambda_{\max}(\mathbf{\Sigma}_0)^2$, which puts the line near the crossover t_*, consistent with what E16 measures.

18.5.4 How E16 used it

E16 (`experiments/e16_edgeworth_oracle.py`) takes the closure protocol of E01: 512 chains from the stream `rng_for("e01-closure", family, 0)` for the Laplace, bimodal, uniform and Student-t_5 chains at $\rho = 0.85$, one common Gaussian draw $\mathbf{z}$ reused across the twelve levels and the two extra levels 3.5 and 4, and the tree-exact reference on the $(801, 8)$ grid. Per family and level it computes the twin's score, the linear and the log corrected scores with the *true* $\boldsymbol{\Sigma}_0$ and the *true* tensor $\boldsymbol{\kappa}_4[\mathbf{a}]$ (`theory.chain_kappa4`, an oracle), and reports the relative score error of each against the reference and the fraction of the squared closure gap closed,

$$\text{closed}(t) = 1 - \frac{\sum_j \|S^{\text{E4}}(\mathbf{x}_j) - S(\mathbf{x}_j)\|^2}{\sum_j \|S_{\text{G}}(\mathbf{x}_j) - S(\mathbf{x}_j)\|^2}, \tag{18.20}$$

which is one when the correction is the whole gap, zero when it does nothing, and negative when it hurts. The predeclared reading is the level above which the log form closes more than ninety per cent, the level below which it is negative, and the pooled error of the three estimators over the schedule. The paper quotes the closed fractions at several levels for the Laplace chain and the ninety-per-cent levels for the bimodal and uniform chains (Section 3 and Appendix A5, macros `audEdge...`). Two cautions the registry and the audit attach: on Student-t_5 the reference on the $(801, 8)$ grid is dominated by the truncated prior above $t \approx 3$ (Chapter 12), so the Student-t fractions at those levels measure the truncation and are not used; and the oracle uses the true fourth cumulant, so it bounds what a moment method built on fourth-order statistics could do rather than being one (a version with the moment estimate of Chapter 16 in place of the true tensor would be the competitor, and the audit recommends it).

Where it is used. The paper, Section 3 ("the gap is the fourth-order term"), Section 5 (misspecification; Figure 5) and Appendix A5 (E16, E17). Code: `experiments/e17_misspec_bias.py` (`kl_project`, `pooled_err`), `experiments/e08_misspecification.py` (`_truth`, `_ar2_cov`, `_sitevar_cov`), `src/markov_score/priors.py` (`GaussianAR1PlusGlobal`, `GaussianLongRange`), `experiments/e16_edgeworth_oracle.py`, `src/markov_score/theory.py` (`chain_kappa4`, `score_gap_constant`), `tools/make_misspec_numbers.py`, `tools/make_audit_numbers.py`.

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Part V

The experiments

Chapter 19

The neural baselines

The paper compares its structured estimator with six neural sequence denoisers. This chapter explains each of them from scratch, for a reader who has never built a neural network: what a layer is, what the six architectures compute, how many numbers each one holds, how the noise level enters, what the network is asked to output and why that output is a score, the single training recipe every architecture shares, the two losses (denoising and implicit score matching), the three ways of computing the divergence that the implicit loss needs, and what experiments E03, E08 and E14 do with all of this. Everything here is implemented in one file, `src/markov_score/nn_torch.py`, and the section headings follow it.

19.1 From a map to a score

19.1.1 What a denoiser network is

A **neural network** here is a function $f_\theta : \mathbb{R}^L \times (0,\infty) \to \mathbb{R}^L$ that takes a noised sequence $\mathbf{x}$ and a level t and returns one number per site, built from a fixed sequence of simple operations whose numerical constants, the **parameters** θ, are chosen by minimising a loss on data. The paper's networks are **denoisers**: their output is used to form an estimate of the posterior mean $\langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$, from which Tweedie's identity (Chapter 7) gives an estimate of the score,

$$\hat{S}(\mathbf{x}, t) = -\frac{\mathbf{x} - e^{-t}\hat{m}(\mathbf{x}, t)}{\Delta_t}. \tag{19.1}$$

The paper's estimand compares $\hat{S}$ with the tree-exact score, so the network is judged as a score estimator whatever it was trained to output.

19.1.2 The two output parameterisations

Write $\alpha = e^{-t}$ and $\mathbf{x} = \alpha\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$. Two ways of turning the network's output into $\hat{m}$ are kept, because the diffusion literature uses both and the screen lets each family pick the one it trains better with:

- the **x0 parameterisation**: the output is the mean itself, $\hat{m} = f_\theta(\mathbf{x}, t)$;
- the **eps parameterisation**: the output is an estimate $\hat{\varepsilon}$ of the standardised noise $\mathbf{z}$, and the mean is recovered by inverting the channel, $\hat{m} = (\mathbf{x} - \sqrt{\Delta_t}\,\hat{\varepsilon})/\alpha$.

(`Denoiser.posterior_mean`.) At the optimum the two are equivalent, because $\mathbb{E}[\mathbf{z} \mid \mathbf{x}] = (\mathbf{x} - \alpha\mathbb{E}[\mathbf{a} \mid \mathbf{x}])/\sqrt{\Delta_t}$ is a bijective affine image of $\mathbb{E}[\mathbf{a} \mid \mathbf{x}]$; they differ in how the training loss weights the levels and in the scale of the output the network must produce (order one for eps at every t; order one for x0 too, but with a factor $1/\alpha$ amplifying its errors in the mean at high t). Substituting the eps form into Equation (19.1) gives the identity $\hat{S} = -\hat{\varepsilon}/\sqrt{\Delta_t}$: an eps network is a score network up to the scalar $-1/\sqrt{\Delta_t}$.

19.1.3 The time features

The level enters every architecture through a small vector, the **Fourier features** of $\log t$ (`TimeFeatures`):

$$\tau(t) = \big(\log t,\ \sin(2^k \log t),\ \cos(2^k \log t)\big)_{k=0,1,2,3} \in \mathbb{R}^9. \tag{19.2}$$

Two choices are deliberate. The argument is $\log t$ rather than t because the twelve levels of the schedule are log-spaced on $[0.05, 3]$ (Chapter 20), so in $\log t$ they are equally spaced and the network sees a uniform grid. The sines and cosines at four octaves give the network smooth, non-monotone basis functions of the level, so that a level-dependent gain (the correct denoiser shrinks less at low noise and more at high) can be represented by a linear combination in the first layer. The MLP concatenates $\tau(t)$ once to its input; every other architecture projects it with a linear map to its channel width and adds the result at every site, so that the level modulates every position identically, which is the natural symmetry of a coordinatewise channel.

19.2 The building blocks

Affine layers and activations. An **affine layer** (`nn.Linear`) maps $\mathbf{h} \in \mathbb{R}^n$ to $\mathbf{W}\mathbf{h} + \mathbf{b}$ with a weight matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$ and a bias $\mathbf{b} \in \mathbb{R}^m$: $mn + m$ parameters. An **activation** is a fixed nonlinear scalar function applied entrywise; the code uses tanh and the **GELU**, $x\,\Phi(x)$ with Φ the standard normal distribution function, a smooth relative of the rectifier. Without activations a stack of affine layers is one affine layer, so the activations are what make a network more than a linear map.

One-dimensional convolution. A **1-D convolution** with C input channels, O output channels, kernel width k and dilation d maps a signal $\mathbf{h} \in \mathbb{R}^{C \times L}$ to

$$(\mathbf{W} * \mathbf{h})_{o,i} = b_o + \sum_{c=1}^{C} \sum_{j=0}^{k-1} W_{o,c,j}\, h_{c,\, i+d(j-r)}, \qquad r = \tfrac{k-1}{2}, \tag{19.3}$$

with $h_{c,i} = 0$ for i outside $\{1, \dots, L\}$ (**zero padding**). It has $OCk + O$ parameters, the same weights at every site i (**weight sharing**), and its output at i depends on the inputs at sites $i - dr, \dots, i + dr$ only: its **receptive radius** is dr. Weight sharing is the network's version of the chain's homogeneity (the same kernel at every transition), and the receptive radius is the network's version of locality. The code implements Equation (19.3) as a gather followed by a matrix product (`FastConv1d`), which is algebraically identical to the framework's convolution and was ten to a hundred times faster on the laptop's CPU; the test suite checks the two agree to round-off.

Residual connections and layer normalisation. A **residual connection** replaces a block g by $\mathbf{h} \mapsto \mathbf{h} + g(\mathbf{h})$, so that the block learns a correction to the identity and deep stacks train stably. **Layer normalisation** rescales a vector to zero mean and unit variance across its channels and then applies a learned per-channel gain and shift ($2d$ parameters for width d); the transformer applies it before each block (**pre-LN**).

19.3 The six architectures

19.3.1 The fully connected network (MLP)

The **multilayer perceptron** sees the whole sequence at once, with no structure imposed: the input is the concatenation $(\mathbf{x}, \tau(t)) \in \mathbb{R}^{L+9}$, followed by two hidden affine layers with tanh and an output affine layer to $\mathbb{R}^L$:

$$f(\mathbf{x}, t) = \mathbf{W}_3 \tanh\big(\mathbf{W}_2 \tanh(\mathbf{W}_1(\mathbf{x}, \tau(t)) + \mathbf{b}_1) + \mathbf{b}_2\big) + \mathbf{b}_3. \tag{19.4}$$

Every output depends on every input through dense matrices, there is no weight sharing across sites, and the network is tied to the length it was trained at (`length_agnostic = False`), so it is excluded from the length study. It is the baseline against which weight sharing is measured.

19.3.2 The weight-shared window head

The **window head** is the smallest local architecture: one convolution of width $2r+1$ from one channel to w channels (zero-padded, so the output has the input's length), the level added through a linear projection of $\tau(t)$ to the w channels, then a **pointwise MLP**, that is, two convolutions of width one (affine maps applied at each site separately) with tanh between and after the first:

$$\mathbf{h}_i = \tanh\big(\mathbf{W}_{\text{conv}} * \mathbf{x}\big|_i + \mathbf{W}_\tau \tau(t) + \mathbf{b}\big), \qquad f(\mathbf{x},t)_i = \mathbf{w}_2^\top \tanh(\mathbf{W}_1 \mathbf{h}_i + \mathbf{b}_1) + b_2. \tag{19.5}$$

Its output at site i depends on $x_{i-r},\dots,x_{i+r}$ only: the receptive radius is exactly r, and the module declares it (`receptive_radius = radius`), which is what the coloured-probe divergence of Section 19.6 relies on. It is length-agnostic, since nothing in it refers to L. The screen chose radius 4 and width 64 for E03; E08 uses radius 2 and width 64.

19.3.3 The dilated temporal convolutional network (TCN)

A **temporal convolutional network** stacks convolutions of width three with dilations $d \in \{1,2,4,8\}$, each in a residual block:

$$\mathbf{h}^{(0)} = \mathbf{W}_0 * \mathbf{x} + \mathbf{W}_\tau \tau(t), \qquad \mathbf{h}^{(\ell)} = \mathbf{h}^{(\ell-1)} + \mathbf{P}_\ell\big(\tanh(\mathbf{W}_\ell *_{d_\ell} \mathbf{h}^{(\ell-1)})\big), \qquad f = \mathbf{w}^\top \mathbf{h}^{(4)} + b, \tag{19.6}$$

with $\mathbf{P}_\ell$ a width-one convolution and $*_d$ the dilated convolution Equation (19.3). The convolutions are non-causal (they look both ways), as they should for a smoother. Each block of dilation d widens the receptive radius by d, so the stack sees $1+2+4+8 = 15$ sites on either side: a long range at a cost that grows linearly in the depth rather than in the radius, which is the point of dilation. It is length-agnostic. The screen chose 32 channels.

19.3.4 The bidirectional gated recurrent unit (bi-GRU)

A **recurrent network** processes the sequence site by site, carrying a hidden state $\mathbf{h}_i \in \mathbb{R}^H$ from $i-1$ to i. The **gated recurrent unit** updates it by

$$\begin{aligned} \mathbf{r}_i &= \sigma(\mathbf{W}_r \mathbf{u}_i + \mathbf{U}_r \mathbf{h}_{i-1} + \mathbf{b}_r), \qquad \mathbf{z}_i = \sigma(\mathbf{W}_z \mathbf{u}_i + \mathbf{U}_z \mathbf{h}_{i-1} + \mathbf{b}_z), \\ \tilde{\mathbf{h}}_i &= \tanh\big(\mathbf{W}_n \mathbf{u}_i + \mathbf{r}_i \odot (\mathbf{U}_n \mathbf{h}_{i-1} + \mathbf{b}_n) + \mathbf{b}'_n\big), \qquad \mathbf{h}_i = (1-\mathbf{z}_i) \odot \tilde{\mathbf{h}}_i + \mathbf{z}_i \odot \mathbf{h}_{i-1}, \end{aligned} \tag{19.7}$$

where σ is the logistic function, $\odot$ the entrywise product, and $\mathbf{u}_i = \tanh(\mathbf{W}_{\text{in}}(x_i, \tau(t)) + \mathbf{b}_{\text{in}})$ the per-site input. The **reset gate** $\mathbf{r}_i$ and the **update gate** $\mathbf{z}_i$ let the state be overwritten or kept, which is what allows information to travel far along the chain without vanishing. **Bidirectional** means two such recurrences, one running left to right and one right to left, whose states are concatenated; a per-site head then maps $(\mathbf{h}_i^{\rightarrow}, \mathbf{h}_i^{\leftarrow}, x_i)$ through one hidden layer to the output. The analogy with the forward and backward messages of Chapter 11 is exact in shape and inexact in content: the GRU's state is a learned vector, not a function on the grid. The recurrence has $6H^2 + 6H$ parameters per direction. Global propagation, shared weights, length-agnostic; the screen chose $H = 32$.

19.3.5 The transformer encoder with relative-position bias

The **transformer** replaces recurrence by **self-attention**: every site forms a weighted average of every other site's representation, with weights it computes itself. With $\mathbf{H} \in \mathbb{R}^{L \times d}$ the current

representation (row i for site i), one head computes queries, keys and values $\mathbf{Q} = \mathbf{H}\mathbf{W}_q$, $\mathbf{K} = \mathbf{H}\mathbf{W}_k$, $\mathbf{V} = \mathbf{H}\mathbf{W}_v$ of width $d_h = d/\text{heads}$ and

$$\text{att}_{ij} = \frac{\exp\left(\mathbf{q}_i \cdot \mathbf{k}_j/\sqrt{d_h} + \beta_{c(i-j)}\right)}{\sum_{j'} \exp\left(\mathbf{q}_i \cdot \mathbf{k}_{j'}/\sqrt{d_h} + \beta_{c(i-j')}\right)}, \qquad \text{out}_i = \sum_j \text{att}_{ij}\, \mathbf{v}_j, \tag{19.8}$$

where $c(k) = \max(-D, \min(D, k))$ clips the offset at $D = 32$ and $\beta_{-D}, \ldots, \beta_D$ are $2D+1$ learned numbers per head, the **relative-position bias** (`RelPosAttention`). The heads are concatenated and projected; the block is $\mathbf{H} \leftarrow \mathbf{H} + \text{Attn}(\text{LN}(\mathbf{H}))$ followed by $\mathbf{H} \leftarrow \mathbf{H} + \text{FF}(\text{LN}(\mathbf{H}))$ with a two-layer feed-forward map of inner width $2d$ and GELU, repeated for two layers; a final layer normalisation and a linear map give one output per site. Why this design is *length-agnostic*: the network never sees an absolute position. It sees x_i, the level, and offsets $i - j$ through the bias, so the same weights apply to a sequence of any length. Why it nevertheless fails to extrapolate in the paper's length study is also visible in Equation (19.8): with training length $L = 32$ the offsets range over $-31, \ldots, 31$, so the clipped entries $\beta_{\pm 32}$ never receive a gradient and stay at their zero initialisation, while at test length 256 most pairs have $|i - j| \geq 32$ and use exactly those entries; and the softmax normalises over eight times more keys, which changes every attention weight. The paper attributes the degradation to the positional encoding rather than to attention as such, with references to the length extrapolation literature. The screen chose $d = 32$, four heads, two layers.

19.3.6 The two-level one-dimensional U-Net

A **U-Net** processes the sequence at two resolutions. An encoder of two width-three convolutions with GELU acts at full resolution (c channels); a strided convolution (width four, stride two) halves the length and doubles the channels; a second encoder acts at half resolution; a transposed convolution (the adjoint of a strided convolution) restores the length; the result is concatenated with the full-resolution encoder output (the **skip connection**) and a decoder of two more convolutions and a width-one head produce the output. The level is added after the first encoder. The coarse branch sees a receptive field twice as wide per layer, so the network combines local detail with a wider, cheaper view; the skip connection keeps the fine detail the downsampling would lose. Multiscale, shared weights, length-agnostic (a one-site padding fixes odd lengths); the screen chose $c = 16$.

19.3.7 Parameter counts

The number of parameters of a network is fixed by its architecture and can be counted by hand; the code's `n_params` sums them. For the six screened configurations the counts are 26,016 (MLP), 5,505 (window head), 17,057 (TCN), 15,169 (bi-GRU), 18,057 (transformer) and 13,713 (U-Net); these are the values in the paper's screen table, obtained there from the code and here from the architecture, which is a useful check that the two agree.

Example 19.1 (Two parameter counts, by hand)**.** *The window head* at radius 4, width 64: the first convolution has one input channel, 64 output channels and width $2 \cdot 4 + 1 = 9$, so $64 \cdot 1 \cdot 9 + 64 = 640$ parameters; the level projection is an affine map from 9 to 64, $9 \cdot 64 + 64 = 640$; the pointwise MLP is a width-one convolution from 64 to 64, $64 \cdot 64 + 64 = 4160$, and one from 64 to 1, $64 + 1 = 65$. Total $640 + 640 + 4160 + 65 = 5505$. *The MLP* with two hidden layers of 128 at $L = 32$: the input has $32 + 9 = 41$ entries, so the layers hold $41 \cdot 128 + 128 = 5376$, $128 \cdot 128 + 128 = 16512$ and $128 \cdot 32 + 32 = 4128$ parameters, total 26016. The MLP has five times the parameters of the window head and, at every sample size in the paper's baseline table, several times its error: what it lacks is not capacity but the weight sharing and locality that match the data.

19.4 The one training recipe

Every architecture, at every interface, trains under the same `TrainConfig`:

- **batch** 64: each gradient step uses 64 sequences drawn with replacement from the training set;
- **per-example level**: each sequence in the batch receives its own level t, drawn uniformly from the twelve frozen levels, so a batch mixes levels and the network learns all of them together;
- fresh **corruption** at every step under interface A: $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ is redrawn for every sequence in every batch (`_draw_batch`);
- **Adam**, a first-order optimiser that scales each parameter's step by a running estimate of its gradient's magnitude, with learning rate 10^{-3} or $3 \cdot 10^{-3}$ (the screen chooses);
- **gradient clipping** at norm 1: if the gradient of the loss over all parameters has Euclidean norm above 1 it is rescaled to norm 1 before the step, which prevents a rare large batch from throwing the parameters far;
- a **checkpoint ladder** $\{500, 1000, 2000, 4000, 8000, 16000, 32000\}$: the parameters are copied at each rung and the rung is chosen afterwards (Chapter 17);
- both output parameterisations screened, one run per configuration in the screen (the pre-data amendment of E03 cut three restarts to one for budget), and a seed for the initialisation and the batch stream derived from the cell's tags by `rng_for`, so a run is reproducible from its row;
- one thread per process; the confirmatory runs used the CPU code path operation for operation.

The recipe is what makes a difference between two rows of the baseline table a difference between architectures. The earlier screen of this model, which the audit found had given the long-range architectures one level per step and one restart against the window head's twelve and eight, is the counterexample the recipe was written against.

19.5 The two losses

19.5.1 Denoising score matching (DSM)

Under interface A the network regresses on a clean target:

$$\mathcal{L}_{\mathrm{DSM}}(\theta) = \mathbb{E}\big\| f_\theta(\mathbf{x}, t) - \mathbf{y} \big\|^2, \qquad \mathbf{y} = \begin{cases} \mathbf{z} & \text{(eps)}, \\ \mathbf{a} & \text{(x0)}, \end{cases} \tag{19.9}$$

averaged over the batch, the level and the corruption (`dsm_loss`). Why this learns the score is a two-line argument. For a fixed $(\mathbf{x}, t)$ the squared loss is minimised by the conditional expectation of the target, $f^*(\mathbf{x}, t) = \mathbb{E}[\mathbf{y} \mid \mathbf{x}, t]$, because $\mathbb{E}\|f - \mathbf{y}\|^2 = \mathbb{E}\|f - \mathbb{E}[\mathbf{y} \mid \mathbf{x}]\|^2 + \mathbb{E}\|\mathbf{y} - \mathbb{E}[\mathbf{y} \mid \mathbf{x}]\|^2$ and only the first term depends on f. For the x0 target this is $\mathbb{E}[\mathbf{a} \mid \mathbf{x}]$, the posterior mean, and Equation (19.1) turns it into the score. For the eps target, $\mathbb{E}[\mathbf{z} \mid \mathbf{x}] = (\mathbf{x} - \alpha \mathbb{E}[\mathbf{a} \mid \mathbf{x}])/\sqrt{\Delta_t} = -\sqrt{\Delta_t}\, S(\mathbf{x}, t)$: the optimal eps network is minus $\sqrt{\Delta_t}$ times the score. This is Vincent's connection between denoising and score matching, and Chapter 8 gives it with more care. The clean chain enters only as the regression target; the interface-B variant of E03 and E14 keeps the same loss but with one fixed $(\mathbf{z}, t)$ per chain.

19.5.2 Implicit score matching (ISM)

Under interface C there is no clean target, and the network minimises Hyvärinen's objective on the noised views alone:

$$\mathcal{L}_{\mathrm{ISM}}(\theta) = \mathbb{E}\Big[\tfrac{1}{2}\|\hat{S}_\theta(\mathbf{x},t)\|^2 + \nabla_{\mathbf{x}}\cdot \hat{S}_\theta(\mathbf{x},t)\Big], \qquad \hat{S}_\theta = -\frac{\mathbf{x}-\alpha\hat{m}_\theta}{\Delta_t}, \tag{19.10}$$

where $\hat{m}_\theta$ is formed from the network output by the parameterisation in use; the functions are `ism_loss` and `_score_from_out`. Chapter 17 derives the identity

$$\mathcal{L}_{\mathrm{ISM}}(\theta) = \tfrac{1}{2}\mathbb{E}\|\hat{S}_\theta - S\|^2 + \text{const} \tag{19.11}$$

that makes this a score-matching loss without the score. Its cost is the divergence term, a sum of L partial derivatives of the network's own output with respect to its own input, which must be differentiated again for the gradient step (`create_graph=True`). The batches are drawn from the fixed set of one-view samples (no redrawing, there is nothing to redraw), each with the level it was observed at.

19.6 Three divergence estimators

Let $\mathbf{J}(\mathbf{x}) = \partial\hat{S}/\partial\mathbf{x}$ be the $L \times L$ Jacobian of the score map at $\mathbf{x}$. The divergence is $\operatorname{tr}\mathbf{J} = \sum_i J_{ii}$. Automatic differentiation provides, for a cost comparable to one forward pass, the product $\mathbf{J}^\top\mathbf{v}$ for any vector $\mathbf{v}$ (one backward pass through the network), but not $\mathbf{J}$ itself. The function `divergence` offers three ways to obtain the trace from such products.

Exact. Take $\mathbf{v} = \mathbf{e}_i$ for $i = 1, \dots, L$: the i-th backward pass returns row i of $\mathbf{J}$, whose i-th entry is J_{ii}. Cost: L backward passes per batch, thirty-two at the paper's length, at every training step.

Hutchinson's estimator. Draw one probe $\mathbf{v} \sim \mathcal{N}(0, \mathbf{I})$ and use $\mathbf{v}^\top\mathbf{J}\mathbf{v}$. Since $\mathbb{E}[v_i v_j] = \delta_{ij}$, $\mathbb{E}[\mathbf{v}^\top\mathbf{J}\mathbf{v}] = \sum_{ij} J_{ij}\mathbb{E}[v_i v_j] = \operatorname{tr}\mathbf{J}$: unbiased at the cost of one backward pass. Its variance is $\sum_{i\neq j} J_{ij}^2 + \sum_{ij} J_{ij}J_{ji}$ for a Gaussian probe, of the order of the squared off-diagonal mass of $\mathbf{J}$, which for a dense map at low noise is large; the estimator adds noise to every gradient step, and the audit's record of E14's first screen (Chapter 20) shows what that noise did to a window head trained with it.

Coloured probes for a banded Jacobian. If output i depends only on inputs j with $|i-j| \le r$, then $\mathbf{J}$ is banded with bandwidth r, and $p = 2r+1$ deterministic probes give the trace exactly.

Proposition 19.2 (Coloured trace)**.** *Let $J_{ij} = 0$ whenever $|i-j| > r$, let $p = 2r+1$ and let $\mathbf{v}^{(k)} \in \{0,1\}^L$ have entries $v_j^{(k)} = \mathbf{1}\{j \equiv k \pmod p\}$ for $k = 0, \dots, p-1$. Then $\sum_{k=0}^{p-1} \mathbf{v}^{(k)\top}\mathbf{J}\mathbf{v}^{(k)} = \operatorname{tr}\mathbf{J}$.*

Proof. $\mathbf{v}^{(k)\top}\mathbf{J}\mathbf{v}^{(k)} = \sum_{i\equiv k}\sum_{j\equiv k} J_{ij}$. Two indices in the same residue class differ by a multiple of $p = 2r+1$, so either $i = j$ or $|i-j| \ge p > r$, in which case $J_{ij} = 0$. Hence the sum is $\sum_{i\equiv k} J_{ii}$, and summing over the p classes covers every i once. □

This is the graph-colouring idea of Curtis, Powell and Reid for sparse Jacobians: sites of the same colour do not interact, so one probe per colour suffices. The zero padding at the ends only removes dependencies and never adds one, so the band structure holds at the boundary sites too. For the window head the code's `receptive_radius` supplies r; for a dense map the function refuses the coloured mode, since a transformer's or GRU's output at i depends on every input and the proposition does not apply. Cost: $2r+1$ backward passes, nine at radius 4, against

thirty-two for the exact loop, with no variance. E14 trains the window head with the coloured probes, the transformer with Hutchinson's estimator, and selects every arm's checkpoint with an exact trace (coloured for the head, the L-pass loop for the transformer); the test suite checks that the coloured and exact traces agree to 10^{-5} and that Hutchinson's estimator averages to them.

Example 19.3 (Coloured probes at length five, radius one)**.** With $L = 5$ and $r = 1$, $p = 3$ and the probes are $\mathbf{v}^{(0)} = (1, 0, 0, 1, 0)$, $\mathbf{v}^{(1)} = (0, 1, 0, 0, 1)$, $\mathbf{v}^{(2)} = (0, 0, 1, 0, 0)$. The Jacobian is tridiagonal. Then $\mathbf{v}^{(0)\top}\mathbf{J}\mathbf{v}^{(0)} = J_{11} + J_{14} + J_{41} + J_{44} = J_{11} + J_{44}$ because $|1-4| = 3 > 1$; likewise $\mathbf{v}^{(1)\top}\mathbf{J}\mathbf{v}^{(1)} = J_{22} + J_{55}$ and $\mathbf{v}^{(2)\top}\mathbf{J}\mathbf{v}^{(2)} = J_{33}$; the three sum to the trace, with three backward passes instead of five. At $L = 128$ and $r = 4$ the saving is nine passes against a hundred and twenty-eight, which is what made the interface-C window head affordable in the longer regime of E14.

19.7 What the experiments do with the networks

E03: the screen, the confirmatory table, the matched arm and the length study. Chapter 17 describes the screen and the confirmation. The *matched* part trains the two families that carry the interface-C comparison, the window head and the transformer, by implicit score matching on the one-view groups with the exact divergence for the head and Hutchinson's estimator for the transformer, and selects on the held-out implicit objective; beside each it trains the interface-B arm (denoising loss on the clean chain with one fixed corruption). The *length* part loads each family's selected confirmatory checkpoint at $L = 32$ (saved under `outputs/E03/ckpts/`) and scores it, unchanged, against the tree-exact score of chains of length 64, 128 and 256; the MLP is excluded as not length-agnostic. The paper's Section 5 reports all four parts (the baseline table, the matched paragraph, the length table) and its Appendix A3 the screen.

E08: one window head against the misspecified chain. The misspecification experiment's network arm is a radius-2, width-64 window head trained under interface A for 16,000 steps with learning rate 10^{-3} and the eps parameterisation, checkpoint on the validation bulk risk (Chapter 18). The paper names its configuration in the figure caption and the protocol appendix; it is not the screen's radius-4 winner, and the audit asked that this be said.

E14: the matched comparison across regimes. The network arms train exactly as E03's matched winners: the window head with the coloured-probe divergence at learning rate 10^{-3}, the transformer with Hutchinson's estimator at $3 \cdot 10^{-3}$, eps parameterisation for both, re-screened inside each (ρ, L) regime over E03's candidate grid on two development seeds for 8,000 steps, then confirmed for 32,000; checkpoints selected on the held-out implicit objective with the exact trace. The arms are `em_bp`, `em_bp_evidence`, `ism_one_view`, `dsm_one_corruption`, `lmmse_true` and `isotropic`, each defined in Chapter 17. The device is chosen at run time (`pick_device`), so the same code runs on the cluster's GPUs; state dictionaries are always saved on the CPU so a checkpoint is portable.

E05: the networks as samplers. The generation experiment drives its reverse SDE (Chapter 9) with each arm's score, and its network arms are the selected confirmatory checkpoints of E03, loaded per cell so that the network is paired with the other arms within a seed.
Where it is used. The paper, Section 5 (baselines under one recipe; the comparison at matched information; length extrapolation), Section 1 (the list of architectures), Appendix A3 (the recipe and the screen table) and the misspecification figure. Code: `src/markov_score/nn_torch.py` (all architectures, `TrainConfig`, `train`, `dsm_loss`, `ism_loss`, `divergence`, `ism_objective`, `Denoiser`), `tests/test_nn_torch.py`, `experiments/e03_baselines.py`,

`experiments/e14_regime_baselines.py`, `experiments/e08_misspecification.py`, `experiments/e05_generation.py`, `tools/make_baseline_numbers.py`.

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Chapter 20

The experimental protocol and the provenance pipeline

A number in the paper is a macro. The macro was written by a generator script that read a CSV file. The CSV file was written by an experiment script under a frozen configuration, on a machine whose identity and code revision it recorded, and it was then pinned by a cryptographic hash in a registry entry that had been written before the experiment ran. Two independent checks stand between the CSV and the PDF: the manuscript check regenerates every generated fragment and refuses any typed decimal, and an audit script recomputes every headline number from the CSVs with independent code. This chapter walks that chain link by link, defines every object in it, and explains the rules the project follows when something has to change. It is the chapter that lets a reader who is not the author decide how much to trust a number.

20.1 Two frozen objects

The protocol separates two things that earlier versions of this work had mixed: the *world* the experiments are run in, and the *comparison* the experiments make. Each is one Python dataclass whose fields cannot be mutated after construction; an experiment that needs a different value must say so at the call site, and the difference is then recorded in its provenance.

20.1.1 The world: `FrozenConfig`

`experiments/frozen_config.py` defines `FROZEN`, an instance of `FrozenConfig`. Field by field:

`rho` the correlation of the chain, $\rho = 0.85$; one value everywhere.

`n_sites` the chain length $L = 32$, which is also the ambient dimension.

`innovation` the default innovation family, Laplace (excess kurtosis 3; its third cumulant vanishes, so the fourth is the leading non-Gaussian coordinate).

grid the fields `n_grid` and `half_width`: the quadrature grid, $N_{\mathrm{g}} = 401$ points on $[-A, A] = [-8, 8]$ (Chapter 12); the field's docstring records the boundary-mass statistics that justified it and a warning about quoting a quantile as if it were a bound.

`t_grid` the twelve-level schedule, log-spaced on $[0.05, 3]$ and rounded to four decimals (Example 20.1 below).

`n_seeds` sixteen replicate seeds, with the reason (three seeds once read a falling curve as flat).

archival sizes the fields `sizes` and `efficiency_sizes`: the archival size lists of the earlier comparison; the registered experiments take their sizes from the protocol instead.

archival held-out the fields `n_heldout` and `heldout_seed_offset`: the archival held-out design.

EM stopping the fields `em_max_iters`, `em_loglik_tol` and `em_shape_tol`: EM's archival cap, its likelihood tolerance of 10^{-9} nats per transition (the one the registered experiments still pass to `fit_em`), and the shape tolerance, with a docstring explaining why convergence is judged on the shape and never on ρ alone.

`n_components` the default mixture size $C = 8$.

`ring_*` the fields of an exploratory model (a ring of frames) that the paper does not use.

`base_seed` the seed from which `seeds()` derives the replicate seeds.

Three methods matter: `innovation_variance` returns $q = 1 - \rho^2$, the variance-matching choice that makes every family share $\mathbf{\Sigma}_0$; `narrowed(**kw)` returns a copy with some fields changed, for a deliberate and reported deviation; `as_dict()` is what every provenance record stores. The module also defines `config_hash`, a twelve-character prefix of the SHA-256 digest of the sorted `key=value` lines of the configuration, and `provenance(config)`, which records the configuration, its hash, the frozen hash, a flag `is_frozen` and the dictionary of deviations from `FROZEN`.

Example 20.1 (The twelve-level schedule, by hand)**.** `_log_grid(0.05, 3.0, 12)` returns $t_k = 0.05 \cdot 60^{(k-1)/11}$ for $k = 1, \ldots, 12$, since $3/0.05 = 60$. The ratio between consecutive levels is $60^{1/11} = e^{(\log 60)/11} = e^{4.0943/11} = e^{0.3722} \approx 1.4509$. Multiplying out: 0.0500, 0.0725, 0.1053, 0.1527, 0.2216, 0.3215, 0.4665, 0.6768, 0.9819, 1.4247, 2.0671, 3.0000. These are the levels every table and figure of the paper is pooled over, and the reason the network's time features take $\log t$ as their argument (Chapter 19). The crossover $t_* = \frac{1}{2} \log \lambda_{\max}(\mathbf{\Sigma}_0)$ at $\rho = 0.85$ falls between the ninth and tenth level.

20.1.2 The comparison: `Protocol`

`experiments/protocol.py` defines `PROTOCOL`, version 2, the registered protocol of the cluster runs. Field by field:

`sizes` the training-set sizes $M \in \{32, 128, 512, 2048\}$.

seeds the fields `seeds` and `dev_seeds`: the confirmatory seeds 0–15 and the development seeds 10000–10003 (Chapter 17).

bundle sizes the fields `n_val` and `n_test`: 1024 validation and 2048 test chains.

`bundle_tags` the stream tags `exp31-train`, `exp31-val`, `exp31-test`; the training stream is keyed by `(tag, seed, M)` and the other two by `(tag, seed)`, so that every experiment at the same key sees identical chains (Section 20.1.3).

`estimand` the schedule-pooled relative score error, written out as a formula in the field's value so that it is part of the hash.

regions the fields `headline_region`, `selection_region` and `bulk_margin`: all sites for the headline, the bulk for selection, four sites trimmed at each end.

`capacities` $C \in \{1, 2, 4, 8, 16\}$.

EM arm the fields `em_checkpoints`, `em_init_rho`, `em_init_var` and `one_view`: the EM ladder to 1200, the initialisation $\rho = 0.3$ and innovation variance 0.8, and the one-view flag.

network arms the fields `net_steps`, `net_checkpoints`, `batch_size`, `lrs`, `parameterizations` and `grad_clip`: the network recipe of Chapter 19.

screen the fields `screen_sizes`, `screen_restarts` and `screen_steps`: the screen budget as registered; E03's amendment narrowed it before the screen ran.

`factors` the table of which experiment varies which single factor, with everything else held at the protocol: the **ceteris paribus** design.

`V1` is the laptop protocol of the first runs (ladder to 400, capacities to 8, two development seeds, one restart of 16,000 steps), kept so that a version-1 output can be described exactly and so that `v1_overrides()` can reproduce it; `protocol_hash()` digests the protocol together with the world. A params file that records the protocol version and the resolved settings lets a reader tell which protocol produced a file without remembering anything.

20.1.3 Keyed random streams

Every random draw in the registered experiments comes from `rng_for(*keys)` (`src/markov_score/utils.py`): the keys, prefixed by a project constant, are hashed by a stable function into a NumPy `SeedSequence`, from which a generator is created. The same keys give the same stream in every process and on every machine. This is the **common random numbers** device of simulation: two arms that call `rng_for("exp31-train", seed, M)` see the same training chains, and a difference between them cannot be Monte Carlo noise. The chapter on model selection explains how the tags separate training, validation and test data; the negative-results record explains how a driver that re-created its generator inside a loop defeated the device (Section 20.2.3).

20.2 The registry

20.2.1 One JSON entry per experiment

Every registered experiment has a file `experiments/registry/<id>.json`. Its fields:

identity the fields `id`, `script` and `script_sha256`: the experiment's identifier, its driver under `experiments/`, and the driver's SHA-256 at registration, so a later edit of the script is detectable.

`hypothesis` the statement the experiment tests, written before it runs, with its prediction.

`design` (or `parts` and `settings`) the factor varied, what is held, the sizes, the arms, the estimator, the grid.

`predeclared_analysis` the generator that will read the output and the statistic it will compute, named before the data exist, including what will be reported as censored or bracketed rather than fitted.

`seeds` the development and confirmatory seed blocks.

`status` a string from the vocabulary PRE-REGISTERED, EXPLORATORY, CONFIRMATORY, DONE, WITHDRAWN, with a FROZEN prefix added by the freezing tool and the cluster job identifiers appended as the run progresses.

history the fields `registered` and `amendments`: the registration date and a list of dated amendments, each with what changed and why (Section 20.2.3).

`hpc_runs` the cluster jobs, their deployments and their fates.

free text the fields `motivation`, `limitations` and `outcome`, present where the experiment was registered during the audit or where it returned a result against its hypothesis.

freeze record the fields `frozen_outputs`, `frozen_at`, `frozen_commit` and `frozen_run_commits`, written by the freezing tool (Section 20.5).

`docs/EXPERIMENT_REGISTRY.md` is the human-readable table of the same entries; `tools/registry.py -sync` writes each output's commit back into it, and `-check` fails if any experiment with rows lacks a clean provenance record or if a frozen file changed.

20.2.2 Pre-registration

Pre-registration means that the hypothesis, the design, the seeds and the analysis are written down and committed before the confirmatory data exist. Its purpose is to make the analysis independent of the outcome: a choice made after seeing the data (which levels to fit a slope on, which arm to call "matched", which seeds to report) can be tuned to the result, and a choice made before cannot. The paper's contract states the rule in four lines: confirmatory settings live in the protocol and the registry entry, frozen before the first confirmatory cell runs; development seeds never enter a reported number; confirmatory seeds never enter a selection; a running confirmatory experiment is never changed.

20.2.3 Pre-data amendments

A registration can be wrong before it runs, and the instrument for correcting it is the **pre-data amendment**: a dated entry in the registry's `amendments` list, made before any confirmatory cell has produced a number, stating what changed and why. The registry holds several, and they are worth reading as specimens of the rule:

- E03's screen budget was cut from four development seeds, three restarts and 32,000 steps to two seeds, one run and 16,000 steps after the single-thread cost was measured, with the confirmatory protocol untouched;
- E04's held-out bundle was reduced from 2048 to 512 chains after the finest grid produced no completed cell in nineteen hours, with the estimand (a paired movement between resolutions) unchanged;
- E08's site-varying strength 0.2 was replaced by 0.14 when the construction was found invalid at $\rho = 0.85$ (a site coefficient exceeds one), with the invalid cells recorded as never having existed;
- E12b's analysis plan was changed from "compare the fitted slopes with 4 and 8" to "compare $M^*(t)$ with the exact information curve" after E21 showed that the asymptotic exponents are not the prediction on the swept levels, while the job was still pending in the queue;
- E14 was amended twice pre-data: to record the whole held-out implicit objective curve so that a diverged fit can be called a divergence, and, on 20 September, to fix and guard its data generator.

The last case is the most instructive and is recorded as entry N-22 of `docs/NEGATIVE_RESULTS.md`. E14's first screen gave the window head, at the frozen point and with E03's own configuration, a test risk several times the value the same network had reached in E03's matched arm, with every held-out objective curve rising from the first checkpoint. The first diagnosis blamed the divergence estimator (the screen had used Hutchinson's one probe where E03 used the exact trace); the driver was changed to E03's per-arm estimators and learning rates, and a re-run failed the same way. What settled it was running E03's training function and E14's on the same data: both gave the bad number on E14's data and the good number on E03's, so the cause was the data. In the driver's `_common` the training chains had been drawn by a comprehension that re-created the generator *inside* the loop,

```
prior.sample(rng_for("e14-train", seed, L, rho), L) for _ in range(n),
```

so all 2048 training chains were one and the same draw, and both arms had fitted two thousand copies of a single chain. The fix creates one generator per set and asserts that the chains are

distinct (`assert len(np.unique(A[:, 0])) > 1`); with it E14's pipeline reproduces E03's value at the frozen point. The void screens and confirmations are kept under `outputs/E14_screen_invalid/`, and a search of the current drivers found no other instance of the pattern.

Why is a pre-data amendment the right instrument for such a fix, rather than a silent edit or a fresh registration? Four reasons, in the order the contract gives them. First, the guarantee that pre-registration buys is that the confirmatory design and analysis were not chosen in the light of a confirmatory result. A runs whose data were degenerate produced no confirmatory result about the registered question at all, so amending before any valid cell exists preserves the guarantee exactly; the amendment's timestamp, placed against the job log, is the evidence. Second, the alternative of editing the driver silently would leave the registry describing a run that never happened as registered, and the alternative of a new registration would suggest that the scientific design had changed, when only its implementation had; the amendment says precisely which. Third, the rule that a frozen or running output is never overwritten still applies: the void runs are named, kept and excluded, so a reader can see what was measured, why it is void, and that nothing was quietly replaced. Fourth, the amendment converts the lesson into code, an assertion that a training set has as many distinct chains as it claims, in the same spirit as the typed-literal gate and the dirty-tree refusal: a defect that was caught once by a person is caught next time by the machine. The audit's own summary of the lesson is the sentence to remember: a fitted parameter that contradicts the generating law is a data check, not an estimator result.

20.3 What a run records: the provenance record

Every part of every experiment writes `params_<part>.json` beside its CSVs (`provenance()` in `experiments/common.py`). Its fields:

machine the fields `timestamp`, `hostname`, `platform`, `python`, `numpy`, `cpu_count`, `blas_threads`, `slurm_job_id` and `slurm_array_task_id`: where and when the run happened and with what.

revision the fields `git_commit`, `git_branch` and `git_dirty`: the revision, read from `git` when a repository is present; `git_dirty` is the porcelain status, so a non-empty value means the commit alone does not reproduce the run.

archive the fields `source_archive_sha256`, `revision_source` and `source_is_clean`: on a cluster node without `.git`, the commit and the digest of the exact source archive that was shipped are read from a `REVISION` file written at deployment (Section 20.8); `revision_source` says which of `git`, the file or the environment supplied them. The archive digest exists because a stamp and a separately synchronised tree once disagreed and an output named a commit at which its own command could not have run.

configuration the field `resolved_config_hash`, a digest of the settings the run actually used, overrides included, so two runs of the same commit under different `-set` values are distinguishable.

settings the fields `settings`, `quick`, `workers` and `frozen`: the resolved settings dictionary, whether the run was a smoke test, the worker count, and the world.

With `NGBP_REQUIRE_CLEAN=1` in the environment a run refuses to start unless it can name a commit, carries an archive digest and reports a clean tree. **Dirty** is judged on the *code path* only: `tools/provenance_gate.py` computes the static import closure of the driver and counts an uncommitted file as contaminating only if it lies in that closure, so an edit to an unrelated script does not void a run and an edit to a module the run imported does.

20.4 Resumable cells

Every multi-cell driver runs through `run_cells` in `experiments/common.py`. A **cell** is one unit of work (one seed at one size, say), identified by a key made from its arguments. The worker that computes a cell writes that cell's rows to `<output-dir>/cells/<csv stem>/<key>.json` by an atomic rename, and the driver rebuilds the CSV from the cell files after every completion. Cells whose file exists are skipped, so an interrupted run resumes by reissuing the same command; a cell that raised leaves `<key>.failed` with its traceback and stops the run; duplicate keys are refused. Nothing but a short key travels through the process pool, because a pool that returned multi-kilobyte results once stopped delivering them after fifteen cells while its workers computed on for five hours; the driver detects completion from the files. The design is what allows a cluster job that hit its time limit to be resubmitted into the same output directory.

20.5 Freezing

Definition 20.2 (Frozen output)**.** A registered output is **frozen** when every file of it that the manuscript may read (`*.csv`, `params_*.json`, `winners.json`, `floor_fits.json`) has been hashed with SHA-256 into its registry entry's `frozen_outputs`, together with the time, the repository commit at freezing and the commits recorded in the run's params files.

`tools/freeze.py <id>` does this and refuses a directory whose params record a dirty code path or no commit. `tools/freeze.py -check` recomputes every digest and fails on any byte change; it runs inside the registry check and inside the manuscript check. **SHA-256** is a cryptographic hash: a fixed function from a file's bytes to a 256-bit digest (sixty-four hexadecimal characters) such that no feasible computation produces two files with the same digest; a single changed byte changes the digest unpredictably. Freezing therefore turns "the file the paper cites" into a value that can be checked rather than a name that can be reassigned. The rules that follow from it: a frozen file is immutable; a re-run is a new artefact under a new registry key or a versioned directory, never over the old one; and a generator's source must be frozen before its numbers may enter the manuscript.

20.6 From a CSV to a macro: the generators

Every number, table and figure in the paper is produced by a script under `tools/` that reads frozen outputs and writes either a fragment `paper/sections/<name>.tex` or a figure `paper/figures/<name>.pdf`. A number fragment is a list of `\newcommand` definitions, one per quantity, with a header line saying it was generated. The macro names are built by a fixed encoding: a prefix for the experiment (`cap`, `base`, `learn`, `mis`, `aud`, ...), a quantity, and the digits of any level or strength spelt out with `.` written as `p`, so that the name is a legal LaTeX command; sizes are lettered $32 \to$ A, $128 \to$ B, $512 \to$ C, $2048 \to$ D. For instance the root-mean-square error of $\hat{\rho}$ at level $t = 0.4$ from E06 is the macro `\learnRmseRhoTzeropfour`, and the level itself is emitted as `\learnLevelTzeropfour`, so that the prose never types a level next to a number computed at it. `paper/main.tex` inputs every fragment, and the body files use only the macros.

Generators refuse rather than warn: on a missing input, on a params file with a dirty code path, on an incomplete design (a confirmatory seed missing a capacity), on an unfrozen source, and in one case on a wrong constant (the theory generator refuses to emit if the measured closure ratio for the Laplace chain lies outside a declared window around one). A cell whose narrowest fitted mixture component is under two grid cells is flagged with a dagger and reported, not dropped; the paper's protocol appendix now says so, after the audit found an earlier sentence that said "refused".

20.7 The manuscript check and the independent audit

20.7.1 `tools/check_manuscript.py`

The manuscript check runs five gates and exits non-zero on any failure:

1. *Regenerate and diff.* Every generator (a fixed list plus an optional list of generators keyed by the CSV they need, skipped if that CSV is absent) is run into a temporary tree that links to everything except `paper/sections`, and each emitted fragment is compared line by line with the committed one. A difference means an input or a generator changed without the fragment being rebuilt, or that a fragment was hand-edited; this gate exists because a generated table once shipped with numbers from a superseded run beside a paragraph that carried the frozen ones.

2. *Typed literals.* Every decimal number, every number in exponent notation and every $\times 10^k$ in `main.tex`, the body files and the non-generated sections is matched by a regular expression, skipping lines that define commands, labels, references, citations or inputs; a match must appear in `paper/LITERALS_ALLOWED.txt` with a justification (an analytic constant, a definition of the protocol, a quoted prior-literature value), or the check fails and names the line. This is what makes "every number is a macro" a property of the build rather than a promise.

3. *Unresolved references.* The build log is scanned for undefined references and citations and for multiply defined labels.

4. *Frozen sources.* `freeze.py -check` runs, and any CSV that exists, feeds a generator and is not frozen fails the check.

5. *Overfull boxes.* The paper is built with `tectonic`; any overfull horizontal box in a generated table or a figure file fails, and any box more than ten points too wide anywhere fails (`docs/FIGURE_TABLE_STANDARDS.md`, rules T1 and F4).

20.7.2 `tools/audit_claims.py`

The generators are the only path by which a number reaches the manuscript, which is a strength for provenance and a weakness for auditing: a bug in a generator is invisible to the manuscript check, which only asks whether the macros match what the generator emits. The audit script recomputes the headline quantities a second way, in short explicit blocks of plain `pandas` on the frozen CSVs with no shared helpers, reads the macros directly from `paper/sections/*-numbers.tex`, and compares. A comparison passes when the recomputed value is within a tolerance of 0.0015, or within half a unit of the macro's last printed decimal, whichever is larger; a block that cannot run is itself a finding. Blocks exist for the capacity table, the baselines and the matched arm (with its own percentile bootstrap), the length study, generation, the misspecification brackets, the regime sweep, the grid envelope, the learning experiment's at-cap rates, the phase ladder (including a re-pooling of the per-level rows against the pooled macro) and the sample-complexity slopes with an exactly enumerated bootstrap. The audit of 18 September found no arithmetic disagreement anywhere; every problem it found was in what a number was said to mean, which is what a second computation cannot catch and a second reader can.

20.8 The cluster runs

Deploy. `hpc/icml_deploy.sh` ships a *committed revision*, never the working tree. It refuses a dirty tree (untracked outputs excepted), writes `git archive HEAD` to a tarball, computes its SHA-256, copies it to the login node, verifies the digest on the other side, extracts it under a

directory named by the short commit and a timestamp, and writes a `REVISION` file (commit, archive digest, deployment time, dirty list) at the package root, which `provenance()` reads on nodes without `.git`. It creates the shared virtual environment once, runs an import check, and writes `hpc/.last_deploy.env` with the host, root, source and output paths and the tag, which the submission script sources.

Submit. `hpc/icml_submit.sh <experiment>` maps a name to the driver command with its `-set` overrides (the same strings that appear in the registry entries, so that the amended designs of E06c, E07b, E07c, E11b, E12b and E14 are reproducible from the script), and submits one SLURM job per experiment: one node, one task, the requested CPUs, memory and time; `-propagate=NONE` and an unlimited CPU time limit, because the login shell's limit had once been inherited and had killed the workers of two jobs; single-threaded BLAS so that the workers do not oversubscribe the node; `PYTHONPATH` pointing at the deployment; the `REVISION` file printed into the log; `-workers` set to the job's CPU count for drivers that declare the flag. Dependencies are wired with `afterok`, so that `e03-confirm` waits for `e03-screen` and `e05` for both E02a and E03's confirmation; `all` submits the whole registered set with those dependencies.

Fetch. `hpc/icml_fetch.sh <E> ...` copies the CSV and JSON files (cell files and checkpoints stay on the cluster) into `outputs/<E>/` and prints the next step: `registry.py -check` and `freeze.py <id>`.

20.9 The chain, end to end

For a reader who wants to verify one number, the path from the PDF back to the data is:

1. find the macro in the body file (every quantitative statement is one);
2. find its definition in `paper/sections/<x>-numbers.tex`, whose header names the generator;
3. read the generator under `tools/` to see which CSV columns it reads and what it computes; the registry entry's `predeclared_analysis` says the same in words and was written before the data;
4. confirm the CSV is frozen: its path and digest are in `experiments/registry/<id>.json`, and `python tools/freeze.py -check` agrees;
5. open the CSV's `params_<part>.json` for the commit, the clean-tree flag, the archive digest, the host and the resolved settings, and check that the settings agree with the registry's design and the protocol;
6. run `python tools/check_manuscript.py` to regenerate the fragment and `python tools/audit_claims.py` to recompute the headline from the CSV independently;
7. if the number is a comparison between arms, confirm the rows were joined on the seed, the size and the bundle hash, which is what makes it paired.

None of these steps requires trusting the author; each is a file, a digest or a script that a reader can run. That is the sense in which the paper's numbers are checkable rather than reported.

Where it is used. The paper, Appendix A3 (frozen configuration, estimand, arms and interfaces, provenance) and every generated table and macro. Documents: `docs/EXPERIMENTAL_CONTRACT.md`, `docs/REPRODUCIBILITY.md`, `docs/EXPERIMENT_REGISTRY.md`, `docs/NEGATIVE_RESULTS.md` (N-22), `docs/FIGURE_TABLE_STANDARDS.md`. Code: `experiments/frozen_config.py`, `experiments/protocol.py`, `experiments/common.py` (`provenance`, `run_cells`), `experiments/registry/*.json`, `src/markov_score/utils.py` (`rng_for`),

`tools/freeze.py`, `tools/registry.py`, `tools/provenance_gate.py`, `tools/check_manuscript.py`, `tools/audit_claims.py`, `paper/LITERALS_ALLOWED.txt`, `hpc/icml_deploy.sh`, `hpc/icml_submit.sh`, `hpc/icml_fetch.sh`, `experiments/e14_regime_baselines.py` (the guarded generator).

References. B. A. Nosek, C. R. Ebersole, A. C. DeHaven and D. T. Mellor, 2018, The preregistration revolution, *Proceedings of the National Academy of Sciences* 115, 2600–2606. R. D. Peng, 2011, Reproducible research in computational science, *Science* 334, 1226–1227. G. K. Sandve, A. Nekrutenko, J. Taylor and E. Hovig, 2013, Ten simple rules for reproducible computational research, *PLoS Computational Biology* 9, e1003285. National Institute of Standards and Technology, 2015, *Secure Hash Standard (SHS)*, FIPS PUB 180-4. A. B. Yoo, M. A. Jette and M. Grondona, 2003, SLURM: Simple Linux Utility for Resource Management, in *Job Scheduling Strategies for Parallel Processing*, Springer, 44–60. P. Glasserman, 2004, *Monte Carlo Methods in Financial Engineering*, Springer (common random numbers, chapter 4).

Chapter 21

Reading the figures and tables

This chapter takes every figure and every table of the paper in turn and says what is plotted, what the axes and units are, how to read the panel, which claim it supports, and which registered experiment and generator produced it. The figures are reproduced at text width from the paper's own files; the tables are described rather than reproduced, because their entries are measured numbers that live in the paper's generated fragments and nowhere else (Chapter 20). Figure and table numbers refer to the draft of 20 September 2026 and follow the order of the captions in the source; the file names do not change when the numbering does, so the file name is the reliable handle.

21.1 Conventions shared by every figure

Four conventions recur, and a reader who has them in mind can read any panel of the paper quickly.

The estimand. Unless a caption says otherwise, an "error" is the schedule-pooled relative score error of Chapter 6: squared score error and reference score energy are summed over the twelve levels and over all sites of all test sequences before the square root. It is dimensionless; zero is the exact score; the isotropic score $-\mathbf{x}$ sits near two thirds and the LMMSE floor of the Laplace chain near one sixth (the paper's oracle table gives the values). A *per-level* error is the same ratio at one level.

Log axes. Errors and information are plotted on logarithmic axes because the theory is about exponential rates: a decay e^{-ct} is a straight line of slope $-c$ against a linear t axis and a curve against a logarithmic one; the theory figure uses log–log axes so that both the low-t plateau and the high-t power-law approach are visible. A guide line labelled e^{-4t} is drawn to be compared in slope, not in level.

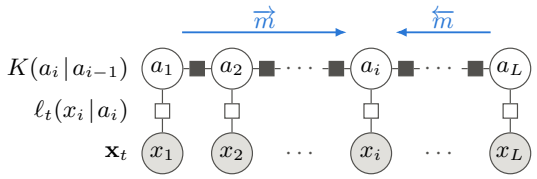
Seeds, bands and intervals. Points are single fits; a line through them is a mean or an RMSE over seeds; a shaded band is the range over replicates; an error bar is a standard error over seeds or a percentile bootstrap interval; a paired ratio is formed within seed before averaging (Chapter 6). Which of these a panel shows is stated in its caption.

Example 21.1 (Reading a rate and a crossover off a panel, by hand)**.** In panel (b) of the theory figure the two guide lines run from $t = 1.2$ to $t = 3$. Over that interval a law e^{-4t} falls by the factor $e^{-4\cdot 1.8} = e^{-7.2} \approx 7.5 \times 10^{-4}$, that is by $7.2/\ln 10 \approx 3.1$ decades, and a law e^{-8t} by 6.3 decades. A measured curve that drops by six decades over the same interval has an average log-slope of $-6 \ln 10/1.8 \approx -7.7$, and one that drops by three decades has -3.8: the two guides are three decades apart in their fall, which is what makes the two coordinates distinguishable by eye.

The crossover tick in panel (a) is at $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$; the Toeplitz bound $\lambda_{\max} \le (1+\rho)/(1-\rho)$ gives, at $\rho = 0.85$, $\lambda_{\max} \le 1.85/0.15 \approx 12.3$ and $t_* \le \frac{1}{2}\ln 12.3 \approx 1.26$, and the finite chain's value is a little lower. Either way the tick falls between the ninth and tenth levels of the schedule (0.98 and 1.42, Chapter 20), which is where the learning figure shows the correlation being lost.

House style. Every figure is drawn by a generator under `tools/` with `tools/figstyle.py`: the text font is the paper's serif, mathematics is set by the math engine, no prose is allowed inside a panel (only axis labels, tick labels, legend entries and short series labels), a marked or excluded datum is a distinct glyph explained in the caption, and a single-column figure is generated at column width and a full-width one at text width. The rules are in `docs/FIGURE_TABLE_STANDARDS.md` and two of them are enforced by the code.

21.2 Figure 1: the mechanism (`fig_mechanism.tex`)

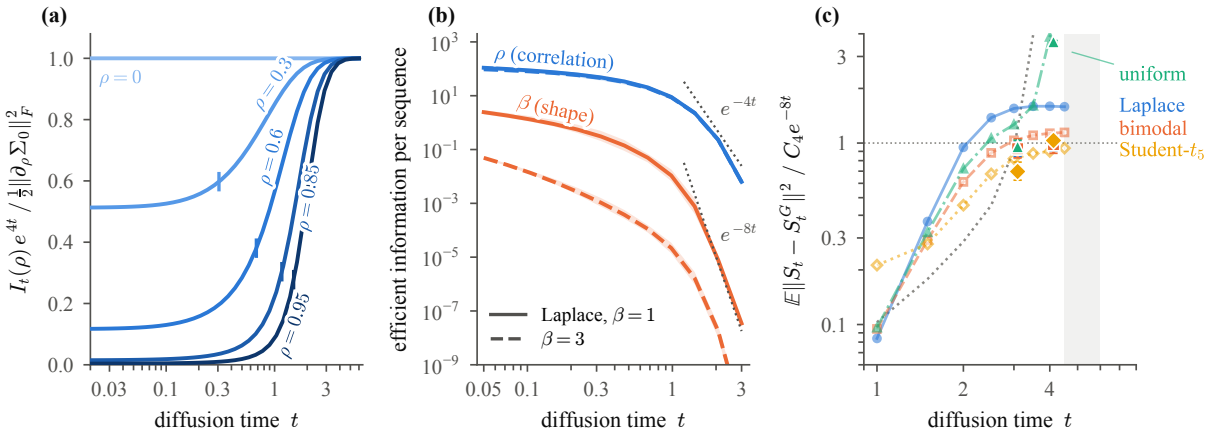


What is drawn. A factor graph (Chapter 11) with three rows: the latent sites $a_1, a_2, \ldots, a_i, \ldots, a_L$ (white circles) joined by the transition factors $K(a_i \mid a_{i-1})$ (filled squares); one unary factor $\ell_t(x_i \mid a_i)$ per site (open squares); and the observed noised coordinates x_i (shaded circles). Two blue arrows are the forward and backward messages $\overrightarrow{m}$ and $\overleftarrow{m}$ meeting at the query site i. The interior is elided and the last site is labelled symbolically, so the picture states the general case.

How to read it. Follow any path between two latent variables: it runs along the top row only. Conditioning on $\mathbf{x}$ attaches the unary factors below each site and adds no edge among the latent variables, so the posterior graph is the prior's chain, a tree, and sum-product on it is exact. Where the two arrows meet, the product $\overrightarrow{m}_i \ell_t \overleftarrow{m}_i$ is the site marginal, whose mean is $\langle a_i \rangle_{x,t}$, and Tweedie's identity (Chapter 7) turns that mean into the score.

Claim and source. The claim is the first sentence of the paper: the score is exact. There is no experiment behind the figure; it is a TikZ drawing whose palette is the same as the plotted figures'.

21.3 Figure 2: what survives diffusion noise (`fig_theory.pdf`)



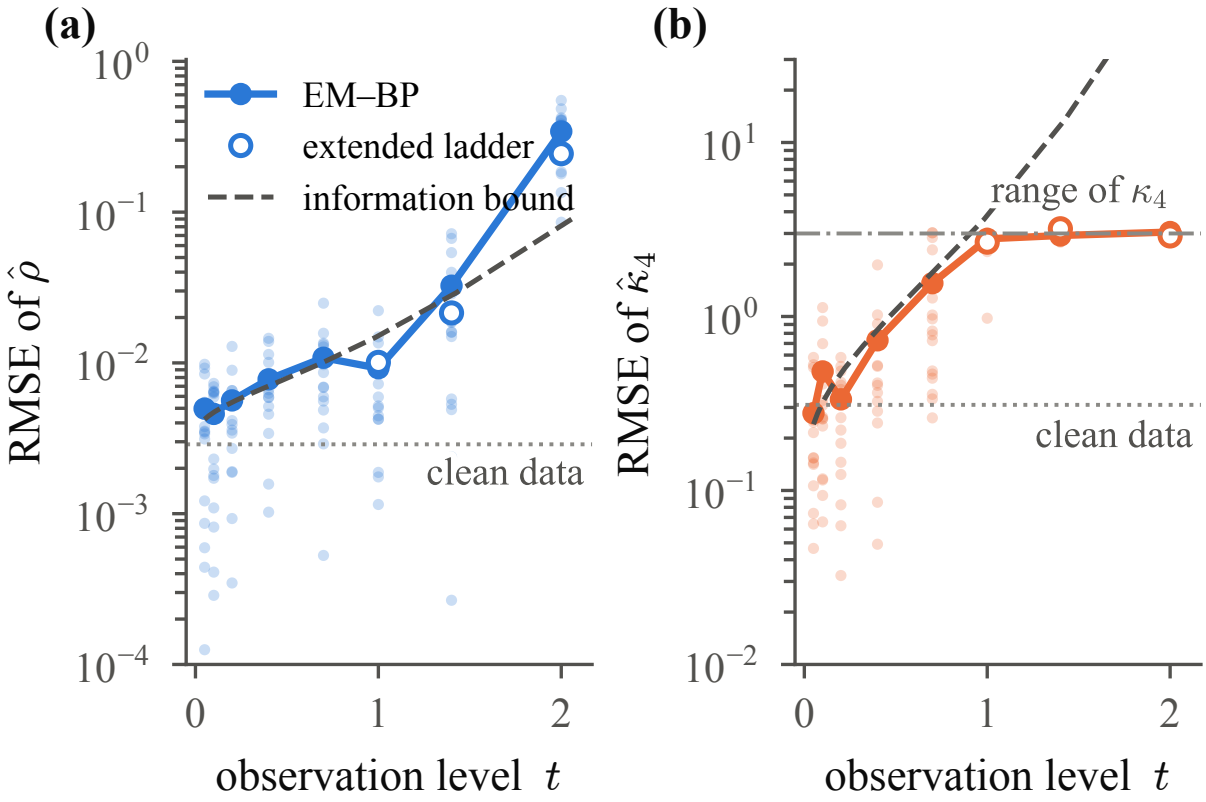
Panel (a). Horizontal axis: diffusion time t, logarithmic. Vertical axis: the exact Fisher information of the Gaussian chain about ρ, divided by its predicted asymptote $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 e^{-4t}$, so that the theorem predicts the value one (dotted line). One curve per correlation, in a blue ramp from $\rho = 0$ to 0.95; the short vertical tick on each curve marks its crossover $t_*(\rho) = \frac{1}{2}\log \lambda_{\max}(\mathbf{\Sigma}_0)$. Reading: every curve rises to one, which is the paper's Theorem 3 for a covariance coordinate, exactly (there is no Monte Carlo in this panel; the information is the closed-form trace of Chapter 5); the more correlated the chain, the later the curve arrives, and the tick shows that the arrival is governed by t_*. At $\rho = 0$ the curve is one at every t. The curves are computed by the `gauss` part of E01.

Panel (b). Both axes logarithmic: t against the efficient Fisher information per noised sequence, for the correlation coordinate ρ (degree two, blue) and the shape coordinate β (degree four, orange) of the generalised-Gaussian chain, at the Laplace member $\beta = 1$ (solid) and at $\beta = 3$ (dashed), estimated from per-sequence scores obtained by Fisher's identity through the tree-exact recursion (Chapter 14); the bands are the range over eight replicates, and the two dotted guides have slopes -4 and -8. Reading: above the crossover the two families of curves run parallel to their guides, three decades apart, which is the order-by-order decay; below it both are shallower, which is the pre-asymptotic regime. The kinked Laplace density reaches the numerical floor of the outer-product estimate near $t = 2.5$, which is why the slopes quoted in the text are fitted on the smooth members. The curves are the `shape` part of E01; the fitted slopes come from `tools/make_theory_numbers.py`.

Panel (c). Both axes logarithmic: t from about one to five against the measured closure gap $\mathbb{E}\|S - S_{\mathrm{G}}\|^2$ divided by its predicted value $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$, so that the theorem predicts one (dotted line). Four innovation families (Laplace, bimodal, uniform, Student-t_5) from E18 on the wide $(3201, 32)$ grid: thin curves with small markers are the ratio from 512 sequences, paired across levels by a common noise draw, so their *shape* is precise while their level carries a common sampling factor; the bold markers with bars at $t = 3$ and 4 are the constant from many more sequences with a chain bootstrap interval, and they are what pins the level. The grey dotted curve is the same Student-t_5 sequences on the operating $(801, 8)$ domain, where the truncated prior drives the reference; the shaded region beyond $t = 4.5$ is where the double-precision floor of the posterior means enters. Reading: every family reaches one within its interval, Student-t_5 included; the departure of the grey curve is a truncation artefact, not the moment condition; the uniform family, with the smallest constant, hits the floor first. Sources: E18 (the `chain` and `constant` parts) and the E01 protocol; drawn by `tools/make_fig_theory.py`.

Claims supported. The rates e^{-4t} and e^{-8t}, the constants $\frac{1}{2}\|\partial_\rho \boldsymbol{\Sigma}_0\|_F^2$ and $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$, the crossover, and the heavy-tailed regime as a statement about expectations rather than samples (Chapter 10).

21.4 Figure 3: learning through noise (`fig_learning.pdf`)

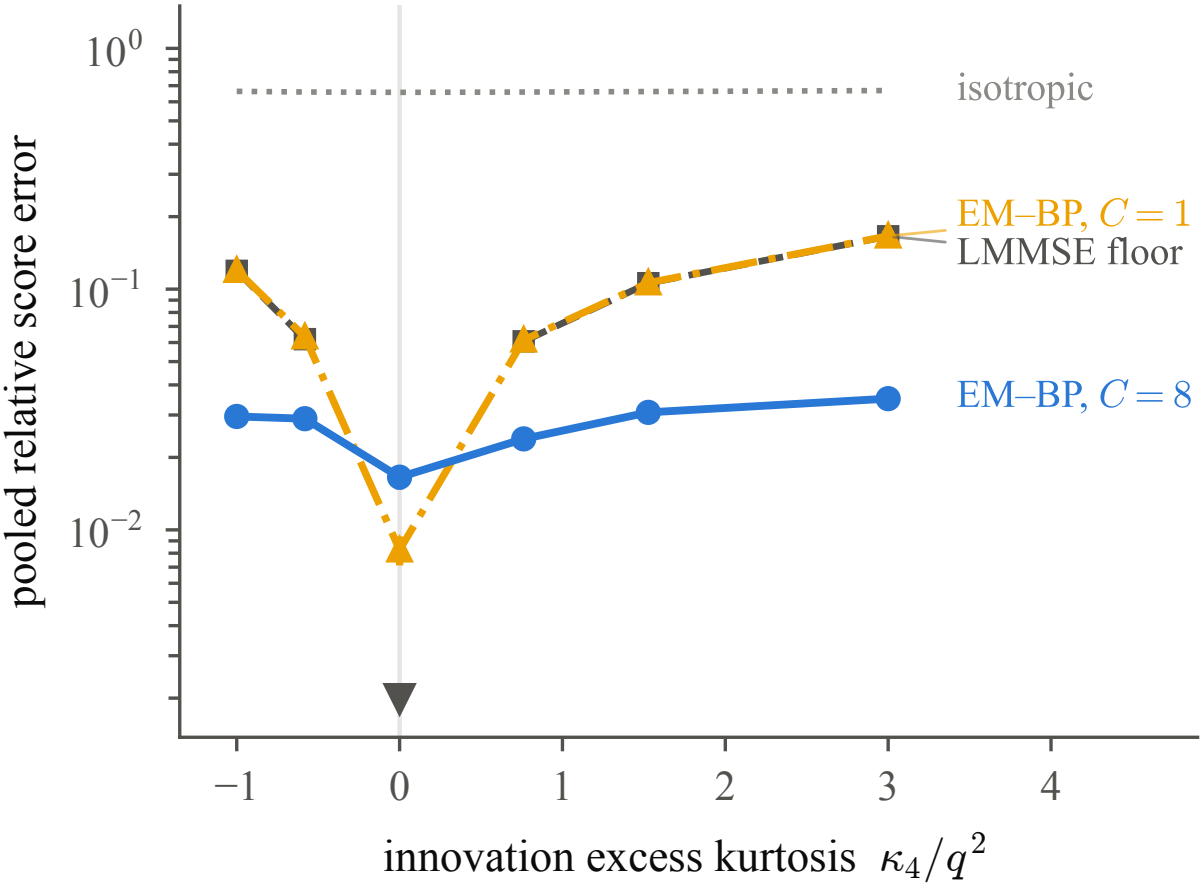


What is plotted. Two panels with a linear horizontal axis, the observation level t at which all $M = 512$ training sequences were noised, and a logarithmic vertical axis. Panel (a): the error of the fitted correlation $\hat{\rho}$; panel (b): the error of the fitted innovation excess kurtosis $\hat{\boldsymbol{\kappa}}_4$. In each, faint points are the sixteen individual fits $|\hat{\theta} - \theta|$ at that level, the solid line is their RMSE, the dashed line is the Cramér–Rao reference $1/\sqrt{MI_t}$ read from the frozen E01 efficient information at the Laplace member (through the chain rule for the kurtosis), the dotted horizontal line is the clean-data fit (the optimiser's own floor at $t = 0$), and in panel (b) the dash-dotted horizontal line at 3 is the full range of the coordinate, the distance from a Gaussian innovation to the truth. In the current draft hollow markers add the same fits with the EM ladder extended (E06c), the budget control (Chapter 17).

How to read it. At the lowest levels the points cluster around the reference in both panels: the estimator is unimodal and at the information reference while the coordinate is identifiable. Moving right in panel (b), the cloud splits into a cluster near the reference and a group of points at the range line: those are fits that collapsed to a Gaussian innovation. Where the dashed reference crosses the range line, the information runs out and every fit collapses; panel (a) shows the correlation surviving to a higher level, and being lost only past the crossover. The vertical distance between the solid line and the dashed line, in decades, is the estimator's inefficiency; where the points sit *below* the reference the estimator is biased (shrunk by early stopping) and the reference, being for an unbiased estimator in a one-parameter family, is a yardstick rather than a bound (Chapter 15).

Claims and sources. The per-level statements of the paper's Section 4 (at the reference while identifiable; collapse where the reference crosses the range; the correlation survives to the crossover; the two highest levels possibly budget-limited). Source: E06 (and E06c), `outputs/E06/levels.csv`; drawn and its macros written by `tools/make_learning_numbers.py`; the collapse fractions and dispersions of the text are from `tools/make_audit_numbers.py`. The moment estimators of E19 (Chapter 16) are quoted in the text as the rate-correct competitor and, in the audit's recommended redesign, would appear as a second pair of curves.

21.5 Figure 4: capacity across the family (`fig_family.pdf`)



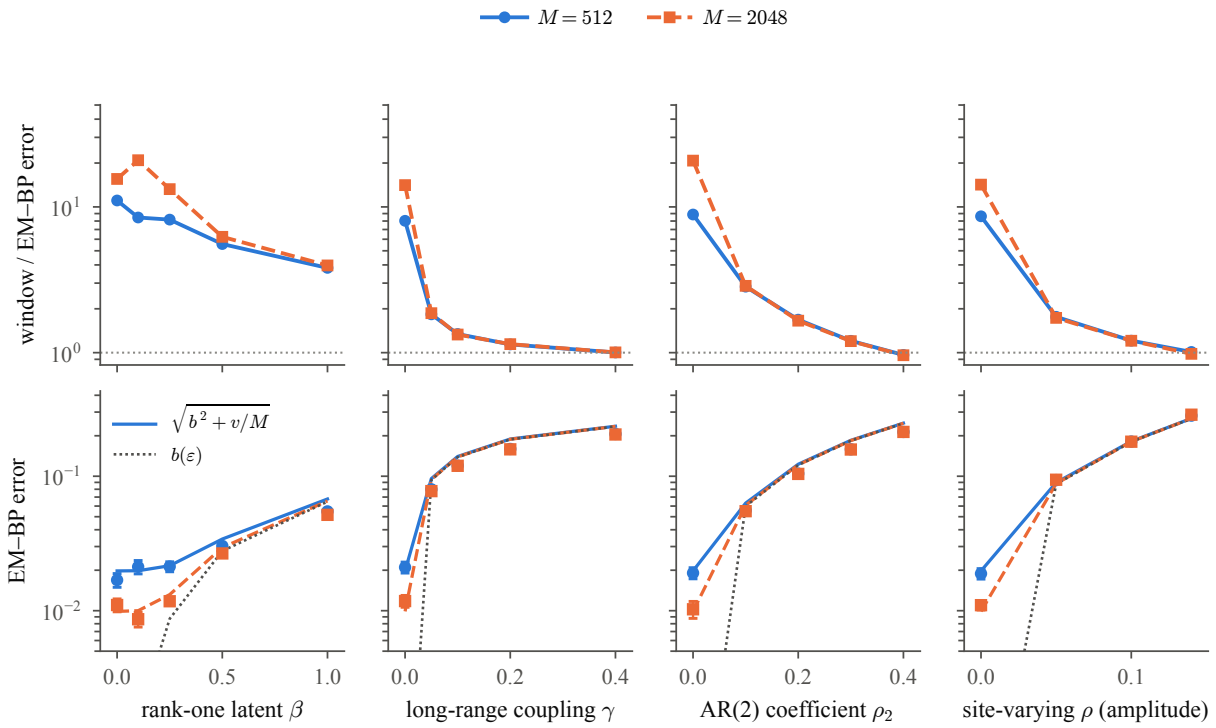
What is plotted. Horizontal axis: the innovation's excess kurtosis κ_4/q^2, from $+3$ at the Laplace member through 0 at the Gaussian member to about -1.1 at the platykurtic end, along the generalised-Gaussian family at fixed ρ and variance; a vertical grid line marks zero. Vertical axis: pooled relative score error, logarithmic. Four curves: the isotropic reference (grey dotted), the LMMSE floor (dark dashed with hollow squares), EM–BP with a single Gaussian innovation $C = 1$ (dash-dotted triangles) and with $C = 8$ (solid circles), the last two with standard-error bars over sixteen seeds at $M = 512$. The floor is exactly zero at the Gaussian member, which a logarithmic axis cannot show, so the dashed line is broken there and a caret at the bottom marks the point.

How to read it. The floor traces a V with its vertex at the Gaussian member: that is the price of second-order closure across the family, largest where the fourth cumulant is largest in either sign. The $C = 1$ curve lies on the V within its bars at every member, which is Proposition 1 read across a family (Chapters 13 and 16). The $C = 8$ curve stays low throughout and meets the

$C = 1$ curve at the Gaussian member, where the gap between them, the paired gain of capacity, is a small loss: with no fourth-order structure to buy, the extra components only add variance.

Claims and sources. Capacity buys exactly what closure discards; the gain is ordered by $\boldsymbol{\kappa}_4$; at the Gaussian member it reverses. Source: E02b, `outputs/E02b/family.csv`; `tools/make_family_numbers.py`.

21.6 Figure 5: where the structure stops helping (`fig_misspec.pdf`)

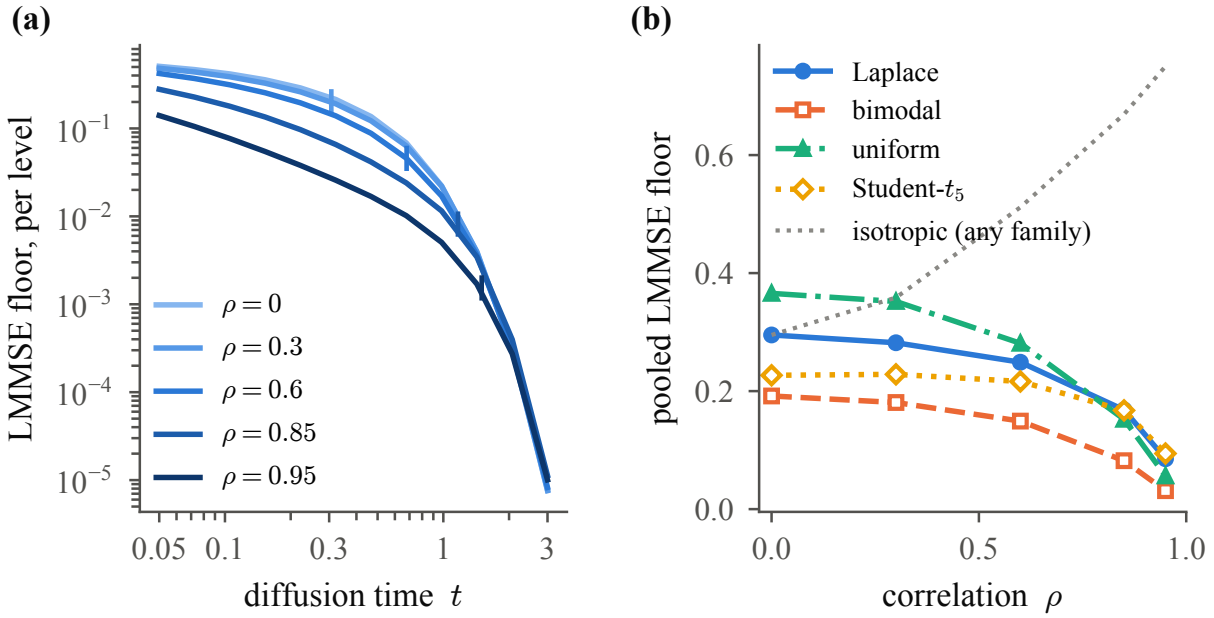


What is plotted. Two rows of four panels, one column per misspecification mechanism of Chapter 18: the rank-one latent (strength β), long-range precision coupling (γ), the AR(2) second coefficient (ρ_2) and the site-varying correlation (amplitude δ). Top row: the ratio of the radius-2 window head's error to EM–BP's, formed within seed and averaged, against the strength, logarithmic vertical axis shared across the row, one curve per sample size ($M = 512$ and 2048; the legend is a single strip above the row); the dotted horizontal line is ratio one. Bottom row: EM–BP's own error at both sizes (points with paired standard errors), the closed-form bias $b(\varepsilon)$ of the Kullback–Leibler projection (dark dotted) and the finite-sample prediction $\sqrt{b(\varepsilon)^2 + v/M}$ for each size (lines), with v read from the well-specified cells.

How to read it. In the top row, a curve above the dotted line means the misspecified chain estimator still beats the network; where it crosses below, the network wins. The crossing is reported as the two bracketing strengths, never as a fitted point. A curve that stays above the line across the whole swept range (the rank-one latent) is a mechanism this estimator absorbs; a curve that reaches the line but does not cross it (long-range coupling, at the largest strength) is a tie. In the bottom row the two sizes separate estimation from bias: at zero strength the points of the two sizes are apart and the larger sample is lower (estimation); at large strength they coincide on the dotted bias curve (bias that no sample size removes). The slope of the bias curve at small strength on these axes is its order: linear for three mechanisms and quadratic for the latent.

Claims and sources. Graceful degradation; mechanism-dependent break-even; the closed-form bias reproducing the large-sample error; the order in the strength explaining which mechanisms bite. Sources: E08 (`outputs/E08/misspecification.csv`) for the points and ratios, E17 (`outputs/E17/bias.csv`) for the curves; `tools/make_misspec_numbers.py`. The network arm is the radius-2, width-64 window head of Chapter 19, not the screen's radius-4 winner.

21.7 Figure 6: the floor across correlation (`fig_phase.pdf`)

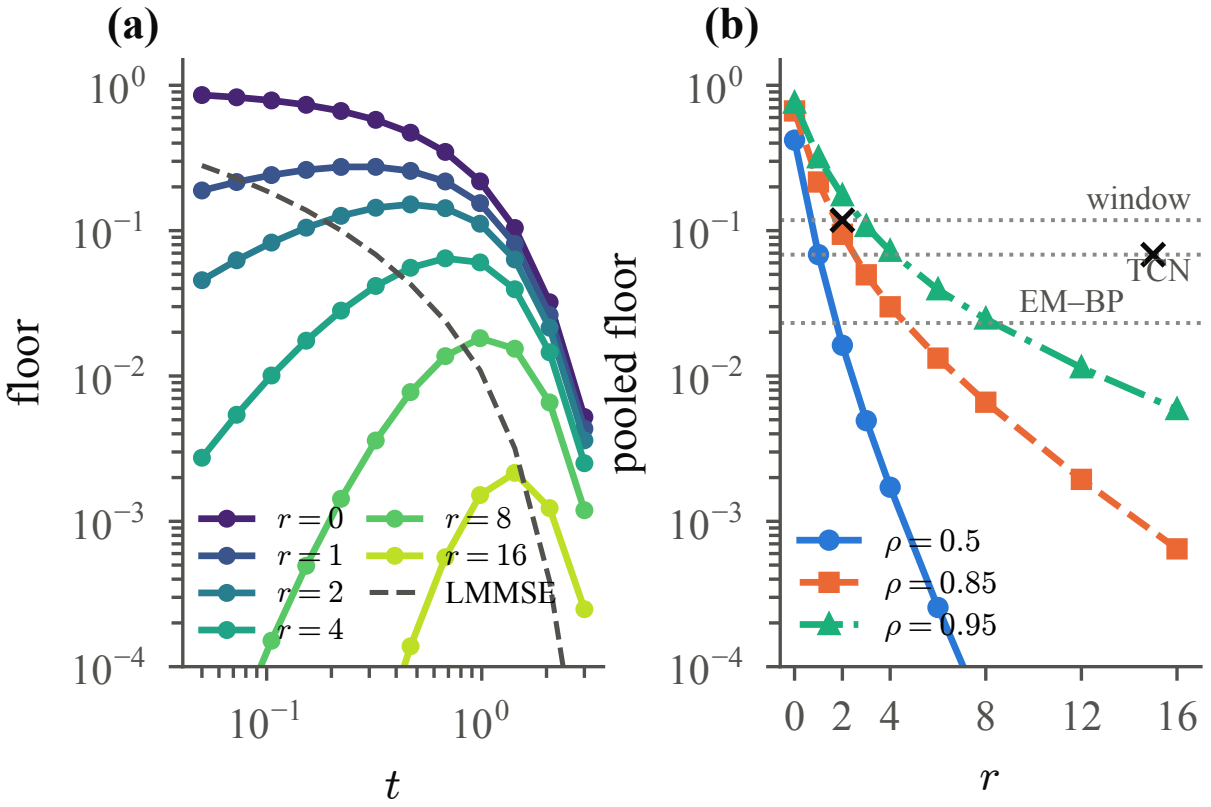


What is plotted. An appendix figure from the deterministic oracle ladder. Panel (a): log–log axes, t against the per-level LMMSE floor on the Laplace chain, one curve per correlation in $\{0, 0.3, 0.6, 0.85, 0.95\}$ (blue ramp), with a tick at $t_*(\rho)$ on each. Panel (b): the schedule-pooled LMMSE floor against ρ (linear horizontal axis) for the four non-Gaussian families, with the isotropic reference as a grey dotted curve (the isotropic floor is the same for every family, so one curve serves). In the current draft the Student-t_5 curve is computed on a wider domain, where its $\rho = 0$ cell is converged (E07b, E07c); earlier drafts ringed that point as excluded.

How to read it. In panel (a) each floor is flat at low t and collapses after its tick: past the crossover the twin's score is nearly the score, at every correlation. In panel (b) the families are ordered by the size of their fourth cumulant at every ρ (Laplace above uniform above bimodal, with the heavy-tailed family where its constant puts it), and every floor falls with ρ while the isotropic reference rises: a strongly correlated chain has a score dominated by its second-order part, so closure costs relatively less and learning nothing costs more.

Claims and sources. The floor is a function of the innovation law and of ρ; the crossover locates its collapse; the ordering by $\boldsymbol{\kappa}_4$. Source: E07 (and E07b, E07c), `outputs/E07/ladder.csv`; `tools/make_fig_phase.py`, which also writes the phase macros.

21.8 Figure 7: locality against its floor (`fig_locality.pdf`)



What is plotted. Panel (a): log–log axes, t against the relative score error of the best estimator that sees only the window $x_{i-r}, \dots, x_{i+r}$ (the exact posterior mean given the window, computed by sum-product on the sub-chain with the exact marginal of the window's first site as initial factor), for radii $r \in \{0, 1, 2, 4, 8, 16\}$ in a colour ramp, on the Laplace chain at $\rho = 0.85$, with the Gaussian twin's full LMMSE as a dashed line for scale. Panel (b): the same floor pooled over the schedule against the radius, one curve per correlation in $\{0.5, 0.85, 0.95\}$; dotted horizontal lines mark the frozen all-site errors at $M = 2048$ of the window head, the dilated TCN and EM–BP, and crosses mark the two networks' radii.

How to read it. In panel (a) the radius-zero curve is the price of using no neighbours at all, and the curves fall quickly with r: at low noise one site is nearly enough, at intermediate levels even a radius of sixteen leaves a small remainder. In panel (b) the floor at the window head's radius is far below the head's own error line: the plateau of the local architecture is not the information ceiling of its radius but a limit of the architecture and its training. The gap between the floor at $\rho = 0.5$ and at 0.95 shows that locality costs more when the chain is more correlated, and that at $\rho = 0.95$ the radius-four floor approaches the LMMSE floor itself.

Claims and sources. The withdrawal of "locality is a ceiling" and its replacement by the floor; the non-Gaussian content of a wider window. Sources: E15 (`outputs/E15/locality.csv`) for the floors, E03 and E02a for the marks; `tools/make_fig_locality.py`. One point to check against the paper: the generator places the window head's cross at radius 2 and the caption calls it the radius-2 head, while the confirmatory head of E03, whose error the mark shows, has radius 4 (`outputs/E03/winners.json`); the text beside the figure compares that error with the radius-4 floor.

21.9 The tables

21.9.1 Table 1: capacity (`tab-capacity.tex`)

Rows: the LMMSE floor and EM–BP with $C \in \{1, 2, 4, 8, 16\}$. Columns: the four sample sizes and a last column, $M = \infty$, which is not a fit but the population reference of E09: the C-component mixture innovation closest in Kullback–Leibler divergence to the Laplace law, scored by tree-exact BP on the same test bundles. Entries are the all-site pooled error averaged over sixteen seeds; standard errors, paired differences and bootstrap intervals are in the macros and the text. A dagger marks a fit whose narrowest component is under two grid cells, reported and not dropped. Reading: the $C = 1$ row equals the floor row to the printed precision at every size, which is the apparatus check of Chapter 16; every $C \geq 2$ row is far below it; the best capacity moves right with the sample size; the ∞ column says what each capacity could express with unlimited data, a reference and not a bound (a Kullback–Leibler-optimal innovation need not minimise the score error). Sources: E02a (`outputs/E02a/capacity.csv`) and E09 (`outputs/E09/floor.csv`); `tools/make_capacity_numbers.py`. The retired reading of the earlier capacity study and its stopping-rule diagnosis are in the paper's appendix on capacity.

21.9.2 Table 2: baselines under one recipe (`tab-baselines.tex`)

Rows: the four sample sizes. Columns: the isotropic reference, the LMMSE floor, the six architectures (MLP, window head, TCN, bi-GRU, transformer, U-Net) under interface A with their screened configurations, and EM–BP with $C = 8$. Entries: all-site pooled error, mean over the sixteen confirmatory seeds, every arm on identical test bundles (the oracles and EM–BP come from the capacity experiment and are joined on the bundle hash). Reading: down a column, how each estimator improves with data; along a row, the ladder isotropic, networks, floor, EM–BP, exact (zero). The fully connected network is far behind the weight-shared ones at every size; the window head is the best network at the smallest size and the worst of the shared five at the largest; several networks cross the floor at $M = 2048$, so the floor is not a barrier for them; EM–BP is ahead everywhere. The paired factors against EM–BP, the at-cap rates and the matched-interface numbers are macros in the text. Source: E03 confirm (`outputs/E03/confirm.csv`); `tools/make_baseline_numbers.py`.

21.9.3 Table 3: length extrapolation (`tab-length.tex`)

Rows: the five length-agnostic networks and EM–BP, all fitted at $L = 32$ and $M = 2048$. Columns: the test length $L \in \{32, 64, 128, 256\}$, each scored against the tree-exact score of chains of that length. Reading: a length-agnostic estimator should improve slightly with length, because longer chains have proportionally fewer boundary sites; EM–BP is flat, since inference at any length is the same exact recursion with the same kernel; the convolutional and recurrent networks improve a little, the window head most (its boundary error is a large share at $L = 32$); the transformer degrades steeply, for the positional-encoding reason of Chapter 19. Sources: E03 length (`outputs/E03/length.csv`) and EM–BP re-evaluated at each length by `tools/eval_em_length.py` (`outputs/E03/em_length.csv`); `tools/make_baseline_numbers.py`.

21.9.4 Table 4: generation (`tab-generation.tex`)

Rows, in two blocks for $M \in \{128, 2048\}$: the reference sample of the true law at $t_{\min}$, then the samplers driven by the exact score, the fitted EM–BP kernel, the LMMSE twin, the isotropic score, and two network checkpoints (bi-GRU, window head) from E03. Columns: $\Delta\ell$, the mean log-likelihood gap in nats per sequence between the generated sample and the reference sample under the true $P_{t_{\min}}$, computed by the exact evidence of the tree-exact recursion (the

exact arm's own gap is the sampler's discretisation floor); $\hat{\boldsymbol{\kappa}}_4$, the excess kurtosis of the innovations of the generated chains; the lag fourth cumulant; and a sliced Wasserstein-2 distance. Every arm integrates the same reverse SDE (Chapter 9) on the same Brownian path within a seed, so the comparison is paired. Reading: the fitted kernel sits close to the exact arm in $\Delta\ell$ and reproduces the innovation kurtosis; the twin generates a Gaussian chain (kurtosis zero by construction) at a much larger gap; the isotropic score is off by tens of nats; the Wasserstein column barely separates the arms, which is why the likelihood gap is the headline. Source: E05 (`outputs/E05/generation.csv`); `tools/make_generation_numbers.py`.

21.9.5 Table 5: the architecture screen (`tab-screen.tex`)

An appendix table. Rows: the six architecture families. Columns: the selected configuration (hyperparameters, learning rate, parameterisation), its parameter count, its pooled bulk risk on the development seeds, and the risk of the family's runner-up. The caption's statement of the screen budget is generated from the screening file itself (number of development seeds, restarts and steps found in it). Reading: the screen ranks configurations within a family and never produces a headline; the parameter counts are those derived by hand in Chapter 19; a family whose winner and runner-up are close was insensitive to the choice. Source: E03 screen (`outputs/E03/screening.csv`); `tools/make_baseline_numbers.py`. A reader comparing this caption with the paper's protocol appendix should note that the two describe the screen budget differently (the appendix text gives the amended two-seed, one-run, 16,000-step budget; the generated caption reports what the frozen screening file contains), and take the generated caption as the description of the file the table was made from.

21.9.6 Table 6: the oracle ladder

An appendix table typed from macros: for each of five innovation families (Laplace, Student-t_5, uniform, bimodal, Gaussian) its excess kurtosis and the pooled relative score error of the isotropic score $-\mathbf{x}$ and of the LMMSE oracle with the true covariance, from 512 chains at the frozen configuration. Reading: the isotropic column is nearly the same for every family (it measures the covariance the estimator has not learned, which every family shares); the LMMSE column is the exact price of second-order closure per family, zero for the Gaussian, and the floor every second-order estimator is bounded by. Source: `tools/make_oracle_numbers.py`, a deterministic computation recorded in `docs/AUDIT_ORACLE_LADDER.md`; its values are cross-checked by the oracle rows of the capacity experiment on the confirmatory bundles.

21.9.7 Generated fragments the current body does not input

The generator suite also emits a regime table (`regime-table.tex`, from E13; the paper quotes its content as macros in the regime paragraph) and several tables of the archival comparison (`tab-efficiency`, `tab-structured`, `tab-aggregation`, `tab-screening`, `tab-hellinger`, `tab-convergence`), which the manuscript check regenerates for consistency but which the current body files do not include. Likewise `paper/figures/` holds figures of the archival study (capacity, screening, sample efficiency, value of structure, the non-Markov study, the grid domain and the toy-model score) drawn by `tools/make_figures.py` from the archival outputs; they are not referenced by the current manuscript and are not discussed here.

Where it is used. Every figure environment and `\input` of a table in `paper/body/*.tex`; the captions quoted above. Generators: `tools/make_fig_theory.py`, `tools/make_learning_numbers.py`, `tools/make_family_numbers.py`, `tools/make_misspec_numbers.py`, `tools/make_fig_phase.py`, `tools/make_fig_locality.py`, `tools/make_capacity_numbers.py`, `tools/make_baseline_numbers.py`, `tools/make_generation_numbers.py`, `tools/make_oracle_numbers.py`,

`tools/make_regime_numbers.py`, `tools/figstyle.py`; `paper/figures/fig_mechanism.tex`; `docs/FIGURE_TABLE_STANDARDS.md`. Experiments: E01, E02a, E02b, E03, E05, E06, E06c, E07, E07b, E07c, E08, E09, E13, E15, E17, E18.

References. E. R. Tufte, 2001, *The Visual Display of Quantitative Information*, second edition, Graphics Press. W. S. Cleveland, 1993, *Visualizing Data*, Hobart Press. J. D. Hunter, 2007, Matplotlib: a 2D graphics environment, *Computing in Science and Engineering* 9, 90–95. T. Tantau, 2023, *The TikZ and PGF Packages*, manual for version 3.1.10.

Appendix A

Glossary

Every technical term used in the paper and in this compendium, in alphabetical order, with one to three sentences and the chapter in which it is defined. Symbols follow the paper's notation (Chapter 1).

Adam A first-order stochastic optimiser that scales each parameter's update by running estimates of the gradient's mean and magnitude; the optimiser of every network arm. Chapter 19.

Affine layer A map $\mathbf{h} \mapsto \mathbf{W}\mathbf{h} + \mathbf{b}$; the basic block of every network, with $mn + m$ parameters for an $m \times n$ weight. Chapter 19.

Amendment, pre-data A dated entry in a registry file recording a change to a registered design or analysis made before any confirmatory cell has produced a number; the instrument by which a registration is corrected without losing its guarantee. Chapter 20.

Amplitude (of a feature) The size of a Hermite moment of degree m, attenuated by the channel as e^{-mt}; the first of the three clocks. Chapter 10.

AR(1) chain A first-order autoregression $a_i = \rho a_{i-1} + \varepsilon_i$ with i.i.d. innovations; the paper's data model, with $a_1 \sim \mathcal{N}(0, 1)$ and $q = 1 - \rho^2$. Chapter 4.

AR(2) A second-order autoregression $a_i = \rho_1 a_{i-1} + \rho_2 a_{i-2} + \varepsilon_i$; one of the four misspecification mechanisms, parameterised by ρ_2 at fixed lag-one correlation. Chapter 18.

Assumption (exponential moment) $\mathbb{E}e^{\lambda|\mathbf{a}|} < \infty$ for some $\lambda > 0$, the hypothesis of the paper's Theorem 3; every family used except Student-t satisfies it. Chapter 10.

Asymptotic relative efficiency The limiting ratio of the Cramér–Rao variance to an estimator's variance; one for an efficient estimator, below one otherwise. Chapter 16.

At cap A cell whose validation-selected checkpoint is the last rung of its ladder, so the reported error is an upper bound on what a longer ladder would give. Chapter 17.

Attention (self-attention) The transformer's operation in which every site forms a softmax-weighted average of every other site's representation with weights it computes from the data. Chapter 19.

Attenuation, Hermite The exact identity $\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)] = e^{-|\nu|t}\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]$, the paper's Theorem 2. Chapter 10.

Audit script `tools/audit_claims.py`, an independent recomputation of every headline number from the frozen CSVs with plain code, compared with the macros. Chapter 20.

Backward message $\overleftarrow{m}_i$, the function of a_i carrying the evidence from sites $i + 1, \ldots, L$; the right-to-left sweep of sum-product. Chapter 11.

Belief (site marginal) The posterior density of one site, $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$; its mean is $\langle a_i \rangle_{x,t}$. Chapter 11.

Belief propagation Sum-product message passing on a factor graph; exact on a tree, hence exact for the paper's posterior. Chapter 11.

Bias (of an estimator) $\mathbb{E}\hat{\theta} - \theta$; in misspecification, the error that remains as $M \to \infty$. Chapters 5 and 18.

Bi-GRU A bidirectional gated recurrent network; one of the six baselines. Chapter 19.

Bimodal innovation The symmetric two-component Gaussian mixture innovation (separation parameter 0.6), excess kurtosis -0.72. Chapter 4.

Bootstrap, percentile Resampling seeds with replacement and reporting the 2.5th and 97.5th percentiles of the resampled statistic; the paper's interval for paired ratios. Chapter 6.

Break-even The misspecification strength at which the paired log-ratio of the network's error to EM–BP's changes sign; reported as the bracketing grid points. Chapter 18.

Bulk region Sites 5 to $L - 4$; the region used for checkpoint selection. Chapter 17.

Bundle A fixed set of chains with their noised views at every level and the reference posterior means under the true kernel; training, validation and test bundles are drawn from disjoint keyed streams. Chapter 17.

Capacity (mixture) The number of components C of the mixture-innovation kernel. Chapter 15.

Cell One unit of work of a driver (one seed at one size, say), written to its own file so that a run is resumable. Chapter 20.

Checkpoint ladder The fixed list of budgets (EM iterations or gradient steps) at which a fit is saved; the selected checkpoint is chosen afterwards among its rungs. Chapter 17.

Clean chain The unnoised sequence $\mathbf{a} \sim P_0$. Chapter 1.

Closure gap $\mathbb{E}_{P_t}\|S - S_{\mathrm{G}}\|^2$, the squared distance between the true score and the covariance twin's; $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2 e^{-8t}$ to leading order. Chapter 10.

Coloured probes The $2r+1$ deterministic probe vectors that give the exact trace of a Jacobian of bandwidth r; the divergence estimator of the window head. Chapter 19.

Common random numbers Keying every random stream by a tag so that compared arms see the same draws and their difference is not Monte Carlo noise. Chapter 20.

Conditional expectation $\mathbb{E}[Y \mid X]$, the best mean-square predictor of Y from X; the posterior mean is the case $Y = \mathbf{a}$, $X = \mathbf{x}_t$. Chapter 1.

Confirmatory seeds Seeds 0–15 (eight in the sweeps), used only for reported numbers, never for selection. Chapter 17.

Consistency Convergence of an estimator to the true (or pseudo-true) parameter as $M \to \infty$. Chapters 5 and 16.

Convolution, one-dimensional A weight-shared linear map of a signal whose output at i depends on inputs within a fixed radius; zero-padded at the ends. Chapter 19.

Correlation coordinate A parameter that moves the covariance $\boldsymbol{\Sigma}_0$, such as ρ; a degree-two coordinate. Chapter 10.

Covariance twin The Gaussian chain with the same ρ and q; its score is $S_{\mathrm{G}} = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$. Chapter 13.

Cramér–Rao bound $\operatorname{Var}\hat{\theta} \geq 1/(MI(\theta))$ for an unbiased estimator; the reference curve of the learning figure. Chapter 5.

Crossover time $t_* = \frac{1}{2}\log\lambda_{\max}(\boldsymbol{\Sigma}_0)$, where e^{2t} passes the largest eigenvalue of the clean covariance; the scale beyond which the asymptotic rates apply. Chapter 10.

Cumulant The coefficients of the log of the moment generating function; additive over independent summands and homogeneous of degree n. Chapter 2.

Cumulant generating function $\log\mathbb{E}e^{s\cdot\mathbf{a}}$, whose Taylor coefficients are the joint cumulants. Chapter 2.

Damped cumulant generating function $\Lambda(s,b) = \log\mathbb{E}_{P_0}\exp(s\cdot\mathbf{a} - b|\mathbf{a}|^2/2)$, the function whose grading gives the paper's Theorem 3. Chapter 10.

Delta method The first-order propagation of an estimator's variance through a smooth function. Chapter 6.

Denoiser A map from a noised sequence and a level to an estimate of the posterior mean; every network arm is one. Chapter 19.

Denoising score matching Regressing the network on the clean chain (or the noise) with fresh corruptions; its minimiser is the posterior mean, hence the score. Chapters 8 and 19.

Development seeds Seeds 10000–10003, used only for architecture and hyperparameter choices. Chapter 17.

Dilation The spacing d between the taps of a convolution; a dilated stack widens its receptive radius by d per layer. Chapter 19.

Divergence $\nabla\cdot s = \sum_i \partial s_i/\partial x_i$, the trace of the score map's Jacobian; the term implicit score matching must estimate. Chapter 19.

Drift (grid) The relative change of the posterior-mean field between two grids on the same chains; the quadrature convergence diagnostic. Chapter 12.

Edgeworth correction The expansion of a law around its covariance-matched Gaussian, whose first term is $\frac{1}{24}\boldsymbol{\kappa}_4$ contracted with the multivariate Hermite polynomials. Chapter 18.

Efficient information The Fisher information of a coordinate after the nuisance coordinates are projected out. Chapter 5.

EM (expectation–maximisation) Alternating an E-step (posterior expectations under the current parameters) and an M-step (maximisation of the expected complete-data log-likelihood). Chapter 14.

EM–BP The paper's structured estimator: EM whose E-step is the tree-exact recursion on the grid and whose kernel is the mixture-innovation AR(1) kernel. Chapter 15.

Eps parameterisation The network predicts the standardised noise $\mathbf{z}$; the mean is $(\mathbf{x} - \sqrt{\Delta_t}\hat{\varepsilon})/e^{-t}$. Chapter 19.

Estimand The quantity a number estimates; the paper's is the schedule-pooled relative score error. Chapter 6.

Evidence (marginal likelihood) $P_t(\mathbf{x})$, the probability of the noised view under the model, returned by the forward sweep with its constants accumulated. Chapter 12.

Evidence rule Selecting EM's rung by the held-out one-view log-evidence per edge; the oracle-free rule of E20 and E14. Chapter 17.

Exceedance count The expected number of sites of a bundle beyond the grid domain, $MLP(|a| > A)$; the right diagnostic for polynomial tails. Chapter 12.

Excess kurtosis $\boldsymbol{\kappa}_4/q^2$, the fourth cumulant divided by the squared variance; +3 for Laplace, 0 for Gaussian. Chapter 2.

Factor graph A bipartite graph of variables and factors representing a product density; the posterior's is the prior's chain. Chapter 11.

Fair screen A screen in which every candidate is trained under the same recipe, budget, bundles, ladder and selection rule on development seeds disjoint from the confirmatory ones. Chapter 17.

Fisher divergence $\frac{1}{2}\mathbb{E}_p\|s_\theta - s_p\|^2$, what score matching minimises. Chapters 8 and 17.

Fisher information $I(\theta) = \mathbb{E}[(\partial_\theta \log P)^2]$; per noised sequence, $I_t(\theta)$. Chapter 5.

Fisher's identity The gradient of the marginal log-likelihood equals the posterior expectation of the complete-data score; on the grid it is read off the pairwise marginals. Chapter 14.

Forward message $\overrightarrow{m}_i$, the function of a_i carrying the evidence from sites $1, \dots, i-1$. Chapter 11.

Forward process The Ornstein–Uhlenbeck noising $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$. Chapter 7.

Fourier features The vector $(\log t, \sin 2^k \log t, \cos 2^k \log t)_{k \le 3}$ by which the level enters every network. Chapter 19.

Frobenius norm $\|\mathbf{A}\|_F^2 = \sum A_{ij}^2$, extended to tensors over ordered index tuples. Chapter 1.

Frozen configuration `FROZEN`, the world every experiment runs in: chain, channel, grid, schedule, seeds. Chapter 20.

Frozen output An output directory whose files are hashed into its registry entry; immutable thereafter. Chapter 20.

Gaussian closure Moment-projected sum-product, which replaces every message by the Gaussian with its mean and variance; equals the LMMSE of the covariance twin. Chapter 13.

Gaussian quasi-likelihood The Gaussian log-likelihood written for data that are not Gaussian; depends on the data only through the sample covariance. Chapter 16.

GELU The activation $x\Phi(x)$, a smooth rectifier. Chapter 19.

Generalised EM An EM whose M-step only increases the expected complete-data log-likelihood (conditional maximisation); its ascent is monotone. Chapter 14.

Generalised Gaussian family Innovations $\propto \exp(-|\varepsilon/s_\beta|^\beta)$ at fixed variance, from Laplace ($\beta = 1$) through Gaussian ($\beta = 2$) to platykurtic members. Chapter 4.

Generator A script under `tools/` that reads frozen outputs and writes the macros, tables and figures of the paper. Chapter 20.

Gradient clipping Rescaling the gradient to a maximum norm before an optimiser step; at 1 in the recipe. Chapter 19.

Grid (quadrature) The N_g uniform points on $[-A, A]$ on which messages are carried with trapezoidal weights. Chapter 12.

GRU Gated recurrent unit; a recurrence with reset and update gates. Chapter 19.

Hájek–Le Cam The local asymptotic minimax theorem: no estimator beats $1/I^{\mathrm{eff}}$ locally, and maximum likelihood attains it. Chapter 5.

Heavy tails Innovation laws with polynomial tails (Student-t), outside the exponential-moment assumption, with an $e^{-\nu t}$ term in the expected gap. Chapter 10.

Held-out Data not used to fit; the validation bundle for selection, the test bundle for scoring. Chapter 17.

Hermite moment $\mathbb{E}[H_\nu(\mathbf{x})]$ with H_ν a product of Hermite polynomials over sites; the quantity the channel attenuates diagonally. Chapter 3.

Hermite polynomial The probabilists' He_n, orthogonal under the standard Gaussian with $\mathbb{E}\mathrm{He}_n^2 = n!$. Chapter 3.

Hutchinson's estimator $\mathbf{v}^\top \mathbf{J}\mathbf{v}$ with a standard normal probe, an unbiased but noisy estimate of $\operatorname{tr}\mathbf{J}$. Chapter 19.

Hyvärinen identity $\mathbb{E}_p[\frac{1}{2}\|s\|^2 + \nabla \cdot s] = \frac{1}{2}\mathbb{E}_p\|s - s_p\|^2 + \text{const}$, by integration by parts. Chapters 8 and 17.

Identifiability Distinct parameters give distinct laws; (ρ, q) is identifiable from $\mathbf{\Sigma}_0$, hence from the noised covariance. Chapter 16.

Implicit score matching Minimising Hyvärinen's objective on noisy samples only; the loss of interface C. Chapters 8 and 19.

Information interface What a learner may see: A (clean chains, fresh corruptions), B (clean chains, one fixed corruption), C (one noised view per sequence and nothing else). Chapter 17.

Innovation The i.i.d. increment ε_i of the chain, with density φ of mean zero and variance q. Chapter 4.

Isotropic score $-\mathbf{x}$, the score of $\mathcal{N}(0, \mathbf{I})$; the reference of a model that learned nothing beyond the unit variance. Chapter 13.

Isserlis' theorem Gaussian moments are sums over pairings of covariances (also Wick's theorem). Chapter 3.

k-statistic Fisher's exactly unbiased sample cumulant. Chapter 16.

Kalman smoother The linear-Gaussian smoother; the Gaussian part of the paper's score. Chapter 13.

Kernel (transition) $K(a' \mid a) = \varphi(a' - \rho a)$; K_θ when parameterised. Chapter 4.

Kullback–Leibler divergence $D_{\mathrm{KL}}(p\|q) = \mathbb{E}_p \log(p/q) \geq 0$, zero only at $p = q$. Chapter 1.

Kullback–Leibler projection The member of a family closest to a law in $D_{\mathrm{KL}}(\text{law}\|\cdot)$; the limit of a misspecified maximum-likelihood estimator. Chapter 18.

Lag-one moment estimator $\hat{\rho} = e^{2t}\,\overline{x_i x_{i+1}}$, unbiased for ρ by Hermite attenuation. Chapter 16.

Laplace innovation The double-exponential density, excess kurtosis 3; the frozen family. Chapter 4.

Layer normalisation Rescaling a representation to zero mean and unit variance across channels with a learned gain and shift. Chapter 19.

Length-agnostic An architecture that refers to no fixed L and can be applied at any length. Chapter 19.

LMMSE The linear minimum-mean-square estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$ of $\mathbf{a}$ from $\mathbf{x}_t$; the best linear map. Chapter 13.

LMMSE floor The relative score error of the LMMSE with the true covariance; the bound of every second-order estimator. Chapter 13.

Locality floor The error of the best estimator that sees only a window of radius r, the exact posterior mean given the window. Chapter 10.

Log-evidence per edge The held-out marginal log-likelihood divided by the number of transitions; the evidence rule's statistic. Chapter 17.

Log-spaced schedule The twelve levels $0.05 \cdot 60^{(k-1)/11}$. Chapter 20.

M-estimator The solution of an estimating equation $\sum_\mu \psi(\mathbf{x}^\mu;\theta) = 0$; its variance is the sandwich. Chapter 16.

M-step The maximisation step of EM; for a Gaussian kernel the belief-weighted Yule–Walker pair. Chapter 14.

Marginal likelihood See evidence. Chapter 14.

Markov chain A sequence in which each site depends on the past only through its predecessor. Chapter 4.

Matched interface Interface C, EM–BP's own; the comparison at matched information. Chapter 17.

Mehler's identity $\mathbb{E}_z \mathrm{He}_n(\alpha a + \sqrt{1-\alpha^2}z) = \alpha^n \mathrm{He}_n(a)$; the proof of Hermite attenuation. Chapter 3.

Method of moments Solving sample moments equal to model moments for the parameter. Chapter 16.

Misspecification Fitting a family that does not contain the truth. Chapter 18.

Mixture-innovation kernel $K_\theta(a' \mid a) = \sum_c \pi_c \mathcal{N}(a' - \rho a - \nu_c, s_c^2)$, the paper's parametric kernel. Chapter 15.

MLP The fully connected network; the baseline with no weight sharing. Chapter 19.

Moment $\mathbb{E}[X^n]$ or its multivariate version. Chapter 2.

Multi-index $\nu \in \mathbb{N}^L$ with $|\nu| = \sum \nu_i$, labelling a product of Hermite polynomials. Chapter 3.

Multivariate Hermite polynomial $H^{\mathbf{\Sigma}}_{ijkl} = \partial_i\partial_j\partial_k\partial_l\phi_{\mathbf{\Sigma}}/\phi_{\mathbf{\Sigma}}$, the tensor version of He_4 for a Gaussian with covariance $\mathbf{\Sigma}$. Chapter 18.

Noised view One draw $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ at one level. Chapter 7.

One-view protocol Each training sequence contributes one noised view at one level, the sequences split evenly across the twelve levels. Chapter 15.

Oracle A quantity that uses the truth (the true kernel's posterior means, the true score); an oracle rule is a selection rule that uses one. Chapter 17.

Oracle selection rule Choosing the rung by the validation risk against the true kernel's posterior means on the bulk. Chapter 17.

Ornstein–Uhlenbeck channel The coordinatewise noising process with decay e^{-t} and variance $\Delta_t = 1 - e^{-2t}$, which keeps the unit variance. Chapter 7.

Outer-product information estimate Estimating Fisher information by the sample covariance of per-sequence scores. Chapter 5.

Overfull box A line LaTeX could not fit in its width; a gate of the manuscript check for tables and figures. Chapter 20.

Paired comparison A difference or ratio formed within seed on identical bundles before averaging over seeds. Chapter 6.

Parameter count The number of trainable numbers of a network, fixed by its architecture. Chapter 19.

Pointwise MLP Affine maps and activations applied at each site separately (width-one convolutions). Chapter 19.

Pooled relative score error $\left(\sum_t \sum_{j,i} (\hat{s} - s)^2 / \sum_t \sum_{j,i} s^2\right)^{1/2}$, the paper's estimand. Chapter 6.

Posterior mean $\langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$; the score is $-(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$. Chapter 7.

Pre-registration Writing hypothesis, design, seeds and analysis before the data exist. Chapter 20.

Provenance record `params_<part>.json`: commit, dirty list, archive digest, host, resolved settings. Chapter 20.

Pseudo-true parameter The minimiser of $D_{\mathrm{KL}}(P^\star \| Q_\theta)$; where a misspecified estimator converges. Chapter 18.

Quadrature, trapezoidal Approximating an integral by a weighted sum on a uniform grid; second order in the spacing for smooth integrands. Chapter 12.

Quasi-MLE The maximiser of a quasi-likelihood; consistent for (ρ, q) here. Chapter 16.

Quasi-score The derivative of the quasi-likelihood; a moment equation in the sample covariance. Chapter 16.

Receptive radius The r such that a network's output at site i depends on inputs within $|j - i| \le r$ only. Chapter 19.

Registry `experiments/registry/<id>.json`: hypothesis, design, analysis, seeds, status, amendments, frozen digests. Chapter 20.

Relative-position bias A learned attention bias depending on the clipped offset $i - j$; what makes the transformer length-agnostic. Chapter 19.

Residual connection $\mathbf{h} \mapsto \mathbf{h} + g(\mathbf{h})$. Chapter 19.

Reverse SDE The time-reversed diffusion driven by the score, used to generate samples. Chapter 9.

RMSE Root mean squared error over seeds of a fitted coordinate. Chapter 6.

Rung One entry of a checkpoint ladder. Chapter 17.

Sample covariance $\mathbf{S} = M^{-1} \sum_\mu \mathbf{x}^\mu \mathbf{x}^{\mu\top}$. Chapter 16.

Sandwich covariance $\mathbf{J}^{-1}\mathbf{V}\mathbf{J}^{-1}$, the asymptotic variance of an M-estimator under a possibly wrong model. Chapter 16.

Schedule The twelve levels over which errors are pooled. Chapter 20.

Score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$. Chapter 7.

Score matching Fitting a score by minimising the Fisher divergence, implicitly or by denoising. Chapter 8.

Screen The selection of a configuration per architecture family on development seeds. Chapter 17.

Seed The integer that determines a random stream; the inferential unit of the paper's comparisons. Chapter 6.

SHA-256 A cryptographic hash producing a 256-bit digest; a changed byte changes the digest. Chapter 20.

Shape coordinate A parameter that leaves $\boldsymbol{\Sigma}_0$ fixed and moves $\boldsymbol{\kappa}_4$; a degree-four coordinate, such as the generalised-Gaussian exponent at fixed variance. Chapter 10.

Site One coordinate i of the chain. Chapter 1.

Sliced Wasserstein distance A distance between samples averaged over random one-dimensional projections; a column of the generation table. Chapter 9.

SLURM The cluster scheduler that runs the registered experiments. Chapter 20.

Sub-chain A window of consecutive sites treated as a chain with the exact marginal of its first site as initial factor; the locality floor's computation. Chapter 10.

Sum-product The message-passing algorithm computing marginals on a factor graph. Chapter 11.

TCN The dilated temporal convolutional network, receptive radius 15. Chapter 19.

Test bundle The 2048 held-out chains on which a selected fit is scored. Chapter 17.

Three clocks Amplitude e^{-mt}, information e^{-2mt}, samples e^{2mt} for a feature first visible at Hermite degree m. Chapter 10.

Time features See Fourier features. Chapter 19.

Transformer The attention-based encoder with relative-position bias. Chapter 19.

Truncation (domain) The error from carrying messages on a finite domain; bounded by the exceedance count, not by the grid spacing. Chapter 12.

Tweedie's identity $S(\mathbf{x}, t) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$. Chapter 7.

Typed literal A decimal typed into the manuscript rather than emitted by a generator; forbidden unless listed with a justification. Chapter 20.

U-Net The two-resolution convolutional network with a skip connection. Chapter 19.

Unary factor $\ell_t(x_i \mid a_i) \propto \exp[-(x_i - e^{-t}a_i)^2/2\Delta_t]$, the likelihood factor attached to each site. Chapter 11.

Uniform innovation The uniform density at unit variance, excess kurtosis -1.2. Chapter 4.

Validation bundle The 1024 held-out chains on which the checkpoint is chosen. Chapter 17.

Validation risk The pooled relative score error on the validation bundle's bulk against the true kernel's posterior means. Chapter 17.

Variance-matched family Innovation laws with the same mean and variance, hence the same $\mathbf{\Sigma}_0$, differing only beyond second order. Chapter 4.

Weight sharing Using the same parameters at every site, as a convolution or a recurrence does. Chapter 19.

Wick product $:x_i x_j x_k:$, the Hermite (orthogonalised) monomial; Wick's theorem gives its Gaussian second moments by pairings. Chapter 3.

Window head One convolution of width $2r + 1$ followed by a pointwise MLP; the smallest local baseline. Chapter 19.

Winner's curse The downward bias of the minimum of several noisy measured risks; the reason selection and reporting use disjoint seeds. Chapter 17.

x0 parameterisation The network predicts the posterior mean directly. Chapter 19.

Yule–Walker equations The linear relations between an autoregression's coefficients and its autocovariances; the Gaussian M-step and the AR(2) covariance. Chapters 4 and 14.

Zero padding Treating inputs outside $\{1, \ldots, L\}$ as zero in a convolution; the source of the window head's boundary error. Chapter 19.